Index Calculus for the Elliptic-Curve Discrete Logarithm:

Theory, Algorithms, and a Measured Frontier

A treatise from a four-month research programme,

July–October 2026

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Abstract

Pollard rho remains the fastest known algorithm for the discrete logarithm on elliptic curves of cryptographic shape. Index calculus could in principle exploit more structure, but it has never beaten rho there. This treatise collects everything a four-month research programme (July–October 2026) learned about why. Every number is drawn from committed evidence and carries its boundary: oracle stage, complete solve, cold one-target cost, or online cost.

Theory. We give the Galois group of Semaev's summation cover. It is $(\mathbb{Z}/2)^{m-2}$ for every curve, field and characteristic, with arithmetic monodromy equal to geometric. We also give an exact factorisation law for its fibres over $\mathbb{F}_q$. Two consequences follow:

- fibre statistics are constant on the factor-base locus;
- at fixed factor-base size the mean decomposition yield is fixed.

So levers that rely on curve-dependent monodromy or on raising mean yield at matched base size are closed without experiment. Coverage, oracle cost and solving difficulty remain free.

On fall degrees, we prove for the binary Weil descent of S_3 that:

- the first fall degree is 2 for every n, including Semaev's chained systems, where 4 was stated;
- the last fall degree is at most $k+2$, or $(t-1)k+t$ for chains.

We also certify that the last fall degree is at least 5 on named instances at $n = 33$–45 and in the $m = t = 3$ chain, with certificates that an independent program checks. Every F4 run on those systems therefore exceeds degree 4, and Semaev's Assumption 1 fails as stated. For an explicit-membership chain we prove $(d_{\mathrm{ff}}, d_F) = (4, 5)$ exactly at $B = 3$ for every $p > 3$, and $\max(5, B+1) \le d_F \le 3B-4$ in general. For PKM Kummer towers we prove a constant first fall degree of 4 and show that the last fall degree depends on the presentation, not the ideal. Throughout, the first fall degree is a constant of the presentation and the last is not.

Algorithms. The programme engineered:

- root-finding and meet-in-the-middle decomposition oracles, two to five orders of magnitude cheaper than Gröbner solves where they apply;
- inherited and gated F4 engines, and an $\mathbb{F}_p$ F4 at msolve parity;
- compact Frobenius-orbit and known-logarithm-closed factor bases;
- FGHR's D_4 symmetry at $k = 4$, cutting one decomposition 370–481×;
- double-large-prime genus-3 index calculus after the cubic GHS transfer, from $q^{1.91}$ to $q^{1.36}$;
- Diem's plane-quartic index calculus on the same family, at exponent ≈ 1.

Frontier. A priced campaign of 4,848 executions found no index calculus below rho on any prime or prime-degree binary curve at any weight of a pricing band applied to every arm. The distance at a matched reference is 4.7–18× on Koblitz curves of 37–47 bits and 10.5–102× on prime curves of 20–40 bits. Only in weak extension-field families does index calculus reach or cross rho:

- genus 3 via GHS: below the strongest rho at one size, unresolved at the others;
- plane quartics: 0.005× a negation rho at $q \approx 2^{21}$;
- Joux–Vitse covers: 1.6–2.0× with Nagao decompositions, and 0.0065× a matched negation rho with the sieve route (validated, $p = 1511$);
- $k = 4$ in characteristic two: unsupported beyond $m = 15$.

Every online win was retired by an equal-advice audit against Bernstein–Lange rho. We record the corrections that matching references, pricing native work and replaying records forced. Nothing here concerns a deployed curve or key recovery.

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Introduction

For a subgroup $\langle G \rangle$ of prime order r on an elliptic curve $E/\mathbb{F}_q$, Pollard rho [34, 40] computes discrete logarithms in about $\sqrt{\pi r/(2A)}$ group operations, where A is the number of efficiently computable automorphisms [19, 44]. No algorithm is known to beat it on curves of cryptographic shape.

Index calculus breaks finite-field discrete logarithms in subexponential time. It reached elliptic curves through Semaev's summation polynomials [35] and the extension-field analyses of Gaudry [20] and Diem [10]. It is the method that should exploit the most structure. The question is *where it crosses rho*: on which curves, at which sizes, under which accounting.

This treatise records what one research programme learned about that question between July and October 2026. The programme spanned three repositories (`crypto`, `crypto-autoresearcher`, `cryptanalysis`), several thousand commits and several hundred preregistered studies. Two companion documents organise the same work differently:

- a programme record [3], chronological and exhaustive, with study catalogues;
- an improvement ledger [2], organised as a ratchet of incumbents.

This document organises it as a treatise:

- the theory that constrains index calculus on elliptic curves;
- the algorithms engineered against those constraints;
- the measured frontier against a fairly priced rho;
- what remains open.

It runs no new experiment. It re-verified one certificate (§6.3), adds no improvement of its own, and cites the committed artefact behind every number.

The argument in brief

1. Where the cost lives (Part I). An index calculus pays for factor-base set-up, relation collection, linear algebra and the target's descent. Relation collection costs (trials per relation) × (cost of one decomposition). Chapter 2 writes this out and maps every lever the programme pulled to the term it moves.

2. What is fixed for every curve (Chapter 5). The Galois group of the summation cover is $(\mathbb{Z}/2)^{m-2}$ for every curve, field and characteristic. Its fibres over $\mathbb{F}_q$ split by an exact law, and they split completely on the factor-base locus. The mean decomposition yield at fixed base size is a double count. Three consequences follow:

- no curve has anomalous monodromy to exploit;
- no fibre census can find an anomalous relation rate;
- no factor-base geometry can raise mean yield.

The design space that remains is the cost of the oracle, the algebraic difficulty of the decomposition system, the linear algebra, and where the problem lives.

3. How hard is the decomposition system (Chapter 6)? Every subexponential claim for prime-degree binary curves assumes that the solving degree stays near the first fall degree. In three presentations we prove that the first fall degree is a constant of the presentation, and show that the last fall degree, which equals the solving degree for unsatisfiable systems, is not bounded by it:

- it is proved to differ for an explicit-membership chain ($(4, 5)$ exactly at $B = 3$);
- it is certified to differ, by independently checked closure certificates, for binary Weil descent (≥ 5 against 2 at $n = 33$–45);
- it is measured to differ for prime-field Kummer towers (6 against 4).

A proof that the last fall degree grows without bound remains the open problem, as Huang, Kosters and Yeo [23] and Kousidis and Wiemers [29] left it.

4. What engineering bought (Part III). Root-finding oracles made the decomposition step two to five orders of magnitude cheaper where they apply. Gröbner engines improved one to two orders in counted work. Factor bases closed under maps of known logarithm cut linear algebra by constant factors.

On the cold one-target frontier none of this changed the conclusion: the stages these changes leave alone (relation yield per probe, table set-up) now dominate.

5. Where index calculus wins (Chapter 25 and Chapter 26). Only in weak extension-field families does the measured cost cross or approach a fairly priced rho:

- $E_0/\mathbb{F}_{q^3}$ through a genus-3 GHS cover, by double-large-prime hyperelliptic index calculus or by Diem's plane-quartic index calculus;
- the Joux–Vitse cover of $E(\mathbb{F}_{p^6})$;
- possibly $k = 4$ over $\mathbb{F}_{(2^m)^4}$.

These are known theoretical results. The contribution here is matched, priced, verified measurement. None of these families is deployed.

6. What the measurements say (Part V). At a matched reference, rho is cheaper on every prime and prime-degree binary curve measured: 4.7–18× on Koblitz curves of 37–47 bits, and 10.5–102× on prime curves of 20–40 bits, with the gap widening as $r^{1/6}$. Every apparent online win disappeared against Bernstein–Lange rho with equal advice [4]. Matching the reference, pricing native work and replaying records each moved recorded gaps upward. Chapter 33 lists every such correction.

Conventions

- *Unit.* Costs are group-addition equivalents per $\sqrt{r}$ (S) unless a family uses its own counted unit, such as $\mathbb{F}_q$ multiplications.
- *Ratios.* "$X\times$" is baseline over candidate when it describes an improvement, and IC over rho when it describes a distance. Each sentence says which.
- *Intervals.* Intervals are 95%, as the source recorded them.
- *Change classes.* Every change is an ADVANCE (an exponent or crossover moves under complete accounting), ENGINEERING (a constant moves), RELABELLING or ACCOUNTING. Against rho, the programme produced no ADVANCE on any prime-order curve of cryptographic shape.
- *Paths.* Artefact paths are relative to the `crypto` repository, unless prefixed `crypto-autoresearcher:` or `cryptanalysis:`.

What this treatise does not claim. No speed-up on a deployed curve, no key recovery, no asymptotic sub-rho algorithm for prime fields, and no statement about the security of ECC2K-130, P-256 or secp256k1. The programme's GPU Pollard-rho work on ECC2K-130 is published separately (IACR ePrint 2026/2469) and is not repeated here.

Part I

Foundations

1 Background

Rho and its floor. With A efficiently computable automorphisms acting on $\langle G\rangle$, a rho walk on orbit representatives has expected cost $\sqrt{\pi r/(2A)}$ group additions. $A = 2$ (negation) on every curve; $A = 2n$ on a Koblitz curve $y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, where the Frobenius $\pi(x, y) = (x^2, y^2)$ acts on $\langle G\rangle$ as a scalar λ [19, 44]. ECC2K-130 is the case $a = 0$, $n = 131$ of this family; its record GPU rho attempt is a separate paper in this repository (`paper/ecc2k130/`).

Index calculus on elliptic curves. Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$. A *relation* is a decomposition $aG + bQ = R = P_1 + \cdots + P_m$ with $P_i \in \mathcal{F}$; once about $|\mathcal{F}|$ relations are found, linear algebra over $\mathbb{Z}/r$ yields $\log_G Q$. The pipeline has five stages (factor base, decomposition oracle, relation yield, rank/linear algebra, verification) and a sixth, the comparison with rho on the same target. The *decomposition oracle* decides whether R splits over $\mathcal{F}$. Semaev [35] reduced it to the roots of the summation polynomial $S_{m+1}(x_1, \ldots, x_m, x_R)$, which vanishes exactly when $\pm P_1 \pm \cdots \pm P_m = R$ for points with those abscissae. Over $\mathbb{F}_{2^n}$ or $\mathbb{F}_{p^k}$ the factor base is chosen as the points whose abscissa lies in a subspace or subfield, and Weil restriction turns $S_{m+1} = 0$ into a polynomial system over the base field; the textbook oracle solves that system with a Gröbner-basis engine (F4/F5, [12, 13]) or a SAT solver [39]. Gaudry [20] and Diem [10] give the asymptotics over extension fields; for prime fields no factor base is known that makes the method competitive, and summation-polynomial systems over $\mathbb{F}_p$ are the subject of the Kummer-tower and rational-map bases of Petit–Kosters–Messeng [33].

Frobenius-invariant bases. Galbraith, Granger, Merz and Petit [18] observe that a factor base with $\pi(\mathcal{F}) = \mathcal{F}$ splits into Frobenius orbits of size dividing n, with $\log \pi^k(P) = \lambda^k \log P$. The unknowns collapse by $\approx n$, the matrix by $\approx n^2$, and ordered decompositions save $m!$—more than rho's $\sqrt{2n}$, but from a base too far behind to overtake it at deployed sizes. Invariant bases come from linearised polynomials attached to the factors of $x^n - 1$ over $\mathbb{F}_2$; the subspace dimension is $\ell = \mathrm{ord}_n(2)$, so the base size is fixed by n (Table 1 of `RESEARCH_KOBLITZ_INDEX_CALCULUS.md`).

Symmetries of the system. The decomposition system is invariant under the symmetric group S_m; rational 2-torsion adds sign changes, and Faugère, Gaudry, Huot and Renault [14] show that in the coordinate Y with $Y(P + T) = -Y(P)$ the invariance group becomes the Weyl group D_m, with a much smaller quotient. We measure this at $m = 4$ in §21.10.

2 Where the cost of index calculus lives

An improvement is only useful if it moves a term that matters. This chapter writes the cost of one index-calculus solve as a sum of terms, and maps each lever of the programme to the term it acts on.

2.1 The pipeline

Fix a factor base $\mathcal{F} \subset E(\mathbb{F}_q)$ of size N. A *relation* is a decomposition $aG + bQ = P_1 + \cdots + P_m$ with every $P_i \in \pm\mathcal{F}$. Once about N independent relations are known, linear algebra over $\mathbb{Z}/r$ yields $\log_G Q$. A heuristic count is enough to place the levers. A uniformly random point decomposes into m base points with probability about $N^m/(m!\,r)$. Collecting N relations therefore needs about $m!\,r/N^{m-1}$ trials. The cold cost of one target is then

$$T \;\approx\; \underbrace{T_{\mathrm{setup}}(N)}_{\text{base, tables}} \;+\; \underbrace{\frac{m!\,r}{N^{m-1}}\,\frac{1}{\rho_{\mathrm{yield}}}\,C_{\mathrm{dec}}(N,m)}_{\text{relation collection}} \;+\; \underbrace{T_{\mathrm{LA}}(N,m)}_{\text{linear algebra}} \;+\; \underbrace{T_{\mathrm{target}}}_{\text{descent, check}} \quad . \tag{2.1}$$

Here C_{dec} is the cost of one decomposition attempt (the *oracle*). The factor $\rho_{\mathrm{yield}} \geq 1$ collects everything that makes a trial more productive than the heuristic: free relations, large primes, symmetry, re-randomised targets. T_{LA} is roughly mN^2 for sparse elimination. With rho's cost $\sqrt{\pi r/(2A)}$ on the other side, the classical balance at $m = 2$ with a constant-cost oracle gives $N \approx r^{1/3}$ and $T \approx r^{2/3}$. That is why two-summand prime-field index calculus loses ground to rho as $r^{1/6}$ (Chapter 32).

2.2 The levers

Table 2.1 lists, for each term of (2.1), the levers the programme pulled and the sections that report them.

2.3 Which term dominates now

Early in the programme, C_{dec} dominated. A Gröbner solve per trial cost milliseconds to seconds. After root finding (Chapter 9), the oracle is a handful of field operations and table probes, and the cost has moved elsewhere.

- On the Koblitz descent of the leaderboard's Table A, reusable set-up is 98.5–99.7% of cold S, and descent is 0.3–1.4% (`docs/ic/LEADERBOARD.md`, phase table).
- On the $n = 83$ cold pipeline with base K1107, a mean 113.755 s process splits into 47.166 s of set-up and 66.486 s of rank stage. The target stages take 0.099 s (`research/n83_cold_base_confirm_20261021/RESULT.md`).

So the relation-collection and set-up terms dominate. Their cost is set by N and by the yield per probe, not by the oracle. This is the main structural fact behind the ledger. Oracle speed-ups of three to five orders of magnitude moved the end-to-end gap by one order at most.

Term	Levers pulled	Where
C_{dec}	root finding of S_3; pair and meet-in-the-middle tables; inherited F4, F5, F6-IC; gate-first order; harvest gates; $\mathbb{F}_p$ F4; D_4 invariants	Ch. 9, 10, 25
ρ_{yield}, N	compact Frobenius orbits; free relations; stride-renewed targets; greedy coverage; $n = 83$ base selection; double large primes	Ch. 15, 21, 26
T_{setup}	stacked producer; cache-blocked pair table; u64 root batching	Ch. 15, 21
T_{LA}	CADO-style filter and Wiedemann; structured Gaussian elimination; block Wiedemann; first-dependency stop	Ch. 15, 21, Ch. 25, 26
the exponent	genus-three transfer (cubic GHS); $k = 4$ extension fields	Ch. 25, 26

Table 2.1: The terms of (2.1) and the levers that act on them.

3 The theory, by field type

3.1 What a decomposition costs

Every index-calculus algorithm for elliptic curves in the literature spends its time in three places: finding about $|\mathcal{F}|$ relations (each a decomposition of a random combination $aG + bQ$ over $\mathcal{F}$), solving the sparse linear system over $\mathbb{Z}/r$, and, in some variants, a descent of the target. The second and third are well understood. The first is the crux: a uniformly random point decomposes over a base of size N into m summands with probability about $N^m/(m!\,r)$, so relations need about $m!\,r/N^{m-1}$ trials, and each trial costs whatever the decomposition oracle costs. Every result below is a statement about one of three things: how large N can be while membership stays algebraically testable, how cheap the oracle is, and how the two trade against the linear algebra.

3.2 Extension fields $\mathbb{F}_{q^k}$ with $k > 1$ fixed or small

Gaudry [20] takes $\mathcal{F}$ to be the points with abscissa in the base field $\mathbb{F}_q$ and decomposes into $m = k$ summands by Weil restriction of S_{k+1}, a system of k equations in k unknowns over $\mathbb{F}_q$. For fixed k this gives $\tilde{O}(q^{2-2/k})$, asymptotically below rho's $q^{k/2}$ for $k \geq 3$; the constant hidden in the Gröbner solve grows very quickly with k, which is what makes the crossover large in practice; the measured handovers at $k = 4$ are in Chapter 25. Diem [10] extends the analysis to families where k grows, proving subexponential complexity over sequences of fields with suitably balanced q and k. Joux and Vitse [27] combine Weil descent with decomposition-based index calculus into one algorithm, well suited to curves over $\mathbb{F}_{p^6}$, and computed logarithms on a 149-bit group over a degree-6 extension in about 110,000 CPU-hours; they also give an $(n-1)$-decomposition variant. FGHR [14] exploit the action of S_m and of the 2-torsion to replace S_{m+1} by its invariant form, cutting the Gröbner work by large factors; our measurement at $k = 4$ (§21.10) is a direct instance.

3.3 Binary fields $\mathbb{F}_{2^n}$ with n prime

Prime n leaves no subfield, so the factor base is taken to be the points whose abscissa lies in an $\mathbb{F}_2$-vector subspace V of dimension ℓ. Weil restriction then gives a quadratic-to-low-degree Boolean system whose difficulty is governed by its degree of regularity. The whole debate about binary curves is a debate about that degree.

The subexponential claims. Petit and Quisquater [32] conjectured, from experiments and heuristic evidence, that these systems have degrees of regularity only slightly above the degrees of their equations (the *first fall degree assumption*), and deduced a heuristic $2^{c\,n^{2/3}\log n}$, $c < 2$, for a variant of Diem's algorithm, beating generic methods only for $n \gtrsim 2000$. Semaev [36] split summation polynomials into cubic Boolean systems, argued for regularity degree at most 4 under the same kind of assumption, and obtained a heuristic $2^{c\sqrt{n \ln n}}$ with $c \approx 1.69$; its first version concluded that several FIPS binary curves fall faster than Pollard rho, a conclusion its revision says does not hold under the weaker assumption that the regularity degree is $o(\sqrt{n/\ln n})$.

The evidence against. Shantz and Teske [37] tested the Petit–Quisquater heuristics experimentally. Huang, Kosters and Yeo [23] introduced the *last fall degree*, independent of the monomial order, gave computational evidence against the first fall degree assumption in characteristic 2, and constructed summation-polynomial Weil descent systems where it is unlikely to hold. Galbraith and Gebregiyorgis [17] improved the variable choice (binary Edwards curves), found symmetry-breaking factor bases and SAT solvers inconclusive, and concluded that rho remains much faster for prime-degree binary fields of reasonable size. The Galbraith–Gaudry survey [16] reaches the same verdict for characteristic-two curves over prime-degree fields.

Where it stands. No proof or refutation of a subexponential algorithm for prime-degree binary curves exists. The open quantity is the solving degree of Weil-descended S_{m+1} as n and m grow; recent work supplies diagnostics (saturation-exponent bounds, multi-graded regularity, tests for where systems stop looking generic) but no bound for these systems: rigorous last-fall-degree bounds for summation-polynomial descents are named as open by Huang, Kosters and Yeo [23] and by Kousidis and Wiemers [29], and Kosters and Yeo [28] show that the first fall degree of the descent of S_3 is usually 2, through a trace identity. Chapter 6 proves and certifies what the programme could add. Our own measurements on toy fields (§16.1) find first and last fall degrees of 3–4 at every reachable size while the solving degree grows with ℓ and stays within one of the semi-regular prediction; closure algorithms refute a degree or two earlier but rise too. That supports no subexponential claim and refutes none.

3.4 Subfield (Koblitz) curves

When the curve is defined over $\mathbb{F}_2$ the Frobenius acts on the group, and Galbraith, Granger, Merz and Petit [18] show that a Frobenius-invariant factor base saves a factor $\approx n$ in relations and $\approx n^2$ in linear algebra—more than rho's $\sqrt{2n}$—while concluding that Koblitz-curve security is not threatened at deployed sizes. Our Koblitz work (§15) builds on this.

3.5 Prime fields

Over $\mathbb{F}_p$ there is no subfield and no subspace, and no factor base is known for which membership is both dense enough and algebraically testable. Semaev's original small-abscissa base leads to systems of very high degree. Petit, Kosters and Messeng [33] construct bases from rational maps (including the Kummer tower μ_{2^t}) that give the summation system extra structure; Amadori, Pintore and Sala [1] build the base from the instance and need only one relation search, improving on both; McGuire and Mueller [30] avoid Gröbner bases altogether by fast evaluation of S_m through iterated resultants, and in a second variant by summing $m-1$ factor-base points and testing membership of the result—the same mechanism as the two-summand root-finding oracle of §9. All of these remain exponential and slower than rho; a 2024 analysis again concludes that no subexponential prime-field method is known.

3.6 Lower bounds

Generic-group lower bounds bound everything that does not use curve structure. With preprocessing, Corrigan-Gibbs and Kogan [8] show $ST^2 = \tilde{\Omega}(\varepsilon N)$ for S bits of advice and online time T; this is the bound against which the retained-set-up (online) results of §15 must be read. A 2026 structured-generic-group bound shows that an algorithm exploiting structure on only a δ fraction of elements needs $\Omega(\min(\sqrt{q}, 1/\delta))$ time—for index calculus, the decomposable set must be dense *and* cheaply testable.

4 Measuring index calculus against rho

The programme's first lesson was that index-calculus numbers from different studies did not compose: one study timed a decomposition, another a relation, a third a whole multi-target batch with shared tables, each on a different host and in a different unit. Four mechanisms, built in this order, fixed that.

4.1 The boundary ledger (10 September)

`docs/ic/BOUNDARY_TARGETS.md` and its machine twin `docs/ic/boundary_targets.json` give each regime (Koblitz, binary, prime) and each pipeline stage (`factor_base`, `decomposition`, `relation_yield`, `rank`, `end_to_end_dlp`, `vs_rho`) a current record, a next target and acceptance gates. A row promotes only when its evidence JSON carries every field the measurement schema requires for that stage—for example, a `decomposition` claim that beats an algebraic frontier must report the first-fall degree, and a `vs_rho` claim needs one target, the identical public point for both methods, verified scalars, five exclusive IC online phases, the automorphism discount, matched resources and a host-isolation receipt. Missing fields fail closed. The primary `vs_rho` question is the online cost of *one* previously unseen target after reusable IC preparation; multi-target and amortised results are secondary and must be declared as such.

4.2 ecbench (from 2 October)

ecbench (`src/cryptanalysis/ecbench/`, `src/bin/ecbench.rs`, documented in `docs/ecbench/README.md`) runs every ECDLP method behind one interface, on the same single-target workloads, in one unit, into sealed records.

Unit.

$$S \;=\; \frac{\text{total group-addition equivalents (GAE)}}{\sqrt{r}}, \qquad S_{\text{floor}} = \sqrt{\pi/(2A)}.$$

An addition or doubling is one GAE; a scalar multiplication is charged as the additions it performs; native work (lookups, row operations, square roots, Artin–Schreier solves, word XORs) is priced at a pinned conversion where the repository has one and is otherwise recorded as *uncharged*, which marks the record's S as a *lower bound*. Plain rho has $S \approx 1.25$.

What a measurement is. One workload is one curve, subgroup, generator and target; nothing survives between executions, so every table a method builds is inside its cost (*cold*). Beside the cold S every record carries the one-target *online window*, split for index calculus into the five exclusive phases `target_query`, `target_PDP`, `target_relation_check`, `target_descent`, `target_recovery_check`. Targets are either planted or *public*—hashed to the subgroup so that nobody knows the logarithm—and the runner verifies $[k]G = Q$ in its own process; `ecbench verify` checks it again from the files. Failed, exhausted and timed-out runs are rows, never dropped.

Identity, sealing and replay. Every object (curve, factor base, method parameters, spec, session, host environment class) is named by a hash of what it is (`ECBS1h...` for sessions, `ECBND1h...` for bounds). Each execution appends one sealed record; `ecbench verify` replays measured records and emits an audit receipt whose SHA-256 is the replay certificate. Operation counts are deterministic, so a replay must match to the last digit; a mismatch is preserved as a failed audit (§4.5).

Isolation. A run earns L0 (recorded under a host capsule), L1 (pinned, affinity read back, no foreign SMT sibling, CPU/wall ≤ 1.05), L2 (quiet: zero steal, foreign busy time $\leq 1\%$, run-queue delay $\leq 0.5\%$, ≤ 50 preemptions/s) or L3 (isolated cores, fixed governor, bare metal). Wall time is admitted only at the spec's required level (default L2) with at least five admitted pairs and an A/A interval; otherwise it is printed as *descriptive*. Operation counts need no isolation. macOS hosts earn L0; a GitHub-hosted runner earned L1 on 72/72 runs and matched a macOS session's operation counts to the last digit. *isolab* prepares lab hosts for L3 and cross-host replay.

Statistics. The comparable statistic is the mean S (the floor is an expectation). Comparisons are paired by (workload, round) with shared seeds, reported as a ratio of totals $\sum S_B / \sum S_A$, never a mean of ratios, with a two-stage cluster bootstrap that resamples workloads, then pairs. An operations A/A is exactly 1 by construction and checks determinism. A claim about scaling fits exponents over at least four sizes.

Bounds and the frontier. `ecbench` also records sealed *bounds*—constant and exponent with intervals, per phase—and a challenge protocol whose verdict says whether a candidate advances, trades, matches or regresses an incumbent and at which level (exponent, constant, primitive). The incumbent for cold one-target Koblitz operations is `rho.signed_frobenius_strong` (bound `ECBND1h05a2ea918f3c`): $1.089\times$ the floor $[0.927, 1.261]$, fitted exponent $\alpha = 0.451$ $[0.399, 0.512]$, 144 runs at six sizes (`research/ecbench_frontier_20261013/`).

Definition 4.1 (comparability)**.** A performance point is keyed by the tuple

$$D = \big(\text{curve and subgroup},\ \text{target law},\ \text{cold} \mid \text{online} \mid \text{stage},\ \text{unit and unpriced work},\ \text{set-up amortisation } q,\ \text{memory envelope},\ \text{correctness/replay/isolation class}\big).$$

Two points are ratio-comparable only when every component of D except the method matches; a method is on the frontier of a domain when no admitted point is no worse on every axis and strictly better on one.

This rules out the four collapses we saw most often: online wall as cold cost; word XORs, probes, field operations and GAE as interchangeable; a lower bound as a complete cost; an L0 wall ratio as an admitted speed-up.

4.3 The pricing band

Almost no cost in a record is complete, and the rho baselines are no exception. Rho, BSGS and the kangaroo leave canonicalisations, partition hashes and table inserts and lookups unpriced. An index-calculus arm is complete only on curves that have a pinned timing calibration (`docs/ic/calibration.json`), which converts lookups, row operations, Artin–Schreier solves, Frobenius maps and solver word XORs to GAE at measured ratios of roughly 10^{-3} to 10^{-1}. A "complete" index calculus set against a lower-bound rho is therefore lopsided, and several earlier comparisons in this repository were of that kind.

Since 15 October every cross-method comparison uses a *pricing band* (repository rule, `AGENTS.md` §5a; native implementation `ecbench band`). For a record with charged cost G, calibration-priced native cost C over n_C operations, and U operations behind its unpriced counters,

$$S(w) = \frac{G - C + w\,(U + n_C)}{\sqrt{r}}, \qquad w \in \{0,\ 0.1,\ 1\} \text{ GAE per native operation},$$

applied alike to every arm, references included. Two arms are ranked only if their order is the same at every weight; otherwise the pair is reported as unresolved. An arm that names an unpriced counter without counting it is a lower bound and cannot be ranked. The band brackets the pinned calibrations from below and above, so a ranking that survives it does not depend on the host the calibration came from.

4.4 Claim hygiene and change classes

Every study directory states a protocol before it runs (frozen inputs are committed first), keeps failures, and closes with a decision in one of the four classes of [3]. A controlled-experiment rule requires a negative control from the right null population, a positive control that proves the detector can see the effect, and the same instrument on every arm before any comparative statement. Blockers are recorded as named problems (`problems/`) with exit criteria rather than silently dropped. Measurement code is native (Rust); Python only lays out recorded values.

Generic baselines. The rho, BSGS and Pohlig–Hellman references were engineered alongside index calculus, which is why each round's distance is quoted against the strongest reference at the time. On Koblitz curves the strong signed-Frobenius rho fell from 1,196 to 679.7 instructions per step over four rounds with no established native-time gain (§31.4). On hyperelliptic Jacobians a negation fold and hash mixing each fell short of their registered gains, and full-width branch coefficients showed that the earlier 8-bit reference walk had been slow at large N (`benchmarks/20261018-hec-rho-negation/`, `benchmarks/20261019-hec-rho-branch-width/`). Generic DLP fixes cut Pohlig–Hellman steps 26.6× and BSGS time 1.3–2.0×, and the candidates run at 0.020–0.167 of PARI's `elllog` time, in shared-host wall time the validation sweep left unvalidated (`research/generic_dlp_memory_parallelism_20261017/`).

4.5 Corrections the accounting forced

Building the harness and the ledger exposed errors in earlier claims, including our own. We list them because they change how earlier numbers must be read.

1. *Results below rho that reversed against a stronger rho* (September; §15.3). A tournament candidate at 0.5347 of rho's instructions lost all five cells once rho used the same canonical form (`research/ic_triple_counted_20260923/`); an orbit-DLP ratio of 0.2403 became 1.923 against a strengthened rho and was withdrawn; a compact-orbit batch lead of 0.309 reversed against a batched-inversion rho.
2. *Online Koblitz ratios relabelled* (7–8 October, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`). Same-point online ratios of 124.85× ($n = 61$), 2,864.1× ($n = 71$), 1,189.7× ($n = 73$) and 8.2× ($n = 83$) had verified answers but no host-isolation receipt; they are *exploratory*. Only the later 64-target pinned panel at $n = 61$ is admitted (§15).
3. *ECC2K-130 extrapolation constants corrected* (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`): the structural model predicts $C = 0.25$ in probes/relation $\approx C\,r/(n^2K^2)$; the fitted C is 0.27 at $n = 61$ but 1.2–1.5 at $n = 71, 73$, an unexplained gap now stated rather than absorbed.
4. *"F4" that was Buchberger* (`research/ecbench_rootfinding_assessment_20261012/REPORT.md`): the NIST benchmark of the root-finding manuscript printed its baseline as F4 but called `groebner_f4::buchberger`; its 3,021–5,802× are ratios to Buchberger. The manuscript's claim that no implementation can use fewer than linear field operations per ideal is too broad once a reusable pair table is allowed.
5. *Unpriced word XORs* (`research/ecbench_rootfind_faster_20261014/`). Inherited F4's Boolean word XORs were counted but not charged, so its $S = 16.7$ at Koblitz $n = 13$ was a lower bound that looked competitive with rho (14.4). Pricing them gives $S = 421.4$.
6. *Cross-host counter drift* (`research/ecbench_frontier_20261013/audits/`): a replay-all audit of the root-finder session on x86_64 matched 120/168 records; all 48 inherited-F4 rows differed in solver word-XOR counts (e.g. 301,961 versus 300,531), with every scalar, trial count and charged GAE unchanged. The failed receipt is kept; the F4 stage comparison is historical until the drift is explained. A replay of the same session on current `main` (20 October) matches only 72/168, with mismatches in all four IC arms, root finding included. The cause was then isolated: an x86-only BMI2 rewrite in inherited F4 counts word XORs differently from the portable path; with it disabled the replay is 168/168 on x86 and on emulated aarch64 (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`). F4 counts are comparable across instruction sets only when the path is stated, which the method identity does not yet record.
7. *A soundness bug* (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md` §2.1): a degree-bounded `f4_fp::solve` could certify solutions of an inconsistent system (over $\mathbb{F}_5$, $\{x^3 + 1, xy - 1, x + y\}$ returned two "solutions"). Fixed by evaluating every candidate on the input.
8. *F6-IC charged cost* (§10): a stage win in word XORs became a 64.6–1,353× loss in charged S once its batched geometry was counted—recorded as an accounting result, not an advance. With inherited F4's own word XORs priced the $m = 13$ loss is about 2.5× (§10).
9. *Lopsided pricing* (§4.3): comparisons of a calibration-priced index calculus with a lower-bound rho. On the frontier campaign's train panel at Koblitz $n = 13$ the compact-orbit arm, earlier quoted at 1.08× rho as a lower bound, is 0.93×, 1.53× and 5.7× the strong rho at $w = 0$, 0.1 and 1 (columns 2, four train targets, `research/ic_frontier_campaign_20261015/select/SELECTION.md`): below rho with native work free, above it otherwise, so the pair is unresolved inside the band rather than a win for either side.
10. *Wall-time figures recomputed* (validation sweep). Recomputing from the committed per-target records gives 2,321× for the $\ell = 12$ binary MITM oracle (quoted 2,187×), 98.1× for its $\ell = 8$ run 3 (quoted

100.7×), and 6.08× and 6.14 s for the quartic solve at $d = 6$ (quoted 6.4× and 5.9 s); all are shared-host wall times left unvalidated, and the counted evidence behind them stands (`research/validation_sweep_20261019/B-benchmarks/VERDICT_TABLE.md`).

11. *An unmatched rho reference* (21–24 October). The Koblitz leaderboard's rule v3 priced its rho as built rather than at the canonical batched step. Re-pricing rho in the same process raised the cold distance 1.76–1.94×, to 9.9–43× (`research/ic_tool_program/rule/v4/README.md`); the ledger's rule v8, matching the reference its own way, raised it 1.30–1.58× and found that an apparent flattening of the gap's slope was mostly this artefact (`papers/ecdlp-ic-improvement-ledger/`). The same week withdrew a volcano-crater claim (IC 22–25% cheaper) whose reference should have been the cubic-automorphism rho, against which IC is 84.9× rho, and marked the plane-quartic S values unverified because their degree-3 reduction merges distinct classes. Canonical Pic^0 keys later separated those classes, and a re-run found that the earlier toy logarithms came from enumerating the subgroup (§26).

Part II

Theory

5 The summation cover: what is fixed for every curve

Every index calculus on an elliptic curve asks one question many times: does a target R decompose as a sum of $m-1$ factor-base points? Semaev's summation polynomial answers it algebraically, and every design lever of the programme acted on this question. This chapter collects three facts about it that hold for *every* curve. They closed, without an experiment, the levers that rely on curve-dependent monodromy, on fibre censuses, or on raising mean yield at matched base size. They also mark what stays free (§5.6).

The statements and proofs are those of `crypto-autoresearcher:papers/semaev-conservation-specialization/` (15 pages, anonymized draft, verification script `verify.py`, seed 20260807, which replays byte for byte), and of the internal finding `KN-FIND-bdcd60`. An integrity audit (`crypto-autoresearcher:research/sheriff/SHF-20261023-7b9227/`) upheld the mathematics within its scope. It blindly re-derived the group orders $2, 24, 192, 1920, 23040$ and checked 2.1 million admissible fibres with zero violations. It also rated as overclaimed a statement that three *independent* derivations agree: they came from one model family. **Novelty is not adjudicated** (§5.7).

5.1 Setting

Let K be a field, E/K an elliptic curve with inversion $\iota = [-1]$, and $\pi : E \to E/\langle\iota\rangle \cong \mathbb{P}^1$ the quotient (the x-coordinate in odd characteristic). The m-th summation polynomial $S_m \in K[X_1, \dots, X_m]$ is the reduced defining polynomial of the image of $\{(Q_1, \dots, Q_m) : \sum Q_i = O\}$ under π^m. Equivalently, $S_m(\xi_1, \dots, \xi_m) = 0$ if and only if there are points Q_i with $\pi(Q_i) = \xi_i$ summing to O [35]. For $m \geq 3$ it is symmetric, absolutely irreducible, and of degree 2^{m-2} in each variable. It satisfies the resultant recursion $S_m = \mathrm{Res}_X(S_{m-1}(\dots, X), S_3(X_{m-1}, X_m, X))$.

Fix $m \geq 3$, put $n = m-1$, and let $t_1, \dots, t_n$ be independent transcendentals. Write $L = K(t_1, \dots, t_n)$ for the function field of $(\mathbb{P}^1)^n$ and $M = K(E^n)$ for that of E^n. The group $\Gamma = \langle\iota\rangle^n \cong (\mathbb{Z}/2)^n$ acts on E^n coordinatewise; $\Delta \subset \Gamma$ is its diagonal. M/L is Galois with group Γ. With P_i the generic points ($\pi(P_i) = t_i$), the roots of $S_m(t_1, \dots, t_n, T)$ in T are the abscissae of the signed sums $\theta_\varepsilon = \pi(\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n)$, $\varepsilon \in \{\pm 1\}^n$, with $\theta_\varepsilon = \theta_{-\varepsilon}$.

5.2 The Galois group

Theorem 5.1 (Monodromy of the summation cover)**.** *For every field K, every elliptic curve E/K and every $m \geq 3$:*

1. *$S_m(t_1, \dots, t_n, T)$ is separable of degree 2^{m-2} over L, with root set $\{\theta_\varepsilon\}$;*
2. *its splitting field is $N = M^\Delta = K(E^n/\Delta)$;*
3. *N/L is Galois with group $\Gamma/\Delta \cong (\mathbb{Z}/2)^{m-2}$, acting simply transitively on the roots;*
4. *K is algebraically closed in N, so arithmetic and geometric monodromy coincide,* $\mathrm{Mon}^{\mathrm{arith}} = \mathrm{Mon}^{\mathrm{geom}} \cong (\mathbb{Z}/2)^{m-2}$.

None of the four statements depends on E, on K or on its characteristic.

Proof outline. A root-identification lemma shows that the roots are among the θ_ε. A generic-separability lemma shows that the 2^{m-2} classes $\{\pm\varepsilon\}$ give distinct values. Γ permutes the roots by $\sigma \cdot \theta_\varepsilon = \theta_{\sigma\varepsilon}$, and the kernel of this action is exactly Δ. So Γ/Δ acts faithfully and regularly on 2^{m-2} roots. The stabiliser of one root is Δ, which gives $N = M^\Delta$. Finally, E^n/Δ is geometrically integral, so K is algebraically closed in its function field. □

For a symmetric (unordered) factor base the relevant group is the wreath extension $(\mathbb{Z}/2)^{m-2} \rtimes S_{m-1}$, of order $2^{m-2}(m-1)!$ for $m \geq 4$. It is primitive if and only if $m-1$ is odd (`KN-FIND-bdcd60`). Statement (4) is the operative one: a cover whose arithmetic monodromy exceeds its geometric monodromy can split differently over different base fields, and hence on different curves. Here no such room exists.

5.3 An exact factorisation law over a finite field

Over $K = \mathbb{F}_q$, define for $a \in \mathbb{P}^1(\mathbb{F}_q)$ the *splitting invariant*:
- $c(a) = 0$ if both points above a are rational;
- $c(a) = 1$ if neither is;
- $c(a) = *$ at a branch point.

For odd q and $E : y^2 = f(x)$ this is the quadratic character of $f(a)$.

Theorem 5.2 (Factorisation law)**.** *Let $a = (a_1, \dots, a_n) \in \mathbb{F}_q^n$ lie outside the degenerate locus, which has $O_m(q^{n-1})$ points. Then $S_m(a_1, \dots, a_n, T)$ is squarefree of degree 2^{m-2}, and:*
1. if $c(a_1) = \cdots = c(a_n)$, it splits into 2^{m-2} distinct linear factors, with roots $x(\pm P_1 \pm \cdots \pm P_n)$;
2. otherwise it is a product of 2^{m-3} distinct irreducible quadratics and has no root in $\mathbb{F}_q$.
No irreducible factor of degree ≥ 3 ever occurs.

Proof. On the étale locus, the factorisation type is the cycle type of Frobenius in Γ/Δ (Dedekind's theorem for specialisations of a Galois cover [15]). The Frobenius element is $((-1)^{c(a_i)})_i$. It is trivial in Γ/Δ exactly when every $c(a_i)$ agrees. Otherwise it is a non-trivial element of order 2 of a regular group, and so has cycle type $2^{2^{m-3}}$. □

The density of totally split fibres is $2^{-(m-2)} + O(m^2/(2^{m-1}q))$. The first-order Hasse terms $\pm t/q$ cancel, so the statistic carries no first-order information about the curve.

5.4 Specialisation to the factor-base locus

Theorem 5.3 (Complete splitting on the factor-base locus)**.** *If each $a_i = x(P_i)$ with $P_i \in E(\mathbb{F}_q)$ and a is non-degenerate, then $S_m(a_1, \dots, a_n, T)$ splits into 2^{m-2} distinct linear factors over $\mathbb{F}_q$, with roots $\{x(\varepsilon_1 P_1 + \cdots + \varepsilon_n P_n)\}$.*

This is Theorem 5.2(1) with every $c(a_i) = 0$. It is the statement behind the root-finding oracle of Chapter 9: given $m - 1$ base points, the admissible last coordinates are the signed sums, computable in $O(2^{m-2})$ group operations. It is *not* an algorithm for relation search. Relation search fixes the target and asks for the base points. The theorem says nothing about how many tuples must be tried, nor about the algebraic difficulty of the system posed in that direction. Its value is negative:

Corollary 5.4 (Census obstruction)**.** *Every statistic of the form "factorisation type, cycle type or Frobenius class of the fibre of S_m" is constant on the factor-base locus, for every curve and every m. A census of generic fibres measures a quantity whose value where relation search operates is known in advance and is the same for all curves.*

The programme ran such censuses before this corollary was proved. The census-as-relation-rate probe was closed by ledger decision `DEC-20261017-435b01` (`reject_scoped`, replicated).

5.5 Conservation of decomposition yield

Proposition 5.5 (Conservation)**.** *Let G be a finite abelian group of order N and $\mathcal{D} \subseteq G \setminus \{0\}$ a set of size B. The number of size-m multisets from $\mathcal{D}$, summed over all targets, is $\binom{B+m-1}{m}$. The mean number of decompositions per target is therefore $\binom{B+m-1}{m}/N$, for every $\mathcal{D}$ of size B.*

The proof is a double count, and no novelty is claimed for it. Its consequences shaped the programme:
- at matched base size, no factor-base geometry can raise the mean yield;
- only the distribution of decompositions is free, and coverage (the fraction of targets with at least one decomposition) is at most $\min(1, \mu)$;
- additive structure that repeats sums *lowers* coverage;
- typing the summands, each from its own sub-base, is a fixed penalty: $\prod B_i \leq (B/m)^m < \binom{B+m-1}{m}$.

Gaudry and Diem's $1/m!$ decomposition probability over extension fields *is* this mean. The extension-field advantage lives in the solve, not in the yield.

5.6 Which levers are fixed and which are free

Together the results above divide the design space as follows:

- *Fixed for every curve:*
  - the monodromy group of the summation cover (Theorem 5.1);
  - the fibre statistics on the factor-base locus (Corollary 5.4);
  - the mean yield at fixed base size (Proposition 5.5).
- *Free:*
  - the coverage distribution (bounded by the mean);
  - the cost of deciding membership (the oracle, Part III);
  - the algebraic difficulty of the decomposition system when it is posed as a polynomial system (Chapter 6);
  - the linear algebra;
  - where the problem lives (transfers, Chapter 26).

Every measured improvement in Part III acts on a free item. Chapter 26 shows that transfers by isogeny conserve the counts as well.

5.7 Prior work and the limits of the novelty assessment

Absolute irreducibility of S_m is classical [20, 35]. Corollary 5.4's ingredient, that the roots are the signed sums, is the defining property of S_m. The action of 2-torsion and of sign changes on summation systems is used by Faugère, Gaudry, Huot and Renault [14] and by Galbraith and Gebregiyorgis [17]. Theorem 5.1's group may be folklore: it is the deck group of $E^n \to (E/\pm)^n$ modulo the diagonal. Theorem 5.2 is in essence a 2-descent statement. The draft could not reach ePrint during its literature search, and says so. Corollary 5.4 and the exact dichotomy of Theorem 5.2 are the parts most likely to be new. Two items must be settled before any of this is submitted:

- an expert check of the separability lemma in characteristic 2, where $E[2]$ is non-reduced;
- a full literature search.

6 Fall degrees of point-decomposition systems

Every claim that elliptic-curve index calculus can be subexponential rests on the cost of solving the point-decomposition system. In practice that means its *solving degree*: the largest degree a Gröbner-basis computation reaches. Petit and Quisquater [32] conjectured that the solving degree stays close to the *first fall degree*, which is small and easy to compute. Semaev [36] assumed it is at most 4 for his chained systems. Kosters and Yeo [28], Huang, Kosters and Yeo [23] and Kousidis and Wiemers [29] gave reasons and experiments to doubt this. All three name rigorous bounds on the *last* fall degree, which equals the solving degree for the systems that matter, as the open problem.

This chapter collects what the programme proved and certified about these degrees, in three presentations of the decomposition system:

- binary Weil descent;
- explicit membership over any field;
- prime-field Kummer towers.

The message is the same in all three. The first fall degree is a constant of the presentation, whereas the last fall degree is not. Where the last fall degree rises above 4, the chapter gives a certificate that a separate program checks without trusting the producer.

6.1 Definitions

Let $F = \{f_1, \dots, f_M\} \subset R = k[x_1, \dots, x_N]$, graded by total degree.

- *Closure.* $V_{F,d}$ is the smallest k-subspace of $R_{\le d}$ that contains $F \cap R_{\le d}$ and is closed under $g \mapsto hg$ whenever $\deg(hg) \le d$ [5, 23].
- *Last fall degree.* $d_F = \min\{d : f \in V_{F,\max(d,\deg f)}\ \forall f \in (F)\}$.
- *First fall degree.* We use the Caminata–Gorla convention [5]: $d_{\mathrm{ff}}(F)$ is the least d at which the top-form syzygies modulo the trivial (Koszul and field) syzygies are non-zero.
- *Boolean closure.* Over $\mathbb{F}_2$ with field equations it is convenient to work in $B = \mathbb{F}_2[x]/(x_i^2 + x_i)$. There $W_d(F)$ is the same closure computed on multilinear representatives, and $M_d(F)$ is the Macaulay space spanned by the products mf with $\deg m + \deg f \le d$.

Remark 6.1 (Definitions matter). Three readings of "first fall degree" are in use. They are:

- Petit–Quisquater: a combination $\sum g_i f_i \neq 0$ whose degree drops below $\max \deg(g_i f_i)$;
- Caminata–Gorla: non-trivial syzygies of the top forms;
- operational: the least c with $V_{F,c} \cap R_{\le c-1} \neq V_{F,c-1}$.

They can disagree. Two documented cases follow.

- Nagao's ePrint 2015/984 [31] defines a "fake" first fall degree d'_F computed modulo the field equations, and states $d_F \le d'_F$ as its Lemma 4. Under the equality reading of its condition (1) the lemma fails over $\mathbb{F}_2$:
  - $\{X^2Y,\ XY + X\}$ has $d'_F = 2 < d_F = 3$, and so does $\{X_1X_2 + X_3,\ X_1X_3 + X_2\}$;
  - 94 of 400 random multilinear pairs in three variables violate it.

  The paper's own applications check the unreduced degrees instance by instance, so they are unaffected (`crypto-autoresearcher:papers/nagao-lemma4-note/`, `KN-FIND-936151`; two implementations, 15 tests).
- A finite descent of S_3 over $\mathbb{F}_{2^4}$ ($n = n' = 4$, seed 32201, $x_R = 1$, basis $[9, 3, 2, 12]$) has operational first fall 3. Its closure dimensions are $\dim V_D = 1, 11, 53$ and $\dim(V_D \cap R_{\le D-1}) = 0, 1, 12$ for $D = 1, 2, 3$. Lean finite certificates check the ranks, the semantics on all 256 assignments, and an explicit witness. The literal Petit–Quisquater reading of the same system gives 2, through a field-equation identity.
  - This refutes an internal claim of a uniform degree-2 first fall, since withdrawn.
  - It contradicts no published theorem: Kosters and Yeo say "usually 2" [28].
  - Source: `crypto-autoresearcher:docs/reports/2026-10-22-semaev-first-fall-replay/`.

Every statement below names its convention.

6.2 General closure facts

The following hold for any Boolean system F, with $\tilde{F}$ its lift to R together with the field equations. They are proved in `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/theorem-dossier.md`, §1.

Proposition 6.2 (Bridges). *1.* $\pi(V_{\tilde{F},d}) = W_d(F)$, *where* $\pi : R \to B$ *is the quotient map. In particular* $1 \in V_{\tilde{F},d}$ *if and only if* $1 \in W_d(F)$.
2. If F *has no Boolean zero, then* $d_{\tilde{F}} = d_{\mathrm{ref}}(F) := \min\{d : 1 \in W_d(F)\}$. *This also equals the solving degree for every degree-compatible order (Caminata–Gorla, reduced basis* $\{1\}$*).*
3. If F *has no Boolean zero, then* every *F4 run on* $\tilde{F}$*, under any selection strategy, produces a polynomial of degree* $\geq d_{\mathrm{ref}}(F)$.
4. If F *has Boolean zero set* S *and* $\operatorname{codim} W_d(F) > |S|$, *then* $d_{\tilde{F}} \geq d+1$, *and every F4 run reaches degree* $d+1$.

Lemma 6.3 (Branching). *If both restrictions* $F|_{x=0}$ *and* $F|_{x=1}$ *satisfy* $1 \in W_d$, *then* $1 \in W_{d+1}(F)$.

Proposition 6.4 (Dual certificate). *Let* $\pi : B_{\leq d} \to Z$ *be linear and onto, with kernel* U. *Suppose:*
(i) $\pi(f) = 0$ *for every* $f \in F$ *of degree* $\leq d$;
(ii) $\pi(xu) = 0$ *for every variable* x *and every* $u \in U$ *of degree* $\leq d-1$.
Then $W_d(F) \subseteq U$, *so* $\operatorname{codim} W_d(F) \geq \dim Z$. *If moreover* $\pi(1) \neq 0$, *then* $1 \notin W_d(F)$. *Conversely, the quotient map* $B_{\leq d} \to B_{\leq d}/W_d(F)$ *is such a certificate whenever* $1 \notin W_d(F)$.

Proposition 6.4 is what makes the lower bounds below checkable. A certificate is a matrix. Checking it needs the curve parameters, an independent descent of the system, and three linear-algebra tests: (i), (ii) and $\pi(1) \neq 0$. Nothing about how the certificate was found has to be trusted. With Proposition 6.2(3), one accepted certificate for $1 \notin W_4(F)$ proves that *no* F4 run on that instance stays at degree 4. This is a statement about the system, not about a solver run.

6.3 Binary Weil descent of S_3

Setting.
- The field is $K = \mathbb{F}_{2^n}$ with trace Tr, and $K_0 = \ker \mathrm{Tr}$.
- The curve is $E : y^2 + xy = x^3 + a_2x^2 + a_6$ with $a_6 \neq 0$.
- The target abscissa is $z \in K^\times$.
- $V \subset K$ is an $\mathbb{F}_2$-subspace of dimension k; Semaev's diagonal is $k = \lceil n/2 \rceil$.
- $X_1 = \sum a_j v_j$ and $X_2 = \sum b_j v_j$ are linear in the $N = 2k$ Boolean variables.
- The system F is the Weil descent of $S = S_3(X_1, X_2, z)$: the coordinates $\mathrm{Tr}(\gamma_i S)$. This is Semaev's system (5) for $m = t = 2$.

Fix ϑ with $\mathrm{Tr}\,\vartheta = 1$, and an Artin–Schreier section G (the half-trace for odd n). Put

$$w = \frac{X_1X_2}{z} + G(s^2) + G(a_6/z^2), \qquad s = X_1 + X_2, \qquad \ell_0 = \mathrm{Tr}(s) + \mathrm{Tr}(a_6/z^2).$$

Theorem 6.5 (Exact reformulation). *For every* n, V *and* z:
1. $S = z^2(w^2 + w + \ell_0\vartheta)$;
2. $\mathrm{span}_{\mathbb{F}_2} F = \{\mathrm{Tr}(\delta w) : \delta \in K_0\} + \mathbb{F}_2\ell_0$;
3. in particular, the affine-linear form $\ell_0 = \mathrm{Tr}(S/z^2)$ *always lies in the span of the equations.*

Proof. (1) Squaring is a ring endomorphism, so $S = z^2(u^2 + u + s^2 + a_6/z^2)$ with $u = X_1X_2/z$, and $h = G(s^2 + a_6/z^2)$ satisfies $h^2 + h = s^2 + a_6/z^2 + \mathrm{Tr}(s^2 + a_6/z^2)\vartheta$, with $\mathrm{Tr}(s^2) = \mathrm{Tr}(s)$. (2) Put $\gamma' = \gamma z^2$. Then $\mathrm{Tr}(\gamma S) = \mathrm{Tr}((\gamma'^{1/2} + \gamma')w) + \mathrm{Tr}(\gamma'\vartheta)\ell_0$, and $\gamma' \mapsto (\gamma'^{1/2} + \gamma', \mathrm{Tr}(\gamma'\vartheta))$ is a bijection $K \to K_0 \times \mathbb{F}_2$. (3) Take $\gamma' = 1$. □

Part (3) is Kosters and Yeo's Corollary 4.11 [28], seen through a different coordinate. The reformulation makes the falls explicit.

Corollary 6.6 (First fall degree 2, for every n). *Suppose every descended equation has degree exactly 2; this held on every instance computed. Then:*
- $d_{\mathrm{ff}}(F) = 2$ *in the Caminata–Gorla sense, for every* $n \geq 3$ *and every* V;
- *if moreover* $\ell_0 \neq 0$, *the Petit–Quisquater first fall degree is also 2.*

The same holds for Semaev's chained system (5) for every $t \geq 2$*: the last equation* $S_3(y, x_t, R_X)$ *always contributes the linear form* $\mathrm{Tr}(Y + X_t) + \mathrm{Tr}(a_6/R_X^2)$.

Semaev states the value 4 for the first fall degree of his systems [36, §4.5]. His argument concerns only the cubic equations of the chain. Under the definition he quotes, the trace relation makes the value 2.

Lemma 6.7 (Explicit degree-3 falls)**.** *Let* $V^\perp$ *be the trace-orthogonal of* V*, and* $\omega = \mathrm{Tr}(\mu w)$ *for any* μ *with* $\mathrm{Tr}\,\mu = 1$.

- *For* $\lambda \in V^\perp$ *and* $i \in \{1, 2\}$, $\mathrm{Tr}(\lambda X_i w)$ *lies in* $M_3(F)$ *and has degree* ≤ 2.
- $\mathrm{Tr}(sw) + \mathrm{Tr}(a_6/z^2)\,\omega$ *lies in* $W_3(F)$ *and has degree* ≤ 2.

These are $2(n - k) + 1$ *explicit degree-3 falls, independent of* n *in form. On random subspaces they span the whole degree-*≤ 2 *part of* W_3*, of dimension exactly* $3n$*; this was checked on six instances with* $n \in \{23, 25, 27\}$*. Polynomial subspaces* $\langle 1, t, \ldots, t^{k-1} \rangle$ *have* $2n - 1$ *further falls, which is why they resist longer.*

Theorem 6.8 (Upper bound)**.** *If* F *has no Boolean zero, then* $1 \in M_{k+2}(F)$*. So* $d_{\tilde{F}} \leq k + 2$*, which is* $\lceil n/2 \rceil + 2$ *on the diagonal. For Semaev's chained system (5) with* t *summands the bound is* $d_{\tilde{F}} \leq (t - 1)k + t$*: each auxiliary variable adds 1, not* n.

Proof for $t = 2$. Substitute $c_j = a_j + b_j$. Then $S = X_1^4 + X_1^2 s^2 + z^2 s^2 + z X_1^2 + z X_1 s + a_6$ is at most linear in a and in c, so each $f_i = \sum_j a_j L_{ij}(c) + M_i(c)$. For each $\sigma \in \mathbb{F}_2^k$ the affine system $f_i(a, \sigma)$ in a is inconsistent, so some combination $\sum_i \lambda_i^\sigma f_i(a, \sigma) = 1$. With the indicators $e_\sigma = \prod_j (c_j + \sigma_j + 1)$,

$$1 = \sum_\sigma \sum_i \lambda_i^\sigma e_\sigma f_i \in M_{k+2}(F). \qquad \square$$

Proposition 6.9 (Lower bound $d_{\tilde{F}} \geq 3$)**.** *Under two explicit genericity conditions on* V *and* z*, which a script checks and which held on every random-subspace instance, together with* $\ell_0 \neq 1$*:*

- $W_2(F) = \mathrm{span}\, F + \ell_0 B_{\leq 1}$*, and its degree-*$\leq 1$ *part is* $\mathbb{F}_2 \ell_0$*;*
- *so an unsatisfiable* F *has* $d_{\tilde{F}} \geq 3 = d_{\mathrm{ff}} + 1$.

The bound is attained at $n = 17, 19$.

Computer-verified instances.

- *Implementations.*
  - Implementation A: a C closure over M4RI, `code/lfdclose2.c`.
  - Implementation B: independent Python sharing no code with A; `verify/b_sweep.py` and `verify/indep_verify.py`.
  - Certificate checker: `verify/cert_check.py`, which rebuilds F from the curve parameters by its own Möbius interpolation of S_3 values.
- *Instances.*
  - Every instance on the diagonal $k = \lceil n/2 \rceil$, other than the two satisfiable ones, was shown unsatisfiable by exhaustive count, and the count was repeated by B.
  - B reproduced all 61 instances with $n \leq 29$: the same solution count, the same dimensions of $W_{d-1} \cap B_{\leq e}$ for every e, and $1 \in W_d$.

Figure 6.1 shows the result: as n grows, the first fall degree stays at 2 while the last fall degree climbs.

Theorem 6.10 (Exact last fall degrees; computer-verified on instances)**.** *On the diagonal* $k = \lceil n/2 \rceil$*:*

- $d_{\tilde{F}} = 3$ *on all six random-subspace instances at* $n = 11, 13$ *and on all ten polynomial-subspace instances at* $n = 17, 19$*;*
- $d_{\tilde{F}} = 4$ *on all 29 random-subspace instances with* $15 \leq n \leq 33$*, and on all 25 polynomial-subspace instances with* $23 \leq n \leq 35$ *or* $n = 41, 43$*;*
- $d_{\tilde{F}} = 4$ *on Semaev's own cell (*$n = 40$*,* $k = 20$*, uniform* z*), matching his reported* $d_{F4} = 4$*;*
- $d_{\tilde{F}} = 5$ *exactly on instance R35a (*$n = 35$*,* $k = 18$*, random* V*):* $1 \notin W_4$*, and both restrictions* $a_1 = 0, 1$ *are refuted at* W_4 *(Lemma 6.3).*

Table 6.1 gives every instance behind Theorems 6.10 and 6.11, one row per subspace family and n, in the layout of Kousidis and Wiemers' Table 1.

Theorem 6.11 (Certified $d_{\tilde{F}} \geq 5$)**.** *For each instance below, a certificate (Proposition 6.4) for* $1 \notin W_4(F)$ *was accepted by the independent checker. Hence* $d_{\tilde{F}} \geq 5$*, and every F4 run reaches degree* ≥ 5.

m	V	n	k	inst.	D_{ff}	$d_{\tilde{F}}$	cols at $D=4$	s at $D=4$	codim $W_4 \geq$
2	P	17	9	6	2	3			
		19	10	4	2	3			
		21	11	4	2	3×2, 4×2	9,109	7.3	
		23	12	4	2	4	12,951	7.3	
		25	13	4	2	4	17,902	11.1	
		27	14	4	2	4	24,158	20.3	
		29	15	3	2	4	31,931	40.9	
		31	16	3	2	4	41,449	51.9	
		33	17	3	2	4	52,956	100.1	
		35	18	2	2	4	66,712	160.9	
		41	21	1	2	4	124,314	1076.1	
		43	22	1	2	4	149,986	2001.3	
		45	23	1	2	≥ 5	179,447	1850.7	12,742
	S	40	20	2	2	4	102,091	537.9	
	R	11	6	3	2	3			
		13	7	3	2	3			
		15	8	3	2	4	2,517	1.0	
		17	9	3	2	4	4,048	1.3	
		19	10	3	2	4	6,196	2.1	
		21	11	3	2	4	9,109	3.2	
		23	12	3	2	4	12,951	4.1	
		25	13	3	2	4	17,902	6.6	
		27	14	3	2	4	24,158	15.4	
		29	15	3	2	4	31,931	24.0	
		31	16	3	2	4	41,449	45.2	
		33	17	2	2	4	52,956	91.9	
		35	18	2	2	≥ 5	66,712	121.0	5,287
		37	19	2	2	≥ 5	82,993	312.9	10,066
		39	20	2	2	≥ 5	102,091	445.9	
	U	33	17	2	2	4	52,956	95.4	
		35	18	2	2	≥ 5	66,712	126.4	5,287
	T	29	15	2	2	4	31,931	38.3	
		31	16	2	2	4	41,449	68.3	
		33	17	2	2	≥ 5	52,956	128.8	4,888

Table 6.1: Binary Weil descent of S_3 ($m = 2$, $k = \lceil n/2 \rceil$): the first fall degree $D_{\mathrm{ff}} = 2$ (Corollary 6.6, proved for every n) beside the last fall degree $d_{\tilde{F}}$, the least D with $1 \in W_D$, for each instance. Subspace families: P, Semaev's polynomial subspace $\langle 1, t, \ldots, t^{k-1} \rangle$; S, the same with uniform z (Semaev's own cell); R, random; U, random with uniform z; T, random inside $\ker \mathrm{Tr}$. "≥ 5" means $1 \notin W_4$; R35a is exactly 5 (Theorem 6.10). Columns: instances; columns of the degree-4 closure matrix; median closure wall time at $D = 4$ in seconds, over the instances run to that degree (host not recorded); certified codimension of W_4 where an independent certificate was checked. Source: `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/results/`, laid out by `figures/make_tab_fall_degrees.py` from `figures/data-falldegree-instances.json`.

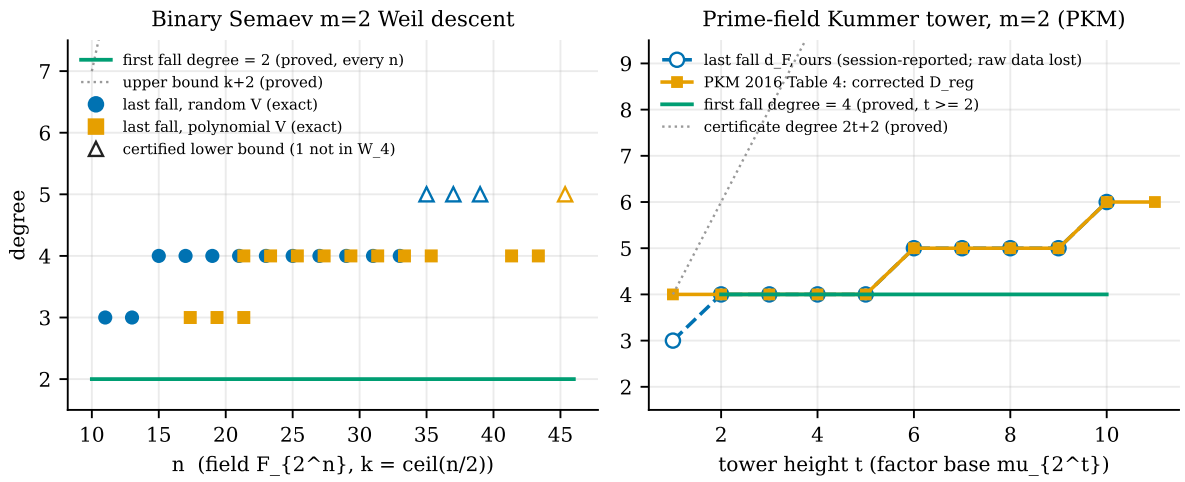


Figure 6.1: *Left:* binary Semaev $m = 2$ descent. The first fall degree is 2 (Corollary 6.6, proved). Filled marks are exact last fall degrees per unsatisfiable instance. Open triangles are certified lower bounds $d_{\bar{F}} \geq 5$ (Theorem 6.11). The dotted line is the proved upper bound $k + 2$. *Right:* prime-field Kummer tower, $m = 2$ (§6.5). The session-reported last fall degree lies on the corrected degree of regularity of PKM 2016, Table 4. The first fall degree is 4 (proved). Values and sources: `figures/data-new.json`.

instance	n	k	*subspace*	*certified* codim $W_4 \geq$
T33a, T33b	*33*	*17*	*random* $\subset K_0$	*4,888*
R35a, R35b, U35a, U35b	*35*	*18*	*random (U: uniform z)*	*5,287*
Q35a, Q35b (satisfiable)	*35*	*18*	*random*	*5,287*
R37a, R37b	*37*	*19*	*random*	*10,066*
P45a	*45*	*23*	*Semaev's default* $\langle 1, t, \ldots, t^{22} \rangle$	*12,742*
C19a, C19b (chain $m = t = 3$*)*	*19*	*7*	*random*	*9,253, 6,817*

Two further instances, R39a and R39b ($n = 39$)*, and C21a and C21b (the chain at* $n = 21$*) have* $1 \notin W_4$ *by implementation A alone.*

For Q35a and Q35b, which have 6 and 2 Boolean solutions, the bound follows from Proposition 6.2(4), since $5{,}287 \gg |S|$. The P45a closure stalls after three rounds at $\dim W_4 \cap B_{\leq d} = (0, 1, 224, 8950, 166705)$, out of 179,447 monomials. Its quadratic part, $224 = 5n - 1$, is exactly the polynomial-subspace count of Lemma 6.7: at degree 4 the closure finds no new falls.

Corollary 6.12 (Semaev's Assumption 1 fails as stated)**.** *Semaev's Assumption 1 [36, §4.5] asserts* $d_{F4} \leq 4$ *for Boolean systems equivalent to his system (5), for* $q = 2^n$, $k = \lceil n/m \rceil$*, any subspace of dimension* k *and any* $2 \leq t \leq m$*. Theorem 6.11 contradicts it in three cells:*

- $(n, m, t, k) = (35, 2, 2, 18)$*, with random subspaces;*
- $(45, 2, 2, 23)$*, with his own default subspace;*
- $(19, 3, 3, 7)$ *and* $(21, 3, 3, 7)$*, inside the parameter range of his tables, where his polynomial subspace gives 4 and random subspaces do not.*

Table 6.2 lists the chain instances. With Semaev's polynomial subspace the closure refutes at degree 4 in every cell; with random subspaces it does not at $n = 19, 21$.

What is new, and what is not. That the degree of regularity reaches 5 has been *observed* before:

- Kosters and Yeo saw Magma step degrees of ≥ 5 at $n = 40$ and $n = 45$ [28];
- Kousidis and Wiemers saw F4 at step degree 5 for $n = 37$–48, $m = 2$, and wrote that a lower bound on the degree of regularity "can potentially answer an open question" [29].

Table 6.3 sets their $m = 2$ rows beside ours at the same n. The quantities differ: D_{reg} is the highest step degree of their F4 runs, while $d_{\bar{F}}$ is the closure degree of the system, and no F4 run refutes below it. So their 5 at $n = 41, 43$ against our 4 on polynomial subspaces is not a contradiction; F4 overshoots the closure there. At $n = 35$ their $D_{\text{reg}} = 4$ agrees with our polynomial subspace and differs from every random subspace we drew, which are all ≥ 5: which subspace a table uses decides the degree it reports. At $n = 45$ both reach 5.

instance	$m=t$	V	n	k	$d_{\tilde{F}}$	cols at $D=4$	s at $D=4$	codim $W_4 \geq$
D17a	3	P	17	6	≤ 4	59,536	294.6	
D17b	3	P	17	6	≤ 4	59,536	297.4	
D19a	3	P	19	7	≤ 4	102,091	1283.7	
D19b	3	P	19	7	≤ 4	102,091	1047.1	
D21a	3	P	21	7	≤ 4	124,314	1270.3	
D21b	3	P	21	7	≤ 4	124,314	1225.4	
C17a	3	R	17	6	≤ 4	59,536	281.3	
C17b	3	R	17	6	≤ 4	59,536	270.9	
C19a	3	R	19	7	≥ 5	102,091	1088.5	9,253
C19b	3	R	19	7	≥ 5	102,091	1184.4	6,817
C21a	3	R	21	7	≥ 5	124,314	1806.2	
C21b	3	R	21	7	≥ 5	124,314	1790.4	

Table 6.2: Chained system, $m = t = 3$, $k = \lceil n/3 \rceil$: closure at $D = 4$ per instance. "≤ 4" means $1 \in W_4$; "≥ 5" means $1 \notin W_4$. Certified codimensions as in Table 6.1; C21a and C21b have no checked certificate: $1 \notin W_4$ by implementation A alone. Source: `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/results/instances_t3.json`.

Kousidis–Wiemers, Table 1 ($m=2$, F4)						this work, $d_{\tilde{F}}$	
n	n'	D_{ff}	D_{reg}	s	GB	P or S	R, U or T
34	17	2	4	188	1.2		
35	18	2	4	1,237	16.1	4	$\geq 5^{\text{R}}$; $\geq 5^{\text{U}}$
36	18	2	4	1,342	16.4		
37	19	2	5	2,542	29.2		$\geq 5^{\text{R}}$
38	19	2	5	2,815	25.2		
39	20	2	5	4,785	45.6		$\geq 5^{\text{R}}$
40	20	2	5	4,858	46.3	4^{S}	
41	21	2	5	7,930	65.3	4	
42	21	2	5	8,901	66.7		
43	22	2	5	16,816	95.5	4	
44	22	2	5	15,690	96.8		
45	23	2	5	38,352	140.0	≥ 5	
46	23	2	5	31,735	140.7		
47	24	2	5	103,200	207.7		
48	24	2	5	86,636	208.2		

Table 6.3: The $m = 2$ rows of Kousidis and Wiemers' Table 1 [29] (F4, averages over ten repetitions; $n' = \lceil n/2 \rceil$; time in seconds, memory in GB), transcribed from the published table, beside the last fall degree of our instances at the same n (Table 6.1). Superscripts give the subspace family. Their $m = 3, 4$ rows ($D_{\text{ff}} = D_{\text{reg}} = 7$ and 13, $n \leq 21$) concern S_4 and S_5 and have no counterpart here.

Those are properties of solver runs. Theorem 6.11 is a property of the systems, carried by a certificate any reader can check. With Corollary 6.6 it makes the separation between first and last fall degree rigorous on named instances. What remains open is the same as before: a proof that $d_{\tilde{F}} \to \infty$. Semaev's asymptotics survive if the regularity degree is $o(\sqrt{n/\ln n})$, so only a growing lower bound bears on the subexponential claim.

Re-verification. For this treatise, the T33a certificate was regenerated from the committed producer (M4RI 20200125, 4 min 40 s on 4 vCPUs) and checked with `cert_check.py`:

- the certificate is byte-identical to the recorded SHA-256 `cf9dcae8...22cf`;
- the checker reports "1 is not in $W_4(F)$, and codim $W_4(F) \geq 4888$" (38.8 s);
- the closure dimensions $(0, 0, 64, 2968, 48068)$ match the record.

Source: `research/paper_opportunities_20261029/REPORT.md`, §6.

6.4 Explicit membership over any field: the guarded chain

A second presentation encodes factor-base membership directly. Let S be a set of B abscissae of rational points, and $g(T) = \prod_{s \in S}(T - s)$. The two-$S_3$ chain with an inverse guard is

$$F_S = \{g(x),\ g(y),\ g(z),\ S_3(x, y, u),\ S_3(u, z, r),\ v(x - y) - 1\},$$

with $r = x(R)$. Degrees are ordinary total degrees with explicit inputs. The first fall degree is the homogeneous-top syzygy degree modulo Koszul syzygies. Source: `research/guarded_fall_degree_20261028/`, which holds written proofs, independent audits, a native exact replay and partial Lean.

Theorem 6.13 (Guarded chain)**.** *For unsatisfiable instances:*

1. *Short Weierstrass curves, $p > 3$, $B = 3$: $(d_{\mathrm{ff}}, d_F) = (4, 5)$,* exactly*, for every p.*
2. *Same presentation, any $B \geq 3$: $d_{\mathrm{ff}} = B + 1$ and $\max(5, B + 1) \leq d_F \leq 3B - 4$.*
3. *Ordinary binary curves $Y^2 + XY = X^3 + a_2X^2 + a_6$, $B = 3$:*
   - *$d_F = 4$ if and only if $g = T^3 + c$, which is possible only for even n;*
   - *$d_F = 5$ otherwise.*

   So for odd n, which covers every SEC 2 binary size, every unsatisfiable instance has $(4, 5)$.
4. *Binary curves, $B \geq 4$: $B + 1 \leq d_F \leq 3B - 4$.*

Proof sketch of (1). Upper bound. The divided difference $Q = (g(x) - g(y))/(x - y)$ enters V_4 through the guard. The pair is then replaced by the missing root $w = s_1 - x - y$. The two quadratics in u are normalised and their resultant extracted in stages inside V_5. Finally the unit $T(w, z)$ is inverted in $K[w, z]/(g(w), g(z)) \cong K^9$.

Lower bound. The space $W = (I_0 \cap R_{\leq 4}) + Kf_1 + Kf_2$ is closed under the degree-4 operations and excludes 1. This uses characteristic > 3, through $-3 \neq 0$. □

Exact degree-5 refutation DAGs were replayed in native arithmetic on public fixtures:

- $p = 101$ and $p = 5$;
- secp256k1, P-256, P-384 and P-521, with $S = \{x(G), x(2G), x(4G)\}$ and $R = 16G$; the DAGs have 413–415 nodes, and OpenSSL rechecks the points.

Twenty polynomial identities used in the proofs are checked in Lean (`FallDegreeIdentities.lean`, `PrimeBitLocal.lean`; axioms `propext` and `Quot.sound` only). The degree-4 lower bound is now machine-checked: `GuardedModel.no_degree_four_refutation` proves that 1 is not derivable at degree four from the six generators, over every field with $6 \neq 0$ and three distinct base abscissae, and `lastFall_lower` states the last-fall consequence under an explicit unit-ideal hypothesis (`research/guarded_fall_degree_20261028/formal/GuardedModel.lean`; six modules, fresh `leanchecker` replays and a transitive axiom audit, merged in #2814). The degree-5 upper construction, the bridge from rational unsatisfiability to the unit ideal, and the first-fall syzygy statement are *not* formalised, so the exact pair $(4, 5)$ is not yet a Lean theorem. A `lastFallDegree` definition and Caminata–Gorla's Theorem 2.8 in `crypto-autoresearcher:formal/CryptoResearch/FallDegree/LastFall.lean` are the bridge a complete formalisation would build on.

Two Boolean upper bounds accompany the guarded chain:

- $d_F \leq 3\ell + 1$ for the fully Boolean binary encoding, with ℓ bits per base coordinate;
- $d_F \leq 3\ell + 2$ for a prime-field bit-restricted lookup encoding.

Both are instances of the standard brute-force refutation in Polynomial Calculus [6]: fix the base bits, and refute the affine residue. For the binary chain at $t = 3$ they are weaker than Theorem 6.8's $2k + 3$.

6.5 Prime-field Kummer towers

Petit, Kosters and Messeng [33] take the factor base $V = \mu_{2^t}$ over $\mathbb{F}_p$ (with $2^t \mid p - 1$). They write membership as a tower $y_0 = x/g$, $y_{j+1} = y_j^2$, $y_{t-1}^2 = 1$, and argue the complexity from the first fall degree. Source for this section: `research/prime_fall_degree_20261027/DERIVED.md`, written proofs D1–D8.

Theorem 6.14 (Kummer towers, $m = 2$)**.** *Let $R = \mathbb{F}_p[y]/T$ be the tower quotient. Items (3)–(5) concern* inconsistent *systems, those whose reduced S_3 is a unit of R; a satisfiable target has no inverse in R, and the refutation bounds do not apply to it.*

1. *T is a Gröbner basis, and $R \cong \mathbb{F}_p^{V^m}$. So every ideal over R is radical, and an $m = 2$ system is inconsistent if and only if the reduced S_3 is a unit of R.*

t (PKM: n_1)	D_{ff} (proved)	d_F (session median)	PKM $D_{\mathrm{reg}}^{\mathrm{corr}}$
1		3	4
2	4	4	4
3	4	4	4
4	4	4	4
5	4	4	4
6	4	5	5
7	4	5	5
8	4	5	5
9	4	5	5
10	4	6	6
11			6

Table 6.4: Prime-field Kummer tower, $m = 2$, factor base μ_{2^t}: the proved first fall degree (Theorem 6.14(3); it needs $t \geq 2$), the session-reported median last fall degree at $p = 2{,}013{,}265{,}921$ (raw data not committed), and the corrected degree of regularity $D_{\mathrm{reg}}^{\mathrm{corr}}$ of Petit, Kosters and Messeng, Table 4 (prime case) [33]. Source: `figures/data-new.json`, from `research/prime_fall_degree_20261027/`.

m	ℓ	n	d_F
3	5	10–15	4
	6	16–24	4
	7		≥ 5 (every draw)
4	3	9–14	4
5	3	11–15	4

Table 6.5: Binary decomposition systems in other presentations: last fall degree d_F by m and base-subspace dimension ℓ, over the measured range of n. At $(m, \ell) = (3, 7)$ every draw is refuted only at degree ≥ 5. Source: [3, §6].

2. *$R = \mathbb{F}_p[z]/(z^K - 1)$ with $K = 2^t$. The degree of a monomial z^k is* popcount(k)*, and degree falls are exactly binary carries.*
3. *If $t \geq 2$, $g \neq 0$, $3x_R^2 + a \neq 0$ and the system is inconsistent, then $d_{\mathrm{ff}} = 4$ (operational convention), for every t. The degree-2 part of f is a bilinear form with $\det M = -2g^6(3x_R^2 + a) \neq 0$, so nothing falls at 3; the monomial $q = y_{10}y_{11}$ gives a fall at 4.*
4. *For an inconsistent system, the no-fall certificate has degree $d_F \leq 2 + \mathrm{popdeg}(f^{-1}) \leq 2t + 2$.*
5. *For an inconsistent system in the naive presentation $\{S_3,\ x_i^K - g^K\}$ of the* same *set, $d_F \geq 2^t$. So d_F is not an invariant of the ideal; it depends on the presentation.*
6. *A single summand has $d_F = 2$ for every t. Growth at $m = 2$ must therefore come from the genus-one curve $f(z_1, z_2) = 0$; no specialisation to finitely many points can show it.*

Measured, and prior. Table 6.4 sets the study's session-reported median last fall degree for $t = 1..10$ at $p = 2{,}013{,}265{,}921$ beside the corrected degree of regularity in PKM's own Table 4 for $n_1 = 1..11$ (also Figure 6.1, right). The raw records were never committed (the study's `RESTART.md`), so the d_F column is session-reported, not re-derivable. From $t = 2$ on the two columns coincide.

So the *growth* is prior art. What Theorem 6.14 adds is the proved constant first fall degree beside it, and the presentation dependence. A prediction, $d_F = 4 + \lfloor (t-2)/4 \rfloor$, was registered before any $t \geq 11$ row and is untested.

6.6 Measured solving degrees in other presentations

The binary programme measured refutation and closure degrees on many presentations; the full record is in [3, §6]. Table 6.5 lists the last fall degree by base-subspace dimension ℓ. It is flat in n at fixed (m, ℓ) and rises with ℓ.

Three further measurements bear on the solver rather than the system:

- *Plain refutation tracks the semi-regular degree.* The plain Macaulay refutation degree stays within one of the semi-regular prediction in all 14 exact cells.

- *Closure algorithms refute one or two degrees earlier.* A MutantXL-style closure refutes all 92 unsatisfiable draws at degree 3–4, where XL needs 5–7.
- *Guessing variables restores degree 4.* Fixing one or two variables restores degree 4 at $(m, \ell) = (3, 7)$, at 1/116–1/260 of the projected degree-5 cost.

Every observed growth is growth in the size of the factor base: in ℓ, k, t, B, or the digit count s of a prime digit presentation (where $d_{\text{ff}} = d_F = 4 + \lfloor s/2 \rfloor$ at $m = 2$, reproduced by a curve-free null). None was observed in n at a fixed base.

6.7 What this settles and what it does not

- *Settled.* In the three presentations the programme studied, the first fall degree is a proved constant of the presentation: 2 for binary Weil descent, $B + 1$ for explicit membership, 4 for Kummer towers. The last fall degree is not bounded by it.
  - This is proved for the guarded chain at $B = 3$ (exactly 5 against 4).
  - It is certified on named binary instances: ≥ 5 against 2, at $n = 33$–45 and in the $m = t = 3$ chain.
  - It is measured for Kummer towers (6 against 4 at $t = 10$).

  Any complexity argument that substitutes the first fall degree for the solving degree is unsupported for these systems.
- *Not settled.* No presentation has a proved last-fall lower bound that grows with n. That is the open problem of HKY [23] and Kousidis–Wiemers [29], and the only result that would bear on the subexponential claims. The upper bounds $k + 2$ and $(t - 1)k + t$ are linear in n, far above every observed value.
- *Missing before submission.* The certificates are committed only as hashes; they regenerate (above) but need public hosting. The sample is small: two instances per n at $n \geq 33$, one polynomial instance at $n \geq 41$. The written proofs need expert review. The Kummer-tower data must be regenerated (`research/paper_opportunities_20261029/`).

7 What index-calculus hardness rests on

Chapter 6 studied one quantity, the solving degree of the point-decomposition system, and showed that its first fall degree does not bound it. A complexity claim for elliptic-curve index calculus needs more than a degree. It needs a bounded solver cost, a decomposition yield with a known distribution, relations of sufficient rank, a usable geometry, an interface that no generic algorithm reproduces, and an end-to-end composition of all of these. Each of the subexponential claims in the literature discharges some of these obligations by proof and the rest by an assumption. This chapter records, assumption by assumption, what the programme proved, refuted, measured or left open between 2 and 4 November 2026.

The work falls into four kinds:

- a typed obligation chain and a reanalysis of committed evidence (`research/ic_weak_assumptions_20261102/`);
- exact measurements of six weak points on Weil-descent systems (`research/ic_weakpoints_20261102/`);
- exact censuses against five named heuristics (`research/ecdlp_weakness_census_20261104/`) and Lean 4 theorems aimed at the same nodes (`research/ecdlp_weakness_lean_20261104/`, `research/lean_attacks_20261102/`, `research/inside_gap_n4_20261102/`);
- a claim graph of the hardness literature and a triage of its weak points (`docs/ecdlp-hardness-corpus/WEAKNESS_TRIAGE.md`).

Standing. None of this work is an ecbench session, a leaderboard row or a validated result. No validator report or replay certificate exists for any census or measurement below. Every measured number in this chapter is therefore an unverified assertion of its producer under the proof-carrying-claims rule; the census README marks its numbers `[pcc:unverified]`. The Lean results are stated with their exact formal status in §7.11: "kernel-checked" is used only where a Lean kernel accepted the declaration and its source contains no `sorry`, `admit`, `native_decide` or user axiom. The corpus records every internal theorem here as `proved_internal_unreviewed`: no external reviewer has read the statements against the literature they address. Nothing in this chapter changes a row of Tables A–C, and none of it is a measured improvement of the index-calculus baseline.

7.1 Six serial obligations

The weak-assumption programme of 2 November (`research/ic_weak_assumptions_20261102/README.md`; literature pass and reanalysis of committed records, no new run) writes an index-calculus security argument as a chain of six obligations. The chain is serial in the sense that matters: failure of any one edge invalidates the complexity claim, whatever the others show. Its machine-readable form, `obligations.json`, is checked by `audit.py`, a fail-closed gate that refuses to promote a route while any field is missing. It covers naive prime-field Semaev, generic pair tables, Kummer (PKM) towers, the proposed arity-three hybrid solver, Wagner filters, the hashed-base tensor route (§24.1) and xedni. The route-by-route reading for prime fields is in §22.7; Table 7.1 lists the obligations with the proxy usually offered for each and the section of this chapter that bears on it.

A stable degree does not certify cost. The reanalysis extracted the committed PKM $m = 4$ random-target F4 rows, two per cell (`research/ic_weak_assumptions_20261102/evidence-summary.json`, key `pkm_m4_random_f4`; `extract_existing.py -check` recomputes it from the JSONL records):

- at $N = 8, 12, 16$ the solving degrees are $6, 7, 7$;
- the maximal column counts are 1,049, 9,452 and 76,837;
- the maximal multiply-add counts are 48,447,128, 85,477,602,933 and 27,048,206,854,510, that is $4.84 \cdot 10^7$, $8.55 \cdot 10^{10}$ and $2.70 \cdot 10^{13}$.

From $N = 8$ to $N = 16$ the columns grow 73.25-fold and the arithmetic 558,304-fold, although the last two degrees are equal. Three cells are not an asymptotic fit. They are a direct counterexample to reading a bounded degree as a bounded cost: the cost of a Gröbner computation at a fixed degree d is governed by the number of monomials of degree $\leq d$ in the variables of the system, and that number grows with N. The same six rows are all `inconsistent=true`: their ideal is the unit ideal. So the `zero_dim` flag on a random target refutes a decomposition and says nothing about the geometry of a successful one. The

Obligation	What it needs	Proxy often used instead	Here
Solution geometry	dimension and radicality of the full system, rational lifts, exceptional charts, target strata	`zero_dim=true` on a random target	§7.2 (W5)
Solver growth	maximal matrix dimensions, nonzeros, arithmetic and memory, with timeouts and out-of-memory exits kept	first fall degree; a stable solving degree	§7.1, §7.3, Chapter 6
Yield and variance	rational decompositions per natural target: distribution, zero mass, multiplicity, solver failures	the mean count B^m/r	§7.7
Relation rank	rank growth, dead columns, full-rank probability	raw row count	§7.6
Non-generic interface	a coordinate predicate whose useful candidates no generic table reproduces at equal cost	"uses coordinates"	§7.9
End-to-end composition	one parameter family charging set-up, failed queries, collection, verification, linear algebra, descent, memory	product of the separate bounds	§7.2 (W6)

Table 7.1: The six obligations of an index-calculus complexity claim, after `research/ic_weak_assumptions_20261102/README.md`. The README's table also states, per obligation, the "strongest next falsifier"; the right column points to the evidence in this chapter.

planted toy rows at $N = 8, 12$ are consistent, with finite bases of 28 and 49 elements, and are the positive control. The README's reading is that the useful theoretical target for PKM is now a lower bound on the last fall or closure degree, or on the matrix width, and not another first-fall plot; the $N = 20$ cell is projected at 240–380 GB and is not to be launched to reconfirm the trend. The solving-degree data of the Kummer towers themselves are in §6.5 and §23.1.

The follow-up. The same day's follow-up (`research/ic_weak_assumptions_20261102/FOLLOWUP.md`) added an arbitrary-precision mirror of the tensor route's D.4 admission guards and a finite-field unit-explanation proof kernel for the arity-three canonical representation, which it describes as validated implementation artefacts. It took no scientific run. Its status line per lane: the tensor route's predicate cost and aggregate complexity remain uncertified; the PKM closure-dual target is an exact nullspace certificate but no growing certificate has been found; the Wagner bucket statistic W has no frozen native producer; for xedni no theorem comparing the selected height with the global minimal discriminant is proved.

7.2 Six weak points measured on Weil-descent systems

The obligations were then measured, as exact counts, on one family where every one of them can be computed: Gaudry–Diem descent for $E : y^2 = x^3 + ax + b$ with random $a, b \in \mathbb{F}_{p^n}$, $n \in \{2, 3\}$, factor base $\{P : x(P) \in \mathbb{F}_p\}$, groups up to 2^{28} for the complete solves (`research/ic_weakpoints_20261102/README.md`; protocol frozen before data, `PROTOCOL.md` with Amendments 1–3a; native Rust tool, standard library only, one 4-vCPU container). The README states that it is not an ecbench session, not a leaderboard row and not admitted by the validation gate, and reports no wall time. Table 7.2 lists the six points in the README's numbering.

The README's reading is that on this family W3, W4 and W5 are cleared, with cheap filters for their failure modes (subfield targets, duplicate rows, spurious solutions), that W1 is a constant and not a growth, and that the factor-base size is a seventh, required lever. Its derived figures, which are models and not measurements:

- *No large primes.* With $\tau = 3.50$ trials per p, $c(3) = 1.06 \cdot 10^6$ and 3.97 nonzeros per row, the best factor-base fraction gives IC/rho $\to 684$ at $w = 0.1$ and 3,761 at $w = 1$, constant for groups of 2^{60} and more. Without large primes $n = 3$ balances at $p^{3/2}$, rho's exponent, so the gap is a constant proportional to $\sqrt{w\, c(3)}$.

Weak point	Measured (exact counts)	Producer's reading
W1 solving degree, matrix growth	$n = 3$: solving degree 10 in 88/88 systems, $p = 31$ to 1,073,741,789 (2^5–2^{30}); Macaulay matrix 252×286 at that degree, $c(3) = 1.03$–$1.07 \cdot 10^6$ $\mathbb{F}_p$ multiplications (dense, an upper bound for F4); no first fall in 87/88. $n = 2$: degree 3, 6×10, about 87 multiplications	a p-independent constant $c(n)$; its size, not its growth, is the question
W2 non-generic filter	not addressable on this family (the subfield base is the structure; $\mathbb{F}_p$ is untouched)	open; the prime-field question
W3 yield mean and variance	20 curves, 2,000 or 6,000 targets: mean/$\lambda_{\text{pred}} = 0.88$–$1.19$ ($\lambda = F^3/6N$); $P(0) = e^{-\text{mean}}$ to three decimals; dispersion interval contains 1 on 19/20 curves; the two heavy tails are subfield targets $x_R \in \mathbb{F}_p$	Poisson; drop subfield targets
W4 relation rank	15/15 complete solves return the correct $\log Q$; rank shortfall 0 on 7 curves, 1–3 on 7, 6 on one; cofactor scaling, duplicated rows and dropped signs as controls	not a blocker at this scope
W5 geometry	zero-dimensional 176/176; quotient dimension 64 (Bézout) on 87/88, 4 at $n = 2$; at $p \le 127$, 24 targets: 32 rational solutions, of which 5 genuine and 27 non-split (84%), 2 singular, no positive-dimensional component	benign; about five rejections per relation
W6 end-to-end cost against matched rho	complete toy solves, $\ell = 2^{12}$–2^{25}; IC/rho (negation map, analytic $\sqrt{\pi\ell/4}$): 0.29–3.4 at $w = 0$, $2.8 \cdot 10^4$–$3.2 \cdot 10^5$ at $w = 0.1$, $1.5 \cdot 10^5$–$1.7 \cdot 10^6$ at $w = 1$; rho ahead on 7 curves, 8 unresolved, IC ahead on none	4–5 orders of magnitude remaining at $w \ge 0.1$

Table 7.2: The six weak points on Gaudry–Diem descent over $\mathbb{F}_{p^n}$, $n \in \{2, 3\}$, as reported in `research/ic_weakpoints_20261102/README.md`. Unverified producer figures; not an ecbench session; not admitted by the validation gate. The eight unresolved curves are unresolved only because $w = 0$ makes the Gröbner work free.

- *Double large primes.* At $|\text{FB}_r| \approx p^r/2$ the fitted exponent of trials to a square system is 1.27 ± 0.04 at $r = 2/3$ (theory 4/3, rho 1.5), from $p \le 2039$; $\log Q$ was correct in 105/105 reached cells. Extrapolated with $c(3)$ and the measured row weights, 2LP reaches IC/rho of 58, 7.7 and about 1 at 2^{96}, 2^{132} and 2^{180} at $w = 0.1$, and 319, 42 and 5.3 at 2^{96}, 2^{132} and 2^{180} at $w = 1$, where it does not cross 1 by 2^{240}. The band therefore ranks IC ahead nowhere up to 2^{240}: above 2^{180} the order flips between $w = 0.1$ and $w = 1$. The README calls this crossover optimistic: the fits are extended by up to 70 bits, and the 2LP slopes at $r = 3/4$ and $5/6$ are probably pre-asymptotic.

For the theory of this chapter the result is that, on the one family where the theorems of Gaudry and Diem apply, the obligations other than W1 and W6 hold at toy size, and the distance that remains is a constant set by the cost of one decomposition. For prime fields, where W2 is the open obligation, the family says nothing.

7.3 Petit–Quisquater Assumption 2

Petit and Quisquater [32] derived their heuristic $2^{c\, n^{2/3} \log n}$ for binary curves (§3) from an assumption that the corpus records as `HEUR-GB-PQ-ASSUMPTION-2`: the degree of regularity, read as the maximal degree a Gröbner solver reaches, of a Weil-descended summation system equals its first fall degree up to $o(1)$. At $m = 2$ the first fall degree of the descended S_3 system is 2 for every n (Kosters and Yeo [28], Corollary 4.11; Corollary 6.6 in our Caminata–Gorla convention), so the literal reading predicts solving degree 2. Kousidis and Wiemers [29] read the assumption as $D_{\text{reg}} = m^2 - m + 1 + o(1)$, which is $3 + o(1)$ at $m = 2$. Degrees are integers, so "$+o(1)$" means eventual equality, and a lower bound that exceeds the predicted value for infinitely many n refutes that reading for the family.

Theorem 7.1 (Subspace criterion; kernel-checked)**.** *Let K be a field, $I \subseteq K[x]$ an ideal with $1 \notin I$, and $\prec$ a degree-compatible monomial order. Suppose I contains k polynomials whose leading monomials are k distinct variables. If a Gröbner basis of I lies in a finite-dimensional subspace S, then $k \le \dim_K\big(S \cap K[x]_{\le 1}\big)$. In particular, if S is the span of the Macaulay rows of degree $\le d$ and $\dim(S \cap K[x]_{\le 1}) < k$, no Gröbner basis of I lies in S.*

n	instances	iterated degree 3 reaches k	d_{iter}	d_{lin}
8–14	21	21	3	4 (20), > 4 (1)
15	3	0 (`iter3` $= 1$)	4	> 4
16	3	3	3	> 4
17–20	12	0 (`iter3` $= 1$)	> 3	> 4
21–44	72	0 (`iter3` $= 1$)	> 3	> 3
45–47	8	not run	–	> 3
17–24 (fresh)	24	0	4 (24/24)	

Table 7.3: `pq2_rank`: exact degree lower bounds for the $m = 2$ descended S_3 system over $\mathbb{F}_{2^n}$, 119 instances in the main sweep. The last row is two separate degree-4 sweeps at $n = 17$–20 and 21–24 on fresh instances. At every n, `low2` $= 1$; `low3` $= 1$ for every $n \geq 9$. Unverified census figures. Source: `research/ecdlp_weakness_census_20261104/pq2_rank/README.md`.

The Lean declarations are `SolvingDegreeBound.linear_count_of_gb_in_subspace` and `SolvingDegreeBound.no_gb_in_rows_of_low_lt` (`research/ecdlp_weakness_lean_20261104/lean/SubspaceCriterion.lean`). Its degree-2 special case, `solving_degree_ge_three` in `SolvingDegreeBound.lean`, adds the rank–nullity count $\dim(\operatorname{span} F \cap K[x]_{\leq 1}) + \operatorname{rank}(\operatorname{top}_2 F) \leq |F|$: for generators of degree 2, field equations included, if $k + \operatorname{rank}(\operatorname{top}_2 F) > |F|$ then the solving degree is at least 3. A non-vacuity instance over $\mathbb{F}_2$ is $F = \{x_0^2 + x_0,\ x_1^2 + x_1,\ x_0x_1 + x_0 + x_1\}$, with $k = 2$, rank 3 and $|F| = 3$. A further declaration, `not_tendsto_of_frequently_gap`, proves that an integer gap of at least 1 infinitely often is not $o(1)$.

Write d_{lin} for the least d at which the degree-$\leq d$ Macaulay row space contains k independent polynomials of degree ≤ 1, and d_{iter} for the same quantity computed on the iterated degree-d closure (every multiple of a newly found polynomial that stays within degree d is added until the rank stops growing). By Theorem 7.1, d_{lin} is a lower bound on the Caminata–Gorla one-shot solving degree (Proposition 6.2 relates it to the last fall degree), and d_{iter} is a lower bound on F4's maximal step degree under the containment property that F4's output at maximal step degree D lies in the degree-D closure. That property is argued in the census README and is not formalised. The number k of linear consequences is computed as $N - \operatorname{affdim}(Z)$ from the enumerated solution set Z; the Boolean Nullstellensatz step behind it is kernel-checked only for $|Z| = 1$ (`BooleanNullstellensatz.linear_mem_of_unique_zero`), and the general affine-span case is not formalised.

The census. `pq2_rank` (`research/ecdlp_weakness_census_20261104/pq2_rank/README.md`; Rust, no dependencies, seed `0x5eed20261104`) computes these quantities exactly over $\mathbb{F}_2$, with no Gröbner run, on the curve $y^2 + xy = x^3 + ax^2 + b$ with a random $\lceil n/2 \rceil$-dimensional factor-base subspace and a target with a non-empty solution set. Each instance checks, on 20 random assignments, that the descended equations agree with a direct S_3 evaluation. The main sweep was stopped during $n = 47$, before the binary wrote its own JSON; a script copied the log lines and recomputes nothing. $n = 48$–56 were not reached. Table 7.3 gives the results.

- *Literal Assumption 2 fails on all 119 instances:* $d_{\text{iter}} \geq 3$ everywhere, against the predicted first fall degree 2.
- *The Kousidis–Wiemers value is not reached* at any of the 87 instances with $n = 15$ or $17 \leq n \leq 44$: the iterated degree-3 closure reaches only one of the k linear consequences. On the 24 fresh instances at $n = 17$–24 the iterated degree is exactly 4.
- *Cross-checks.* A pure-Python closure by naive saturation, sharing no code with the Rust one, agrees on five dumped instances at $n = 14$–18, at degrees 3 and 4. A same-seed rerun for $n = 8$–40 reproduces the log line for line; it also gives $|F'|/|V|$ between 0.84 and 1.31 (mean 1.021) over 99 instances, and 0.997–1.004 for $n \geq 30$.

What this does and does not settle. A finite range cannot refute an $o(1)$ statement. It refutes the literal reading on every instance computed, and shows that the asymptotic value 3 is not reached anywhere from $n = 17$ to 44. The bound on F4's step degree rests on the unformalised containment property; the bound on the one-shot solving degree is a kernel-checked criterion applied to an unverified computation. The measured degree sits above the value the subexponential binary heuristics assume, which removes support from them; it does not support a subexponential algorithm. The corpus node that records the census, `MEAS-PQ2-DLIN-CENSUS`, obstructs `HEUR-GB-PQ-ASSUMPTION-2`, which remains a heuristic in the corpus. The certified binary instances of Theorem 6.11 reach $d_{\tilde{F}} \geq 5$ at $n = 33$–45 on

random subspaces of dimension $\lceil n/2 \rceil$; together the two give a lower bound of 4 at $n = 17$–24 by census and of 5 at $n \geq 33$ by certificate. What remains open is the same as in §6.7: a proof that the bound grows with n.

7.4 Semaev's Assumption 1

Semaev [36] states two degree assumptions for his chained systems over $\mathbb{F}_{2^n}$. Assumption 1 (§4.5) asserts $d_{F4} \leq 4$ for any subspace of dimension $k = \lceil n/m \rceil$ and any $2 \leq t \leq m$, supported, as the corpus records, by 2,300 Magma systems with $t = m$, $n \leq 21$, $m \leq 6$. Corollary 6.12 refutes it as stated, in the cells $(35, 2, 2, 18)$, $(45, 2, 2, 23)$ with Semaev's own default subspace, and $(19, 3, 3, 7)$, $(21, 3, 3, 7)$ inside his tables' range. The revision of the paper (ePrint 2015/310, 10 April 2015) replaces it with a growth condition on the regularity degree that the corpus records as $o(\sqrt{n/\ln n})$ (`HEUR-SP-SEMAEV15-WEAK-REGULARITY`), sufficient for the $2^{c\sqrt{n \ln n}}$ asymptotic and not for the FIPS-curve conclusions. Nothing the programme proved bears on that weaker condition: only a growing lower bound would.

His first-fall-degree assumption $d_{F4} \leq d_{\mathrm{ff}}$ (§4.4), applied to his own $t = 2$ cells with his statement that the first fall degree "is always 4", was tested on 200 instances of the ten cells of his Table 2 (`research/semaev_t2_trace_20261102/REPORT.md`; protocol frozen; exact standard-library arithmetic). The first fall degree is 2 on 200/200, each with an explicit trace certificate: with $\lambda = z^{-2}$,

$$\mathrm{Tr}\big(z^{-2} S_3(x_1, x_2, z)\big) = \mathrm{Tr}(x_1) + \mathrm{Tr}(x_2) + \mathrm{Tr}(B z^{-2}),$$

a constant combination of the coordinate equations of degree ≤ 1, the Kosters–Yeo identity in Semaev's coordinates. None of the 200 systems has a solution in $V \times V$ (brute force). The degree at which a Macaulay computation proves this is 2 on 154 instances and 3 on 46, never 4. So the statement "first fall degree always 4" is refuted, while the inference that these rows refute $d_{F4} \leq d_{\mathrm{ff}}$ splits: the sharp assumption fails on 46/200, where 3 exceeds 2, and Semaev's reported $d_{F4} = 4$ exceeds what the systems require. The corpus inference that had read the rows as a refutation everywhere is corrected in Chapter B. The `RT-SP-SEMAEV-A1-N45` target of the triage, an exact F4 run in the $n = 45$ cell, is still listed as open.

7.5 Nagao's Lemma 4: four cells and three repairs

Nagao's ePrint 2015/984 [31], which the corpus records as "polynomial-time binary ECDLP under FFDA" (status conditional), bounds the first fall degree d_F by a "fake" first fall degree d'_F computed modulo the field equations (Lemma 4: $d_F \leq d'_F$). Remark 6.1 gives the equality-reading counterexample over $\mathbb{F}_2$. The lemma's verdict depends on two choices: the reading of Definitions 5 and 6 ("equality": the maximal degree of the products is exactly d; "literal $\geq$") and the placement of the field equations (outside or inside the generating set). Table 7.4 lists the four cells and the three candidate repairs tested after the refutation.

The fourth cell. `NagaoLiteralInside.lemma4_literal_inside` proves, for any field, any family F and any finite family E of "field equations" placed among the generators, that $d_F \leq d'_F$ whenever a fake fall exists. The proof lifts a fake fall: replace the sum by a least-degree representative modulo $\langle E \rangle$ and pay the difference with multiples of E, which is allowed because E is inside. The file's comment records why the argument fails outside, or under the equality reading, where the added multiples can push the maximal degree above d. The reading axis is now complete, refuted in three cells and proved in the fourth. Which cell Nagao meant is a question of the source text: the triage lists a fresh PDF-level transcription of Definitions 5 and 6 as owed.

A1: outside, the fake degree bounds nothing. Modulo the field equations $X^k Y \mapsto XY$, so every W_k has the reduced system of the original witness and $d'_F = 2$. With the field equations outside, $X^{k-1}(XY + X) + X^k Y = X^k$ is a fall at $k + 1$, and no fall exists below it, so $d_F = k + 1$: the native table gives $3, 4, \ldots, 8$ for $k = 2, \ldots, 7$, and 3 for every k with the field equations inside (`research/lean_attacks_20261102/results/family.txt`). `IcNegative.FakeDegreeGap.gap_two_witnessW3` kernel-checks $k = 3$ ($d'_F = 2$, $d_F = 4$) in core Lean with `decide +kernel`; the repository proof gate passed it in run 41399938870 (target `papers/ic-negative-results/lean` PASS, 40 theorems, 0 trusted). The all-k statement is the hand proof plus the table.

Statement	Result	Formal status
Lemma 4, equality, outside	refuted: $\{X^2Y, XY+X\}$, $d'_F = 2 < d_F = 3$ (Remark 6.1)	two implementations
Lemma 4, equality, inside	refuted	corpus record
Lemma 4, literal $\geq$, outside	refuted ($d_F = 3$, $d'_F = 2$)	corpus record
Lemma 4, literal $\geq$, inside	*theorem*: every fake fall is a true fall at the same degree, so $d_F \leq d'_F$, over any field	kernel-checked (Mathlib)
A1: $d_F \leq f(d'_F)$, outside	refuted: $W_k = \{X^kY, XY+X\}$ has $d'_F = 2$, $d_F = k+1$	$k=3$ kernel-checked; all k by hand proof
A2: $d_F \leq d'_F + 1$, inside	refuted: $\{YZ+1, XZ+X+1\}$, $d'_F = 1$, $d_F = 3$; 10,752 of 521,731 pairs have gap 2	native, two implementations
Inside gap, $n \geq 2$	$d_F - d'_F \leq 3$, tight ($\{XYW+XY+W, XZW+XW+1\}$); unbounded for $n = 1$	key identity kernel-checked in two variables
A3: $d'_F \leq d_F$ when the reduced generators are independent	survives: 0 counterexamples in 25,536 and 322,896 independent triples	census only

Table 7.4: Nagao's Lemma 4 over $\mathbb{F}_2$ by reading and placement, and the candidate repairs. Sources: `docs/ecdlp-hardness-corpus/WEAKNESS_TRIAGE.md` (W4), `research/ecdlp_weakness_lean_20261104/lean/NagaoLiteralInside.lean`, `research/lean_attacks_20261102/REPORT.md`, `research/inside_gap_n4_20261102/REPORT.md`.

A2 and the inside gap. The census of all 521,731 pairs of degree ≤ 2 in three variables (`research/lean_attacks_20261102/results/p3gap.txt`) finds gap 2 inside on 10,752 pairs; the witness $\{YZ+1, XZ+X+1\}$ has a fake fall at 1 and, inside, the true fall $Z(XZ+X+1)+X(Z^2+Z) = Z$ at 3. Two implementations agree (a Rust census and a Python brute force over every admissible multiplier tuple); the witness has three variables and is outside the two-variable Lean encoding. The open question this left was answered the same day (`research/inside_gap_n4_20261102/REPORT.md`; protocol frozen): for $n \geq 2$ variables, $d_F(F \cup S_{\text{fe}}) \leq 4$ for every system, because $Y^2(X^2+X)+X^2(Y^2+Y) = XY^2+X^2Y$ is a fall at 4 built from the field equations alone, so the inside gap is at most 3; a gap-3 witness in four variables was found in a sample of 2,000,000 pairs, and for $n = 1$ the family $\{X^k, X^k+1\}$ has gap $k-1$. `InsideGap.lean` states the field-equation fall in two variables in core Lean; the gate passed it in run 41484047855 (44 theorems, 4 new, 0 trusted). The step "zero multipliers on further generators leave the fall intact" is argued, and the four-variable witnesses are native only. The consequence is that inside-placement d_F under the equality reading is degenerate: it never exceeds 4 in two or more variables, through the trivial falls among field equations that Ding–Hodges and Petit–Quisquater exclude from their first fall degree. No estimate can take its degree from that quantity without first removing them.

A3 and Nagao's Proposition 5. The reverse repair, $d'_F \leq d_F$ outside when the reduced generators are linearly independent, has no counterexample among the triples tested: 266 counterexamples among the dependent triples of degree ≤ 2 in two variables, none among the 25,536 independent ones, and none among the 325,500 multilinear triples in three variables (`research/lean_attacks_20261102/REPORT.md`). Nothing here touches Nagao's Proposition 5 or the paper's own applications, which check unreduced degrees instance by instance; only a bound through d'_F alone is ruled out, and only outside.

7.6 Relation rank and the left kernel

The corpus node `HEUR-RELATION-RANK` conditions sparse linear algebra and, through it, marked three externally proved results as resting on a heuristic: Theorem 1 of Gaudry, Thomé, Thériault and Diem [22], Diem's small-genus theorem [9] and the GHS Weil-descent attack [21]. These are 4 of the 17 status inflations of §7.10. Galbraith and Gaudry [16] (§6.2) already note that target-dependent relations are easier to analyse rigorously than relations reused across targets. The triage wrote the argument out (W2) and the Lean round of 4 November formalised its finite core.

Theorem 7.2 (Left-kernel relations; kernel-checked)**.** *Let K be a field and $M \in K^{n \times B}$ a relation matrix with $a_i + b_i x = \sum_j M_{ij} f_j$ for $i = 1, \ldots, n$, where f_j are the factor-base logarithms and x the unknown logarithm.*

1. *If $n > B$, a nonzero $\lambda \in K^n$ with $\lambda M = 0$ exists, for every M.*
2. *For any such λ, $\sum_i \lambda_i a_i + x \sum_i \lambda_i b_i = 0$; hence $x = -\sum \lambda_i a_i / \sum \lambda_i b_i$ whenever $\sum \lambda_i b_i \neq 0$.*
3. *Let K be finite with q elements and let Λ be any function of the vector of relation logarithms $\ell = (a_i + b_i x)_i$ alone whose value is always a nonzero vector. Exactly q^{2n-1} of the q^{2n} pairs (a, b) give $\Lambda(\ell) \cdot b = 0$: the failure density is exactly $1/q$.*

The declarations are `LeftKernel.exists_left_kernel`, `left_kernel_identity`, `left_kernel_recovers` and `card_bad_pairs` (stated without subtraction as $|\text{bad}| \cdot \text{q} = \text{q}^{2\text{n}}$) in `research/ecdlp_weakness_lean_20261104/lean/LeftKernel.lean`. Two steps are not formalised: that a decomposition of $R_i = a_i P + b_i Q$ gives the row M_i (true by the definition of the factor-base logarithms), and the reduction from uniform (a_i, b_i) to the counting statement, which is the uniform-measure statement itself. The consequence for the corpus: for a single target, with relations drawn from target-dependent points, index calculus needs no rank condition at all, and the four inflation rows are edge errors in the graph, not fragile theorems. Removing the heuristic *strengthens* the closures that rest on it, such as the GHS and two-summand closures. The triage's own toy check (12 seeds at $p = 10007$, every matrix 2–7 short of full rank, left-kernel recovery correct 12/12) was run in scratch space and is labelled there as an illustration, not evidence.

The heuristic stays load-bearing where relations are reused across targets: right-kernel amortisation, multi-target batches, and PKM's surplus of $\Delta = 10$ relations (`HEUR-PF-PKM-RANK-DELTA`). `rank_deficit` (`research/ecdlp_weakness_census_20261104/rank_deficit/README.md`; results in `results/tables.md`) measures that regime exactly: ten prime-order curves, two per $p \in \{1009, 2003, 4001, 8009, 16001\}$, small-$x$ bases with B closest to $\sqrt{r}$ ($B = 32$ to 126), rows $aG = P_1 + P_2$, five samples each.

- With negation pairs merged into $B/2$ unknowns, $\Delta = 10$ gave full rank in all five samples on 2 of the 10 curves; on the other eight the largest deficit at $\Delta = 10$ is 1–3. At $\Delta = 40$ four curves still have a deficit in some sample, at most 2. On four curves no tested $\Delta \leq 40$ gives full rank in all five samples.
- On the unmerged B columns no tested $\Delta \leq 40$ gives deficit 0 in all five samples on any curve.
- Free relations $P_1 + P_2 = P_3$ among factor-base points span only 3 to 15 of the $B/2$ merged unknowns.

At these sizes PKM's "$\Delta = 10$ suffices" is contradicted (`MEAS-RANK-DEFICIT-CENSUS` obstructs `HEUR-PF-PKM-RANK-DELTA`). The census README also states that the unmerged deficit is "at least about $B/4$ at every Δ"; its own table does not show this (at $\Delta = 40$ the minimum unmerged deficit over five samples is 0 or 1 on seven curves), so that sentence is not used here. The measured complete solves agree with the theorem: on the descent family 15/15 solves returned the correct logarithm with rank shortfalls up to 6 (§7.2), and the 842 prime $m = 2$ rows of the reanalysis report rank 842, a correctness control and not a rank law, since each episode stopped before a full matrix was collected (`research/ic_weak_assumptions_20261102/evidence-summary.json`).

7.7 Yield: the mean is a theorem, the shape is a heuristic

The corpus node `HEUR-DECOMPOSITION-PROBABILITY` has the largest impact on the frontier of the attack graph (13). Its mean, B^m/r, follows by counting: `MeanCount.sum_card_reps` kernel-checks that the ordered representation counts over all targets sum to $|F|^m$, the ordered form of Proposition 5.5. Only the shape is heuristic: Poisson zero mass, variance, independence between targets, rationality and visibility to the solver. The triage (W3) splits the node in two. Over $\mathbb{F}_{q^n}$ with n fixed, Diem's rigorous results bypass the heuristic. Over prime fields it is genuinely heuristic. Under the left-kernel formulation the cost needs only the fraction of points that decompose, a set size, exact on small fields.

Prime field, $m = 2$. `pf_yield` (`research/ecdlp_weakness_census_20261104/pf_yield/results/table.md`) enumerates every target on 18 prime-order curves, p from 2003 to 80021, with three base shapes (small x; random; random closed under negation) at $\mu = B^2/r$ near 0.5, 1 and 2: 162 rows. With ordered pairs the decomposable fraction is 0.55–0.78 of $1 - e^{-\mu}$. With unordered pairs $U(R)$, of mean $\mu/2$, $P(U \geq 1)$ matches $1 - e^{-\overline{U}}$: the mean absolute deviation of the ratio is 0.011, 0.007 and 0.011 for the three shapes. The variance-to-mean ratio of U is not Poisson for the bases index calculus uses: averaged over 54 rows it is 1.455 for small x and 1.447 for the random negation-closed base, against 0.981 (range 0.91–1.01) for the random base that is not closed under negation. The overdispersion follows closure under negation, through $U(R) = U(-R)$, which couples targets in pairs, and not the small-x shape. The heuristic's zero class is right once the pairing convention is stated; its variance is off by a factor of about 1.45.

The descent family and the $1/n!$ proportion. On the Gaudry–Diem family of §7.2 the yield is Poisson: mean/$\lambda_{\text{pred}} = 0.88$–$1.19$ and $P(0) = e^{-\text{mean}}$, the heavy tails being subfield targets. `ext_decomp` (`research/ecdlp_weakness_census_20261104/ext_decomp/results/table_primary.md`) measures the n-fold decomposable fraction over $\mathbb{F}_{q^n}$ with factor base $x \in \mathbb{F}_q$ against the survey value $1/n!$: 0.350–0.413 at $n = 2$ ($q = 101, 211, 401$) and 0.101–0.165 at $n = 3$ ($q = 23, 41, 61$), that is 0.70–0.83 and 0.61–0.99 of $1/n!$ (the README's summary, 0.70–0.99 for both, overstates the lower end at $n = 3$). The $n = 4$ rows have factor bases of 10–18 points and scatter widely. For control curves defined over $\mathbb{F}_q$ the premise fails: the factor base is $E(\mathbb{F}_q)$ or lies in a small subgroup, and the fraction is 0.0003–0.002 at $n = 3$. The $1/n!$ is the mean of Proposition 5.5 and carries the hypothesis that the curve is not defined over the subfield.

A withdrawn closure. The triage first argued by a dimension count that the trace pull-back (base-change $E/\mathbb{F}_p$ to $\mathbb{F}_{p^n}$, decompose over points with x in an $\mathbb{F}_p$-subspace V, pull back by the trace) cannot give a small factor base, and withdrew the argument on 4 November: the count bounds $|F'|$, not $|\mathrm{Tr}(F')|$. `trace_pullback` (`research/ecdlp_weakness_census_20261104/trace_pullback/results/summary_table.md`) computes the trace image of every one-dimensional subspace for ten curves, $p = 103$ to 2003, over $\mathbb{F}_{p^2}$: the smallest image is 0.319–0.389 of $\#E(\mathbb{F}_p)$, attained at $V = \mathbb{F}_p \cdot i$ on every curve, and two-point decomposition over it covers every target. The lead `GAP-PF-TRACE-PULLBACK-FACTOR-BASE` is narrowed, not closed: higher extension degrees and higher-dimensional V are not covered.

7.8 Waiting times: marginals suffice, hazards are not implied

The Jacobson–Koblitz–Silverman–Stein–Teske obstruction to xedni, audited in Lean on 1 November (`research/jksst_xedni_audit_20261101/`; see Chapter B for the correction of the treatise's earlier "closed" wording), derives a linear expected waiting time from a conditional-hazard recurrence $(1 - \varepsilon)s(n) \le s(n + 1)$ on the survival masses $s(n) = P(\tau > n)$, and listed the recurrence for repeated adaptive trials as open. The triage (W7) promoted that free-text assumption to a node, `ASM-XEDNI-HAZARD-RECURRENCE`.

Theorem 7.3 (Marginals suffice; kernel-checked)**.** *Let μ be a probability mass on a finite set, τ a stopping time, and S_t the event that trial t succeeds, with $\tau \le n$ implying some S_t, $1 \le t \le n$. Trials may be correlated or adaptive.*

1. *If $\mu(S_t) \le \varepsilon$ for every $t \le N$, then $s(n) \ge 1 - n\varepsilon$ for all $n \le N$.*
2. *If moreover $N\varepsilon \le 1/2$, then $\mathbb{E}[\min(\tau, N)] \ge N/2$.*
3. *The converse inference fails: on three equally likely outcomes, outcome k succeeding at trial $k + 1$, every trial has marginal success mass at most $1/3$, the recurrence with $\varepsilon = 1/3$ fails at $n = 1$ ($\frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9} > \frac{1}{3} = s(2)$), and still $\mathbb{E}[\min(\tau, 3)] = 2$.*

The declarations are `survival_ge_of_marginals`, `truncated_expectation_ge_half_horizon_of_marginals`, `marginals_U`, `marginals_do_not_give_hazard` and `wait_U` in `research/ecdlp_weakness_lean_20261104/lean/HazardSeparation.lean`. Part 2 is the conclusion of the JKSST audit's `truncated_expectation_ge_half_horizon` without its hazard hypothesis; part 3 refutes the corpus inference `CLAIM-MARGINAL-IMPLIES-HAZARD`. The open bridge for xedni is therefore a per-trial marginal bound for the adaptive trial actually run (`ASM-XEDNI-TRIAL-MARGINAL`: each trial succeeds with marginal probability at most the one-shot bound C_0/p), a weaker and more natural premise than a history-uniform hazard bound. This strengthens the obstruction; it does not reopen xedni, and it leaves the other conditional links of the JKSST chain (heights and discriminants, Lang's conjecture, selected lifts, useful relations, cofactors) as they were.

7.9 Generic bounds and what they cover

The non-generic-interface obligation exists because a generic algorithm cannot beat $\sqrt{r}$ (§27.1). Two Lean results this autumn fix the scope of that statement.

Shoup's bound is relative to random encodings. `TransparentEncoding.dlog_identity_encoding` and `zero_operation_dlog` (`research/ecdlp_weakness_lean_20261104/lean/TransparentEncoding.lean`) prove that with the identity encoding of $\mathbb{Z}/r$ the logarithm is the encoding, read with

zero group operations. This refutes the corpus claim `CLAIM-SHOUP-ALL-ALGORITHMS` ("every algorithm needs $\Omega(\sqrt{r})$") and qualifies the bound and "BSGS is optimal" as generic-model statements. It is a scope statement, not an attack: it is the reason every subexponential claim must name an encoding property, and the reason the obligation of Table 7.1 asks for a coordinate predicate that no generic table reproduces. The generic-interface audit of 1 November (`docs/ecdlp-hardness-corpus/GENERIC_INTERFACE_AUDIT.md`; an internal argument, not a reviewed theorem) sorts the programme's oracles: subtract, meet-in-the-middle, k-sum and pair-enumeration oracles are generic table algorithms once their set-up is charged; the root-finding oracles return the same relations as group subtraction, with representation-specific set-up; Frobenius-orbit methods need an augmented interface with a cheap known-scalar automorphism; and F4/F5/SAT, Weil-descent, PKM-tower and small-root lattice oracles are coordinate-algebraic, so Shoup's and the Corrigan-Gibbs–Kogan [8] bounds do not exclude them. Only the last group can discharge the obligation. The weak-assumption reanalysis credits the PKM/F4 and compact Boolean predicates with a genuinely coordinate-algebraic interface; none of them yet has a solver cost that discharges the obligations after it (§7.1, §22.1).

Rho is conductor-blind in distribution. The conductor-gap programme asked whether the depth of $\mathrm{End}(E)$ in an isogeny volcano changes the cost of rho. The theorem that answers it in distribution is `research/conductor_gap_rho_large_sample_20261018/lean/RepresentationHashInvariance.lean`.

Theorem 7.4 (Representation-hash invariance; checked by the producer)**.** *Let $\varphi : G \to H$ be a bijection compatible with binary operations on G and H. For an r-adding walk $w = (h, c)$ on G, with partition $h : G \to \{0, \dots, r-1\}$, multipliers c and step $x \mapsto x \cdot c(h(x))$, let $\varphi_* w = (h \circ \varphi^{-1}, \varphi \circ c)$. Then the trajectory of $\varphi_* w$ from $\varphi(x)$ is φ of the trajectory of w from x; the rho length (first repeat, capped at the budget) is the same at every budget; and $w \mapsto \varphi_* w$ is a bijection of walk spaces.*

No group axiom is used. Read outside Lean, the push-forward of the uniform distribution on walks under a bijection is uniform, so over a uniformly random partition the cost distribution of rho is the same on the crater and on the floor, and on every curve of an isogeny class of prime order (abstract groups of equal prime order are isomorphic). The conductor, the j-invariant or any invariant that is not an invariant of the group can enter only through the one fixed hash an implementation uses; whether ecbench's fixed partition does so is an empirical question, and the measurements are in §29.4. Formal status: 11 theorems in core Lean 4.34.1 without Mathlib or a Lake project; the committed `build.log` of 18 October reports `[propext, Quot.sound]` for the three audited declarations and no `sorryAx`. The file is not among the targets of proof-gate run 41399938870. Nothing about curves, isogenies, endomorphism rings or conductors is formalised, and the file says nothing about index calculus or about the cost of computing the connecting isogeny. It complements `GenericInvariance.lean` of `research/conductor_gap_cm_pairs_20261010/lean`, which covers only algorithms whose control flow depends on equality answers (the gate run above passes that project with 60 theorems); see also §28.7.

7.10 The hardness corpus

`docs/ecdlp-hardness-corpus/` is a typed claim graph of what supports elliptic-curve discrete-log security: definitions, assumptions, reductions, algorithms, model lower bounds, conditional attacks, heuristics, measurements, obstructions and open obligations, each with its model and scope, and every edge typed (`supports`, `conditions`, `obstructs`, `qualifies`, ...). The triage of 3 November (`docs/ecdlp-hardness-corpus/WEAKNESS_TRIAGE.md`) read the assembled version of 1 November, 583 claims, 950 edges and 236 sources, and marks its own findings `[pcc:unverified]`. The generated `stats.tsv` on the current main branch counts 608 nodes, 32 of them refuted and 68 `proved_internal_unreviewed`; the corpus README's "current snapshot" still gives an older 525. Counts below are the triage's.

The triage's findings, W1–W7, are the targets of the censuses and Lean modules of this chapter:

- *W1: the first-fall pillar is obstructed and still load-bearing.* `HEUR-FIRST-FALL-DEGREE` has four incoming `obstructs` edges from established claims and still conditions seven claims, among them the ECC2K-130 2^{86} projection, Nagao's complexity, the PKM algebraic route and binary Weil descent. Its siblings `HEUR-GB-SEMIREG-ASSUMPTION` (six `obstructs` edges) and `HEUR-GB-PQ-ASSUMPTION-2` (four) head the attack queue with it. Because the graph keeps them at "heuristic", the seven conditional estimates read as one discharged condition away from usable, which overstates them (§7.3).
- *W2: relation rank* (§7.6).

Check	What it finds	Count
Status inflation	proved, algorithm or measured claims whose support closure reaches an open, heuristic or assumed claim	17
Unwired dependence	a claim's assumption text invokes a heuristic node with no edge to it	11
Live obstruction	an established claim obstructs, qualifies or does not cover a claim that still supports others	49
Route closure on weak support	an `obstructs` edge whose obstruction rests on an assumed, heuristic or open claim	16
Hidden shared assumption	the same assumption text on 3 or more nodes, never a node itself	18 clusters
Source contagion	a primary source with a refuted claim and other live claims	21
Model-scope leap	a generic-model or measured claim supporting a broader one without a model bridge	9

Table 7.5: Structural checks of the hardness corpus (version of 1 November: 583 claims, 950 edges, 236 sources), as counted by `weakness.mjs` and reported in `docs/ecdlp-hardness-corpus/WEAKNESS_TRIAGE.md`. It then ranks 246 attackable claims by impact times breakability and keeps a separate hygiene queue of 100; the triage states that the scores are a heuristic, not evidence.

- *W3: yield*, two regimes in one node (§7.7).
- *W4: the Lemma 4 refutation rests on a reading choice*, and two internal papers disagreed about it; the wording of `papers/ic-negative-results/` was corrected on 4 November to the four-cell table (§7.5).
- *W5: eleven unwired dependences* understate the impact of the top frontier nodes; refuting the heuristic would not reach them.
- *W7: route closures on weak support.* Xedni is closed twice, both softly; lifted index calculus is closed by three heuristics; the naive prime-field Semaev method is closed by a lower bound conditional on a statistical conjecture validated only to about 25-bit primes; the quasi-subfield routes are closed by results resting on the yield heuristic. Rank-only closures are not reopenings, by W2.

The triage states that W1–W7 change no claim's status by themselves; the Lean and census rounds added nodes and edges. Its recommended order puts corpus and proof work first (re-wire the Weil-descent chain to the left-kernel theorem, add the eleven edges, transcribe Nagao's definitions) and lists the remaining measurements (YYTK's conjecture past 25 bits, Semaev's $n = 45$ cell, the per-trial xedni marginal) as runs that go through the experiment route and the validation gate before any number enters the corpus.

7.11 Formal status

Table 7.6 gathers the formal artefacts of this chapter with what a kernel has checked and what is only stated or argued. The degree-four lower bound of the guarded chain (`GuardedModel`, Theorem 6.13) and the other fall-degree identities are in §6.4; the seven theorems of the Semaev-alternatives audit, which correct an oracle gate to $s < m - 3$, are in §22.1.

For the 4 November project the exact status is as follows. Its README says each module was compiled with `lake env lean -o` against the Mathlib v4.34.1 cache, that a full local `lake build` was not possible for lack of disk, and that CI's `lean-proof-gate` runs the full build and the transitive axiom audit. The triage records that every theorem audited after round 3 (35) depends only on `propext`, `Classical.choice` and `Quot.sound`. A search of the committed sources finds no `sorry`, `admit`, `native_decide` or `axiom`. No proof-gate run identifier is recorded for this project in its directory or in the corpus fragments, which say the gate "runs on the landing PR" or "on main after merge"; the pcc claim records are described as pending CI signing. The corpus marks the nodes `kernel_checked` with status `proved_internal_unreviewed`. The theorems are general statements of linear algebra, counting and finite probability; their bearing on the literature is through the bridges named in the right column, which are argued, not proved.

7.12 What the literature's subexponential claims now rest on

Table 7.7 sets the claims of §3 and the two recent prime-field claims against the obligations of Table 7.1, with what is proved, refuted, conditional and open. Statuses in the second column are the corpus's.

Artefact	Kernel-checked	Not formalised
`ecdlp_weakness_lean_2026 1104` (Lean and Mathlib v4.34.1): `LeftKernel`, `HazardSeparation`, `SolvingDegreeBound`, `SubspaceCriterion`, `MeanCount`, `BooleanNullstellensatz`, `NagaoLiteralInside`, `TransparentEncoding`, `ProtocolCountermodels`	Theorems 7.1, 7.2, 7.3; the fourth Lemma 4 cell; the mean count; the $\lvert Z\rvert = 1$ Nullstellensatz; 45 `#print axioms` lines in `AxiomAudit.lean`	F4 containment; $k = N - \operatorname{affdim} Z$ for $\lvert Z\rvert > 1$; that PQ's "maximal degree" is the Caminata–Gorla solving degree; relation-to-row step
`IcNegative.FakeDegreeGap`	W_3: $d'_F = 2$, $d_F = 4$ (gate run 41399938870)	W_k for all k (hand proof)
`IcNegative.InsideGap`	the field-equation fall at 4 in two variables (gate run 41484047855)	zero multipliers on further generators; four-variable witnesses
`RepresentationHashInvari ance` (core Lean 4.34.1)	Theorem 7.4, 11 theorems; producer's `build.log`	curves, isogenies, conductors; the fixed hash

Table 7.6: Formal status of the results of this chapter.

Claim	Corpus status	What this programme established
Diem 2011: subexponential ECDLP over $\mathbb{F}_{q^n}$ for n growing with $\log q$ [10]	proved (external)	Untouched. Its scope, as the corpus records it: for fixed n the bound is exponential, and for $q = 2$, n prime it is $e^{O(n^2)}$, far above rho. For the rigorous Weil-descent chain (GTTD, Diem 2006, GHS) the rank step needs no heuristic for single targets (Theorem 7.2).
Petit–Quisquater 2012: $2^{c\, n^{2/3} \log n}$ for binary curves [32]	heuristic premise	Literal Assumption 2 fails on 119/119 instances; the $m = 2$ Kousidis–Wiemers value 3 is not reached at $n = 17$–44; $d_{\bar{F}} \geq 5$ certified at $n = 33$–45. The $o(1)$ form is not refuted: no growing lower bound.
Semaev 2015: $2^{c\sqrt{n \ln n}}$, $c \approx 1.69$ [36]	conditional	Assumption 1 ($d_{F4} \leq 4$) refuted as stated in four cells; "first fall degree always 4" refuted at $t = 2$ (200/200 have 2). The revised growth condition is open.
Nagao 2015/984: polynomial time under FFDA [31]	conditional	Lemma 4 refuted in three reading–placement cells, proved in the fourth; A1 and the +1 form of A2 refuted; inside gap bounded by 3 because inside $d_F \leq 4$ is degenerate; A3 survives on the triples tested. The complexity claim rests on the first-fall heuristic of W1.
PKM 2016: prime-field algebraic route [33]	conditional	$m = 4$ solving degrees $6, 7, 7$ above the threshold 4 and work $\times 558{,}304$ from $N = 8$ to 16; $\Delta = 10$ contradicted at $p \leq 16001$ in the right-kernel regime; no cheap root-finding algorithm.
Hashed-base Boolean tensor contraction (Akshima et al. v0.2)	conditional	The only inspected route with an explicit end-to-end subexponential bound; conditional on the pinned contraction theorem; predicate cost and aggregate complexity uncertified (§24.1).
Xedni and lifted index calculus	obstruction conditional	JKSST is a chain of conditional links; the waiting-time link needs only per-trial marginals (Theorem 7.3); the marginal bound for the adaptive trial is open.

Table 7.7: The subexponential claims and the obligations they discharge. Corpus statuses from `docs/ecdlp -hardness-corpus/`; the right column summarises §§7.1–7.9. Census figures are unverified; Lean results as in Table 7.6.

Read by kind:

- *Proved here.* Five general statements, all kernel-checked and unreviewed: the subspace criterion, the left-kernel theorem with its exact $1/q$ failure density, the marginal waiting-time bound with its separation, the fourth Lemma 4 cell, and representation-hash invariance of rho. None is a lower bound on an index-calculus algorithm. Two of them *remove* a heuristic: the rank heuristic from the rigorous Weil-descent chain, and the hazard recurrence from the xedni obstruction.
- *Refuted.* Literal readings, all at the level of a stated assumption: Semaev's Assumption 1 and his $t = 2$ first-fall statement; Nagao's Lemma 4 in three cells, and the repairs A1 and A2 (+1); Petit–Quisquater Assumption 2 read literally, on every instance computed; PKM's $\Delta = 10$ at toy sizes; the corpus inferences `CLAIM-MARGINAL-IMPLIES-HAZARD` and `CLAIM-SHOUP-ALL-ALGORITHMS`.
- *Conditional.* Every subexponential claim for binary or prime fields remains conditional: on an asymptotic degree assumption (Petit–Quisquater's $o(1)$, Semaev's revised growth condition), on a first-fall heuristic (Nagao, PKM), or on an external theorem (the tensor route).
- *Open.* A last-fall or solving-degree lower bound that grows with n for any presentation (§6.7); a per-trial marginal bound for adaptive xedni lifts; the trace pull-back beyond one-dimensional V over $\mathbb{F}_{p^2}$; a non-generic prime-field predicate with a simulator-or-separation proof; and whether ecbench's fixed rho hash sees the conductor.

The direction is consistent. Each refutation removes support from a subexponential claim, and none supplies one; each theorem either confines a heuristic to a narrower regime or shows a closure to be stronger than its stated support. On the only family where every obligation was measured, the Gaudry–Diem descent, the distance to matched rho at toy size is 4–5 orders of magnitude at $w \geq 0.1$, and IC is ahead nowhere in the band.

8 Obstructions: what moving the problem cannot buy

The programme kept 148 leads against ECDLP in a lead atlas (`crypto-autoresearcher:knowledge/leads/ecdlp/`, reviewed 2026-10-24). Of these:

- 73 are closed with a named obstruction;
- 51 are open with a decided next step;
- 16 are blocked;
- 8 are unexamined.

Most obstructions are textbook results applied to the case at hand: the Shoup–Nechaev generic bound, the Corrigan-Gibbs–Kogan preprocessing bound $ST^2 \gtrsim N$ [8], isogeny invariance of the group order, and the Balasubramanian–Koblitz embedding-degree bound. This chapter records the few that the programme *proved* itself and that constrain index calculus specifically. Each states a lever that cannot work, and why.

8.1 Transport conservation

A homomorphic transfer $\varphi : E \to E'$ with rational kernel κ changes the number of ordered m-decompositions of a target by exactly $|\kappa|^{m-1}$ (Lemma 26.1 in Chapter 26). So an isogeny buys only three things:

- the kernel symmetry, which is trivial on prime-order curves;
- the factor-base geometry available on the destination curve;
- the cost of arithmetic there.

Chains of isogenies compose to one homomorphism. A loop is an endomorphism acting as a known scalar. Multi-curve factor bases glued by isogenies are single-curve bases in disguise. Measured across sizes at fixed cofactor, fibre-aware index calculus loses ground to rho as $\#E^{1/6}$; the full measurements are in Chapter 26.

8.2 The summation cover is the same everywhere

Chapter 5 proved:

- the monodromy of the summation cover is $(\mathbb{Z}/2)^{m-2}$ for every curve;
- its fibres split completely on the factor-base locus;
- mean yield at fixed base size is a double count.

Together these close three families of leads: curve selection by anomalous monodromy, fibre censuses as a relation-rate probe, and factor-base geometries claimed to raise yield.

8.3 Oracle-augmented index calculus stays generic

Several leads proposed giving index calculus an extra oracle on the group:

- a jet or p-adic lift;
- an endomorphism;
- an incidence or halving test;
- an elliptic-net evaluation.

The programme proved, at derivation tier, that each of these is simulable in the generic group model, so it cannot beat $\sqrt{r}$ (`KN-FIND-002`, `KN-FIND-b7e091`, `KN-FIND-194294`; halving is equivalent to an x-coordinate oracle).

The minimal non-simulable order-based identifier is a threshold oracle C_t (`KN-FIND-982fdf`). Table-based decomposition then obeys the preprocessing bound $ST^2 \gtrsim N$. Balanced generic index calculus costs about $r^{2/3}/n^{4/3}$, consistent with the measured growth of the Koblitz gap to rho as $r^{0.165\,[0.117,0.213]}$ (`LEAD-ICB-generic-table-oracle`). These are known instances of generic-group reasoning [8]; they are recorded because each retired a concrete proposal.

8.4 Lifting sections does not produce relations

Silverman's xedni calculus lifts points to a curve over a global field, in the hope that rank forces relations. On a frozen family of rational elliptic surfaces (with $\deg b = 6$ and monic quadratic x), the programme derived the exact lift probability

$$P_{\text{lift}}(p) = \frac{1 - 2/p + p^{-3}}{2p^3},$$

so the exponent is $\alpha = 3$, far above the $\alpha < 1/2$ that a relation source would need. This was checked against 596,268,874 enumerated slots with no disagreement, and a parameter count reproduces $\alpha = 3$ (`LEAD-ICP-xedni-function-field-sections`, `EV-XEDN-001`). The scope is one family ($\deg b \leq 10$, $p \leq 31$). A dimension-count theorem over all section shapes would be needed to close the route in general.

The classic obstruction to xedni is also conditional. A kernel-checked Lean audit of Jacobson, Koblitz, Silverman, Stein and Teske's analysis (`research/jksst_xedni_audit_20261101/`) does three things:

- it repairs their conditional theorem;
- it refutes several overbroad readings of it;
- it leaves open the premises it rests on: Lang's height conjecture, a uniform height–discriminant comparison, and the algorithm-specific bridge from a one-shot relation probability to the actual attack.

The audit calls the result a strong conditional asymptotic obstruction for classic xedni, not a proof that the route is closed.

8.5 Twist points carry no relations

A curve and its quadratic twist share x-coordinates, and therefore share summation polynomials. Could twist points supply relations for the curve? They cannot. Frobenius acts as $+1$ on $E(\mathbb{F}_q)$ and as -1 on the embedded twist, so the twist part B of any sum satisfies $2B = 0$.

The study `research/twist_obstruction_20261026/` checked this exactly on toy binary curves in the ECC2K-130 model (odd n, twist $a \mapsto a + 1$):

- 0 of 356,530,036 mixed sums are rational;
- 0 violations on 4,954 decomposition targets at $m = 2, 3$;
- the off-parity solution sets equal the predicted ones on 1,154 of 1,154 targets;
- positive controls found all 18,042 and 95 planted solutions.

A heuristic that ignored the obstruction predicted that 74–89% of solutions would contain a twist-lifting coordinate; 0% did at $m = 2$.

A follow-up asked whether a factor-base subspace can *avoid* the twist altogether (`research/lift_dense_factor_base_20261027/`; 11 Lean theorems, standard axioms only).

- *Composite degree $N = dn$, $n \geq 3$: yes.* On suitable $\mathbb{F}_{2^d}$-lines the lift condition collapses to the constant $\mathrm{Tr}(a)$. When $\mathrm{Tr}(a) = 0$ every abscissa lifts, giving 2^n relations per Gaudry–Diem trial. This applies to c2pnb208w1 and c2pnb368w1, but the modelled gain is $+0.50$ bits, a model output, not a measurement.
- *Prime degree (ECC2K-130): this construction does not exist.* A prime-degree field has no proper intermediate subfield, so there are no $\mathbb{F}_{2^d}$-lines to use. This does not rule out other twist-avoiding subspaces. At the small prime degrees tested, all-lift subspaces of dimension 4–5 exist, though fewer than for density-paired random sets (exact all-lift 4-subspace counts 0.921 ± 0.018 of the null at $N = 11$, permutation $p = 2 \cdot 10^{-5}$). This is toy evidence. Whether useful twist-avoiding subspaces exist at cryptographic prime degree is unresolved.

8.6 What the obstructions leave open

None of the obstructions above concerns the cost of *solving* the decomposition system, nor the linear algebra. Chapter 6 shows that the solving cost is genuinely open. The algorithms of Part III attack the remaining free terms.

Part III

Algorithms and measurements

9 Root-finding decomposition

9.1 Origin: a suggestion in an audited manuscript

The textbook decomposition oracle builds Semaev's polynomial system and solves it with a Gröbner-basis engine. On 12 October a 13-page manuscript, *Point Decomposition by Root Finding*, was supplied for assessment. We froze it (`research/ecbench_rootfinding_assessment_20261012/source-paper.pdf`, SHA-256 `743d3d3e...7c01`) and audited it against its source branch before measuring anything. Its central suggestion is simple: for two summands, fixing one factor-base point turns S_3 into a quadratic in the remaining variable whose roots are already known.

Proposition 9.1 (roots of S_3)**.** *Let $E : y^2 = x^3 + ax + b$ over a field K of characteristic $\neq 2, 3$, and let $R, F \in E(K)$ be affine with $x_R \neq x_F$. Then*

$$S_3(x_R, x_F, X) = (x_R - x_F)^2 \, (X - x_+)(X - x_-), \qquad x_\pm = x(R \pm F),$$

so $S_3(x_R, x_F, X) = 0$ iff $X \in \{x(R+F), x(R-F)\}$.

Proof sketch. With $\lambda_\pm = (\pm y_F - y_R)/(x_F - x_R)$ the chord slopes, $x_\pm = \lambda_\pm^2 - x_R - x_F$. Expanding $(X - x_+)(X - x_-)$ and eliminating y_R^2, y_F^2 with the curve equation gives exactly Semaev's $S_3/(x_R - x_F)^2$. The identity is machine-checked over an arbitrary field in Lean 4 with Mathlib (`papers/rootfinding-decomposition/lean/S3Roots.lean`, theorems `SemaevS3.S3_factor` and `SemaevS3.S3_root_iff`; no `sorry`; axioms `propext`, `Classical.choice`, `Quot.sound`). □

Hence the two-summand system has a parametrised variety: $R = \pm F_i \pm F_j$ iff $x(R \mp F_i) \in x(\mathcal{F})$. No polynomial system is solved; the oracle is one subtraction and one table probe per base point.

Prior art. Neither the identity nor the oracle is new. The identity is a direct consequence of how S_3 is defined: it is the polynomial whose roots are the abscissae of $P_1 \pm P_2$. The oracle mechanism—sum $m-1$ factor-base points in the group and test whether the result lies in the base—is the second algorithm of McGuire and Mueller [30], who also evaluate S_m by iterated resultants instead of solving it, building on the instance-derived bases of Amadori, Pintore and Sala [1]. What this section adds is engineering and measurement: a machine-checked proof, a fail-closed verification protocol, binary pair-sum tables for up to four summands, GPU, Kummer-tower and dense-index variants, and complete-cost measurements in one harness against rho.

What the audit kept and corrected. The identity is exact under the stated generic affine condition, and a returned witness can be re-added in the group for soundness. Completeness of the exhaustive enumeration holds only with an explicit sign and factor-base convention (the binary table stores full point keys over a base containing both signs; the manuscript's schematic x-only table needed that convention stated). The manuscript's NIST baseline was Buchberger, not F4 (§4.5), and its statement that no implementation can use fewer than linear field operations per ideal fails for repeated queries against a reusable pair table. Larger-summand enumeration has a growing exponent in $|\mathcal{F}|$, and the manuscript did not benchmark it end to end. All of this is recorded in `research/ecbench_rootfinding_assessment_20261012/REPORT.md`.

9.2 The algorithms

Prime fields (Algorithm 1, `src/cryptanalysis/prime_fast.rs`): $\Theta(N)$ field operations and $2N$ lookups per target. *Binary fields* (`src/cryptanalysis/koblitz_index_calculus.rs`, `DecompositionStrategy::PairTable`, made the default): a precomputed table of pair sums $F_i + F_j$ turns an m-decomposition into 1, N or N^2 lookups for $m = 2, 3, 4$—the meet-in-the-middle half of a birthday search, exact and complete like exhaustive search, at the price of $\Theta(N^2)$ set-up and memory. Both are verified fail closed: every decomposition is re-added in the group, S_3 is checked to vanish, Gröbner-variety membership and agreement on refutation are checked against F4, and planted positive controls must be found.

Algorithm 1 Prime two-summand oracle by direct subtraction

Require: target R, base $\mathcal{F} = \{F_1, \ldots, F_N\}$, hash set H of abscissae (or full points) of $\mathcal{F}$

$d_i \leftarrow x_R - x_{F_i}$ for all i; invert all d_i with one field inversion and $3(N-1)$ multiplications (Montgomery's trick)
for $i = 1..N$ **do**
  for $\varepsilon \in \{+1, -1\}$ **do**
    $\lambda \leftarrow (y_R + \varepsilon y_{F_i})\, d_i^{-1}$; $x_3 \leftarrow \lambda^2 - x_R - x_{F_i}$ ▷ $x(R - \varepsilon F_i)$
    if $x_3 \in H$ **then** recover the sign of F_j from y_3; emit $R = \varepsilon F_i + \tau F_j$
    end if
  end for
end for

Step	What it adds; artefact
Prime subtraction, binary pair table	Prop. 9.1; batched inversion; $m \leq 4$ tables. `research/semaev_root_vs_groebner_20261012/`
NIST P-192…P-521	standard parameters, 6- and 64-point bases; dedicated P-256 field 11–12× over generic big integers. `research/semaev_rootfind_nist_20261012/`
End to end vs msolve	complete DLPs with only the oracle swapped. `research/semaev_rootfind_e2e_20261012/`
PDP oracle family	`rf-mitm`, `mitm3`, `mitm4` (2-sum table); `rf-pairs`, `pairs3` (pairs then solve by the group law); `subtract2`; binary rungs $\ell = 8..14$, prime Kummer towers $m = 2, 3, 4$. `research/pdp_root_finder_20261012/`
CUDA kernel	one thread per (target, slice); per-thread Montgomery inversion; SoA base; shared-memory hash for $\lvert\mathcal{F}\rvert \leq 2048$; 64-bit and P-256 fields. Validated in host emulation only (no GPU available). `RESEARCH_GPU_ROOTFIND_DECOMPOSITION.md`
ecbench engines	`pdp3-koblitz:m=2|3,engine=rootfind`, `rootfind-subtract`, later `rootfind-window` and `rootfind-negfold`. `research/ecbench_rootfind_engine_20261012/` and successors
Kummer-tower base	PKM's μ_{2^t} base with the root-finding oracles, 20–28-bit primes. `research/pkm_tower_ecbench_20261013/`
Dense root index, $n = 83$	packed five-byte S_3 root table, prefix widths K1271–K1798, 8 GiB of slots. `research/n83_dense_root_20261013/`
Prime tuning	pair-table size, arity, batched table inversions, synchronised lanes. `research/ecbench_rootfinding_\{tuning,arity,batchtable,lanes\}_2026101*/`

Table 9.1: Root-finding generalisations, 12–14 October. Every row is ENGINEERING: no slope fell below its generic floor.

9.3 Generalising the observation

Over three days the identity was carried through every layer of the pipeline. Table 9.1 lists the steps; each is its own study with a protocol registered before it ran.

9.4 Results

Oracle stage. Per decomposition, root finding is 383× ($\mathbb{F}_{2^9}$, $\lvert\mathcal{F}\rvert = 55$, 254 targets) and 1,535× ($\mathbb{F}_{2^{13}}$, $\lvert\mathcal{F}\rvert = 4{,}005$, 256 targets) faster than matrix F4; 3.5–$4.7 \cdot 10^4$× faster than `groebner_f4` on 16–24-bit prime curves ($\lvert\mathcal{F}\rvert = 6$, 40 targets each); and 3,021–5,802× faster than Buchberger on the decomposition ideal of the five NIST curves (Figure 9.1). Every binary verdict agreed with F4; every prime solution was in F4's variety and every prime refutation was F4's unit ideal. In the PDP study, `rf-mitm` refutes a binary target in 3.1 μs at $\ell = 8$ and 482 μs at $\ell = 14$, against the PS-0 baseline's 9.3 ms and 55.5 s (3,007–115,160×), costs $5\lvert\mathcal{F}\rvert - 3$ multiplications and $2\lvert\mathcal{F}\rvert$ lookups a target, and sits on the pair-table floor at 1.000 ± 0.027 bits per rung; at prime $m = 3$, $N = 18$ it uses 197 multiplications where F4 uses $8.99 \cdot 10^{13}$ multiply-adds; 707/707 brute-force agreements (`research/pdp_root_finder_20261012/`).

Complete solves. End to end, the gain is two orders of magnitude smaller and not one-sided: 50–105× against msolve on 16–24-bit prime curves; 122× on the one binary instance ($n = 23$) large enough

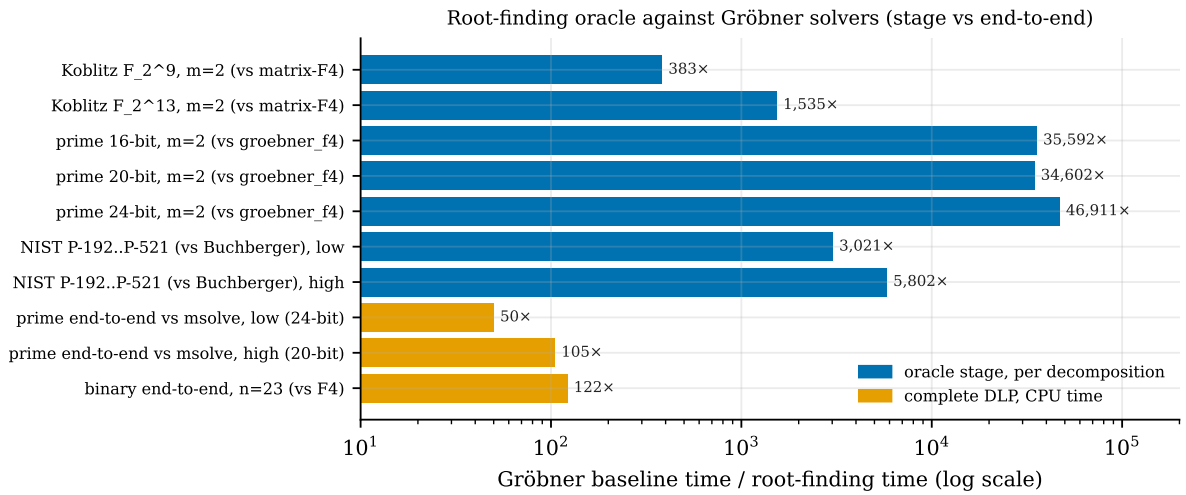


Figure 9.1: Root finding against Gröbner solvers on identical targets. Blue: one decomposition (oracle stage only). Orange: complete discrete-log computations with the oracle swapped and everything else fixed (prime: CPU time against msolve, 20/24-bit totals composed from sampled per-query cost; binary: one cold solve including table build). Sources: `RESEARCH_SEMAEV_ROOTFIND_VS_GROEBNER.md`, `papers/rootfinding-decomposition/README.md`.

to amortise the pair table; and F4 *faster* at $n = 5, 9, 13$, where a solve needs too few decompositions to pay for the table. Inside ecbench, with every group operation charged, the root-finding IC beats inherited F4 in solve wall by about 350× at $n = 13$ (admitted) and about 10^4× at $n = 23$ (descriptive), yet stays 2.9–25,116× the strong rho in S on binary curves and 29–141× rho on prime curves, with the gap widening (`research/pdp_root_finder_ecbench_20261013/`).

The Koblitz loops. Figure 9.2 follows the best complete-count root-finding arm at Koblitz E_0, $n = 13$ through four registered improvement loops (8 public targets × 3 rounds each, L0 operation counts). The first engine had $S = 38.82$ ($m = 3$) and 28.38 ($m = 2$). A windowed third-summand scan cut $m = 3$ by a paired 0.828 $[0.783, 0.863]$; a fold over negation (`rootfind-negfold`) took $m = 2$ to 23.22 and then 22.97. Two further cycles of denser and sparser bases, windowed negfold and plain root finding were dead ends (`research/ecbench_rootfind_vs_rho_20261014/`). With F4's word XORs priced, the paired ratio negfold/F4 is 0.0545 $[0.047, 0.064]$, a complete-against-complete comparison; against strong signed-Frobenius rho ($S = 14.82$) the best complete arm is still 1.55×. A compact-orbit base with a counted Frobenius pair table reaches $S = 15.55$, below the then-unpriced F4 (16.75), but it is itself a lower bound (factor-base construction natives unpriced), and it is 1.08× rho. Amortised over 64 targets that arm reaches $S \approx 11.2$, below rho; by the ledger's rules that is a secondary, multi-target result, not a cold claim.

Prime tuning in ecbench. On a 20-bit prime curve, the pair table's counted S is 0.018861 $[0.018547, 0.019242]$ of direct subtraction (24 pairs; prime native work unpriced, so a bounded ratio), and 0.057047 $[0.041159, 0.092069]$ on a 17-bit Koblitz curve, fully priced, where the table set-up is about 68% of the arm's cold cost. Doubling the table from 32 to 64 abscissae gives 0.830 $[0.697, 0.962]$; three summands instead of two costs 1.431× $[1.225, 1.716]$; batching the table inversions leaves counted GAE unchanged (1.000; 4,096 versus 64 inversions, set-up wall 243 versus 128 μs, descriptive); and a train-set lane selection did not resolve a hold-out winner (0.979 $[0.919, 1.038]$; 1,088/1,088 verified executions) (`benchmarks/README.md`, rows of 12–14 Oct).

Dense root index at $n = 83$. Widening the prefix base from K1500 to K1798 cut mean probes per relation from 300,021.9 to 169,069.8 (paired difference 130,952, 95% [52,058, 209,991]) and gave a one-point online ratio of 18.528 over same-point rho (IC 63.3 ms, rho 1,173.3 ms), with 339.0 s of reusable preparation and an 8 GiB table outside the window. The host was shared, so the result is exploratory; two K1500 runs hit the 900 s timeout and are retained. The probe reduction stands; the online ratio does not: it was taken against rho with an empty table, and rho with a precomputed table matched to the index calculus's set-up is faster in every matched $n = 83$ pair (validation sweep, 20 October; §31.4).

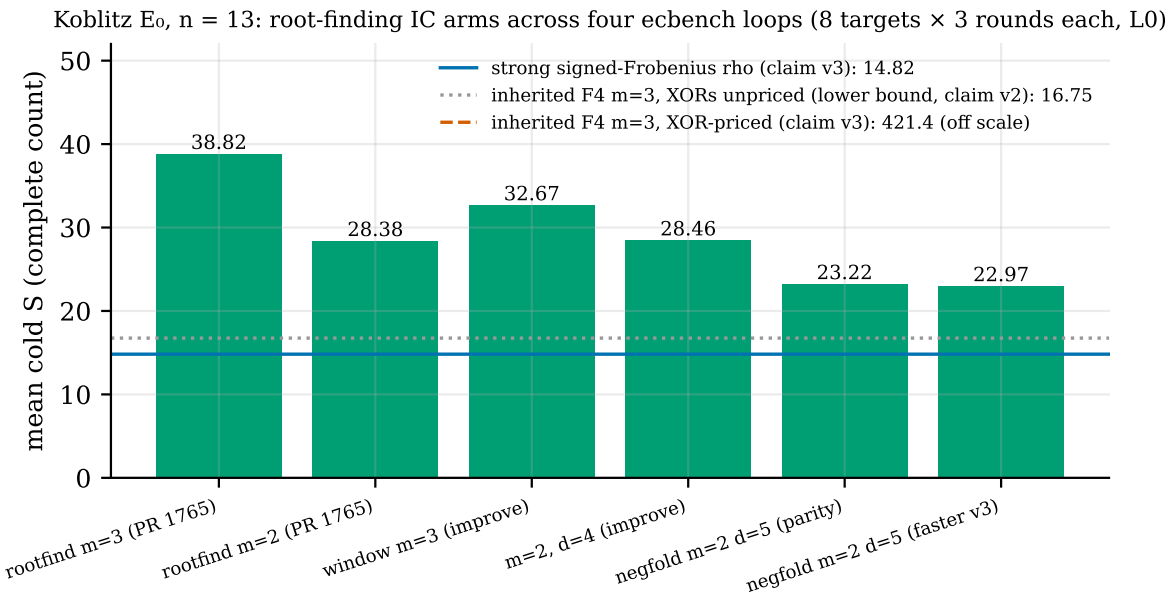


Figure 9.2: Mean cold S of root-finding IC arms at Koblitz E_0, $n = 13$, $d = 5$ unless stated, across the engine, improve, parity and faster loops. All bars are complete counts. Sources: `research/ecbench_rootfind_engine_20261012/`, `research/ecbench_rootfind_improve_20261013/`, `research/ecbench_rootfind_parity_20261013/`, `research/ecbench_rootfind_faster_20261014/`. Each loop had its own seeds and rho arm (rho 14.12–14.82).

Reading. Root finding is the largest constant-factor change of the programme at the oracle stage, and the one with a proof behind it. It does not change the asymptotic count of index calculus: its cost is $\Theta(|\mathcal{F}|^{m-1})$, every measurement is at $m \leq 4$, and it leaves relation yield, factor-base size and linear algebra untouched. Its effect on the cold frontier is to move the decomposition stage from dominant to minor, which exposes the stages that now are.

10 Gröbner-basis engines

Before root finding, the decomposition oracle was a Gröbner solve, and for $m \geq 3$ over binary fields it still is. The repository grew a family of engines (`docs/ic/GROEBNER_F4F5F6_AUDIT_20261010.md`, §1; file names below are under `src/cryptanalysis/`):

Matrix F4 and F5 over $\mathbb{F}_2$
(`koblitz_groebner.rs`, `matrix_f5_f2.rs`): Macaulay matrices of the Weil-restricted Boolean system, reduced by a shared Four-Russians kernel (`gf2_elim.rs`, four 8-bit tables, AVX-512 or AVX2).

Inherited F4
(`inherited_f4.rs`, from 22 September, the default algebraic oracle): a DPLL-style tree over partial assignments in which each child *inherits* its parent's echelon form and re-reduces only what the assignment changed, instead of building a fresh Macaulay matrix per node.

F6-IC
(opt-in): inherited F4 plus a factor-base domain gate (a fully assigned summand whose abscissa is not a base point refutes; partial support pruning refutes earlier), and meet-in-the-middle leaves (with $m-1$ summands fixed the last is looked up; at $m = 3$ and $|\mathcal{F}| \leq 256$ every compatible second point is added and the third looked up).

$\mathbb{F}_p$ F4
(`f4_fp.rs`, `f4_fp_tower.rs`, `sig_fp_tower.rs`): degree-bounded F4 with sparse-by-reducer then dense reduction, a Faugère–Lachartre block for the Kummer-tower oracle, and a signature variant.

Correctness. The audit pinned every engine against every other and against brute force (88+15+9+14+13+12+21+6 passing library tests in the touched modules), and found one soundness bug (§4.5, item 6), two contract defects and several latent holes, all fixed with regression tests.

Engineering results. Figure 10.1 collects the headline ratios and Table 10.1 lists them all; all are ENGINEERING with identical outputs.

Matrix and inherited F4 can return different decompositions, so their splitting trees may differ; only matrix F4 and F5 share a tree by construction. F6-IC's S ratios, once its batched group additions are charged, are an accounting loss, not a frontier move. Those ratios used inherited F4 before its word XORs were priced (#1774); priced, F6-IC is still a loss at K13, while the R13 and $m = 19$ cells remain lower bounds.

The $\mathbb{F}_2$ engine campaigns (about 90 directories, 22 September–17 October; the programme record [3], clusters "boolean", "gf2-elim", "f5-kernel", "f5-support-cut", "f4-f5-shared") gated each micro-change against the strongest reference, and most failed. Kept as defaults: the branchless strip, the bitmap seen-set, batched pair inserts, five-column tables, and direct packing in F5 (default since 17 October). About thirty unpack, pivot, layout and packing variants were refuted, and graded reuse in F5 fell short of the campaign's 2× goal. These are stage speed-ups of the decision engine; none changes an index-calculus distance to rho. Every reduced basis of `f4_fp`, after Gebauer–Möller criteria, packed monomials, interned symbolic preprocessing and a parallel symbolic pass, agrees with Singular or msolve. The Boolean comparison is a difference of task (decide versus compute the basis), not of engine quality.

What did not work. Degree-2 early-abort probes after a 2025–26 proposal: word-XOR ratios 1.00083–1.00378 (no change) and 0.99714–0.99928 (worse). Gate-first geometry: 1.57–1.78× fewer word XORs but lane-batched F4 roots cost 2.6–4.0× a sequential control. Convolution splitting of the summation polynomial: 1.21–2.07× *more* word operations despite lower degree; Frobenius auxiliary variables: 57–701× more (`research/lit2026_w\{4,5,8\}_*_20261011/`). Symmetry-breaking clauses in SAT did not help (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). A registered law for the $m = 3$ refutation degree resolved all 68 units but ended with the verdict "neither": one variant met its generic degree window on 32/32 draws, the other fell below it on 8, and the heaviest degree-10 Macaulay solves were $2.4 \cdot 10^5 \times 2.0 \cdot 10^5$ matrices at 42–51 minutes each (`research/ic_semiregular_law_20261005/`).

Change	Case	Measure	Ratio or value
Matrix / inherited F4	K13	counted matrix work	198.6×
	random-13		222.1×
	$m = 19$	wall, matrix F4/F5 vs inherited F4	timed out at 1,800 s vs about 6 s
F5 / F4	K13	word XORs	0.6465
	random-13		0.6414
	K13, random-13	wall	0.79–0.86
F6-IC / inherited F4	$m = 13$	word XORs	about 0.82
	$m = 19$		0.77–0.78
	—	relation wall	1.7–1.9× faster
	$m = 13$	S, group additions charged	64.6–68.8×
	$m = 19$		1,303–1,353×
	K13	S, word XORs priced (#1774)	1,572.1/634.12 ≈ 2.5×
	R13, $m = 19$		lower bounds
	$m = 31$	first target	both hit 3,600 s cap
Every echelon call to the Four-Russians kernel	oracle's own Macaulay matrices	speed-up	1.75–3.22×
Word-level `pext`/`pdep` rewrite plan	$n = 23$	F4 and F6-IC faster	10–14%
Eager / lazy rewrite of kept rows		speed-up	1.40–1.62×
Fused construction and readback	F4	instructions	−5.28%
	F5		−11.02%
Native hashing	F5		−3.715% [3.714, 3.726]
	standalone criterion		−45.1%
Bounded direct indexing in the F5 packer	F5		−3.947%
Row allocation		requested bytes	42–50% fewer
Quotient-curve model / symmetrised control	$m = 2$, five interleaved cells	F4 word operations cut	1.340–1.642×
Branchless AVX2 strip	one thread	speed-up	1.210×
Bitmap seen-set in Boolean F4	full F4		1.199×
Combined F5 options			1.530×
Fixed support cut	one complete call		3.02–3.07×
Graded reuse in F5			1.78–1.84×
`f4_fp` / its predecessor	square generic systems	speed-up	29.8–54.1×
msolve / ours	single-threaded	ratio	0.96–1.22
Ours / msolve, blocked pair reduction	two largest square systems	time	0.75–0.78×
	four others		1.11–1.49×
msolve faster, a day earlier		time	12–24×
Boolean decomposition systems, $n = 17$, 8 targets	our oracle (decides)	time per target	7–24 ms
	msolve, PolyBoRi (full basis)		40–52 s
	Singular		exceeds 120 s

Table 10.1: Gröbner-engine engineering results, all ENGINEERING with identical outputs. *Change*: candidate / baseline where a slash is shown; *Case*: target set or size (K13, random-13 and R13 as named in the sources; m the number of summands; "—": no case given); *Measure*: the counted or timed quantity; *Ratio or value*: as recorded. Groups, top to bottom, with sources: inherited versus matrix F4 (`research/f4_f5_f6_cross_family_20261011/`); F5 versus F4; F6-IC versus inherited F4 (priced and lower-bound cells: `research/f4_pricing_jump_20261018/`); kernel routing and rewrite (`research/inherited_f4_eager_rewrite_20261011/`); instruction and allocation work, by paired A/B/A′ instruction counts on a frozen eleven-system suite, elapsed time not claimed; the quotient-curve model (`research/lit2026_w8_summation_models_20261011/`); $\mathbb{F}_2$ campaign results (strip: `research/gf2_branchless_strip_20260929/`; seen-set: `research/boolean_f4_bitmap_20260926/`; combined options: `research/f5_combined_2x_20260928/`; support cut: `research/f5_support_cut19_confirm_20261003/`; graded reuse: `research/boolean_f5_graded_batch_20261002/`); $\mathbb{F}_p$ F4 to msolve parity (`research/f4_fp_msolve_parity_20261011/`; blocked pairs: `research/f4_fp_blocked_pairs_20261016/`; the day-earlier msolve reading: `research/gb_cas_comparison_20261011/`).

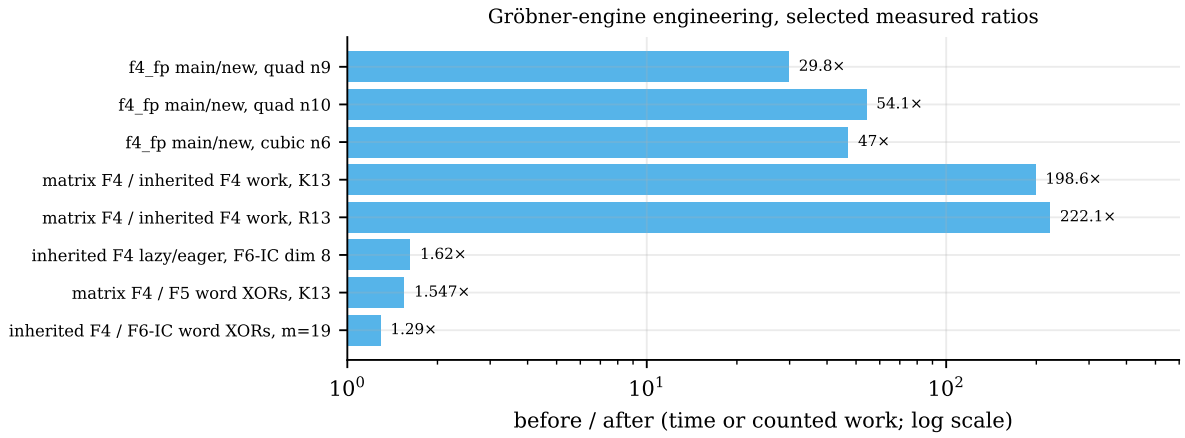


Figure 10.1: Selected engine ratios (before/after or baseline/candidate): `f4_fp` successive versions (single-threaded wall, L0); matrix versus inherited F4 (counted matrix work); lazy versus eager inherited rewrite (median wall); F4 versus F5 and inherited F4 versus F6-IC (word XORs, stage only). Sources in `figures/data-program.json`.

11 Measuring an engine

Chapter 4 describes the instrument as it was built: the unit S, sealed records, replay, isolation levels, the paired statistic and the pricing band of §4.3. Chapter 28 describes how the rho references inside that instrument were calibrated and corrected. This chapter is about the other side of the same instrument: what it takes to measure a decomposition engine, a linear-algebra stage or a whole index-calculus pipeline so that the number survives a replay and a second reader. Between 2 October and 5 November the harness changed its accounting twice, its statistics once, and its notion of a passing audit three times. Each change moved numbers that earlier chapters quote, and each is stated here as what it is: a correction of the instrument, not an improvement of any method.

Three distinctions run through this part of the treatise.

- A *component speed-up* lowers the cost of one stage (a Macaulay build, an echelon, a lookup) with identical outputs. It is measured in the stage's own unit—word XORs, multiply-adds, instructions, stage wall time—and says nothing about the distance to rho until the stage is priced inside a whole solve.
- An *end-to-end change* moves the cold one-target S of a complete pipeline, paired against the same pipeline before the change. It is an improvement under the programme's success criterion when it is verified and measured; it is a frontier move only when it also changes the order against the matched rho across the band of §4.3.
- An *accounting change* moves a number without changing any algorithm: a counter that was not charged is charged, a reference is resized, an interval is recomputed. It is never an improvement, even when it moves the distance to rho in the method's favour.

Every result in Chapters 11–14 carries one of these three labels.

11.1 What the calibration does and does not cover

The calibration study of 2 October (§28.1, Table 28.1) is the harness's only test against a closed-form constant: seven generic methods reach their analytic constants up to $r = 2^{25.35}$ on prime curves and 2^{39} on Koblitz curves. For an engine this has two consequences.

- The counted charge of every generic reference used below is calibrated in its own unit up to those sizes. A ratio of an index-calculus S to a rho S therefore has a calibrated denominator at toy size, and an uncalibrated one above 2^{39}.
- No index-calculus method has a closed-form constant, so no engine in this part is calibrated in the same sense. An engine's charge is checked only by replay (the same counts on a second run, host or instruction set) and by the independent recomputations named `VALIDATION.json` below; neither says that the unit is the right price for the work.

The second point is why the band of §4.3 exists: native work (word XORs, conflicts, lookups, matrix entries) is priced at 0, 0.1 and 1 GAE per operation because no calibration can price it once for all hosts, and two arms are ranked only where the order is the same at all three prices.

11.2 Accounting version 2

PR #2465 (23 October) gave every frozen method identifier a versioned twin and changed what each twin charges (`research/ecbench_accounting_v2_20261023/REPORT.typ`; preregistered, four dated amendments). The study's own first sentence classifies it: an accounting and instrument correction, not an algorithmic improvement and not a speed result. Its effect on the strong Koblitz reference is recorded in §B.3; its effect on engines is recorded here.

- *The verification phase.* Every v2 identifier charges one final check $[d]G = Q$; `ic.large_prime_v2` also charges the membership check $[r]Q$. On matched (workload, round, seed) pairs the v2 total minus the frozen total equals the verification phase exactly on 40 of 40 pairs for each of six generic methods and 16 of 16 for `ic.large_prime`, and every other phase is equal. The preregistered outcome that the phase costs $1.5 \log_2 r$ failed for every record: it is the exact double-and-add count of the planted scalar (19 to 29 GAE here), so its size follows the scalar. The post hoc replacement held on 240 of 240.

- *What it costs.* On these toy curves a fixed 19–29 GAE sits on totals of 235–1,500 GAE, so v2 identifiers cost 1.7–8.8% more S (for example 1.022 $[1.021, 1.024]$ for `bsgs.textbook` and 1.0003 for `ic.large_prime`). The study states that this is a sizing statement for these curves and not for larger r, since one scalar multiplication is $O(\log r)$ against $O(\sqrt{r})$.
- *No frozen count moved.* The same frozen-only specs run on the branch binary and on the main binary agree on every count field of 504 of 504 matched records. Two committed sessions fail a replay identically on both binaries, in their root-finding and inherited-F4 arms only (96 and 16 problems): a pre-existing condition, not caused by the change, that CI's `-replay 0` audit does not see.
- *The distance moves the other way.* The index-calculus arms of the committed claim specs are unchanged; their reference became cheaper. The best arm across the band, `ic-rootfind-m2`, reads 10.6–11.0× the v2 strong reference at $n = 13$ (against 2.0–2.3× the frozen one) and 34–49× at $n = 19$ (against 10–11×). The report states that the corrected reference makes the measured gap larger.

The v2 floor rule (`automorphisms/v2`) also counts six automorphisms on $j = 0$ and four on $j = 1728$; under the first rule a cubic-automorphism rho read 0.720 $[0.591, 0.867]$ of a floor computed with two automorphisms, which is impossible for a correct floor. That correction belongs to the rho side (Chapter 28); for engines its lesson is that a floor is part of the instrument and is versioned with it.

11.3 Statistics version 2: censoring and the wall statistic

PR #2442 replaced the comparison statistic; §28.8 gives the interval widths, the coverage simulation and the censored-rho rows. Three of its changes bear on engines specifically (`research/ecbench_statistics_v2_20261105/README.md`; no new runs, the old and new binaries run on the same committed records).

- *Censored engine runs are rows.* An engine that times out or exhausts its budget now enters the ratio as a censored lower bound. On the yield sweep the two descent-SAT arms over strong rho move from 1.351 (9 of 16 pairs) to 1.429 $[1.241, 1.646]$ with 7 censored, and from 1.892 to 2.486 $[1.442, 4.288]$ with 4 censored; both are lower bounds. The point estimate over verified pairs is unchanged in every row, so the old binary and the new agree where both define a number.
- *Wall time per operation, not per solve.* The v1 headline wall statistic was the median per-pair solve-wall ratio, whose interval was often a single point. The v2 headline is the ratio of nanoseconds per GAE. The two answer different questions and can disagree in direction: on the yield sweep, `sub8-mitm` over strong rho reads 20.26 as a median wall ratio and 0.627 $[0.489, 0.804]$ in ns per GAE, because the arm does 45× the operations at a lower cost per operation. The descent-SAT arms read $5.1 \cdot 10^7$ and $8.8 \cdot 10^6$ ns per GAE: their solver work is unpriced, and the harness now flags them as upper bounds on that statistic.
- *The tail flag needs runs.* A heavy-tail flag (largest run above 25% of an arm's total) fired on 71 of 71 table rows with four runs and on 6 of 6 with two: with n runs the largest share is at least $1/n$, so at $n \leq 4$ the flag carries no information. The same PR made the threshold depend on n.

The engine-side lesson is the first item. A decomposition engine that times out on a hard target is not cheaper on the targets it finished; a comparison that drops it reads the engine as cheaper than it was. Every engine ratio quoted from a session with censored runs before 5 November is a lower bound on the engine's cost, and some (`ic-sub6-mitm`, $6.98 \to 648$) were understated by two orders of magnitude.

11.4 The allocator confound

A runner change (PR #2455) reported its index-calculus online window "about 15% lower". A preregistered study of the same day (`research/ecbench_allocator_confound_20261023/REPORT.md`; ABBA order, three scratch ablations) measured what the change did.

- *Counts are identical.* Ten base-versus-variant session pairs match on every count-bearing field of every record (72 of 72 and 66 of 66 per pair); a tampered copy is flagged by the same script.
- *The small-set arms' wall time collapses.* Eight rho, BSGS and kangaroo arms on a 20-bit prime curve read pooled medians of 5.26–6.77 ms on the merge-base binary and 0.137–0.330 ms on the PR binary, 18.4–38.8× shorter; the index-calculus `ic-mitm` arm (about 1.9 ms) does not move (ratio 1.03).
- *The online window is not distinguished.* On the Koblitz $n = 17$ claim spec the IC online median moves from 127.1 to 119.1 μs (ratio 1.067; 1.135 and 0.983 by order). The "15% lower" is not reproduced; the preregistered verdict is inconclusive.

- *Attribution.* Removing `malloc_trim(0)` alone, or the new pre-solve `/proc` reads alone, leaves the PR binary fast; removing both returns every arm to the merge-base level (0.75–1.05). Either consolidates the allocator before the clock starts.

The study measured the distance to rho on the same claim spec, which no binary changes: the band gives the IC arm 16.85, 17.28 and 21.16 against the strong rho's 4.00, 4.14 and 5.36 at $w = 0, 0.1, 1$, so IC is 4.21, 4.17 and 3.95× the strong rho, and a plain signed-Frobenius rho (1.54–2.48) is cheaper than both at this size because the strong walk's set-up dominates at $r = 2^{16}$. For engines the lesson is that wall time on sub-millisecond solves measured the allocator; the PR's own grader now refuses wall time below 1 ms of solve.

11.5 Sessions that cannot be cited in full

Committed sessions are evidence and are never edited. Two registries record those that cannot pass the audit as committed (`docs/ecbench/`).

- *Incomplete sessions* (`docs/ecbench/INCOMPLETE_SESSIONS.tsv`, since 18 October): four frozen scratch sessions of the $k = 4$ quartic studies, committed without spec and host capsules. Each row pins the SHA-256 of `session.json`, `plan.json` and `records.jsonl`; the CI audit reports them as incomplete evidence, never as audited passes, and fails if a byte changes or a capsule appears.
- *Defective sessions* (`docs/ecbench/DEFECTIVE_SESSIONS.md`, one row, owner-approved 5 November): the interrupted timing smoke of the Curve25519 algebraic ladder. Its `session.json` claims 7 records and a stop at sequence 6, but `records.jsonl` holds sequences 0–3, and the sealed records hash is that of the 4-line file, so the records were lost on the producer's host before sealing. Recovery from Git and from the S3 checkpoints found nothing. The row pins all five capsule hashes and the exact four problem checks (`record_count`, `status_counts`, `last_seq`, `stop`); the audit passes the row only if verify fails with exactly those checks. The session is citable as its four sealed records only (§21.7).

A third class is reported but not fatal: a session whose audit receipt says `records_bound: false` (no whole-file hash binds its records, for example a runner killed before finalisation) is listed as an unbound prefix, not integrity-passed (`.github/scripts/ecbench-audit-sessions.sh`).

11.6 The registry, inventoried

By 1 November the harness exposed 79 top-level methods and 24 nested algorithms behind them: 11 system solvers, 9 three-summand decomposition (PDP3) engines and 4 prime Kummer-tower engines (`research/ecbench_registry_optimization_20261101/README.md`). A native audit (`ecbench optimize inventory` and `audit`) admits a development roster of 114 specs covering 79 of 79 methods and 24 of 24 nested algorithms, with no structural errors or seed leakage.

- *Execution coverage.* 38 fresh L0 sessions (1,095 records, 1,042 verified outcomes) exercised 73 of 73 runnable native methods and every nested engine; 854 of 854 eligible deterministic measured replays reproduced with zero audit problems. Six entries are external CM wrappers without a hash-bound backend and are not executable.
- *A replay that found a defect.* The first pair-index F6 replay exposed an accounting-cache regression; resetting the cache at the independent-run boundary makes the corrected session reproduce 16 of 16.
- *What the count does not include.* A standalone $n = 83$ chained-S_3 lookup (§12.8) and a `chained-x` decomposition oracle registered in `ic bench` are outside the 79+24, because neither is ecbench-selectable; the README lists wiring and pricing them as a coverage gap, not an assumed result.

The inventory changes no distance. Its use is negative: an engine that is in the registry has been run at least once under the same accounting as rho, and an engine that is not has not.

11.7 The end-to-end frontier refresh

The refresh of 31 October and 1 November (`research/ecbench_e2e_frontier_20261031/README.md`, `research/ecbench_e2e_frontier_20261031/REPORT.md`) bound the frontier to one source commit, inventoried every live method and every nested engine, and prepared a 92-shard isolated campaign that did not run (no checkpoint credential; the blocker is recorded). Its answer is a boundary statement, with the domains kept apart as Definition 4.1 requires.

- *Generic prime, random binary and medium Koblitz ladders are rho-led.* No admitted cold one-target index-calculus result in those panels is below its strongest matched reference. The closest markers are 2.28× rho for inherited F4 on one tiny random-binary cell, 2.59× for the prime residual collector at 36 bits (§22.3) and 3.43× on the admitted Koblitz ladder; the report states that these are different studies and units, not entries of one table.
- *Engines on the current binary.* The pair-index F6 reduces inherited-F6 work to 0.2792 $[0.246, 0.317]$ and is still 139.30 $[85.48, 227.02]\times$ strong rho end to end; the prime PKM F4, F5, F6 and enumeration engines are 4.62–5.03× negation rho; the best matched $n = 17$ system solver is wide FES/Gray code at 4.28× strong rho. All are one-target L0 operation comparisons with unpriced work.
- *Historical engine counts are bound to their binary.* A fresh replay of the F4/F5/F6 cross-family sessions on the current binary reproduces 72 of 144 deterministic rows, while the optimised-rho session reproduces 720 of 720. The engine claims of Table 10.1 are therefore statements about the binary that produced them.
- *One restricted family is IC-led.* The plane-quartic attack on the GHS-weak family $E_0/\mathbb{F}_{q^3}$ with $\#E_0 = 4\ell$ is below matched rho in its own comparison (§26); the report keeps it out of the generic ranking because it is a different problem class.

The $n = 29$ crossover is Frobenius-folded BSGS. One panel does cross below rho, and its classification is the instructive part (`research/ecbench_generator_orbit_20261022/README.md`; protocol frozen before data). The study measured an orbit-layout pair table and a generator-orbit factor base on seven Koblitz curves ($n = 13$–37). The layout table finds the same relations as the lazy table on 960 of 960 run pairs at 1.02–1.09× lower cost; with the generator-orbit base and stride the net gain over the lazy baseline is 1.26–3.20×, with the order holding across the band—an end-to-end improvement over an index-calculus baseline. At $n = 29$ and one factor-base column the IC arm reads $S = 0.546$, 0.611 and 1.196 at $w = 0, 0.1, 1$ against 1.569 for `rho-frob`, a paired ratio of 0.366 $[0.241, 0.523]$. The directory carries a `VALIDATION.json` with a replay certificate and a passing validator report, and its own scope statement says what the crossing is.

- At one column the pipeline is a Frobenius-folded baby-step giant-step with a known generator-orbit column, not a factor base with unknown logarithms; in the language of §27.2 it has $F = 0$.
- The crossing rests on `rho-frob`'s fixed 133-GAE set-up at $\log_2 r = 15.4$. The gap to rho grows with r: 1.15× at $n = 31$, 1.93× at $n = 23$ and 3.89× at $n = 37$; at $n = 17$, 23, 31 and 37 `rho-frob` is ahead at every weight, at $n = 13$ the pair is unresolved, and at $n = 19$ the band cannot re-price the IC natives (no pinned calibration row).
- In the validation file's like-for-like band, which prices the same native operation types on both arms, rho rises from 1.503 to 1.58 at $w = 0.1$ while the IC arm reads 0.413, 0.478 and 1.063; the crossing is robust to that choice at this size and does not survive growth in r.

The refresh therefore keeps the panel as a fixed-set-up engineering frontier and does not pool it with the uniform-target ladder; the 1 November current-binary snapshot reads the same cell at 0.4405 $[0.363, 0.534]$ of signed-Frobenius rho and calls it a holdout candidate, not a family-level or leaderboard result. It is the clearest example in the programme of a validated number below rho that is not an index-calculus advance.

11.8 The cost of the harness itself

A measured child is a fresh `ecbench exec` process, so the harness's own start-up sits inside every wall time it records. Two engineering studies measured that share without changing any counted unit.

- *Registry and alias parses* (`research/ecbench_child_overhead_20261024/README.md`; Callgrind instructions, frozen protocol). Before #2184 every child parsed the whole 4.7 MB curve registry, about 190 M instructions or 98% of a rho or BSGS child; after it a `rho.negation` child runs 4.26 M instructions (4.64 M on `main`). The index-calculus child had meanwhile grown a parse of the 2.8 MB alias table to match one curve name against a 25-row calibration table: 169.47 M instructions on `main`, 9.86 M after the fix. Median spawn-to-exit wall time per child fell from 37.20 to 2.92 ms after #2184 (0.0786 $[0.0769, 0.0802]$); the alias fix's own wall ratio, 0.9910 $[0.9754, 1.0083]$, is not distinguished.
- *Inversion in the default IC oracle* (`research/ecbench_ic_asm_euclid_20261103/README.md`). A 32-bit extended-Euclid inversion in x86-64 assembly lowered instructions per GAE of the default `mitm:negation_folded=1` oracle from 537.0 to 476.8 (−11.2%) and of the prime-batch arm from 427.2 to 397.7; the target had been a halving. Every counted field of every record is identical to

`main`'s.

Neither changes an S. Both matter for the wall-time column: an index-calculus child that spends most of its instructions finding its curve name, or a rho child that spends 98% parsing a registry, makes an admitted wall-time ratio a ratio of start-up costs. This, with §11.4, is why wall time is admitted only above a minimum solve length and with an A/A interval.

11.9 What a measured engine result must carry

The corrections above add up to a contract that every result in the following chapters is read against. An engine result is citable as a component speed-up when it carries identical outputs on frozen inputs, a same-binary A/A, a preregistered gate, and its own unit. It is citable as an end-to-end change when, in addition, it is a sealed ecbench session (or an equivalent native record) with every phase charged or explicitly unpriced, censored runs kept, and a paired comparison against the pipeline it replaces. It is citable against rho only when the rho arm is the strongest available for the curve (Chapter 28), measured in the same process and unit, and the order is stated per band weight. It is *validated* only where its directory names a `VALIDATION.json` whose validator report passed; everything else in this part is a producer-reported figure, which is how the text marks it.

12 Boolean decision engines

Over a binary field the decomposition oracle receives the Weil restriction of a summation polynomial: a Boolean system in N unknowns whose solutions are the decompositions of one target over the factor base. The index calculus needs a *decision*—a witness or a proof that there is none—not a Gröbner basis, and almost every target has none. Chapter 10 describes the four engine families and Table 10.1 lists their headline engineering ratios. This chapter gives what that chapter compresses: the generic Boolean search campaign that preceded the algebraic engines, the gate methodology of the $\mathbb{F}_2$ micro-studies and the variants it refuted, the inherited-F4 rounds after 5 October, the F6-IC engine's convergence to the meet-in-the-middle oracle, the crossbred and SAT engines, and the $n = 83$ engine campaign. One observation recurs: as an engine is made faster on index-calculus systems it tends to turn into an enumeration with a table, and the table is the generic oracle whose cost Chapter 27 bounds from below.

12.1 Generic Boolean search, 22 September to 3 October

Before the descended Semaev systems were fed to an algebraic engine, a campaign of standalone studies measured complete generic Boolean solving on generated public systems of up to 36 variables: exact specialisation, affine propagation, fixed branch selection (most frequent variable, zero first) and exhaustive search below a node cap, with node and resource limits returning UNKNOWN and never UNSAT. None accepts a curve or a target; every result is a component result on its own fixture distribution. Each study froze its protocol, held out seeds, rotated arm order, and compared a candidate against the *pointwise fastest* of all retained methods rather than against the method it replaced. Most preregistered a universal 2× gate (every size and family group with a paired 95% lower bound above 2), which almost nothing passed. Table 12.1 lists the outcomes.

Two results of this campaign shaped what followed. First, at 24 variables the degree-3 Macaulay kernels were slower than plain search: on the packed residual study's planted holdouts the flat kernel took 253.1 ms per solve, the sparse-bucket kernel 1,050 ms and search-only 53.8 ms, against 12.0 ms for packed residuals (`research/boolean_quadratic_frontier_20260923/CONCLUSION.md`). On small systems a closure that sees further is more expensive than a search that looks less far. Second, every accepted gain was a faster representation of the same search tree: node counts, logical counters and canonical trace digests are identical across arms, so the campaign moved constants and never an exponent. The lead atlas records the same reading for every Boolean engine (`crypto-autoresearcher: knowledge/leads/ecdlp/ic-engine/LEAD-ICE-boolean-f4-f5-f6-engines.yaml`).

12.2 The $\mathbb{F}_2$ micro-studies and their gate

About thirty matrix-F5 and Boolean F4 variants were tested between 26 September and 3 October against one fixture: the $n = 24$, degree-4 complete F5 call of a seven-case suite (`f5_n24_m24_d4`), and the `n20_m30` Boolean F4 workload. The gate was the same in every study, and it is why most variants failed.

- The protocol and an unmodified baseline build were committed before candidate code or timing; the candidate was opt-in and was removed again if it failed.
- One release binary ran both arms. Each selected block had two warm-ups, five reference/reference A/A pairs and five alternating A/B pairs, 22 processes and all seven cases per process; the first isolated exact block per seed was selected before its timing was read.
- Every arm had to match raw and canonical row fingerprints, rank, terms, columns, builder and criterion counts and reduction word operations on every case.
- Ratios are medians of the five paired ratios, with exact 3,125-resample bootstrap intervals; a candidate advanced only with a frozen one-thread complete-call ratio of at least 1.05 and a lower bound above 1.02, and no smaller case below its own A/A minimum.

Table 12.2 lists the refuted variants with the ratio that refuted them; the accepted ones are in Table 10.1.

What passed, and its size. The one complete-call result above 2× is the support cut at column 19: a CI recheck on the same four seeds passed every cell, with one-thread medians of 3.006–3.049× (lower bounds 2.970–3.024) and two-thread medians of 2.415–2.476×, the candidate's counted reduction XORs

Study	Result (paired, against the fastest retained method)	Verdict
Packed direct construction	construction only: 3.48–4.53× (primary), 3.55–4.90× (confirmation)	passes 2× on construction; no solve claim
Construction plus reduction	charging the full RREF, the staged path is 1.38–1.50× faster at $n = 36$ than either streaming candidate; candidates pass 0 of 9	both rejected
Fixed quadratic state	2.40–2.51× on 9 of 9 primary comparisons; confirmation 2.10–2.55×, 7 of 9; 6,144 verified solves	engineering; same search tree
Packed residual coefficients	1.56–1.67× against the fastest of 8 methods, 18 of 18 lower bounds above 1; 5,832 observations on 72 systems	incremental gate passes; 2× rejected
Linear tail ($W \cap L$ kernel)	3.59× [3.57, 3.62] on the large-active mixture; four small cells 0.964–0.975	guarded portfolio rejected
Whole-solve traces	degree-3 hybrid kernel slower than the flat kernel with identical traces; search-only often faster with more nodes; specialisation gain 1.04–1.14×	2× rejected
Row-space transport	full-RREF paired ratios 0.199–2.667 (3 of 18 groups); affine-tail 0.235–0.952 (0 of 18)	rejected
Affine substitution, SIMD Gray	16/18 and 15/18 groups above 2	universal gate rejected
Sibling pivot-trace screen	0 of 64 eligible $n \geq 12$ branch pairs keep a replayable pivot trace	rejected at the structural gate

Table 12.1: Generic Boolean search on generated systems, September–October. All complete-solve timings, correctness pass on every row. Sources, top to bottom: `research/boolean_packed_construction_20260922/`, `research/boolean_construction_reduction_20260922/`, `research/boolean_quadratic_state_20260923/`, `research/boolean_quadratic_frontier_20260923/`, `research/boolean_linear_tail_20260923/`, `research/boolean_solve_trace_20260923/`, `research/boolean_basis_transport_20260923/`, `research/boolean_affine_substitution_20260923/`, `research/boolean_pivot_screen_20261003/` (each `CONCLUSION.md` or `README.md`).

being 28.08–28.10% of the reference's (`research/f5_support_cut19_final_recheck_20261003/README.md`). The README calls it a recheck, not an independent second holdout. Its predecessor, the support split, cut reduction word operations to 33.7% but its ARM64 timings were non-promoting, with one seed's lower bound at 0.966 (`research/f5_support_split_call_20261002/RESULT.md`), and the mainline rank port read 1.450× with an interval of 0.582–1.664 (`research/f5_mainline_port_20261002/RESULT.md`).

Defaults, 17 October to 29 October. Three switches that had passed their incremental gates on x86 CI were left opt-in because their protocols tied a default change to the 2× gate. A decision rule committed before the deciding run made one of them a default: direct F5 packing (1.006–2.173× on every F5 cell); direct unpack stayed opt-in at 0.986× on one four-thread cell, and table reuse at 0.955× on a matrix-F4 cell, and with all three on the largest F5 cases ran 1.11–1.27× faster than the previous defaults (`research/f5_gf2_defaults_20261017/README.md`; L0, round-to-round variation up to ±25%). On GitHub's Arm runners (Neoverse-N2 and a virtual Apple M1) every x86-chosen default stayed: the lazy rewrite with the bit scatter, Arm's path before #2043, takes 1.08–1.69× more CPU on Linux; turning gate-first F6-IC off costs 1.69–2.10× on Linux and 1.62–2.15× on macOS; and no Gray-code table count beats four (`research/arm64_f4_defaults_20261022/README.md`). The follow-up of 29 October tried four more candidates and landed none: a four-row multiply-accumulate pass cut the $\mathbb{F}_p$ kernels from 976 M to 633 M instructions with no wall gain (0.88–1.08 per site), because the loop is bound by reading pivot rows; a column-tiled parallel B' build was 0.87–1.02× the control; and F5 direct unpack "where it wins" passed its rule on one output mode while identical code on the degree-3 cases read 1.087 against 0.933 across runs, a disagreement larger than the effect, so it stayed opt-in (`research/f4_f5_followups_20261029/README.md`).

Reading. The campaign's own standard is the right one: a stage gain is an engine result, the complete call is the unit that matters for the oracle, and the oracle is a minor share of an index-calculus solve whose relation phase is dominated by the number of targets tried. None of these studies, accepted or refuted, changes a distance to rho.

Variant	Complete-call ratio (reference/candidate)	Why it failed
AVX2 gather unpack	0.9652 [0.9630, 0.9874] (1 thread, Intel), 0.9140 (2 threads, AMD)	slower
AVX2 nibble compaction	0.912 (0.906–0.925)	slower
Nibble-subset unpack	0.649 (0.646–0.650)	slower
AVX-512 unpack	new over prior 0.996–1.016	no gain
POPCNT unpack	1.0005 (0.9944–1.0067)	below 1.05
Compact paired unpack	0.9998 (0.9986–1.0045)	below 1.05
Bounds-free unpack	1.004, 0.982 (ARM64 screen)	below 1.05
One-pass unpack	0.992, 0.953	below 1.05
x86-guarded direct unpack	0.9984 (0.9927–1.0226)	below 1.05
Contiguous (flat) rows	0.978, 1.002	below 1.05
Byte-colex packing	1.0229–1.0262; build phase 1.46×	below 1.05
Initial-density row order	1.006, 0.987	below 1.05
Bounded-weight pivots ($K = 8, 16$); stratified pivots	terms 0.964–0.984 of reference (bounded)	structural gate (both)
Always-echelon output	large case 1.50–1.55; `n12_m12_d4` 0.821–0.830	smaller-case regression
Fused row build, 4 threads	build 1.382, complete call 0.865 (0.745–1.095)	4-thread gate
Boolean F4 unsorted products	0.857, 0.876	slower
Boolean F4 degree-bucket queue	0.993–1.008	below 1.05
Boolean F4 six-column tables	primary 1.074; holdout `n20_m20` 0.881	smaller-case regression

Table 12.2: Refuted $\mathbb{F}_2$ engine variants, 26 September–3 October: complete F5 call (`n24_m24_d4`) unless marked F4 (`n20_m30`), frozen seed unless a range is given. Each source is `research/<study>/RESULT.md`: `f5_avx2_gather_unpack_20260929`, `f5_avx2_nibble_unpack_20260929`, `f5_nibble_subset_unpack_20260929`, `f5_avx512_unpack_20260929`, `f5_popcnt_unpack_20260929`, `f5_compact_unrolled_unpack_20260929`, `f5_bounds_free_unpack_20260930`, `f5_one_pass_unpack_20260929`, `f5_x86_guarded_unpack_20260930`, `f5_flat_rows_20260929`, `f5_byte_colex_20260930`, `f5_initial_density_order_20260929`, `f5_bounded_weight_pivots_20260930` and `f5_stratified_weight_pivots_20260930`, `f5_echelon_20260927`, `f5_row_build_20260928`, `boolean_f4_unsorted_products_20260926`, `boolean_f4_degree_queue_20260927`, `boolean_f4_wide_tables_20260926`.

12.3 Inherited F4 after 5 October

Inherited F4 has been the default algebraic oracle since 22 September (Chapter 10). Its later rounds were of four kinds.

Instruction rounds. Two preregistered rounds lowered the machine work behind the counted unit without moving the unit. The overhead round ran 0.847× the baseline's instructions on the confirmatory $m = 19$ inputs (threshold 0.90), with 48 of 48 frozen outputs identical; inherited F4 went from about 120 to about 95 instructions per counted word XOR (`research/inherited_f4_overhead_20261005/RESULT.md`). The remap round read 0.923× (threshold 0.97; about 95 to 89 per word XOR) but was superseded before merge by a more general word-wise rewrite on `main` (`research/inherited_f4_remap_20261007/RESULT.md`). The eager rewrite through shift runs then brought hosts without BMI2 onto the eager path: 1.22–1.70× lower median CPU on six of nine cells against the bit scatter, and the two architectures charge the same units again (`research/inherited_f4_eager_runs_20261016/README.md`). Because an eager rewrite charges every kept row, it charges *more* word XORs than the lazy one while using less CPU: a reminder that word XORs are a stage unit, not a cost.

A batching idea that lost. Replaying one F4 trace on several specialisations at once, after Demin's Groebner.jl tracer, was exact but used 2.2–4.0× more stage units than the sequential root and more CPU (`research/lit2026_w5_lane_batched_f4_20261011/README.md`). The same study found the gate-first order for F6-IC (1.57–1.78× lower stage unit), which became a default after an x86 check on nine cells (1.336–1.733× fewer word XORs, 80 of 80 targets decided identically; `research/f6_gate_first_x86_20261016/README.md`).

Online, against rho. The first public-target panel (one registered point, prepared 62-point base, three fixed process orders) put F5 at a median 208.49× (181.29–223.86) inherited F4's online time and inherited F4 about 315× slower than the rho reference online; both engines spent over 99.9% of the

Curve	$\dim V$	default/off	95% interval	targets up/down	S/strong rho, off→default
$n = 11$, K1	4	1.0958	$[1.0940, 1.0980]$	24/0	$7.17 \to 7.86$
$n = 13$, K0	5	1.0699	$[1.0646, 1.0759]$	24/0	$36.9 \to 39.5$
$n = 17$, K1	6	0.9969	$[0.9955, 0.9985]$	6/18	$424.1 \to 422.8$
$n = 19$, K1	7	0.9593	$[0.9576, 0.9611]$	0/24	$1{,}248 \to 1{,}198$

Table 12.3: Inherited F4 at $m = 3$: total S with the root linear-row harvest on (the default since #1862) over off, 24 paired public targets per curve. At $n = 19$ (K0) the word XORs are unpriced and S is flat; priced at 0.1 or 1 GAE the ratio is 0.9755. Source: `research/f4_harvest_remeasure_20261024/README.md`.

online window in `target_PDP` (`research/ic_solver_online_20261003/RESULT.md`). A capture defect (the wrapper's `time` exited 1 on the sandboxed host) makes the panel diagnostic only. Its useful datum is the split count: F5 used 2,546 splits on the first, unsuccessful target attempt against inherited F4's 68, which is why a matrix-kernel gain cannot be projected onto the online path.

The harvest default. PR #1862 made the root linear-row harvest the default; on the one workload a pricing bisect saw it raised counted word XORs by about 7%. A preregistered re-measure on five curves (`research/f4_harvest_remeasure_20261024/README.md`; five sessions of 144 records, 144 of 144 replays identical, counts only) found that the sign depends on size (Table 12.3). The $m = 2$ negative control reads 1.0000 on all five curves.
The default is a regression at $n = 11$ and 13 and a 4.1% engineering gain at $n = 19$; at $n = 17$ the whole gain comes from a different gate (#1891's root-rank gate). Every priced F4 arm sits 7 to 1,250 times the strong rho, and the harvest moves that by at most 10%. The correction is listed in Chapter B.

Determinism. The cross-host drift of §4.5 (item 6) was first searched for within one host: thread count, a second process with new hash seeds and a replay under a different thread count all gave byte-identical word counts on 48 of 48 rows (`research/inherited_f4_replay_determinism_20261018/README.md`), which excluded the within-host sources and left the instruction-set path, later confirmed as the cause.

The priced distance. With its word XORs priced, inherited F4 is about 22–33× the strong rho at $n = 13$, and the complete pair table is 17–22× cheaper than F4 on the same cells; the atlas records the whole Boolean-engine line as *closed, dominated* by the pair-table oracle on Koblitz and random binary curves up to $n = 31$, $m \leq 4$ (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-engine/LEAD-ICE-boolean-f4-f5-f6-engines.yaml`). A later diagnostic on a `chained-x` oracle read inherited F4 at 24.71, 144.09 and 794.37× matched rho at $m = 2, 3, 4$ on Koblitz $n = 13$; matrix F5 cut matrix F4's work at $m = 3$ to 0.6276 and still cost 16,296.76× rho, and both $m = 4$ matrix arms were censored (`research/ecbench_registry_optimization_20261101/README.md`, `research/ecbench_e2e_frontier_20261031/REPORT.md`). The atlas also records that on the S_3 chain F4 and F6-IC grow faster in n than the meet-in-the-middle floor at $m = 3$ and $m = 4$, with the slope-difference interval above zero.

12.4 F6-IC: from closure to lookup

Its results for binary decomposition are in §19.6.

F6-IC adds to inherited F4 a factor-base domain gate and meet-in-the-middle leaves (Chapter 10). Its record after 5 October is a single argument in three steps.

Step 1: the closure is charged, and loses. On the E_0 ladder (eight public targets per size, 72 verified runs per arm and size, word XORs counted but not charged, geometric additions charged) F6-IC's S was 54.75× (37.81–84.22) the same-target rho at $m = 13$ and 2,091× at $m = 19$, against inherited F4's 1.105 and 1.582; at $m = 23$ the two IC arms were 2.72 and 2.09×, and at $m = 31$ both hit the 3,600-second cap on the first target, so the protocol reached no decision (`research/f6_ic_ecbench_ladder_20261005/RESULT.md`). The frozen strong rho of that ladder carried its set-up (§11.2), and the inherited-F4 figures are the unpriced lower bounds that §4.5 (item 5) corrects.

Step 2: the time is in the matrix build, not the gate. Online on the prepared $n = 17$ curve, rho was faster on both frozen public points: rho/IC online 0.0219–0.0221 on T1 and 0.000515–0.000530 on T7 (`research/f6_ic_rho_online_gap_20261006/RESULT.md`). A nested profile of T7 found the Macaulay stage at 81–84% of target PDP for all three arms, of which the matrix construction was 98.5%, and the F6 oracle callback at 1.6–2.3%; eliminating every nanosecond inside the oracle would save about 1.6% of the online window (`research/f6_ic_phase_profile_20261006/RESULT.md`).

Step 3: the few-candidate closure makes the algebra vanish. On 5 November the gate was changed to run the exact one-fixed closure as soon as a summand's assigned bits leave at most 64 base codes, deciding the whole subtree (`research/f6_few_candidates_20261105/README.md`; nine Koblitz cells, 5×3 rounds, L0).

- Verdicts and indices are identical in both arms; F6-IC process CPU falls 2.67–10.29× (median of five; worst round pairing at least 1.66), word XORs 1.84–10.13×, and nodes from 204–1,049 reductions per cell to 5–16. Inherited F4, with no gate, is unchanged (0.87–1.05× CPU).
- The gate's pair lookups and group additions are *identical* with and without the closure. On these cells the algebra was not ruling out a single candidate abscissa: F6-IC with the pair index was the pair-index meet-in-the-middle scan wrapped in Gröbner work.
- The README states the conclusion, and marks it unverified: the change does not beat ecbench's `mitm` oracle, because in the limit it is that oracle.

Two companion items of the same study bound what remains. Re-reducing a node's displaced rows (36–61 median per node, 7.9–24.5 pivots crossed per row) by Four-Russians batching would spend about 32 row operations per pivot on its tables, several times the roughly 1,500 row XORs per node it replaces, so nothing was changed. A parallel split of one target's tree into 2^k subtrees is 1.59–2.88× faster in wall time on refuting cells and 0.94–1.11× on cells where most targets decompose, at 1.12–1.49× (refuting) and 2.67–2.86× (decomposing) the counted work, and stays opt-in because it changes the count.

On the current binary the pair-index F6 is at 0.2792 of inherited F6 and still 139.30× strong rho (§11.7). The F6 line is therefore an engine that became a lookup, and a lookup is priced by §27.2.

12.5 MutantXL closure as an engine

The closure instrument `dreg_closure` (Chapter 14) is also a decision engine: MutantXL-style, it multiplies the fallen lower-degree polynomials by variables and re-closes, and it refuted all 92 unsatisfiable draws of 23 cells at degree 3–4 where plain XL needed 5–7 (§16; `research/dreg_closure_mutantxl_20261016/RESULTS.md`). As an engine its costs are those of a dense degree-4 closure: at $(m, n, \ell) = (3, 22, 8)$ one branch has 124,314 degree-4 columns, a scan of 124–331 s and a peak of 10.6 GB (`research/dreg_closure_guess_20261022/RESULTS.md`). At $(3, 19, 7)$ a hybrid guess that restores degree 4 still costs $5.3 \cdot 10^{11}$ to $1.5 \cdot 10^{12}$ word operations per draw (`research/dreg_hybrid_guess_20261024/RESULTS.md`). The closure's value is as an instrument that measures a degree, which is how Chapter 14 uses it; no study priced it as an oracle against rho.

12.6 Crossbred, exhaustive search and engine suites

A matched engine suite (22 September) froze the contract every algebraic engine is compared under: identical targets in one process, systems fingerprinted, timeouts never read as UNSAT, units named, missing values null (`research/ic_framework_engines_20260922/README.md`). Under it the first end-to-end panel with the strategy as the only variable put enumeration far ahead of every algebraic route: at $n = 9$, $m = 3$, enumerate collected 19,696 relations per second against 51.7 (SAT), 33.2 (Gröbner) and 11.7 (crossbred), and the panel notes that relation collection stays $\Theta(2^n)$ with any oracle polynomial in the factor base (`research/crossbred_20260920/e2e_panel.txt`). A cost-sensitive solver selector fitted on matched queries chose pairs every time; the hindsight oracle and the best constant policy had identical cost, so the portfolio offered zero routing headroom (`research/ic_solver_selection_20260915/README.md`). The current-binary refresh read the best matched $n = 17$ system solver as wide FES/Gray-code enumeration at 4.28× strong rho, with SAT-CDCL, crossbred, exhaustive search, every F4 variant, F5, F6 and every root-finding variant rho-led in their cells (§11.7).

Arm	mean cold S	S/strong rho	class
Strong signed-Frobenius rho	3.998	1.000	reference; native work unpriced
IC, meet in the middle	387.277	96.880	baseline
IC, Frobenius meet in the middle	17.073	4.271	engineering
IC, native-XOR SAT	3.413 (lower bound)	0.854 (diagnostic)	accounting incomplete

Table 12.4: Native-XOR SAT against strong rho, Koblitz $n = 17$, 40 measured runs per arm. The SAT row omits 2,755,966 conflicts in 749 solver calls. Source: `research/ecbench_sat_pdp_20261009/README.md`.

12.7 Native-XOR SAT

The ceiling these engines place on learned guidance is in §19.5.

SAT engines with native XOR reasoning (CryptoMiniSat-style Gaussian elimination, WDSat [39]) are the literature's fastest point-decomposition deciders. The matched pipeline study (`research/ecbench_sat_pdp_20261009/README.md`) needed three protocols to produce one clean panel, and its history is part of the result.

- The first pilot's Buchberger arm timed out and one SAT replay differed in the low bits of its GAE, exposing a cost-replay defect in the historical `ic.pipeline` method; the failed receipt is kept and the pilot is not used.
- The counted method `ic.pipeline_counted` reconstructs charged cost from counts. Its pilot replayed exactly but hit 56 per-call conflict-budget exits in the measured CNF runs, a missed stop gate; the main panel started in error was stopped and is retained outside every comparison.
- The third protocol dropped only the budget-hitting CNF arm.

Its session (Koblitz $n = 17$, $r = 65{,}587$, eight public targets, five measured rounds; 240 of 240 verified, 200 of 200 measured replays identical) gives Table 12.4.

The SAT row's counted S is below rho, and the README says in so many words why that is not a result: rho's native table work is unpriced too, the SAT row also omits the solver, and a ratio of two lower bounds bounds nothing. The phase split makes the point concrete: SAT charges zero oracle set-up and 31,930 GAE of relations, the same order as the MITM arms' 30,383, while its exclusive online window summed to $4.16 \cdot 10^{10}$ ns against rho's $1.22 \cdot 10^{6}$ ns (L0, descriptive). The atlas adds a derived identity: the CNF-XOR model of the point decomposition is a complete enumerator of V^m modulo S_m, with no pruning above the leaves, so its next step is to price conflicts and fit their growth against $2^{m\ell}/m!$ (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-engine/LEAD-ICE-sat-xor-pdp-oracle.yaml`; lead disposition *open, next step*). A separate preregistered study measured that identity directly and the most a learned component could gain from it (`research/pdp_ml_sat_ceiling_20261027/README.md`; symmetrised S_4, $m = 3$, native CDCL+XOR, 4,200 labels all agreeing with exhaustive search). A refutation costs 0.963 ($n = 15$, $\ell = 5$) and 0.998 ($n = 17$, $\ell = 6$) conflicts per sorted factor-base triple, every refutation within 4.1% of its family median: the search is an enumerator. A perfect predictor of unsatisfiable targets would raise relations per conflict 5.2–9.0×, but no feature group reached a test AUC distinguishable from 0.5, while a positive control with labels flipped at rate 0.25 reached 0.72–0.79; the study closed by its rule: the ceiling is real and the signal is not. Until conflicts are priced the row is unranked, and statistics version 2 flags such arms as upper bounds on the wall statistic (§11.3). Prime-field SAT and SMT presentations are in §23.6.

12.8 The $n = 83$ engine campaign

The decomposition results at $n = 83$ are in §18.2.

Degree 83 was the largest Koblitz target with a working pipeline (§15.5). Its engine history has three layers.

October 4–5: F6 components on the $a = 0$ field. About two dozen preregistered micro-studies kept kernel components and failed every ordinary solve (§15.5). The retained components are stage results: a sign-folded pair index stores one representative per pair-sum abscissa and is 1.261–1.618× faster on bases of 258–4,054 points (`research/f6_n83_sign_fold_20261004/RESULT.md`); an ARM64 PMULL implementation of the degree-83 arithmetic made exact queries 6.11–6.87× faster and the index build 1.40–3.05× faster at 258–1,048 points but 0.912× at 4,054 (`research/f6_n83_pmull_xonly_`

Step	Result	Study
All-summand ordering	$m = 2$: conflicts 0.50–0.60× (public); median 12.8 → 6.2 s	`order`
Root-finder tables in front of SAT	$m = 3$: every public target refuted with 0 conflicts (0.38 s); $m = 4$: median 156 conflicts, 0.98 s; $m \geq 5$ capped	`order`
Meet-in-the-middle lookup ($\lvert B\rvert^d$ build)	public targets refuted with 0 conflicts at $m = 5$–8; build 0.68 s ($d = 3$) to 958.8 s ($d = 5$); $m = 9$ capped	`meet`
Witness rebuilt from the lookup hit	planted 5 of 5 at $m = 5$–8 with 0 solver calls (0.91–81.5 s)	`rebuild`
Indexed lookup, rebuild by peeling	$m = 8$: 78.4 → 39.7 s; $m = 9$: 3/3 refuted, 5/5 verified in 54 s; 372,406,144 depth-5 fingerprints	`indexed`
Parallel back enumeration (4 threads)	$m = 8$: 41.7 → 11.4 s; $m = 9$: 57.0 → 13.7 s	`parallel`
Parallel depth-5 build	1,006.8 → 287.7 s, 3.50×	`parbuild`

Table 12.5: Chained-S_3 decomposition on the $n = 83$ fixture, decomposition stage only. Sources: `research/n83_chain_sat_<step>_<date>/README.md` for `order` (31 October) and `meet`, `rebuild`, `indexed`, `parallel`, `parbuild` (1 November).

`20261005/RESULT.md`). The same days found a correctness barrier: the union-of-charts SAT path held each allowed abscissa in a `u64` and extracted only the first 64-bit word, so abscissae differing above bit 63 were indistinguishable and an UNSAT answer could not prove absence of a decomposition (`research/f6_n83_domain_20261004/RESULT.md`). It was fixed in that path.

The algebraic gates of the same days ended open or negative. On the registered K0 curve a standard dimension-18 subspace gives 261,447 geometric points (130,722 sign-folded columns), and the direct five-summand S_3 chain is exact on a planted control with 339 Boolean variables and 332 cubic equations, smaller than the six-summand chain's 428 and 415, but its root-pruning gate fails (`research/f6_n83_fivesum_d18_20261005/RESULT.md`). A recursive carry-less (Karatsuba) circuit for the two dense 83×83 products of the middle links passed its encoding gate and the solver timed out (`research/f6_n83_karatsuba_sat_20261005/RESULT.md`). No natural relation came from F6 at $n = 83$.

October 24: every descent engine had been wrong. A summand sweep with every engine on the pinned $n = 83$ fixture (53-bit subgroup) found the same truncation one level lower: `FieldStructure` truncated the field table to one 64-bit word, and every Weil-descent engine—matrix F4, matrix F5, inherited F4, F6-IC, the F6-IC pair index, crossbred, SAT descent in three encodings and MQ-FES—returned "proved UNSAT" on *planted* $n = 83$ relations (`research/n83_msweep_20261024/README.md`). The descent builders now refuse such a field, a regression test pins it, and every earlier $n = 83$ engine refutation is void (Table B.1). On the corrected build:

- chained-S_3 full-width SAT is the only decomposer that runs at every m, and within its 180-second cap it decides only $m = 2$: 17.3 s per public query (4 of 4 refuted) and 2 of 2 planted witnesses recovered. Its derived projection is $2^{75.9}$ s, $2^{75.3}\times$ rho;
- the $m = 4$ root finder solves the complete DLP at $K = 1107$ in 188.5 and 175.2 s against a strong-rho mean of 1.509 s, 116–125× from rho;
- on the degree-61 companion field at $m = 2$, native-XOR SAT is the fastest decider at $D \geq 12$ (0.79–2.41 ms per query) and inherited F4 the fastest Gröbner engine (6.2–33.4 ms); $m \geq 3$ is refused by the 64-variable descent mask.

October 31 to November 1: chained SAT becomes a lookup. Six frozen-protocol studies on the same fixture (L0, 4-vCPU host, 0 soundness flags in every study) extended the chained-S_3 SAT encoding to $m = 9$ (Table 12.5).

By the last step no decided target makes a single SAT solver call: the decisions are made by a $\lvert B\rvert^d$ table and a back enumeration of cost about $\lvert B\rvert^{m-d}$ per query, which is a generic meet in the middle. Every README states that this is a decomposition-stage measurement with no complete DLP and no matched-rho comparison, and the registry counts the lookup outside its 79+24 (§11.6). The matched strong rho on this fixture solves the whole DLP in about 1.5 s online.

12.9 What the Boolean engines settle

Across the four families the record is consistent.

- Every accepted engine change is a component speed-up of the same search: identical verdicts, identical trees or traces, a faster representation. The largest complete-call factor is the F5 support cut (3.0×); most preregistered candidates failed a 1.05 gate.
- Priced inside a whole solve, the algebraic engines are 22–33× (inherited F4, $n = 13$) to 10^2–10^4× rho, and the gap grows with m (§12.3).
- Where an engine became much faster on index-calculus systems—F6-IC with the few-candidate closure, chained SAT with tables and lookups—it did so by becoming the meet-in-the-middle oracle, whose relation cost is bounded below by §27.2.
- Two field-width defects (4 and 24 October) made engines report refutations that were not proofs; the second was found by planted relations and voided every earlier $n = 83$ engine refutation.

The open engine question the atlas leaves is the one the closure measurements of Chapter 14 address: a descended cell where the pair table is infeasible and the algebraic degree stays bounded.

13 Prime and extension-field Gröbner engines

Over $\mathbb{F}_p$ and $\mathbb{F}_{p^k}$ the decomposition system is not Boolean, and the engine is an $\mathbb{F}_p$ F4: `f4_fp` for generic systems, `f4_fp_tower` with a Faugère–Lachartre block for the Kummer-tower systems of §21.2, a signature variant, and for $k = 4$ the weighted Macaulay solves of Chapter 25. Chapter 10 lists `f4_fp`'s headline ratios (29.8–54.1× over its predecessor, msolve parity 0.96–1.22). This chapter gives the arithmetic rounds behind them, the external references, the one study that priced a faster `f4_fp` end to end, the Kummer-tower engine at large primes, the chain-solve levers on the GLS $\mathbb{F}_{p^4}$ line, and F4 trace replay. As in Chapter 12, an engine gain is a component result until a whole solve is priced against rho, and on every line in this chapter rho stays ahead.

13.1 Arithmetic of the $\mathbb{F}_p$ F4

The $\mathbb{F}_p$ rounds used the gate of §12.2: one binary, paired A/B blocks with an A/A floor, exact basis and solution fingerprints, step counts and matrix shapes on 13 cases, holdout seeds, and a preregistered promotion threshold on the complete F4 call of `quad_n8_p65521`.

- *Narrow 32-bit rows* (`research/fp_f4_narrow_20260926/RESULT.md`). The branch-free 32-bit path read 1.113 [1.111, 1.121] and missed its 1.2× gate. Deferring reductions across each row's pivots (mode 2) read 1.641 [1.631, 1.649] on one thread and 1.580 on four, with 2.101× on the overdetermined $n = 12$ case; a default-on confirmation (1.640 one thread, 1.480 four threads) made mode 2 the default for $p \le 65{,}536$.
- *Lazy vector accumulation.* An AVX2 lazy accumulator read 1.059× [1.055, 1.062] over the deferred default and 1.756× over the original 64-bit arithmetic, below its 1.15× promotion gate and with two smaller-case regressions (`research/fp_f4_lazy_avx2_20260928/RESULT.md`); the AVX-512 version read 1.047× and 1.513× on an Intel host and stays hardware-gated and opt-in (`research/fp_f4_lazy_avx512_20260928/RESULT.md`). The two hosts differ, so the two results are not multiplied.
- *Where the time went by 29 October.* On a shared 4-vCPU host, one thread, `f4_fp` was level with msolve 0.6.5 on `quad_n8` (47.8 against 47.4 ms, msolve including start-up) and faster on seven other systems; 44% of instructions on `quad_n9` were in one multiply-accumulate kernel, and at four threads the reduce phase was still 66–84% of the run, with one in-block dependency loop running on a single thread (`research/f4_f5_followups_20261029/README.md`). The two candidates aimed at those findings were exact and did not save time (§12.2).

An independent campaign in `cryptanalysis` reported symbolic preprocessing 2.34× ($fb = 3$) and 1.99× ($fb = 4$) faster on Semaev systems and Boolean F5B-family bases 1.36–1.49× faster, on a shared host under load averages above 40; its README calls this local progress, not a speed record (`cryptanalysis: experiments/groebner-perf-20260924/README.md`).

13.2 External references

Two external comparisons frame what an in-house engine ratio means.

- *msolve.* A day before the parity result of Table 10.1, msolve was 12–24× faster than `f4_fp` on square systems; after the rewrite the ratio is 0.96–1.22 single-threaded, and the follow-up above found parity or better on eight small and medium systems. The parity is on generic square systems, not on the decomposition systems.
- *Magma.* A protocol registered on 13 October, before any Magma output, exports the exact systems behind the repository's Gröbner claims: 26 Kummer-tower systems at $p = 2{,}013{,}265{,}921$, five $k = 4$ systems and eight Boolean root systems of the $n = 17$ search, every one a unit ideal in the repository's runs (`research/magma_f4_comparison_20261013/PROTOCOL.md`). The directory holds the export bundles and the repository's own timings; no Magma output is committed, so the comparison has not been made.

The desk review of 24 October (§B.3) puts these in proportion: on the PKM systems the in-house F4 needed 10^3–10^7 more operations than a brute grid on the same base, so ratios of a root finder or a table

to that F4 (such as those of §9) say nothing about published Gröbner solvers. Solving *degrees* measured with msolve are unaffected.

13.3 A faster `f4_fp`, priced end to end

One study measured what the `f4_fp` rewrite (PR #1717) did to a complete index calculus in which every decomposition is one `f4_fp` solve: Nagao decompositions on a genus-2 cover of $E(\mathbb{F}_{p^6})$, the same seeds and relations in both arms, every logarithm correct (`research/f4_fp_ic_effect_20261018/README.md`). Per decomposition test F4 used 7.7× fewer multiplications and the cover's counted cost 5.0× less (at $p = 53, 101, 251$). End to end:

p, seed	S before/after	S/rho before	S/rho after
53, 1	5.05	4,297	850
53, 2	4.99	1,843	370
101, 1	5.06	981	194
101, 2	5.08	1,897	373

This is the one case in the programme where an $\mathbb{F}_p$ engine gain passed through to the whole solve almost unchanged (5.0× of a per-test 5.0×), because the cover's decomposition is the solve. The S/rho columns are a lower-bound comparison: neither arm emits its unpriced native work, so no band applies. The distance after the change is 194–850×; the cover itself is the subject of §26.

13.4 Kummer-tower engines at large primes

The tower F4 engine held coefficients in `u32`, which left the Kummer tower at 2-adic valuations $v_2(p-1) \geq 40$ unmeasured. A generic-field engine (`research/f4_generic_field_20261015/README.md`; protocol frozen before any measured cell; shared host, L0) ports the same algorithm to any prime field.

- At `Fp32` it reproduces `f4_fp_tower` field for field: 182 rows of every round-8 series with $N \leq 18$, zero differences in degree, width, steps, multiply-adds and every counter; and it is not slower (paired $u32$/generic 1.091 and 1.058).
- Above 2^{31} it is 2.18× ($m = 2$, $N = 16$) and 2.24× ($m = 3$, $N = 12$) faster than the `u32` engine with eager Barrett reduction, and at 45 bits an AVX-512 IFMA backend is 2.57× the 64-bit one; five of five preregistered claims pass.
- The solving degree is independent of v_2 and of the prime's size in every valued cell: Kummer towers at $v_2 = 40, 48, 56$ and at the P-224 prime, and the Dickson torus at P-192 and P-256, give the 31–32-bit degree, width and step count. At $m = 2$, $N = 18$ the multiply-add count is 45,952,850,022 at 45, 52, 61 and 224 bits.

The last point is the engine-side complement of §21.2: the tower's degree is a property of the system shape, and the prime only changes the cost of a multiplication. In ecbench the four PKM tower engines (F4, F5, F6 and geometric enumeration) all run and read 4.62–5.03× negation rho on the current binary (§11.7); the presentations that change the system shape are in §23.1.

13.5 The chain-solve levers on the GLS $\mathbb{F}_{p^4}$ line

Chapter 25 gives the $k = 4$ engineering ladder to 15 October. In parallel a second line of work moved the same problem onto a Klein-four (V_4) invariant line on GLS curves over $\mathbb{F}_{p^4}$ and ran pre-registered experiments E17–E42 (the V_4 line from E22, 18 October), each recorded in the research note `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md` §9 with its own report directory. On this line the decomposition solve is 97.7–98.5% of S and the linear algebra at most 0.006% (E32, §9.36), so the solve's Macaulay matrix—393×448 at degree 11, 176,064 dense cells per call—is the whole cost. The levers below act on that matrix. Each is priced in the line's own unit,

$$S(g) = \text{counted } \mathbb{F}_p \text{ multiplications} + 97g\,(\text{Macaulay entries} + \text{selection work}), \qquad g \in \{0, 0.1, 1\},$$

which is the band of §4.3 with a native operation priced at $97g$ multiplications, and each is ranked against its predecessor only where the order holds at all three weights on every row.

	Lever	S_A/S_X at 0 / 0.1 / 1 GAE	Rank against A	Standing
E33	B with its selection counted ($1.05 \cdot 10^6$ ops/call)	1.82–1.86 / 0.193–0.197 / 0.149–0.152	unresolved (ahead at 0 only)	validated
	B″ (Markowitz estimate, $\approx 1.27 \cdot 10^4$ ops)	1.538–1.558 / 1.046–1.048 / 0.9465–0.9480	unresolved (behind at 1)	validated
E35	T: B's pivots learned per curve, replayed	1.093–1.252 / 1.020–1.055 / 0.998–1.003	unresolved (6 of 20 rows behind at 1)	producer
E39	F: first-arrival pivots, no selection work	1.457–1.537 / 1.094–1.106 / 1.0115–1.0130	**ahead** on 32 of 32 rows	validated
	S: static order	1.402–1.449 / 1.083–1.091 / 1.0077–1.0086	**ahead**	validated
	H: F with static fallback	1.497–1.542 / 1.096–1.103 / 1.0076–1.0083	**ahead**	validated
E40	V: rows stored from their lead	1.0000 / 1.2518–1.2618 / 1.3629–1.3807	equal at 0, ahead at 0.1, 1	validated
E42b	W: V's tails reserved to the fill (vs V)	1.0000 / 1.022–1.023 / 1.032–1.034	equal at 0, ahead at 0.1, 1	producer

Table 13.1: Chain-solve levers on the GLS/V_4 line, end to end, ranges over the paired rows (20 rows for E33 and E35, 32 for E39–E42b); above 1 the lever is cheaper. E42b is against V, not A. Rho is ahead of every row at every weight. "Validated": the study directory or its `benchmarks/` copy carries a `VALIDATION.json` with a passing independent validator report. Sources: `research/e33_solver_b_pivot_count_20261031/REPORT.typ`, `research/e35_chain_trace_20261102/REPORT.typ`, the research note §9.49 and `research/e39_pivot_rules_confirm3_20261102/`, `research/e40_macaulay_entries_20261102/REPORT.typ`, `research/e42_macaulay_tails_20261103/REPORT.typ`.

The baseline: chain solve and handover ladder. E31 replaced the existing solve (E25's) by its P_2-chain solve with identical decompositions on every call: solver A cut the multiplications per call 2.62–2.69× and solver B (sparsest pivots) 4.99–5.32×. End to end, S_{E25}/S_A is 2.50–2.56× at 0 GAE, 1.39–1.43× at 0.1 and 1.034–1.053× at 1, so A is ranked ahead across the band; B, with its selection pass priced, falls behind E25 at 0.1 and 1 GAE and is unresolved (`research/e31_chain_solve_20261024/`; validated, `VAL-20261024-24ef8b`, `benchmarks/20261024-e31-chain-solve/VALIDATION.json`). E32 then measured the distance over eleven sizes, $\log_2 n = 22.6$–42.4, 33 paired curves: rho is ahead on every curve at every weight, the best curve is 1,143× from rho at 0 GAE, the fitted slope of the distance is -0.243 $[-0.279, -0.205]$ in $\log_2 n$ (Gaudry's $-1/4$), and the extrapolated crossover order is $2^{85.5}$ $[2^{79.2}, 2^{94.3}]$ at 0 GAE, $2^{91.2}$ at 0.1 and $2^{97.6}$ at 1 (`research/e32_handover_ladder_20261031/`; validated, independent replay of 66 of 66 runs). The crossover is an extrapolation 37–66 bits past the top size, not a measured crossing.

The levers. Table 13.1 lists the solver and storage levers that followed, each against the ranked solver A.

Read across the table, the band does the ranking work it was built for.

- *A gain that exists only at 0 GAE.* Solver B multiplies 1.82–1.86× less than A, but its pivot selection reads the matrix about six times per call. Counted, B and its cheaper twin B′ fall behind A from 0.1 GAE, and the Markowitz estimate B″ falls behind at 1 GAE, where its $1.2 \cdot 10^6$ priced multiplications of selection exceed its $2.3 \cdot 10^5$ saving (E33). The validator report states that E33 claims no improvement across the band.
- *A pivot order that is not a property of the curve.* If B's choices were fixed per curve they could be learned once and replayed, as in Joux–Vitse F4 trace reuse [27]. E35 found that no call's pivot sequence equals its curve's trace and that 997–1,000 of 1,000 sequences per curve are distinct; in contested columns B's row agrees with the trace in 69.8–88.4% of cases, and another curve's trace

agrees as well. The replay keeps 18.8–36.8% of B's saving and is break-even with A at 1 GAE.

- *A gain at every weight.* Taking pivots in arrival order (F) removes the selection pass entirely and still cuts A's per-call cost 1.47–1.55×. E36 and E37 showed F, S and H ahead of A at every weight, but each time a pre-registered kill criterion fired on a control's positive check, so neither ranked them; E38's kill fired at its pilot. E39, the third confirmation, used a control that alters a read pivot row after its last use, no kill fired, and F, S and H are ranked ahead of A and E25 at 0, 0.1 and 1 GAE on all 32 rows, under three alternative pricings as well (validated, `VAL-20261102-E39`; `benchmarks/20261102-e39-pivot-rules/`). At 1 GAE the margin is 0.8–1.3%.
- *Gains that exist only at positive weights.* Storing each row from its lead (V) leaves the arithmetic untouched—41.6% of the 176,064 cells lie left of each row's lead and are never read by the echelon—so V equals A at 0 GAE and is ahead only where a stored cell has a price (E40; validated, `VAL-20261102-E40`). W reserves each row only to the block after its build end and is ahead of V by 2.2–2.3% at 0.1 and 3.2–3.4% at 1 GAE (E42b). The report classifies it as a data-structure constant; its size is set by the per-cell price.
- *A kill by the study's own control.* E42 registered a range of $[900, 960]$ re-allocations per call for its growth control, derived from the wrong counter; G0 made 245.6, the kill fired, and W stayed unranked. E42b re-registered the question, with the range derived from W's own overflow rule ($[200, 300]$; measured 246.3), before any ranking data existed (Table B.4). E42b has no validator report.

Distance. None of these levers is an end-to-end change of the distance in kind. The best rows of the table sit at S_X/rho of $5.15 \cdot 10^3$ (H, 0 GAE) to about $6 \cdot 10^3$ (V and W, 0 GAE), rising to $5.97 \cdot 10^4$–$9.4 \cdot 10^4$ at 1 GAE; the worst rows reach $2.3 \cdot 10^5$ at 0 GAE and $7.0 \cdot 10^6$ at 1 GAE. Measured against the success criterion E31, E39 and E40 are verified improvements of the index-calculus baseline (E39 across the whole band, E40 at positive weights), each a constant on one stage; measured against rho they leave the gap where E32's ladder put it, closing as $n^{-1/4}$ with an extrapolated crossover near 2^{85}–2^{98}.

13.6 Trace replay as a component speed-up

Joux and Vitse's observation [27] is that the F4 runs of one decomposition family share their sequence of useful rows, so one run can be recorded and later runs replayed. The programme tried it on four objects.

- *Binary S_3 chain.* Learn/apply trace reuse gave 1.4–2.3× in `cryptanalysis`, and the whole Boolean-engine lead is closed as dominated by the pair table (§12.3).
- *The $k = 4$ weighted solve (E21, 18 October).* Over 400 targets at $p = 53$ and 101 the replay agreed on every decomposition set but fell back on 13–23.5% of targets, because which rows reduce to zero is not stable across targets of one curve; the total cost with fallbacks was 0.95–0.98× of the full solve, a net loss (research note §9.11).
- *The chain solve's pivots (E35).* No per-curve order exists (§13.5).
- *The PKM–Semaev oracle on Curve25519 miniatures (2 November).* `f4_fp_tower` gained `record_trace` and `replay`: a recording run keeps, per F4 step, the S-rows whose residue gave a new pivot and the reducers they reached; a replay forms only those rows and stops at the first step whose leads differ. Only a refutation is a result of a replay—every row lies in the ideal, so reaching 1 is sound whatever the trace was learned on—and a non-refuting replay falls back to plain F4, charged (`research/curve25519_f4_trace_20261102/README.md`).

The fourth object is the one where the idea works, and its numbers show the size of a trace-replay gain (Table 13.2).

Replay costs 0.29–0.45 of plain F4, learning costs one plain run per branch system, and it pays back after about two targets. The preregistered prediction that replay would cost at most 0.25 of plain at $G = 0$ failed (0.29–0.41). The gain is a component gain in the oracle's own unit; the README states that nothing is claimed for wall time, other sizes or against rho. Priced end to end, a $t = 4$, $G = 3$ arm with trace replay costs 0.2837 $[0.2413, 0.3335]$ of the previous Curve25519 winner at $w = 0.1$ and 1.567× more at $w = 0$, so the order flips inside the band, and it is still 27.9, $1.83 \cdot 10^4$ and $1.13 \cdot 10^5$× rho at $w = 0, 0.1, 1$ (§21.7; `research/curve25519_t4_candidate_ecbench_20261104/validator-report.json`, status `pass_with_errata`).

Across the four objects the pattern is the one E21 and E35 state: trace replay pays where the useful-row structure is a property of the system shape (the PKM tower, the binary chain) and fails where it depends on the target (the weighted $k = 4$ solve, sparsest-pivot choices). Where it pays it is a constant between about 1.4× and 3.4× on the oracle, which the relation phase's yield then divides.

t	G	plain/target	trace/target	trace/plain	break-even n^*	fallbacks/replays
6	0	$5.55 \cdot 10^7$	$2.25 \cdot 10^7$	0.41	1.68	0/100
6	4	$7.10 \cdot 10^5$	$3.23 \cdot 10^5$	0.45	1.83	0/1,600
7	0	$7.46 \cdot 10^8$	$2.35 \cdot 10^8$	0.32	1.46	1/100
7	4	$9.70 \cdot 10^6$	$4.13 \cdot 10^6$	0.43	1.74	0/1,600
8	0	$6.98 \cdot 10^9$	$2.32 \cdot 10^9$	0.33	1.50	4/100
8	4	$8.64 \cdot 10^7$	$3.85 \cdot 10^7$	0.45	1.86	6/1,554

Table 13.2: F4 trace replay in the PKM–Semaev oracle, Curve25519 $k = 20$ miniature, $m = 2$, 100 targets per cell, in F4 elimination multiply-adds per target (learning excluded; n^* is the number of targets after which learning has paid back). Selected cells of nine. Source: `research/curve25519_f4_trace_20261102/README.md`.

14 Degree instruments, linear algebra and planners

The engines of Chapters 12 and 13 solve; the tools of this chapter measure, finish or choose. A degree instrument measures the degree at which a system refutes, which sets the exponent of an algebraic oracle long before any end-to-end run is affordable. The linear algebra finishes the index calculus once relations are collected. Planners and autolabs choose factor bases, oracles and parameters. A GPU line and a hashed-ECDL engine measure how far explicit relations reach in practice. Each kind of tool has its own way of being wrong, and the record below is mostly about those.

14.1 Closure and guess instruments

What these instruments measured at $\ell = 8$ is in §19.1.

§16 reports the closure-degree measurements to 24 October: a MutantXL-style closure refutes the chained $m = 3$ descent systems at degree 3–4 where XL needs 5–7, stays at degree 4 up to $\ell = 6$, and rises past 4 at $(m, \ell) = (3, 7)$; guessing one or two variables restores degree 4 there at 1/116–1/260 of the projected degree-5 cost; and the ECC2K-130 $m = 4$ row is +8.43 bits over rho if the number of guesses k stays flat in ℓ and +58.27 if it grows linearly, "which only a run at $\ell = 8$ can separate". The $\ell = 8$ runs have since been made (Chapter B lists the staleness). This section reports them, with the instrument's own controls.

The instrument. `dreg_closure` builds the Macaulay closure of one draw's system degree by degree, records the closure and XL ranks, the columns and the word operations at each degree, and scores a draw as refuting at the first degree whose closure contains the constant. `dreg_score` scores the records against a preregistration. Every study below fixes a cell, draws unsatisfiable and satisfiable systems from a seed, and runs three controls: an identity control that reproduces an earlier committed record field for field, a positive control (a guess set known to refute), and a negative or null control that must *not* refute.

Four guesses suffice at $\ell = 8$. At $(m, n, \ell) = (3, 22, 8)$, with $N = 46$ unknowns and 44 equations, guessing variables 7, 15, 23 and 24 (one coordinate per summand and one intermediate coordinate) made all 64 branches of four unsatisfiable draws refute at degree 4 (`research/dreg_closure_guess_20261022/RESULTS.md`; 68 processes, peak 10.6 GB, 124,314 degree-4 columns per branch). The registered verdict favours the rule $k = m(\ell - 7) + 1 = 4$ over $k = m(\ell - 6) = 6$. The study states its limits: $k = 4$ is an upper bound (the smallest count the container could fit under its column cap), it is one guess set, and the unspecialised $\ell = 8$ system was not measured at degree 4.

Three do not, for that set. Guessing only the top coordinate of each summand (variables 7, 15, 23) refuted on one of four draws; on the other three the first branch saturated degree 4 without the constant (closure ranks 128,213–129,893 of 136,698 columns) (`research/dreg_closure_guess3_20261023/RESULTS.md`). Whether the fourth variable or merely a fourth guess mattered is not separated.

The null holds. The unspecialised $(22, 7)$ system does not refute at degree 4 on 4 of 4 draws, while its guessed versions do (`research/dreg_closure_null22_7_20261028/RESULTS.md`). A scorer defect found in review (it scored only the files it was given) was fixed and the verdict regenerated unchanged.

Invariant guesses. A guess need not fix a raw coordinate; it can fix a linear invariant of the summand-swap symmetry.

- The invariant guess $a_7 + b_7$ is never worse than the raw guess, but does not reliably suffice at $\ell = 8$; three of the satisfiable draw's branches ran out of memory, and the soundness rule was replaced after the data, a disclosed post hoc amendment (`research/dreg_invariant_guess_20261029/RESULTS.md`).
- On eight fresh unsatisfiable draws, preregistered, the invariant guess beats the raw guess on 5 of 5 discordant draws ($p = 0.031$) (`research/dreg_invariant_guess_more_20261029/RESULTS.md`).

- A second invariant on the pair, in place of c_7, was never better on any of 12 draws (`research/dreg_invariant_pair_guess_20261029/RESULTS.md`); the summand-swap invariant alone does not refute below degree 4 at $\ell = 6$ (`research/dreg_pair_invariant_20261028/RESULTS.md`).
- An invariant guess on the second link matches the raw guess on x_3: 6 against 7 of 12 draws, no detected difference (`research/dreg_second_link_guess_20261105/RESULTS.md`). This result carries claim `PCC-20261106-607061` and is an unverified assertion until a pcc verdict exists.

What the $\ell = 8$ results price. The guess study derives the consequence for ECC2K-130's $m = 4$ row, at the cell where that row is priced ($\ell = 33$, $N = 2 \cdot 131 + 4 \cdot 33 = 394$, $w = 1$): the ratio of the degree-5 oracle to the guessed degree-4 oracle, $\binom{394}{\le 5}/(2^k\binom{394-k}{\le 4})$, is 78.2 at $k = 0$, 5.09 at $k = 4$, 1.30 at $k = 6$ and 0.66 at $k = 7$, so the hybrid beats degree 5 only if $k \le 6$ at $\ell = 33$. Both registered growth rules give $k \approx 105$–108 there. The measurements—$k \approx 1$ at $\ell = 7$ on 5 of 8 draws, $k \le 4$ at $\ell = 8$ on 4 of 4—do not exclude a bounded k, and the invariant guess lowers k on some draws. Even a bounded k only recovers the degree-4 row, $2^{+7.62}$ over rho in unconverted units, and the free-oracle floor of §27.2 still binds every row (`research/dreg_closure_guess_20261022/RESULTS.md`). The $\ell = 9$ cell ($N \ge 49$) that would show whether k grows at all needs a larger host; a high-arity discriminator has a frozen protocol (version 4) and a blocked launch (`research/dreg_high_arity_20261101/README.md`).

A null control that failed. The specialisation study of 17 October is the counter-example that justifies the controls. Fixing one variable brought most $\ell = 7$ draws back to degree 4, but the null control refuted on 6 of 8 draws where at least 7 of 8 had to fail, so the registered verdict was "uninformative" (`research/dreg_closure_specialise_20261017/RESULTS.md`). Without the null, "one fixed variable restores degree 4" would have read as a structural finding.

14.2 The linearization-excess rule

For two summands there is a structural number that predicts the solving degree before any solve. On the ECC2K-130 family $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^n}$, with factor base the points whose abscissa lies in a subspace V, let $V^{(2)}$ be the span of products of pairs of elements of V and $e = n - \dim V - \dim V^{(2)}$ the *linearization excess*. Over 656 factor bases (seven families, nine fields) the closure solving degree is

$$D = 2 + [e < -3.5] + [e < -8.5],$$

with mean absolute error 0.06 degrees, 0.08 when each field size is held out in turn (`cryptanalysis: experiments/pdp-degree-heuristics/README.md`). For prime n, $\dim V^{(2)} \ge 2\ell - 1$, attained below $n - 1$ only by geometric progressions $c \cdot \mathrm{span}\{1, g, \ldots, g^{\ell-1}\}$; where the rule puts a random base one degree higher, progressions are 15–40× cheaper per query and per relation, at the same yield. The study also finds that the homogeneous degree of regularity is exactly $\ell + 1$ (the top parts are bilinear), so it is the wrong proxy, and that for three summands the formulation, not the base, decides whether the base matters. None of it is an index-calculus result: every row is a decomposition-stage profile on small fields. The atlas lead that would transfer it—rank candidate bases of equal size by e inside the $m = 3, 4$ closure cells above and measure degree and charged cost with a random control—is *open, unexamined* (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-engine/LEAD-ICE-linearization-excess-fb-prior.yaml`).

14.3 Presentations as an instrument

The same decomposition can be written as different systems, and a system's solving degree, not the decomposition, is what an engine pays for. Two studies of late October measured that freedom directly.

Tree-glued encodings. A tree-glued, cherry-symmetric encoding T of the prime-field point decomposition—each summand pair encoded by its symmetric functions with a degree-2 remainder tower, pairs glued by S_3, every input of degree at most 4 for every arity—was compared with the plain chain C and the full symmetric arm F under msolve 0.6.5 on identical Kummer-base targets (`research/ic_tree_gluing_20261103/README.md`; stage diagnostic, no S, contended timings).

- Against C, T is better in every completed cell: at $m = 2$ its degree stays 4 for $t = 3$–7 while C rises to 5 at $t = 6, 7$, with matrices 1.8–3.6× narrower; at $m = 3$, $t = 5$ it finishes at $D = 6$ while C is cut off in degree 7; at $m = 4$, $t = 4$ it finishes at $D = 7$ in 697 s.
- Against F it loses on cost at $m \leq 4$: at $m = 3$, F reaches $D = 5$ against 6 at $t = 4$, and at $m = 4$ the recorded F row ($D = 8$, from a different engine) was 59× faster.

Six further rounds (`research/ic_tree_gluing_20261103/STATUS.md`, all `[pcc:unverified]`, four round-5 statements under claim records) measured a coupling excess $E = D_T - 4$ of 0, 2, 3 at $m = 2, 3, 4$ and at least 5 at $m = 5$ (memory exhausted at 14.5 GiB), about +1.5 per summand and +0.5 per tower level; no encoding, symmetry or branching tried moved either slope. The line is recorded as closed at cryptographic sizes under the measured laws and open asymptotically on whether E grows by at most one per summand, a question its deciding cells cannot answer under 15 GB.

Contraction order on $\mathbb{F}_{p^3}$. The atlas lead on planning the contraction order of a Semaev network (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-engine/LEAD-ICE-gag-contraction-planner.yaml`) was tested at $n = m = 3$ over $\mathbb{F}_{p^3}$, $p \in \{1009, 4099, 32749\}$, 30 planted instances, under a frozen `experiment.protocol/v1` (`crypto-autoresearcher:research/gag_presentation_matched_20261031/README.md`). All arms returned identical verified relations. The monolithic S_4 presentation used 0.110× the chained presentation's F4 matrix cells at every prime (verdict *supported within scope*, descriptive); the tree presentation used 1.095× and is inconclusive, because one charged retry set the $p = 1009$ A/A floor. An independent `review-run` recommends replication, and a filed correction fixes two statements of the report (one instance, not three, has two decompositions) without changing a verdict. Matrix cells are a stage unit; the study prices no pipeline and makes no wall-time or rho statement.

Reading. Both studies say the same thing in different units: a presentation can lower the solving degree or the matrix size of a fixed decomposition by a constant factor, and in the tree line the slope in m and t stays. Chapter 6 gives the degree theory these measurements test.

14.4 The degree-of-regularity forecast

A factor-base sweep decides one decomposition system per probe target and learns the cell's degree of regularity for certain only once the relation phase has run. The forecast `dreg_forecast/v1` (`src/cryptanalysis/dreg_forecast.rs`) estimates it from the solves so far.

- A decided solve observes the solving degree D; an exhausted solve observes only $D > c$ for the highest completed degree c, a right-censored observation. The Kaplan–Meier estimator of $P(D > d)$ with Greenwood's standard error gives the predicted degree (the median) and a 95% band; a weak prior centred half on the semi-regular degree keeps it defined before the first decided solve.
- The semi-regular degree of the system shape is the yardstick: a cell whose band lies below it is extracting structure.
- The cost side projects the remaining relation phase from the observed per-degree work and yield, charges the relation matrix from its own row operations, and reports the rho floor over that work at 1, 10 and 100 multiplications per group addition; advice is given after four real solves and ranks only where the order holds across that band.

The `dreg-triage` skill wraps it as a loop (plan, run, triage, refine), and both the module and the skill state its limits: it is advisory, never stops or shortens a candidate, leaves `certified_degree_of_regularity` null, and is a stage diagnostic, never a complete-DLP result (`.agents/skills/dreg-triage/SKILL.md`). Its estimator is the same censoring logic that statistics version 2 applies to engine costs (§11.3), applied to degrees.

14.5 Linear algebra

The model of Chapter 2 lists the linear-algebra levers (structured Gaussian elimination, a CADO-style filter, Wiedemann's algorithm [43] and its block form [7], a first-dependency stop). The measurements say that those levers are available and are almost never the bottleneck.

Against CADO-NFS. On synthetic $m = 3$ relation systems over a 62-bit prime, in one session on a shared host (`research/la_engines_20261014/README.md`; protocol and five amendments, each before its data):

- the linear algebra is 0.11–1.9% of 27 real `ic logs` precomputes, and the largest real index-calculus matrix in the repository (N73, 600 columns) took 5.1 h to collect and milliseconds to solve;
- the repository's block Wiedemann beats CADO-NFS `bwc` on the identical matrix at every size measured, 1.86× at 131,072 columns;
- CADO-NFS's filter left a 5–8× smaller matrix than the repository's; the port of that filter runs 86.57 → 8.35 s at $D = 262{,}144$ (10.40× [9.99, 11.06]) and is 1.93× [1.74, 1.96] faster than CADO-NFS's own pipeline on the same matrices (7.83 against 14.73 s); below about 4,096 columns it is slower (0.40× at 1,024, both in milliseconds);
- ecbench's default `IncrementalGauss` grows 6–9× per doubling, 19.1 s at 8,192 columns.

These are component results in wall time at L0. The filter port is the default of `ic logs` since.

Inside the pipeline: a null. When the four sparse solvers were made reachable from `ic.pipeline`, a paired study found that every one costs *more* end to end than the baseline: 1.206× in total on an $E(\mathbb{F}_{29^4})$ $k = 4$ ladder and 5.72× falling to 1.030× over 128–4,096 columns on a scaling curve (`research/sparse_la_timing_20261018/README.md`). The cause is the completion rule: the deferred solvers answer only at full column rank, while `structured-gauss` and `incremental-gauss` stop as soon as the target's column is pinned, so the full-rank arms collected 1.75× and 5.5–10.5× more relations. A positive control with the same full-rank rule and dense elimination reproduced the sparse arms' S. In counted row operations the filtered solvers lose at every size; the unfiltered Wiedemann and Lanczos count 10^3–10^4× the row operations of structured Gauss, growing as about columns$^{2.2}$.

Where it matters. The exception is the GHS-weak plane quartic of §26, where double large primes shrink the factor base and put linear algebra and relations both near $q^{1.0}$. There, structured elimination (B1) was a stage gain of 0.81–0.91 of the B0 cost up to $q = 65{,}479$ and a regression (1.53) at $q = 262{,}111$, recorded as a trade and not adopted; double large primes (B2) are the measured improvement, 0.297 $[0.295, 0.299]$ of B0 at $q = 262{,}111$ (`research/quartic_n4_la_20261021/README.md`). On the GLS $\mathbb{F}_{p^4}$ line the linear algebra is at most 0.006% of S (§13.5). The relation phase, not the matrix, sets the cost on every line measured.

14.6 Planners

`ic plan` chooses a factor base, oracle and parameters for a curve from a cost model. Its validation on six curves is in §21. Two later versions added weight rules—`ic-planner/2` a base of $1.41N_0$ at $w = 1$ on prime curves, `ic-planner/3` $0.71c_0$ on Koblitz curves at $w \geq 0.3$—and an exhaustive re-test of 26 October on 14 new curves (3,328 records, 2,428 band verdicts restated in integers and checked by Lean's `decide`) withdrew all three weight rules in `ic-planner/4`: the prime rule failed its $w = 0$ clause, the binary rule failed at $w = 1$, and the Koblitz rule was dearer at every weight (`research/ic_planner_exhaustive_20261026/README.md`; §22.5, Table B.1). The same re-test found that the residual collector is cheaper than the plan across the band from 36 bits—as a generic collision search, whose curve-free control costs the same and whose exponent matches rho's (§22.3). A planner, in short, is a model of the engines it plans for, and its weight rules were the part of the model the band refuted.

14.7 Autolab loops

The autolab is the programme's loop for proposing and screening candidates without letting the screen promote them. Its contract, used by the tournament since September and by the native `ic_autolab` loop since 4 November, has six steps: propose, build a portfolio, screen in sample, select one challenger per size, confirm on fresh targets with the incumbent, an A/A copy and the matched rho, and record the delta on a ratchet (`.agents/skills/ic-autolab/SKILL.md`). The tool enforces what earlier loops left to discipline: a round is frozen once, target seeds are disjoint from every exposure, the in-sample screen never promotes, an improvement needs the paired interval below 1 at $w = 0$ and the band order at every weight, and every arm joins the trial history so it is not proposed again. Its first family, the Curve25519 miniatures, ran 35 rounds; the generic incumbent settled at the collision floor derived in §21.7, which is

the loop working as intended: the ratchet found the floor and could not pass it. Earlier cycles (17–19 October) on the native GF(2^{83}) norm predicate are in §16. The loop's rows are development evidence until the validation gate.

14.8 Explicit relations on a GPU

A separate line measured how far a complete prime-field recovery with explicit relations reaches on one GPU: factor-base construction and generic meet-in-the-middle relations on the device, elimination, scalar extraction and certificate replay on the CPU, on public fixtures (`research/explicit_relations_e2e_20261030/RESULTS.md`).

- Nine pinned versions recorded every launch, failures included. Version 8 completed 12 of 12 recoveries through 60-bit fields on an RTX PRO 6000, with cold solves of 770.3 and 845.2 s at 60 bits; version 9 recorded one 61-bit recovery and one memory refusal.
- On an RTX 4090 a full-cache query kernel lowered the 63-bit cold recovery from 1,183.8 and 1,185.8 s (baseline before and after) to 1,053.9 s, 11.05% lower with a 0.164% baseline repeat spread, certificates byte-identical (`research/explicit_relations_fullcache_20261101/RESULTS.md`); linear algebra took 0.11 s of the 1,054.
- Block normalisation by a product tree uses 765 source-level field multiplications against 4,608 for dual scans, with no cold end-to-end gain established (`benchmarks/20261030-explicit-relations-gpu/`).

Every clock is L0 with `wall_valid=false`; there is no matched rho and no fastest-external-tool comparison, and the benchmark copy says it is not an admitted result. The relations are generic meet-in-the-middle, so the method is bounded by the collision floor of §27.2; what the line measures is engineering reach, not a distance.

14.9 Hashed-ECDL engines

The hashed factor base and the conditional tensor-contraction route are in §24.1. Their CPU engine (`crates/hashed-ecdl-cpu/`) recorded two matched panels of 120 executions with byte-identical certificates across arms; with persistent meet-in-the-middle tables and batched inversions the complete known-answer solve fell from 45.99 to 3.90 ms ($p = 65521$, $m = 3$) and from 80.54 to 4.12 ms ($m = 4$), and from 0.316 to 0.283 ms at $p = 211$ (`research/hashed_ecdl_cpu_20261030/README.md`; L0, one EKS VM). The larger-field ladder expired with zero executions. The README states that these establish exact software behaviour on fixed fixtures and no speed-up, matched-rho advantage or large-curve attack.

14.10 Component speed-ups and end-to-end changes

Table 14.1 sorts the main results of this part by the three labels of Chapter 11.

Three conclusions follow, each with its scope.

- *The machinery works.* Engines reproduce each other and brute force, sessions replay to the last digit, defects are found by planted controls and pinned rather than hidden, and the band ranks levers whose sign depends on a price. That is a statement about the instrument, made by the instrument's own audits.
- *Engine gains are real and small against the gap.* The largest verified end-to-end engine change in this part is a factor of about 5, against distances to rho of 10^2 (Koblitz, binary, prime) to 10^3–10^6 ($\mathbb{F}_{p^4}$). On the binary and prime lines no engine change moved the distance's slope.
- *Where an engine got fast, it became generic.* F6-IC's closure and the $n = 83$ chained SAT both turned into meet-in-the-middle lookups; the GPU and hashed-ECDL lines measure meet-in-the-middle relations; the one validated crossing below rho is a Frobenius-folded BSGS. Chapter 27 explains why each of these is bounded by the square-root floor.

Result	Effect	Against matched rho
Component speed-ups (identical outputs, stage unit)		
F5 support cut 19	complete call 3.0×	not priced
Few-candidate F6-IC closure	CPU 2.67–10.29×	reduces to MITM
Chained SAT with lookups, $n = 83$	m up to 9 decided	no complete DLP
F4 trace replay (PKM–Semaev)	0.29–0.45 of plain F4	$27.9–1.13 \cdot 10^5\times$
CADO filter port	10.40× (filter)	LA $< 2\%$ of solves
Generic-field tower F4	2.18–2.57×	not priced
End-to-end changes over an IC baseline		
Faster `f4_fp` in the cover pipeline	S 4.99–5.08× lower	194–850× (lower bound)
E31 chain solve (validated)	S 1.03–2.56× lower across band	$4.42 \cdot 10^3\times$ at best
E39 arrival-order pivots (validated)	ahead of A at every weight	$5.15 \cdot 10^3\times$ at best
E40 lead-stored rows (validated)	ahead of A at $w > 0$ only	$6.00 \cdot 10^3\times$ at best
Generator-orbit base, Koblitz (validated)	1.26–3.20×	below rho only at $n = 29$, one column (BSGS)
Harvest default, inherited F4	+9.6% to −4.1% by size	7–1,250×
Accounting and instrument changes		
Accounting v2	references resized	gap larger
Statistics v2	censored runs kept	engine ratios lower bounds
Allocator confound	wall time of small arms	counts identical
64-bit field truncation, $n = 83$	refutations voided	—

Table 14.1: The part's results by class. "Validated": a `VALIDATION.json` with a passing validator report; every other row is producer-reported. Sources in §§11.2–14.8.

15 Koblitz curves

Koblitz curves $E_a : y^2 + xy = x^3 + ax^2 + 1$ over $\mathbb{F}_{2^n}$, $a \in \{0, 1\}$, carry the Frobenius endomorphism τ, which acts on the prime-order subgroup as a known scalar λ; ECC2K-130 is the $a = 0$, $n = 131$ member. They were the programme's main family: the compact-orbit producer, the $n = 83$ campaign, leaderboard Table A and the ECC2K-130 cost model all live here. Random and prime-degree binary curves without τ are in §16, and the Weil-descent degree studies that price ECC2K-130's algebraic route are there too, because their systems use random subspaces rather than Frobenius structure. Every study in this section is listed in the programme record [3].

15.1 Frobenius-invariant and compact-orbit bases

The Koblitz implementation (`src/cryptanalysis/koblitz_index_calculus.rs`) builds GGMP's invariant base from the linearised polynomial of a factor of $x^n - 1$, checks $\pi(\mathcal{F}) = \mathcal{F}$ point by point, rewrites each relation into orbit columns with known λ^k weights, and solves the cofactor-corrected system $ha + hbd \equiv \sum \lambda^{k_i} x_{o_i}$. On small fields the orbit collapse is as predicted: $|\mathcal{F}|/\#\text{orbits}$ of 7.86, 10.89, 12.96 at $n = 9, 11, 13$ for relations and 61.7–168.0 for linear algebra, against rho's 4.24–5.10 (`RESEARCH_KOBLITZ_INDEX_CALCULUS.md`). Because $\ell = \text{ord}_n(2)$ fixes the invariant base's size, the programme moved to bases sized to the instance: divisor bases, SAT-optimised bases, a rank-guided base at $n = 53$ (220 orbits, a complete implicit S_5 formula of 22,887 clauses), and finally the *compact signed-Frobenius orbit* base, which stores one representative per orbit and no pair table (`docs/ic/COMPACT_ORBIT_PRODUCER.md`). In the ecbench factor-base suite it is the best measured Koblitz construction: $S = 9.114$ at $n = 17$ (272 points, 8 columns), 9.985× matched signed-Frobenius rho, against full orbit 18.7× and subspace bases 47× (`research/ecbench_frontier_20261013/`).

15.2 The oracle sequence

The binary oracle went through SAT (XOR-native CDCL after Trimoska et al. [39]; decides the $n = 19$, $\ell = 6$ corpus but does not scale), matrix F4 (whose refutation range was the real frontier), pairs-and-solve for S_4 (validated to $n = 21$, $\ell = 7$, still $O(2^{2\ell})$), meet-in-the-middle pair tables, inherited F4 and F6-IC (§10), and root-finding engines (§9). The boundary ledger's current Koblitz decomposition record is an $n = 31$, $m = 2$ F4 cell with an exact block-6 M4RI acceleration; at $n = 53$ an exact Frobenius-orbit S_5 compression removes the coordinate ROM but every unrestricted solver arm remains UNKNOWN.

The September toy studies that fed this sequence are catalogued as campaigns. At $n = 13$ and $n = 19$, rotated subspace bases covered 1,591 of 2,003 targets with five spaces against 61 with one, and joint rank recovered 58/64 targets against 29/64, without passing a 10% yield gate at $n = 19$ (`research/notes/ecc2k130/`, clusters "rotated-n13" and "rotated-n19"). Finite-fibre S_3 semantics were proved exact and an O-aware CNF made 252.7× smaller at $n = 13$, but no existing solver interface was admitted as an arm; three SAT engines decided 5 satisfiable and 27 unsatisfiable targets at $n = 13$, with the five negatives DRAT-certified. At $n = 131$ a symbolic full-point DAG replayed a single-edge DIMACS instance but could not be admitted past the 64-bit ceiling, and the exact Frobenius census of the balanced $(m, d) = (10, 13)$ rotated base has 1,048,134 columns, the only shape with as many columns as the field. A frozen test of the Frobenius quotient found that toy fields saturate and so test nothing; an independent SAT reproduction solved $n = 13$ and none of four planted targets at $n = 19$ in 1,200 s (`research/frobenius_quotient_sat_20260930/`). At $n = 13$ the presentation of the curve does not matter: yield spread is explained by cofactor balance (correlation 0.850) and there is no curve effect at $n = 19$ (intraclass correlation ≤ 0.006; `research/koblitz-isogeny/`).

15.3 Against a stronger rho: the September reversals

Three September results that put Koblitz index calculus at or below rho were reversed when rho was strengthened, which is why every later row is quoted against the strongest available rho (§4.5). Table 15.1 gives each result as recorded and against the stronger rho.

Result	Case	Measure	Recorded	Stronger rho
Tournament	$n = 43$, $a = 1$, best	retired instr.	0.5347	
	$n = 43$, $a = 1$, triple table		0.409	
	five cells[e]			1.19–2.88×
Orbit DLP	$n = 53$, 1,024 targets	retired instr.	0.2403	
	four rungs			$0.240 \to 0.317 \to 1.294 \to 1.923$
	sweep, $n = 37$			0.794
	sweep, $n = 41$			1.520
	sweep, $n = 53$			1.923
	sweep, $n = 61$			2.297
	single target		0.8392[a]	44.0× [39.4, 45.4]
Compact orbit	$n = 41$, $L = 1{,}024$	corrected rho	0.309×	
	$n = 37$, $L = 1{,}024$	32-walk rho[b]		2.081
	$n = 53$, $L = 1{,}024$			3.086
	$n = 41$, $L = 1{,}024$	cold CPU[c]		1.314
	$n = 53$, $L = 1$			22.675
	$n = 37$, $L = 1{,}024$	instructions[d]		0.5165×
	$n = 41$, $L = 1{,}024$			0.6221×

Table 15.1: IC/rho of the three September results as first recorded and against a strengthened rho. Tournament (15 September): IC against a worker rho's retired instructions; giving rho the index calculus's own canonical form made rho win all five cells (`research/ic_triple_counted_20260923/`; the tournament README, `research/ic_candidate_tournament_20260915/`, carries no correction note). Orbit DLP: the recorded rho used software carry-less multiplication on x86-64; rho was strengthened in four rungs, then the preregistered sweep read rising with n (`research/notes/index-calculus/RESEARCH_STRONG_RHO_LADDER_20260929.md`); [a]the selected-panel crossover withdrawn by the direct arm's cost against a strong single-target rho (`research/notes/index-calculus/RESEARCH_SINGLE_TARGET_STRONG_RHO_N53_20260930.md`). Compact orbit: the recorded value is a point panel; [b]a 32-walk batched-inversion normal-basis rho (`research/notes/ecc2k130/compact_orbit_strong_rho_20260929/RESULT.md`); [c]compact/rho CPU of the orbit-disjoint cold confirmation, which ended with six eligible no-gos (`research/notes/ecc2k130/disjoint_cold_v2_outcome_20261001/RESULT.md`); [d]batched compact IC (`research/notes/ecc2k130/compact_ir_ledger_20260930/RESULT.md`). [e]Rho given the index calculus's own canonical form.

Batched compact IC does use fewer instructions than rho while its cold CPU stays above rho, a gap traced to memory behaviour. Some sixty notes on S3 batch inversion, swap quotients, Bloom prefilters, window screens and K grids each moved a constant and none crossed rho (cluster "compact-orbit-cold"). The boundary ledger of the same weeks reached the same place from the other side. Re-pricing every row against a matched negation rho raised the vs-rho ratios 2.1–6.5× (`research/ic_rho_reference_20260923/`); the best batch row was 4.86× [4.21, 5.55] at $r = 2^{39}$, and a page $r^{1/6}$ law was falsified at the small end (`research/ic_exponent_20260926/`); single-target online IC beat table-less rho at all six sizes (8.84–17.5×) while cold rho won at all six (1.98–31.6×; `research/ic_single_target_20260930/`). At $n = 37$ in ecbench the cold IC/rho was 3.64× in instructions and 13.6× [10.6, 18.7] in counted S, and the $n = 37$ parameters did not transfer to $n = 41$ or $n = 53$ (`research/ecbench_n37_shared_rank_20261004/`, `research/ecbench_n41_n53_shared_rank_20261005/`). An $m = 4$ exponent audit measured a cost slope of 0.985–1.060 bits per unit of n at $n \leq 19$, against the 0.25 the lead needed (`research/ic_m4_exponent_audit_20260928/`).

Other Koblitz lines. The remaining Koblitz campaigns moved constants inside index calculus and are catalogued in the programme record [3]. Instruction-level rounds on the Gröbner stage left the whole relation log at 964× rho on K_0 over $\mathbb{F}_{2^{13}}$ (`research/groebner_instructions_20260925/`); a September autolab campaign cut cold instructions by 47.7% and left the candidate at 4.98× matched rho's instructions (`research/ic_autolab_next30_20260915/`); at $n = 37$ a full-rank compact-orbit gate passed with IC at 22.6× rho (`research/notes/ecc2k130/full_rank_compact_orbit_ic_20261003/RESULT.md`). Pipeline kernels gave laned canonical keys 1.09–1.31× (`research/canon_laned_keys_20261008/`), u64 root batching 3.26× whole-process wall (rank cell corrected to 2.47×), and aimed gathering 1.037× set-up (`research/ic_aimed_gather_20261023/`). Transporting a base through ECC2K-130's degree-263 and degree-7 isogenies was built and certified, but on toy fields the transported and descendant bases cost 3.6–4.0× the original (`research/ecc2k130_factor_base_pilot_20260924/`);

n	Base	Columns	Points	Memory	Rho/IC online	Status
53	shared-log[a]			9.34 GB		
61	compact orbit	600	73,200	2.73 GB	124.85×	exploratory[b]
71			85,200		2,864.1×	exploratory[c]
73			87,600		1,189.7×	exploratory[d]
83	$a = 1$				8.2×	exploratory[e]
61	orbit, $a = 0$	$K = 500$	61,000		13.94× [9.81, 19.13]	admitted[f]
					12.87× [7.30, 19.08]	admitted[g]

Table 15.2: The one-target online ladder: rho online time over IC online time on the same point, reusable IC preparation excluded; peak memory where recorded (peak RSS at $n = 53$). [a]Guided shared-log precompute with an explicit pair table; earlier, a 1,024-target shared-log batch beat batched rho in all 15 paired blocks (secondary, multi-target). [b]Fresh reproduction. [c]First rung past 64-bit packing (u128 words); median of three repeats of one point. [d]The guided-rank precompute took 5.1 h at 57–70 million probes per relation. [e]IC online 5.6–7.9 s against rho 43–71 s; re-measured below. [f]Admitted 12 October: 64 fresh public one-target workloads per host, identical candidate manifest on two hosts; median online speed-up with 95% interval; pinned x86_64, claim-check 64/64. [g]The same, unpinned arm64.

and τ-adic known-scalar arithmetic is 0.413 of the binary cost at $m = 131$ with no measured rho benefit (`research/notes/ecc2k130/tau_adic_arithmetic_20260929/README.md`). On the rho side, GPU iteration redesigns for the ECC2K-130 walk did not beat the shipping distinguished-point walk (`research/ecc2k130_iteration_redesign_20261012/`, `research/dp32_sigma_table_20261012/`).

15.4 The one-target online ladder

The ledger's primary `vs_rho` question is: with reusable IC preparation already done, how long does one unseen public target take, against rho with the automorphism discount on the same point? The ladder (Figure 15.1) is in Table 15.2.

Re-measured on 12 October against a default strong rho on five repeats of the same target, the $n = 83$, $a = 1$ IC was *slower* online, 1.382× [1.036, 1.882], and 218.0× rho in fresh-process wall time (`research/n83_evaluation_20261012/`); the later dense-root reading of 18.5× is in §15.5. For the admitted $n = 61$ point, set-up of about 28 s is outside the window; with it charged, IC costs a median 83.4× rho.

The exploratory ratios are large and irregular because a one-point online window measures the luck of one relation search against one rho walk; the admitted panel's range over 64 workloads is 1.06–648.6×. The admitted point is a constant-factor online result at $n = 61$ (47-bit subgroup) with preparation retained; it is not a cold result, not a scaling law, and not an asymptotic claim.

Retired as an index-calculus win (14 October). Every rung of this ladder compares index calculus with a retained table against rho with none. Given a distinguished-point table of equal or smaller size, Bernstein–Lange preprocessing rho [4] is 53–317× faster online than the compact-orbit arm on the same 64 targets at $n = 41$, 53 and 61 (§31.3). At $n = 83$ (§15.5) rho with a table matched to the index calculus's set-up time or memory won every pair that was run. The ladder therefore measures what a table buys, and generic rho buys more per byte. The validation sweep later found `docs/ic/BOUNDARY_TARGETS.md` still printing the 13.94× as a win; it is superseded (`research/validation_sweep_20261019/B-benchmarks/VERDICT_TABLE.md`).

15.5 The $n = 83$ campaign

Degree 83 was the programme's largest Koblitz target with a working pipeline: 145 study directories between 4 and 21 October (the programme record [3], clusters beginning "n83" and "f6-n83"). In order:

- *F6 micro-studies, 4–5 October* (24 directories, almost all on the $a = 0$ curve). They kept kernel gains (queries up to 2.998× faster, resident memory −52%) and shrank SAT encodings, but every ordinary solve timed out; four-summand coverage was at most $4.662 \cdot 10^{-12}$ of the subgroup and no natural relation was found. On the 81-bit $a = 0$ gate curve, SAT and XCNF decompositions succeed only with a known witness, and the prebuilt-base question is an open problem record.

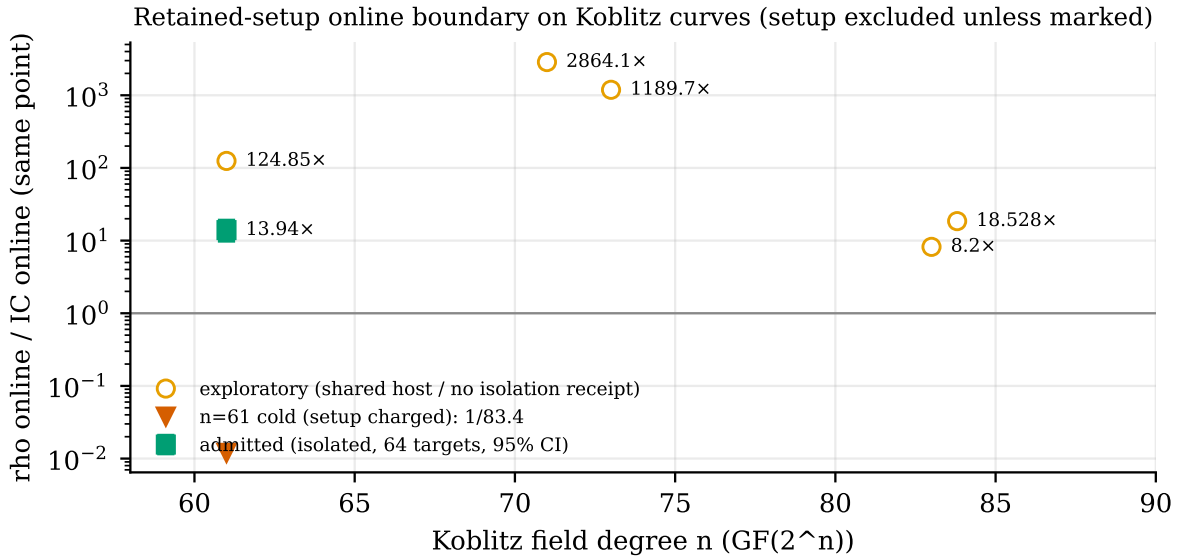


Figure 15.1: Rho online time over IC online time on the identical public point, reusable IC preparation excluded. Open circles: exploratory (verified answers, no qualifying isolation receipt; three repeats of one point or a single point). Filled square: the admitted 64-workload panel with its 95% bootstrap interval. Triangle: the same $n = 61$ candidate with set-up charged (cold). Sources: `docs/ic/boundary_targets.json`, `docs/ic/PLAN_IC_ACCOUNTING_FIXES_20261007.md`, `research/n83_dense_root_20261013/`.

- *The $a = 1$ point, 7–12 October.* The archived K600 point read 8.2× online and was re-measured slower (above).
- *Root-index capacity, 13–16 October* (25 directories). Root-index layouts grew the $a = 1$ base from K512 (84,992 points) to K3596 (596,936 points) within 14 GiB. The dense root index cut probes per relation from 300,021.9 to 169,069.8 (K1500 to K1798; paired difference 130,952 [52,058, 209,990]), a within-IC result that stands; its one-point online 18.528× is retired (`research/n83_dense_root_20261013/`).
- *Factor-base selection, 14–16 October* (56 directories). Root4, root11 and root21 layouts, Bloom filters, sparse and canonical variants and K screens from 100 to 20,000 points selected within index calculus; the best means are about 26–29 ms online (K3425, K3450, K3500), and many screens end BOUNDED_UNKNOWN. A three-hash 512 MiB Bloom filter cut a K3300 online mean from 34.77 to 31.32 ms (`research/n83_root11_bloom_results_20261015/`); a K3350 win of 7.18% did not replicate.
- *Tier-1 online readings, 14–16 October* (14 directories): median rho/IC of 54.1, 210.3, 113.0, 104.2, 35.1, 413.3 (one point) and 208.7× (K3072, five fresh points, [38.3, 2,510]; `research/n83_tier1_online_20261016/`), all against a rho with no table and with set-up excluded.
- *The control, 16 October.* Bernstein–Lange rho given a distinguished-point table matched to the index calculus's set-up time or memory was faster online in every pair run: IC/rho 3.74 [2.65, 11.05] (K3300, time-matched), 40.61 (K3300, memory-matched), and 81.5–395.5 (K600) (`research/20261016-n83-precomputed-rho-control/`). Only the K3300 and K600 bases were paired; the K3072 reading was not, so its retirement rests on the same mechanism rather than on its own measurement. The validation sweep and the ledger audit retired all the online readings on 19–20 October.
- *Cold, 16–21 October* (11 directories). The cold one-target process fell from about 150 s (K1024) to 119.8 s (K1120, holdout) and 113.8 s (K1107; `research/n83_cold_base_confirm_20261021/`), against a rho online time of about 2.2–2.6 s on the same hosts. No matched cold rho was run, so no cold ratio is admitted; the hosted single-target benchmark's 133.7× is the nearest cold figure (`benchmarks/20261014-n71-n83-hosted-single-target/`).
- *A second pipeline in `cryptanalysis`* (K600, $a = 1$; [3]). A Tier 1 ratchet of rank-probe, closure, index-build and rank-finder changes brought its latest row to 3.92× a tuned rho end to end, cold, and 0.093× online with set-up excluded (`ca:experiments/koblitz-n83-rankfinder-20261021/README.md`). The study's own limits apply: one run on one host class, a point estimate with no interval, wall time, factor-base construction charged to no arm (so the cold cost is a lower bound), no computable

(a) Rules: cold IC/rho on one unseen public target

Rule	Date	IC/rho, cold	Status
v3		5.23–21×	printed in Table A; unvalidated (pre-gate)
v6		3.78–12.06×	rho reference 0.57–0.85 of strong walk[a]
v7		3.43–11.51×	rho reference 0.57–0.85 of strong walk[a]
v8	19 October	4.67 [4.34, 5.07] to 18.07 [15.98, 20.58]×	matched GFNI rho; archived as incomplete[b]
v4 recipe	24 October	9.9–43×	validation file withheld[c]

(b) Pipeline baselines, 16–17 October: cold speed-up

Baseline	Change	Cheaper cold
v4	head-of-main	
v5	scan-key	1.20–1.24×
v6	scan-fold	1.33–1.40×
v7	build streams	1.07–1.13×
v8	admitted prefetch	1.06×
v4→v8	together	1.90–1.98×

Table 15.3: Leaderboard Table A, Koblitz $n = 41$–61. (a) Each rule's cold IC/rho range over its sizes, against rho on the same point (rules v6 and v7: `benchmarks/20261016-ic-rule-v7/`; v8: `benchmarks/20261019-ic-matched-rho-v8/`; v4 recipe: §31.5). [a]Per step, so the distance was understated. [b]1.30–1.58× the v7 control, fitted slope 0.168 per $\ln r$; its row is listed although the run is archived as incomplete. [c]A paired retuning improved one size of six; rho is priced in the same process at the canonical batched step, and the validation file is withheld because that reference may not be the strongest, so it is not an admitted row either. (b) Each baseline's recorded speed-up, and the combined cold factor over v4 (`benchmarks/20261017-ic-r12-admitted-prefetch/`).

band, and gaps that are not comparable across runs (the same producer read 4.42× on another host). Against Bernstein–Lange rho with equal advice its online time is 22.55× [16.23, 31.44] worse (`ca:experiments/koblitz-n83-bl-rankprobe-20261019/`). It is the closest cold Koblitz figure at $n = 83$ and is still above rho; the two pipelines' numbers are not comparable with each other.

An instrument correction (24 October). Every Weil-descent engine the programme ran at $n = 83$ had silently returned "proved UNSAT" on planted relations: matrix F4 and F5, inherited F4, F6-IC and its pair index, crossbred, three SAT-descent encodings and MQ-FES. The field table is truncated to one 64-bit word, so earlier $n = 83$ engine refutations are void. The descent builders now refuse such a field, under a regression test. After the fix, chained-S_3 full-width SAT is the only decomposer that runs at $n = 83$. It decides only $m = 2$, at 17.3 s per public query, which projects to $2^{75.3}$× rho. The $m = 4$ root finder solves the complete logarithm in 175–189 s against strong rho's 1.509 s, which is 116–125× rho (`research/n83_msweep_20261024/`).

The open pull requests at the time of writing include further $n = 83$ work ($m = 4$ queries on a retained 16,627-column base, AWS replays, shared-binary panels); none is on `main` (the programme record [3]).

15.6 Leaderboard Table A: the Koblitz rules

Table A of `docs/ic/LEADERBOARD.md` prices one unseen public target on Koblitz curves at $n = 41$–61 ($\log_2 r = 36.6$–47.2), cold, against rho on the same point. Its history is a sequence of rules, each a pipeline baseline plus a rho reference (`research/ic_tool_program/`). Table 15.3 lists the rules and the pipeline baselines.

Under rules v6 and v7 cold rho was still faster at every size (Table 15.3(a)). Seven further baseline candidates (prefetch gates, filter slots, huge pages, a wider base) did not clear their rounds. Table C prices a 32-target batch on nine Koblitz curves ($\log_2 r = 18$–47.2) against batch rho on the same targets: 3.62–14.9×, a historical diagnostic under the one-target rule. All nine failed the 24 October validation gate, and Table D, the Joux–Vitse historical study, is withheld.

The validation sweep found that Table A understates its gap to rho by 1.7–2.0×, and the 24 October

Route	Setting	Cost	Above rho
Decomposition oracle	family bottom	$2^{132.58}$	
	Frobenius-orbit unions, $m = 3$	$2^{124.99}$	$2^{64.18}$ times
Relation sweep	best single relation, $m = 4$		+4.98 bits
	best honest $m = 8$, Frobenius quotient		+13.59 bits
Linearisation	plain		$70.19 - b_{\max}$ bits
	best bilinear XL, $\omega = 2$		71.82 bits
	same, $\omega = 2.81$		78.15 bits
	trilinear		87.16 bits
Weil descent	$m = 3$, free oracle	$2^{68.58}$	
	$m = 4$, degree 4	$2^{68.43}$	+7.62 bits[a]
	$m = 4$, degree 5, $w = 1$	$2^{71.13}$	+10.32 bits
	$m = 4$, degree 5, $w = 2$	$2^{106.52}$	+45.71 bits
	$m = 4$, hybrid guess		+8.43 to +58.27 bits

Table 15.4: Priced algebraic routes to ECC2K-130. *Setting*: the variant priced (m summands, w as in the source, ω the linear-algebra exponent); *Cost*: the route's cost as recorded; *Above rho*: its distance to rho, as a factor or in bits. The decomposition bottom needs a materialised base of $2^{44.53}$ points at $m = 3$; the $m = 4$ single relation needs $2^{70.79}$ bytes; linearisation gaps charge each Macaulay solve; the descent rows are in unconverted column touches. [a]Now contradicted. Sources: `research/notes/ecc2k130/RESEARCH_ECC2K130_DECOMPOSITION_TARGETS.md`, `research/notes/ecc2k130/RESEARCH_ECC2K130_RELATION_SWEEPS.md`, `research/linearization_reach_20260930/`, `research/xl_bilinear_core_20261006/`, `research/xl_trilinear_t_unknown_20261006/`, `research/notes/ecc2k130/RESEARCH_ECC2K130_DESCENT_DEGREE.md`; the hybrid guess is in §16.1.

validation gate did not check Table A (`research/ic_validation_gate_20261024/`). Rules v8 and v4 measure the same sizes against differently built rho references and are kept apart (Definition 4.1).

15.7 ECC2K-130

ECC2K-130 is the $a = 0$, $n = 131$ member of the same family, with a 130-bit prime subgroup and rho at about $2^{60.81}$ group operations. The compact producer already solves $n = 41, 53, 61, 71, 73$ of the family; about 300 notes under `research/notes/ecc2k130/` price the routes to $n = 131$ itself. No route prices below rho. Table 15.4 lists the priced algebraic routes.

- *Extrapolating the producer.* The fitted probes-per-relation law $C\,r/(n^2K^2)$ with $C = 1$ gives $1.1 \cdot 10^{18}$ core-years at $K = 600$ and $6.8 \cdot 10^8$ at $K = 10^{12}$ columns (`RESEARCH_ECC2K130_IC_FEASIBILITY.md`, with the unexplained factor-of-5 gap in C of §4.5).
- *Decomposition oracles.* Decompositions exist, but no oracle finds one for less than 2^{131} per family; unions of Frobenius orbits lower the family's bottom but not to rho. Crossbred, RR-solver and WDSat oracles are engineering only: Crossbred is 0.023 of F4's bit operations at $m = 3$ but slower end to end, and the RR solvers' advantage over Semaev at $d = 6$ is worthless against brute force.
- *Linearisation and Weil descent.* Neither reaches rho (Table 15.4). The studies behind the descent rows, and the hybrid guess for the $m = 4$ row, are in §16.1.
- *Covers.* Over $\mathbb{F}_{2^{131}}$ every ordinary curve has GHS magic number 1 or 131, so GHS descends ECC2K-130 to genus 1 (a zero map on the subgroup) or to genus about 2^{130}, and the Hess–Menezes–Teske walk has nowhere to land. One question is open: $\langle G \rangle$ is the group of a simple 130-dimensional abelian variety, and if that variety were isogenous to a Jacobian with an evaluable isogeny, index calculus on the genus-130 curve would cost $2^{37.17}$, $2^{23.64}$ times *less* than rho; no such curve is known, the Torelli codimension at genus 130 is 8,128, CM constructions with prescribed Frobenius stop at genus 3, and the largest published Weil-descent discrete logarithm is at genus 31 over $\mathbb{F}_{32}$ (§26) (`research/notes/ecc2k130/RESEARCH_ECC2K130_HYPERELLIPTIC.md`, with the magic-number correction of `docs/audits/GHS_MAGIC_NUMBER_MMT_CORRECTION_20261013.md`). The $N = 130$ counterfactual is genus 255 at a modelled $2^{55.60}$ against rho's $2^{60.31}$, and measuring one relation trial turns that margin negative (§31.5). The best explicit curve construction reaches genus $2^{28.92}$ (`research/notes/ecc2k130/RESEARCH_ECC2K130_CURVE_CONSTRUCTION.md`). The isogeny class has 262 curves of conductor gap 263 and no measured level effect (§26); rho on the degree-263 leaves keeps only the negation map, an 11.45× penalty (`research/notes/ecc2k130/isogeny_rho_speed_20260930/`).

- *Rho itself.* The separate GPU rho estimate is about 4.3 RTX PRO 6000 card-years (`paper/ecc2k130/`); on one CPU core the ECC2K-130 client walks 13.2 million steps per second (`research/certicom131_rho_ic_bench_20261022/`). Signed σ steps cut modelled GPU time to 0.961–0.970 (`benchmarks/20261014-ecc2k130-signed-sigma/`).

Index calculus is not a route to ECC2K-130 on this evidence: every route that has been constructed or priced is at least about 5 bits above rho (the $m = 4$ single-relation row, at $2^{70.79}$ bytes), most by tens of bits, and the one hypothetical route below rho needs a genus-130 curve that no known method can construct.

15.8 Factor bases closed under $\mathbb{Z}[\tau]$

The $\mathbb{Z}[\tau]$ small-norm multiplier base of 23 October (`research/ic_improve_20261023/README.md`) is the latest Koblitz change: 1.9–4.4× child CPU at $n = 41$–61 and 6.2–10.5× at $n = 71$, against the programme's own producer, exponent unchanged; it is discussed with the other known-logarithm bases in §31.5.

16 Random and prime-degree binary curves

This section covers binary curves $y^2 + xy = x^3 + ax^2 + b$ over $\mathbb{F}_{2^n}$ with n prime, attacked with random subspace bases that use no Frobenius structure, whether the curve is random or Koblitz: the Weil-descent route and the degree question behind the subexponential claims (§3), the decomposition oracles, the random-binary leaderboard rows, and the 131-bit ECC2-131 target. Many of the descent studies were priced for ECC2K-130 because it was the programme's named target, but their systems are built on prime-degree fields with random subspaces and use no Frobenius structure (the systems at $n = 19, 21$ say so explicitly), so they belong here. The Gröbner and $\mathbb{F}_2$ engines they run on are in §10; every study is listed in the programme record [3].

16.1 The degree question, measured

The subexponential claims for prime-degree binary curves rest on the first-fall-degree heuristic: that Weil-descended summation systems have a solving degree close to their first fall degree (FFD), which stays small. The programme measured both, on toy fields, in about thirty preregistered studies from late September to 24 October.

First fall degree stays low; the solving degree does not. Every Weil-descended S_3 system for $n = 3$–7 has first fall degree 3 (`research/notes/index-calculus/RESEARCH_FFD_MEASUREMENT.md`), and the HKY last fall degree d_F is 3 or 4 at every reachable size—$m = 2$ to $n = 32$, $m = 3$ at $\ell \le 6$, $m = 4$ at $\ell \le 3$, $m = 5, 6$ to $N = 54$ unknowns—while the plain refutation degree rises from 5 to at least 7 and random controls rise with it (`research/fall_degree_bounds_20261008/`). At $m = 2$ the Semaev d_F sits one below the semi-regular degree at $n \le 18$, and the gap does not grow over $n = 10$–18.

The refutation degree tracks ℓ, within one of semi-regular. For chained $m = 3$ systems at fixed surplus the refutation degree reads 5 at $\ell = 2$, 6 at $\ell = 3, 4$, and exactly 7 at $\ell = 5$ on all four draws of the $(10, 5)$ cell, with FFD 3 throughout (`research/dreg_ell_grid_20260925/`, `research/dreg_surplus_control_20260925/`, `research/dreg_degree7_ell5_20260930/`). At $m = 4$ it reads 5, 6 and at least 7 at $\ell = 1, 2, 3$, flat in n at fixed ℓ (`research/dreg_m4_chain_20261006/`). In 10 of 14 exact cells the semi-regular D_{reg} equals the measured degree and in all 14 it is within one. With a dense in-tree engine at $\ell = 6$ ($n = 19$, both Koblitz curves) the x_4 formulation refutes at 11 and the RR formulation at 8 on every draw, 4 above the sharp first-fall bound of 7, rising one degree per rung from $\ell = 3$ (`research/ic_dense_ladder_20261004/`). A registered semi-regular law then fitted none of seven discriminating cells, with x_4 inside $[d_{\mathrm{reg}}, d_{\mathrm{reg}} + 1]$ on all 32 draws (`research/ic_semiregular_law_20261005/`).

Closure algorithms refute earlier, but rise. A MutantXL-style closure refutes all 92 unsatisfiable draws at degree 3–4 where XL needs 5–7, and none of 345 satisfiable draws (`research/dreg_closure_mutantxl_20261016/`). It stays at degree 4 to $N = 42$ at $m = 4$, $\ell \le 4$, but rises past 4 at $m = 3$, $\ell = 7$ on 8 of 8 draws (`research/dreg_closure_scale_20261016/`); the number of unknowns is not the driver, and of two candidate explanations only "$\ell = 7$" survives a run at $(m, \ell) = (4, 6)$, $N = 54$, which refuted at degree 4 (`research/dreg_closure_driver_20261016/`, `research/dreg_closure_m4l6_20261018/`). Guessing one or two variables restores degree 4 at $(3, 7)$ on 10 of 10 draws at 1/116–1/260 of the projected degree-5 cost (`research/dreg_hybrid_guess_20261024/`).

What the measurements price. Applied to ECC2K-130's $m = 4$ row, the closure results contradict the degree-4 floor (+7.62 bits over rho) and leave degree 5 as the optimistic case, +10.32 bits at $w = 1$ and +45.71 at $w = 2$; the hybrid gives +8.43 bits if the number of guesses stays flat in ℓ and +58.27 if it grows linearly, which only a run at $\ell = 8$ can separate (§31.5). The $m = 4$ cost slope on this engine is 0.985–1.060 bits per unit of n against the 0.25 an exponent lead would need, and a symmetrised refutation degree rises about one per unit of ℓ (`research/ic_m4_exponent_audit_20260928/`, `research/ic_m4_head_engine_20260929/`, `research/ic_symmetry_lever_slope_20260929/`).

ℓ	Speed-up
8	94–277×
9	2,474×
10	4,244×
11	928×
12	2,321×[a]

Table 16.1: The meet-in-the-middle pair table over Gaudry's pairs-and-solve on $y^2 + xy = x^3 + zx + 1$, by rung ℓ: wall-time speed-up on a shared host, unvalidated; 128 targets per rung, no disagreement. [a]The validation sweep's recomputation from the committed per-target medians; the study quotes 2,187×. Sources: `research/binary_mitm_oracle_l8_20261015/`, `research/binary_mitm_oracle_l9_l10_20261015/`, `research/binary_mitm_oracle_l11_l12_20261016/`.

Levers that did not move the degree. Decomposition trees cut on the curve's own structure have the same sampled cut-out degree as random sets in all 102 cells (`research/ic_tree_split_structured_v2_20261011/`); nonlinear factor-base labels refute nowhere through degree 6 (`research/nonlinear_factor_labels_20261002/`); Hamming-weight ideals answer correctly but resolve only after 7–18 of n coordinates are assigned (`research/hamming_ideal_pdp_20260930/`); the one measured subspace-shape effect on the degree was a trace artefact (`research/notes/index-calculus/RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md`), and frozen shape cells were killed at a gap of zero (`research/shape_search_g1_g6_20261015/`). The offensive first-fall-degree thread found no lever within 9 bits of the 2^N boundary (`research/notes/index-calculus/RESEARCH_DEGREE_REDUCTION.md`). A census over $\mathbb{F}_{2^7}$ found non-subfield quasi-subfield polynomials below the HKPYY Lemma 4.1 bound, which answers a tightness question but moves no cost (`research/notes/index-calculus/RESEARCH_QUASI_SUBFIELD.md`).

The same question in crypto-autoresearcher. The companion repository measured it independently ([3]). msolve's solving degree of the descended $m = 3$ system is 7 at every n from 11 to 23, a lower bound from $n = 17$ on, where the run walls on matrix size (`ar:experiments/EXP-ALBIN-008/analysis.md`); the study reads this as evidence on the bounded side in a narrow range that cannot distinguish bounded from slowly growing. A degree-4 reuse closure refutes all 400 unsatisfiable S_3 descents at $n = 19$ that plain degree-4 Macaulay refutes none of (`ar:experiments/EXP-CERTBIN-060020/`), the same effect as the MutantXL closure above. A crossbred solver's total exponent is 0.891 $[0.869, 0.914]$ against rho's 0.498 over $n = 10$–24 (`ar:experiments/EXP-ICI-001/analysis.md`); on its older range, $n \leq 20$, the fit reproduces the repository's earlier 0.863 as 0.864.

Reading. On toy fields, the evidence says the solving degree of Weil-descended summation systems grows with ℓ and stays within one of the semi-regular prediction, while the first and last fall degrees stay at 3–4. That is the behaviour of a generic system, and it supports no subexponential claim. It does not refute one: the sizes are small, the closure algorithms bend the curve by a degree or two, and the $m = 4$ question is still open. The $(m, \ell) = (4, 6)$ cell has since run and refutes at degree 4 on 2 of 2 unsatisfiable draws (`research/dreg_closure_m4l6_20261018/`). At $\ell = 8$, only $m = 3$ has run, with guessed variables (§31.5). The $m = 4$, $\ell = 8$ cell has not run. The PKM analogue over prime fields (§21) is in the same state.

16.2 Decomposition oracles

The pair-table floor. On the non-Koblitz curve $y^2 + xy = x^3 + zx + 1$ the meet-in-the-middle pair table is the floor every oracle is measured against. Table 16.1 gives its speed-up over Gaudry's pairs-and-solve, with no disagreement on 128 targets per rung. These are wall times on a shared host and unvalidated; the counted evidence—8,068 and 16,092 point additions against 2.1 and 8.4 million quartic solves—carries the result. An earlier pairs-and-root-find oracle was 926× faster than triple enumeration at $\ell = 8$ and decided $\ell = 12$ (11.5 billion triples) in 15 s (`research/notes/index-calculus/RESEARCH_SEMAEV_DECOMPOSITION.md`).

Algebraic oracles against the floor. No algebraic arm beats the pair table: the cheapest is 303× the floor at $n = 13$ and $3.8 \cdot 10^4$× at $n = 31$ for $m = 3$, and the slopes in n are steeper (F4 0.534, F6-IC

0.583, floor 0.207; at $m = 4$ a wide chained solver 0.805 against 0.333; `research/ic_m3_algebraic_vs_mitm_20261014/`).

SAT, SMT and WDSat are enumerators. Every SAT, SMT and WDSat back end spends about one conflict per candidate tuple (CDCL 0.97–1.02 per triple), so the best enumerate-and-root-find oracle beats every solver by 10^2–10^4 (`research/notes/index-calculus/RESEARCH_PDP_SAT_SMT_NAGAO_SURVEY.md`). The SAT pipeline decides all 11 satisfiable $n = 19$, $\ell = 6$ corpus instances in 17.8 s; SMT runs at 0.44–2.63× the native CDCL+XOR solver (`research/pdp_smt_oracle_20261008/`); WDSat solves $n = 5$ (47 against 51,425 conflicts with and without XOR-Gauss) and crashes at $n = 9$ on a static allocation limit, where CryptoMiniSat takes 0.4–2.6 s (`research/wdsat_rr_20260920/`). Run unmodified on the 60 published S_4 instances of Trimoska et al., WDSat is the fastest solver, 2.37 s on the unsatisfiable $n = 19$, $\ell = 6$ cell against 124.6 s for CryptoMiniSat with XOR clauses (`ar:experiments/EXP-ICPERF-66fd51/`; [3]); that is a solver comparison on toy instances, with no relation, linear-algebra or rho column. The SAT route is closed as an exponent lead. Isogenous and random-b neighbours give identical Macaulay ranks and FFD 3 (`research/pdp_isogeny_probe_20261008/`).

The PDP oracle programme. The point-decomposition speed-up programme (`research/pdp_oracle_speedup_20261011/`), run over $\mathbb{F}_{2^{3\ell}}$, measured the oracle in steps before root finding replaced it: a baseline (PS-0; 338–452 ns per pair across five rungs, A/A spread $\leq 8.8\%$), a gate-count model of the group-law form (PS-1; 7,410–14,608 fused ternary-logic issues against 70,695–190,275, 9.5–13.0×), a scalar group-law kernel (PS-2; 29.8–40.8 ns per pair, 0.067–0.123 of baseline, 400 corpus root sets agreeing), and a bit-sliced kernel (50–62× the baseline on two hosts, about equal to the scalar kernel). The root-finding oracle then ran 3,007–115,160× the PS-0 baseline at $\ell = 8$–14, at the generic floor's slope (`benchmarks/20261012-pdp-root-finder/`). A split/hybrid solving-degree study (Koblitz $b = 1$) projected its closest relation phase at 22.8× strong rho, 391× signed-Frobenius rho (`research/pdp_split_dreg_20261011/`); with Frobenius closure the priced S fell to 0.35–0.54 of the standard base but the best arm stayed at 36× strong rho at $n = 11$ and 889× at $n = 19$ (`research/pdp_split_slots_20261012/`). A shape-replay switch cost more on all 9 cells and stays off (`research/pdp_shape_replay_20261015/`).

Norm and circuit searches at $n = 83$. The 17–19 October autolab cycles built a native norm predicate over GF(2^{83}): a cutoff-8 circuit cuts AND gates 60.5% and clauses 32.9%, an affine support limit of 2 cuts internal coordinates from 58,176 to 45,427, and 1,237 of 1,800 positive-dimensional cells were certified empty. Every hard search returned UNKNOWN with zero verified witnesses, and a primary-source check found the hybrid ANF/CNF approach already established in Bosphorus, SMFF and PolyBoRi (`research/autolab_cycle_20261019_v1/`, `research/autolab_cycle_20261019_v3/`). The companion draft `papers/norm-rootfinding/` reports the same: no measured speed-up.

Toy end-to-end. On toy fields the fastest index calculus, a cached Nagao arm, was 2.33× plain rho, and S_4/S_3 with CryptoMiniSat 14.5–14.7× (`research/toy_ic_e2e_candidates_20260922/`). Frobenius transport preserves zero and lift sets; a constant term c changes zero counts only; neighbouring curves show no consistent F5 advantage; symmetry quotients help three summands (0.26–0.31 of the ordered cost) but not two (1.06–1.32) (`research/toy_binary_elimination_20260924/` and the toy cluster).

16.3 Leaderboard rows

Table B of `docs/ic/LEADERBOARD.md` has five random-binary rows, $\mathbb{F}_{2^{15}}$ to $\mathbb{F}_{2^{27}}$ ($\log_2 r = 14$–24.4), one target, $m = 2$, folded pair table with walked targets: IC is 13.1, 10.6, 18, 23 and 38.2× matched rho. All five passed the validation gate (commit `a8d8e3cd38`; `research/ic_validation_gate_20261024/`), which the ledger paper, written a day earlier, records as no row passing. The cofactor-projected base of LEAD-19 cuts three of these rows by 1.29, 1.58 and 3.36× in cold S, leaving 6.1–14.4× to matched rho (`research/lead19_projected_base/`); it is not yet a leaderboard row. The integer-multiple closure of 23 October cuts S 3–4× against plain seeds at degree 26–28 and remains 6–12× the framework's rho reference (§31.5).

16.4 ECC2-131

The Certicom ECC2-131 target ($\log_2 r = 130.0$, rho $2^{64.83}$ GAE with the negation map) has a cost model, gates and two falsifiers (`docs/ic/CERTICOM131_IC_PROGRAM.md`), all on toy instances. With a free oracle the $m = 3$ floor is $2^{65.2}$–$2^{67.2}$, above rho from linear algebra alone; $m = 4$ is the first usable arity and needs a solving degree of at most 3 (generous pricing) or 1 (plain). On random-b curves at $m = 3$, $n = 5$–11, the measured degree is $D = 4, 5, 6$ at $\ell = 1, 2, 3$, equal to the semi-regular reference with zero spread over 60 draws, and equal to the Koblitz $b = 1$ value on 52 of 52 paired draws; extrapolated to $n = 131$ it is between about 6–7 (if it plateaus) and 39 (semi-regular). One CPU core walks ECC2-131 rho at 8.50 million steps per second with a batched pclmul walk, and 11.46 million bit-sliced (`research/certicom131_rho_ic_bench_20261022/`, `research/certicom131_rho_parity_20261023/`); a CUDA port compiles but has not run on a GPU. The bench's IC ladders on smaller random-binary curves read 11–24× matched rho; it is not an admitted row.

sect113r1. The 113-bit SECG curve is weak by size alone ($2^{56.33}$ generic work); its two degree-5 isogeny routes transfer the problem without an established speed-up, and the GHS and Semaev routes reached no conclusion (`research/sect113r1-weak-curve-20261006/`).

17 Koblitz structure and pipeline constants

Chapter 15 records what the programme built on Koblitz curves $E_a : y^2+xy = x^3+ax^2+1$ over $\mathbb{F}_{2^n}$: the compact-orbit producer, the online ladder, Table A and the priced routes to ECC2K-130. This chapter records why the Frobenius structure buys a constant and nothing more, and what each constant inside the pipeline was measured to be after the leaderboard rules of §15.6. It has nine parts:

- the orbit-fold closure statement and the measurements that bound it (§17.1);
- non-linear Frobenius-stable bases at prime degree (§17.2) and the unrun τ-Horner chain (§17.3);
- the matched rho reference as it applies to Koblitz rows (§17.4);
- the routing between the one-target descent and the pipeline plan (§17.5);
- large primes (§17.6), cofactors and torsion (§17.7), free relations (§17.8) and batch index calculus (§17.9).

Table 17.1 lists the constants in one place.

Standing of the numbers. Under the 4 November rule that every producer may be sabotaging the work, a figure here is either cited with its evidence standing or is a producer-reported, unverified assertion. One study in this chapter carries a `VALIDATION.json`: the large-prime component result of §17.6, whose file, validator report and replay certificate were written by the producer in the same commit and have no independent review. No other study in the chapter has a validation file or a proof-carrying claim, so every other number is a producer-reported measurement or derivation, labelled as such where the source does. Costs are in counted $S = \mathrm{GAE}/\sqrt{r}$ unless stated, against the strongest Koblitz rho the source ran (`rho.signed_frobenius_strong` unless named otherwise), with the §5a band $w = 0, 0.1, 1$ where the source prices it.

17.1 The orbit fold is exactly n

Chapter 15 gives the construction of the invariant and compact-orbit bases. The obstruction that ends the lead is recorded in the lead atlas as a proof, with the ledger hypothesis `H-FROB-6774fb` marked supported (`DEC-20260923-e6cd5d`). It is stated here as the atlas states it; the treatise does not re-prove it.

Proposition 17.1 (Orbit-fold closure; recorded in the ledger, not re-proved here)**.** *Let $E/\mathbb{F}_q$ be attacked over $\mathbb{F}_{q^n}$, let σ be the q-power Frobenius of $\mathbb{F}_{q^n}$ and π the Frobenius endomorphism of E.*

1. *The base $\{P : x(P) \in V\}$ for an $\mathbb{F}_q$-subspace V is π-stable exactly when V is σ-stable.*
2. *The σ-stable subspaces form a lattice fixed by the factorisation of $X^n - 1$ over $\mathbb{F}_q$.*
3. *Folding the base into π-orbits cuts the relations needed by exactly n and linear algebra by n^2, and by no more.*
4. *For prime n with $\mathrm{ord}_n(2) = n - 1$ (as for $n = 131$ and 163) the only stable subspaces over $\mathbb{F}_2$ have dimension 0, 1, $n-1$ or n, so no stable subspace of useful dimension exists.*

Part 4 is the replicated ledger hypothesis `H-BINSTD-b1f016`. Rho receives the matching $\sqrt{2n}$ from the same automorphism group, so the fold changes no gap between index calculus and the matched rho (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-frobenius-orbit-factor-base.yaml`).

The decomposition step gets nothing extra. A pilot in the companion repository (exploratory, "not program evidence" by its own label) enumerated every decomposition of 4,000 targets per cell at $m = 2$ and 1,500 at $m = 3$, on Koblitz curves at $n = 11, 13, 17, 19, 23$, for three bases: the programme's polynomial base V_{poly}, a window of conjugates of a normal element V_{nb}, and a union of random τ-orbits V_{orb} matched in size (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/tau-factor-base-pilot/README.md`). It split the gain into a counting term (orbits per point, $\log_2 g$) and a yield term (distinct projected relations per attempt over the birthday value $\binom{|\mathcal{F}|}{m}/r$).

- V_{orb} has $g = n$ in every cell, so it buys exactly $\log_2 n$ bits of structure. Its yield at matched $|\mathcal{F}|$ is within ± 0.1 bit of the birthday value in every $m = 2$ cell with at least 11 orbits ($+0.01$, -0.09, 0.00 at $n = 23$).

Lever	Measured effect	Distance to matched rho
Frobenius-orbit fold	exactly n on relations, n^2 on linear algebra; $\log_2 n$ bits at matched $\lvert\mathcal{F}\rvert$	IC/rho 2.96 ($n = 13$) to 18.2 ($n = 41$)
Orbit fold, priced as a generic base	$D_{\text{nofold}} = 0.56$ bits $[-0.51, 1.52]$	22.6–33.0 bits (formula)
Trace-spectrum stable base	10.2–12.5× a random subspace per attempt	closed
Hamming ball B_3	0.27–0.43 of random subspaces per attempt; a sparsity effect	not decided
Descent vs pipeline	descent 1.31–2.04× the pipeline	pipeline 9.5–24×
Two large primes, $m = 3$ table	0.604 ($n = 53$), 0.559 ($n = 61$) of the plain $m = 3$ table	30–64×
Two large primes vs `ic-compact`	0.985–1.183 at $w = 0$; null	4.9–28×
Orbit-aware large primes	more verified relations at $n = 17, 29, 31$; none at 23	toy
Torsion anchors	none at $n = 41$; 13.3× dearer at $n = 31$; 7.5× dearer at $n = 37$	$\sim$ 9,700× at $n = 41$
Free relations	0.960× (S3 root pipeline); +5–9% (pair-table pipeline)	unchanged
Batch, 32 targets	Table C 3.62–14.9× batch rho	pre-gate

Table 17.1: Koblitz constants measured after the leaderboard rules, with the distance each leaves to the matched rho. Every row is producer-reported; the two-large-prime row on the $m = 3$ table carries a producer-written `VALIDATION.json`. Sources are in the section of each row.

- V_{nb} buys about 1 bit, not $\log_2 n$: $g = 1.93$–2.02 at $n = 23$, because a window and its shift share a $(k-1)$-dimensional subspace.
- V_{poly} has a free coincidence of 0.03–0.5 bit that shrinks with n (0.06–0.10 bit at $n = 23$).
- At $m = 3$, few-orbit cells lose yield to duplicated projected rows (decompositions that differ by the kernel identity $\tau^2 - t_1\tau + 2 = 0$): 65% of $m = 3$ decompositions duplicate with one orbit, 0.2% with 22–45 orbits. This is the only place where the τ-structure of a base changes its yield, and it lowers it.

Over the populated cells the V_{orb} gain is 3.81 bits mean ($[2.89, 4.52]$) at $m = 2$ and 3.08 ($[1.89, 4.38]$) at $m = 3$, against $\log_2 n = 3.46$–4.52.

The 28 October recheck. An external toy study (`fb-koblitz-orbit`, $n = 15$ and 31, four resolved cells; its repository is not among the sources read here, so the figures are as the atlas records them) compared the best stable subspace base with the cost-model choice of non-stable base. The stable base wins by $D = 4.73$ bits $[3.44, 6.10]$, all of it the orbit credit ($\log_2 n = 3.91$ and 4.95); priced as a generic base the difference is $D_{\text{nofold}} = 0.56$ bits $[-0.51, 1.52]$, whose interval contains zero. The atlas notes that the protocol was frozen by timestamp only, its stored hash matching no text it holds. The stable arm's formula gap to matched rho is 22.6–33.0 bits.

The constant in counted cost. The fully priced ecbench ladder of 14 October (`research/ecbench_frobenius_orbit_vs_rho_20261014/README.md`; 2,816 records verified, 2,112 replayed identically, every A/A ratio exactly 1.000) prices every native counter of the orbit pipeline and of two signed-Frobenius rho arms. Table 17.2 gives the panel. The paired ratio IC/rho grows with r at slope $\beta = 0.165$ $[0.117, 0.213]$ per $\ln r$ once n is in the fit, against the model value 1/6; the study's class is "engineering, constant-only". Index calculus is below the signed-Frobenius rho on no curve; on two curves ($n = 17$, 29) the interval reaches 1, where rho's own spread is wide. Against the negation-folded subspace base the orbit base costs 0.065 $[0.059, 0.071]$ at $n = 23$, which is the constant the fold buys inside index calculus.

17.2 Non-linear stable bases at prime degree

Part 4 of Proposition 17.1 rules out stable subspaces at $n = 131$, but non-linear stable sets exist: trace-spectrum sets $V_E = \{x : \mathrm{Tr}(x^e) = 0,\ e \in E\}$, Hamming balls in a normal basis, multiplicative subgroups, τ-adic patterns over an orbit union, and $\mathbb{F}_2$-points of low-dimensional abelian varieties (Couveignes–Lercier). If one kept the n-fold orbit gain with a membership test no dearer than a subspace's, it would restore the Koblitz lever at $n = 131$. The lead is open (`crypto-autoresearcher:knowledge/le`

a	n	$\log_2 r$	cofactor	S rho-frob	orbit/rho-frob	orbit/negfold
0	13	10.97	4	2.025	2.956 [2.552, 3.242]	0.511
1	17	16.00	2	1.524	1.719 [0.810, 2.980]	0.234
0	19	17.00	4	0.835	2.805 [2.295, 3.236]	0.148
0	23	21.00	4	0.416	3.847 [3.035, 4.819]	0.065
1	29	15.37	12,646	1.665	1.566 [0.992, 2.347]	—
0	31	20.46	1,492	0.394	3.664 [2.907, 4.476]	—
0	37	27.78	596	0.253	5.048 [4.335, 6.086]	—
0	41	39.00	4	0.133	18.16 [15.67, 21.55]	—

Table 17.2: Frobenius-orbit index calculus against the signed-Frobenius rho, every native operation priced; 16 fresh public targets × 3 rounds per curve, L0. Orbit/rho-frob: paired ratio with cluster-bootstrap 95% interval; orbit/negfold: against the negation-folded subspace base where it was run. Source: `research/ecbench_frobenius_orbit_vs_rho_20261014/README.md`.

`ads/ecdlp/ic-binary/LEAD-ICB-nonlinear-frob-stable-bases.yaml`). Its Hamming-ball and trace-spectrum measurements live in an external repository (`aburan28/ml-cryptanalysis`, pull request 53, open when the atlas was last reviewed) that is not among the sources read for this treatise, so every figure below is as the atlas records it. All are per-attempt CryptoMiniSat conflict ratios at matched $|\mathcal{F}|$ on toy fields, with no orbit credit applied, no complete logarithm and no rho comparison.

- *Trace-spectrum bases: closed.* V_E costs $R = 10.2$ $[8.6, 11.8]$ (curve RC-1, $n = 17$), 11.6 $[8.3, 15.4]$ (Koblitz $n = 19$) and 12.5 $[10.7, 14.4]$ (Koblitz $n = 17$, $a_2 = 1$) times a random subspace, above the lead's 2× band.
- *Hamming ball B_3: not decided.* Against the mean of 8 matched random subspaces ($T = 16$ per workload) $R = 0.27$ $[0.13, 0.46]$ and 0.32 $[0.14, 0.55]$ at $n = 19$ ($a_2 = 0, 1$), 0.43 $[0.20, 0.76]$ (RC-1) and 0.29 $[0.11, 0.54]$ at $n = 17$. All eight point estimates lie at 0.20–0.52, but only three intervals exclude 0.5, so the lead's rule does not decide. An earlier $n = 19$ value of 1.10 came from one noisy subspace denominator. The saving sits in the satisfiable stratum (satisfiable targets about 5× cheaper, unsatisfiable ones the same).
- *Stability against sparsity.* A second run ($T = 32$, weights $w = 2$–4) found the per-attempt saving to be a sparsity effect: balls in a random basis read $R = 0.22$–0.40, and the normal-basis ball against the random-basis ball is $D = +0.20$ bits $[-0.05, +0.45]$, with the null gate failed. The exact yield P_{sys} is higher for the stable ball at $w = 2$ (×1.76 in 8 of 8 units, 48 bases per class); the derived per-relation benefit of -0.66 bits $[-1.10, -0.24]$ is post hoc.
- *Encoding at $n = 131$, size only.* $|B_6| = 2^{32.6}$; a Sinz cardinality counter needs 780 auxiliary variables and 1,684 clauses per block (a totaliser 485 and 1,817), against 98 native XORs for a 33-dimensional subspace. No solver price at $n = 131$ exists.
- *Other stable families.* Cyclotomic-sparse bases are yield-neutral at $n = 23$ (§31); an orbit-union base buys exactly $\log_2 n$ (7.03 bits at $n = 131$); a τ-adic base over an orbit union is a Γ_w-orbit quotient that reaches rho only at $w \geq 13$ (`EV-BINSTD-1e1815`, neutral, preliminary); for Couveignes–Lercier abelian varieties, dimensions 1–4 are excluded and dimensions 5–6 hit an impediment (`EV-FROB-a85eaf`, inconclusive; `EV-BINSTD-2d41c3`, preliminary support at dimension 5).

The lead's own next step is a frozen run with at least 20 normal and 20 random bases per class at $w = 2$, cost per relation as the primary; until then the only non-linear branch that is still open is the Hamming ball, and its measured saving is not yet attributable to stability.

17.3 The τ-Horner chain: an accounting test that has not run

Writing a target as a τ-adic Horner sum $R = \tau^{m-1}P_1 + \cdots + \tau P_{m-1} + P_m$ over one base makes ordered tuples distinct relations. The approved ledger hypothesis `H-FROB-855915` claims $m!$ times the distinct-relation yield of the plain chain at an unchanged descended shape. It has never run (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-tau-horner-chain.yaml`). The discriminating comparison (`IDEA-20261018-f2cd4d`) prices the cost per distinct relation of the plain, symmetry-broken and Horner chains at matched core size, $n \in \{17, 19, 23\}$, $m \in \{2, 3\}$, with a size-matched ordinary-curve null. Its reading is fixed in advance: Horner cheaper than the symmetry-broken chain is a constant $m!$ advance for collection; equal cost means the Horner $m!$ is the symmetry-breaking $m!$ relabelled, which the success criterion counts as accounting, not an improvement.

17.4 The matched rho reference on Koblitz rows

Chapter 28 records how the strong signed-Frobenius reference was calibrated, engineered and checked, and Chapter 29 the walk variants and the volcano levels. Four Koblitz-specific facts are not there.

The rule-v4 reference is not provably the strongest. Rule v4 (§31.5) prices rho in the same process on the same point at the canonical batched step. Its admissibility check ran the parallel walk `ParallelRho` against the strong fixture on targets T01–T08 at six sizes (`research/ic_tool_program/rule/v4/README.md`): per step the parallel walk was 1.02, 1.20, 1.08, 0.75, 0.92 and 1.08 times the fixture at $n = 47$, 57, 41, 53, 59 and 61, so admissibility failed at four sizes. On a host without GFNI its 16-lane canonicalisation is slower, and at three sizes its processes ran beside other jobs. The validation gate (`research/ic_validation_gate_20261024/README.md`) wrote no `VALIDATION.json` for rule v4 because `strongest_available` could not be set true: on 46 readable pairs the strong fixture needed 0.845 ± 0.08 of `ParallelRho`'s steps. The canonical-step gap does not depend on either walk; it reads 9.89, 9.20, 16.57, 27.17, 24.25 and 42.12 at those sizes, 1.76–1.94× the as-built gap, reproducing the validation sweep's 1.67–1.96.

The Frobenius fold in rho is measured. The volcano-layer study (`research/volcano_koblitz_layers_20261018/RESULT.md`, all sessions audited, L0) ran rho on Koblitz craters and their level-1 neighbours at $n = 19, 23, 37, 41, 61$. Level-1 over crater for the negation walk contains 1 at every size (1.089 $[0.993, 1.195]$ at $n = 19$ to 0.837 $[0.625, 1.073]$ at $n = 61$); the plain walk excludes 1 once in ten intervals ($n = 41$, 1.152 $[1.021, 1.304]$). On the crater the signed-Frobenius solver over the negation solver is 2.153 at $n = 19$ and 0.864 at $n = 23$, where fixed overhead dominates, 0.264 at $n = 37$ and 0.150 $[0.132, 0.170]$ at $n = 41$, against the ideal $1/\sqrt{n} = 0.156$: about a $\sqrt{n}$ gain from $n \geq 41$, the share of the automorphism group that the atlas credits to rho against the fold of Proposition 17.1. The study retracts an earlier "±8%" statement as wrong. At $n = 83$ the same tool built $\Phi_{6473}(X, j_0)$ from 9,864 CRT primes (606,365-bit coefficients, 8,590 s on three threads), reduced it modulo 2 and factored it into 78 Frobenius orbits of size 83, finding all 6,474 level-1 j-invariants (`research/volcano_koblitz_n83_20261018/RESULT.md`).

No counted 131-bit reference. The rho lead records that no 131-bit signed-Frobenius walk exists in the counted harness, so no ECC2K-130 index-calculus lane has a matched counted rho (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-koblitz-rho-reference.yaml`). The throughput client of §15.7 (13.2 million steps per second on one core) is a wall-time walk, and every ECC2K-130 distance in this treatise is a model against $2^{60.81}$.

The step price, cross-checked. The cryptanalysis repository's normal-basis canonicalisation makes signed-Frobenius rho 4.1× cheaper per step, which by itself reverses an apparent index-calculus win there (the atlas cites `cryptanalysis:papers/frobenius-orbits/sec-abstract.tex`, not read for this chapter). The strengthening is accounting: it widens every Koblitz gap and changes no index-calculus count.

17.5 Routing: the one-target descent against the pipeline plan

Table A prices the ledger §23 one-target Koblitz pipeline at 4.7–11.8× rho, but in mixed units: counted index-calculus work against rho in canonical step units, which charge rho its canonicalisation. ecbench gained `ic.koblitz_descent`, the §23 pipeline charged in counted group operations, with logs and native counts identical to `ic price -single-target` on the same seeds, so the descent and the planner's own recipe (`compact-orbit-scan` with `mitm-frobenius-counted:m=2`) can be run in one unit against `rho.signed_frobenius_strong` (`research/ic_koblitz_descent_routing_20261026/README.md`; protocol frozen in `cd3af7078`; 288 of 288 runs verified, 12 exact replays per audit). The six-size retest (`research/ic_planner_exhaustive_20261026/README.md`; protocol frozen in `0ca110399`; 3,328 records verified, band verdicts restated in integers and checked by Lean's `decide`) is in Table 17.3.

- *Routing.* The descent costs 1.31–2.04× the pipeline at all six sizes, every interval above 1, so the planner keeps `ic.pipeline` for Koblitz curves at $\log_2 r \geq 36$.

Curve	$\log_2 r$	descent/pipeline	pipeline/rho	descent/rho
k47 ($a = 1$, $n = 47$)	36.64	1.313 [1.199, 1.437]	10.8 [8.5, 13.7]	14.2 [11.4, 17.6]
k57	38.00	1.626 [1.544, 1.712]	12.1 [8.5, 17.2]	19.6 [14.4, 26.7]
k41 ($a = 0$, $n = 41$)	39.00	1.471 [1.420, 1.524]	14.1 [11.0, 18.0]	20.7 [16.0, 26.9]
k53	44.26	1.462 [1.410, 1.515]	20.0 [16.6, 24.1]	29.3 [23.9, 35.8]
k59	44.52	1.443 [1.382, 1.507]	22.9 [17.8, 29.5]	33.1 [26.5, 41.3]
k61	47.21	2.039 [1.977, 2.103]	21.4 [17.9, 25.6]	43.6 [36.4, 52.1]

Table 17.3: The §23 descent against the `ic.pipeline` plan, counted S at $w = 0$, 24 pairs per curve, cluster 95% intervals; rho is `rho.signed_frobenius_strong`. The first study (k47, k41, k53) read descent/pipeline 1.421, 1.489, 1.409 and pipeline/rho 9.46, 16.19, 23.99. Source: `research/ic_planner_exhaustive_20261026/README.md`.

- *Where the descent's cost goes.* At k41, relation collection is about 2.05 M GAE and the pair table 0.27 M, while the online descent is 1.0 k–184 k GAE per target; the precomputation is reusable, so a many-target comparison needs batch rho (§29.7).
- *Why Table A read lower.* `rho.signed_frobenius_strong` folds by $2n$ and is charged only group operations, so its counted S is 0.11–0.14; in that unit the Table A descent figures read 13–34× rho (first study), not 4.7–11.8×. The study left re-pricing Table A to the owner, and no row moved.
- *Growth.* Descent/rho rises at every step, 14.2 to 43.6; pipeline/rho rises from 10.8 to 22.9 and reads 21.4 at k61, a literal failure of the preregistered "grows with $\log_2 r$" with overlapping intervals.
- *Band.* `ecbench band` cannot price `ic.koblitz_descent` above $w = 0$: four native categories (field arithmetic outside the group law, hashing, the modular combination, declined AVX-512 lanes) have no count, so the descent is "not ranked" at $w = 0.1$ and 1.
- *Weight rule.* The same retest withdrew the Koblitz planner rule (base $0.71c_0$ at $w \geq 0.3$): held out at 36.6–47.2 bits, the smaller base was dearer at every weight on every curve, 1.06–1.17× at $w = 1$, although it had been cheaper (0.80–0.88×) on 18–26-bit hold-outs. `ic-planner/4` keeps the counted optimum c_0.

17.6 Large primes on Koblitz curves

Large primes appear in this treatise on the genus-3 and extension-field routes (Chapters 25 and 26). On Koblitz curves there are four studies, and the two late-October ones appear to disagree.

Two large primes beat the plain $m = 3$ table. The two-large-prime collector (`mitm-frobenius-counted:m=3,large_primes=2,lp_domain_factor=c`, already on `main`) was run on the classic single-target compact-orbit pair-table pipeline at $n = 53$ and 61, against the plain $m = 3$ table over the same envelope and against the best plain table in the bracket (`research/icb_large_prime_n53_n61_20261029/README.md`, `research/icb_large_prime_n53_n61_20261029/report.html`). Every run verified; the A/A arm and the collector switched off (`large_primes=0`) reproduce the plain arm counter for counter; `ecbench verify -replay-all` replayed every session (612 replays). The benchmarks entry carries a `VALIDATION.json` whose rho reference is the session's own `rho.signed_frobenius_strong` arm, marked `strongest_available` and `order_holds_across_band` (`benchmarks/20261029-icb-large-prime/VALIDATION.json`); it, the validator report and the replay certificate were written by the producer in one commit, so the row is producer-validated, not independently reviewed. The validation file calls it a component result, not a leaderboard row. Table 17.4 gives the best arms.

The order survives the band for the best arms; others win at $w = 0$ and lose at $w = 1$ (`lp2-k20-c10` at $n = 53$: 0.61/0.78/1.28), because their native work exceeds the group operations they save. One large prime is worse than the plain table at equal envelope (1.73 [1.64, 1.82] and 1.47 [1.42, 1.52] at $n = 53$). The optimal large-prime base stays at 40–60 columns while the plain optimum moves from 440 to 880. Memory is not in S: at $n = 53$ both best arms use about 315 MiB; at $n = 61$ the two-LP arm uses 604 MiB against 1,149 MiB. The $n = 61$ interval is over three workloads, and the column optimum was found on the same curves as the confirmation.

Tuned double large primes do not beat the compact baseline. The preregistered tuning study of 23–28 October (`research/ic_koblitz_dlp_tuning_20261023/REPORT.md`; protocol frozen in

n ($\log_2 r$)	best plain $m = 3$	best two-LP	paired ratio	ratio $w = 0/0.1/1$	arm/rho $w = 0/0.1/1$
53 (44.3)	440 cols, $S = 6.38$	`lp2-k40-c10`, 3.86	0.604 [0.599, 0.610]	0.604/0.635/0.730	30/31/35
61 (47.2)	880 cols, $S = 7.62$	`lp2-k60-c10`, 4.26	0.559 [0.535, 0.578]	0.559/0.623/0.831	47/52/64

Tuned double large prime against `ic-compact`, holdout (96 targets per curve)

n ($\log_2 r$)	selected cell	$w = 0$	$w = 0.1$	$w = 1$
37 (27.8)	$K = 1$, $c = 10$	1.001 [0.945, 1.061]	0.793 [0.751, 0.838]	0.570 [0.536, 0.603]
43 (32.1)	$K = 1$, $c = 20$	0.985 [0.928, 1.051]	0.937 [0.885, 0.995]	0.866 [0.819, 0.919]
41 (39.0)	$K = 1$, $c = 160$	1.183 [1.155, 1.213]	1.112 [1.084, 1.143]	0.955 [0.922, 0.989]
47 (36.6)	$K = 3$, $c = 20$	1.095 [1.060, 1.132]	1.067 [1.032, 1.102]	1.015 [0.980, 1.052]

Table 17.4: Two large primes on Koblitz curves. Top: against the plain $m = 3$ Frobenius pair table, 16 workloads at $n = 53$, 3 at $n = 61$ (`research/icb_large_prime_n53_n61_20261029/README.md`). Bottom: the tuned double-large-prime arm over the frozen `ic-compact` baseline, paired ratio of sums with two-stage cluster bootstrap; every verdict "unresolved inside the band" (`research/ic_koblitz_dlp_tuning_20261023/REPORT.md`).

`327afb04f`, Amendments 1–3 before the data they affect) tuned the same oracle family's (K, c) on a train panel and compared it on 96 held-out targets per curve with the frozen baseline `ic-compact`. Its class is *null*: no curve beats the baseline at every weight, and at $n = 41$ and 47 the arm is worse at $w = 0$ and 0.1 (Table 17.4). The $w > 0$ gain at $n = 37$ and 43 comes from the arm leaving fewer unpriced native operations, not from a lower count at $w = 0$. The positive control shows the instrument sees a real double-large-prime effect: at $n = 41$ with $K = 10$ columns, double-LP over the plain $m = 3$ oracle is 0.0316 $[0.0267, 0.0374]$ at $w = 0$, 31.6× cheaper. The only train-side win ($n = 43$, 0.893) shrank to 0.985 on holdout, a winner's-curse loss of about 9 points. Both arms stay 4.9–28× the matched rho at every weight.

Why the two studies disagree. The two studies do not state a reconciliation. Their own protocols and the routing study supply one, and it is an inference from unpaired sessions, not a measurement:

- *Different baselines.* The large-prime study's comparator is the plain $m = 3$ oracle `mitm-frobenius-counted:m=3` (`research/icb_large_prime_n53_n61_20261029/PROTOCOL.md`). The tuning study's comparator `ic-compact` is `compact-orbit-scan:columns=Kb` with the $m = 2$ oracle `mitm-frobenius-counted:m=2` (`research/ic_koblitz_dlp_tuning_20261023/PROTOCOL.md`). The double-LP collector is an $m = 3$ oracle in both.
- *The $m = 3$ plain table is the weaker arm.* On the same curves the $m = 2$ pipeline is cheaper than the best two-LP arm. At $n = 53$ (`icv1-f2m53-tm56619371-dac20a85` in both studies) the pipeline reads $S = 2.728$, 24.0× rho, in the routing study and 20.0× in the retest, against 3.856 and 30× for `lp2-k40-c10`; at $n = 61$ (`icv1-f2m61-t158598901-ab42b6c5`) the pipeline reads 21.4× rho against 47× for `lp2-k60-c10`. The plain $m = 3$ table reads 49× and 85× there.
- *So both results hold.* The double large prime roughly halves the cost of the $m = 3$ table, which is the same direction and mechanism as the tuning study's positive control (31.6× at $K = 10$), and leaves it at or above the $m = 2$ compact pipeline, which is what the tuning study found by direct pairing at $n = 37$–47.

The large-prime study itself says its result does not touch Table A, whose aimed $m = 3$ collection with an $m = 2$ descent reads 7.03× ($n = 53$) and 11.5× ($n = 61$) of its own reference; the collector has not been combined with that recipe, and stage 0 of the experiment that would do so (`EXP-BINSTD-e066d8`) has not run (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-large-prime-variants.yaml`). The tuning study's own next step is to draw the large-prime domain from a structured set (small $\mathbb{Z}[\tau]$-norm multipliers) rather than raw orbits, since its selected cells sit at the $K = 1$ edge where the domain does all the work.

Orbit-aware large primes. Quotienting the large-prime vertices by signed Frobenius orbits, with the $\pm\lambda^k$ phase kept, gave more independently verified relations than the plain double large prime from the same relation stream at $n = 17$, 29 and 31 on every seed, a small and not clearly nonzero gain at $n = 19$, and none at $n = 23$ (`research/orbit_aware_large_prime_20261011/RESULTS.md`; measured 26 October). The gain tracks the number of orbit collisions among the fixed large primes, which a random model predicts at $n = 17$, 19, 29 and which is far above the model at $n = 31$ for a reason not found. All 17,865 emitted relations passed both replays. Online CPU does not separate the arms, so CPU per

relation falls only through yield; at $n = 17$, for example, 9.2 independent relations per cell against 6.2 (10.8 against 6.4 emitted), at 0.181 against 0.275 ms of CPU per independent relation. The scale is toy (r between 2^{15} and 2^{21}).

The 131-bit Certicom curves. Symmetry breaking and double large primes at $m = 2$ on ECC2-131 and ECCp-131 are in §20.2: the binary symmetric oracle steps 28.7× faster than the per-point baseline and the distance to rho stays about 2^{22} steps by the $m = 2$ exponent.

17.7 Cofactors, torsion anchors and quotients

The cofactor-projected base of LEAD-19 and its Table B cuts are in §16; the prime-field cofactor-clearing A/B and its `invalid_controls` verdict are in §22.4. Two further Koblitz results are a measured reversal and a closed quotient.

Torsion anchors: a generalisation withdrawn twice. The $m = 2$ cycle-stop study (`research/ic_m2_cycle_leads_20261030/REPORT.md`, exploratory, no frozen protocol) stated that torsion-randomised anchors make large-cofactor curves workable, solving $a = 0$, $n = 37$ (cofactor 596) in 26 and 31 s where the ladder stopped at 99 s, and that they cost 1.5× ($n = 23$) to 16× ($n = 31$) on cofactor-4 curves. Two frozen follow-ups of 1 November corrected both readings:

- $n = 41$, *cofactor 4* (`research/ic_m2_torsion_n41_20261101/REPORT.md`; protocol frozen in `baa2fcfa8`; eight paired targets, 16 of 16 runs verified). The paired geometric mean of trials, torsion on over off, is 0.98 $[0.64, 1.43]$; the per-pair ratios run from 0.30 to 2.60. No difference is detected. A post-hoc replicate at $n = 31$ (32 pairs) gives 13.3× more trials with torsion on, $[10.3, 17.5]$. The $n = 31$ figure is real; the generalisation to cofactor-4 curves is not.
- $n = 37$, *cofactor 596* (`research/ic_m2_torsion_n37_20261101/REPORT.md`; protocol frozen in `7893863417`; eight pairs). Torsion anchors make the solve about 7.5× dearer: off over on is 0.134 $[0.082, 0.205]$ in trials, with all eight pairs below 1. The relation rate is the same (about 17,000 trials per relation in both arms); without torsion the relation graph stays in components of 3–5 columns and a cycle closes after about 126 relations, with torsion components grow to about 117 columns and the first equation needs about 885. The ladder's failure at $n = 37$ came from its 90% coverage stop, not from the cofactor.

So torsion anchors helped nowhere they were measured. The family's distance is set by the method, not by anchors: against the same-process strong rho the $m = 2$ cycle variant is about 9,700× slower at $n = 41$, and 7,700× in mean wall at $n = 37$ without torsion. Both follow-ups are exploratory, on one shared four-vCPU container, with no replay certificate.

Cofactor and 2-torsion quotients: closed. The atlas records a proof (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-cofactor-torsion-quotients.yaml`). On ECC2K-130 $\gcd(4, \ell) = 1$, so every full-translation homomorphic cofactor quotient reduces to the order-ℓ projection rho already uses; exactly sum-compatible filters on $E(\mathbb{F}_{2^{131}})$ carry 2 bits (the cofactor), so Wagner's k-tree has no quotient to work in; and the corrected ι-stable classification ($V_d = b^{1/4}\mathbb{F}_{2^d}$) empties the cell (subspace base, 2-torsion quotient) on every prime-degree standardised curve. The production 2-torsion census added no relations. A September study measured the same bound from the other side: the invariants used by translation-invariant bases on $y^2 + xy = x^3 + 1$ are pullbacks by $1 + \tau$, $1 - \tau$ and [2], held on all 10,238 points at $n = 7, 11, 13$, and each kernel has at most 4 points, so they change a decomposition by at most 2 bits per summand (`research/cheap_closures_20261007/README.md`).

17.8 Free relations

Pair-table self-collisions give relations with right-hand side zero: four base points that sum to O, verified as exact group identities (`RESEARCH_FREE_RELATIONS_CENSUS_20261012.md`). The relations exist and are highly correlated (Table 17.5): the 588 nontrivial rows at $n = 61$, $K = 600$ touch 168 of 600 columns, and the 72 at $n = 83$ span six dimensions.

- *Rank law.* Rank grows about as K^4 with a small constant, so a free rank stage needs $K \approx$ 1,500–1,900 at $n = 61$ and $K \approx$ 2,600 at $n = 83$, where the census scan and the K^2 linear algebra already exceed

n	K	collisions	verified nontrivial	homogeneous rank	needed
61	600	10,188	588	49	599
61	1,200	12,732	4,800	746	1,199
61	1,500	10,924	6,000	1,197	1,499
83	600	9,672	72	6	599

Table 17.5: Free relations from pair-table self-collisions: the rank they reach against the $K - 1$ the rank stage needs. Source: `RESEARCH_FREE_RELATIONS_CENSUS_20261012.md`.

the guided rank stage (13.1 s at $n = 61$, 4.8 min on 12 threads at $n = 83$), before 36 GB of memory at $n = 83$.

- *In the S3-root pipeline.* On by default for $n \leq 63$, the harvest costs $0.960\times$ $[0.946, 0.976]$ of the tuned incumbent pooled over $n = 41$, 53, 61, in S3 root solves per complete solve; wall time agrees, $0.941\times$ (`research/ic_improve_20261015/02-freerel-production/README.md`).
- *In the pair-table pipeline.* Under rule v4 the harvest is correct (6,187 relations, none rejected, in 510 runs) but yields 2–11 relations per solve against 58–327 needed, cuts trials 0.5–9%, and makes the build 1.4–$1.55\times$ slower: the cold cost rises by 5.3–8.8% at every size (1.053–1.088 over 72 pairs per size). The 0.960 does not transfer, because that pipeline costs 59 s per solve at $n = 61$ against 3.7 s here (`research/ic_tool_program/rule/v4/README.md`).

17.9 Batch index calculus and restart portfolios

The many-target accounting correction and its batch-rho reference are in §29.7; Table C is in §15.6. The lead records two statements that bound the route (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-multi-target-and-portfolio.yaml`):

- *The generic multi-instance bound covers set-up.* L logarithms cost at least about $\sqrt{Lr/|\mathrm{Aut}|}$ for a generic-oracle index calculus including its set-up, and Kuhn–Struik batch rho attains it; the L^{-1} amortisation applies only to the online part, after a set-up that obeys the same bound. This is the multi-target form of the Corrigan-Gibbs–Kogan bound [8] of §3.
- *A restart portfolio cannot beat pure rho* when the index-calculus cost concentrates at a floor ratio of at least 1. The ledger hypothesis `H-BINSTD-32867d` is marked supported as a scoped observational hold, strength preliminary (`EV-BINSTD-2d34cb`); its instrument was a deterministic closure, a controlled null rather than a solver.

The four-target union column of the September batch thread is a rare-event linearisation, not a batch speed-up. Table C's 32-target rows stay 3.62–$14.9\times$ batch rho, with the reference taken from a different session (S20) at the canonical step, and all nine failed the 24 October gate.

18 The $n = 83$ campaign after 21 October

§15.5 covers degree 83 to 21 October and the 24 October field-table correction, after which every Weil-descent engine's earlier $n = 83$ refutation is void. Forty-one study directories were added from 24 October to 1 November. None of them admits a new IC/rho ratio, and none is a leaderboard or `benchmarks/` row. They fall into eight groups (Table 18.1), which this chapter takes in four parts:

- factor-base selection, index capacity and controls on the pipeline curve (§18.1);
- decomposition at $n = 83$ after the correction (§18.2);
- certified degree bounds for the descended systems (§18.3);
- the genus-3 image transfer (§18.4).

Two curves, three families. Degree 83 means two different targets in these studies, and their numbers must not be mixed.

- *The pipeline curve* `EC1N83Ckb1a1haacb69ee9f37`, $y^2+xy = x^3+x^2+1$ ($a = 1$), has a 53-bit subgroup of order 8,569,786,107,849,059 and a cofactor of about $1.1 \cdot 10^9$. Every K-series base, every online and cold time and the matched precomputed-rho control are on this curve.
- *The primary curve* $y^2 + xy = x^3 + 1$ ($a = 0$, modulus $z^{83} + z^{45} + z^2 + z + 1$, trace $-6{,}151{,}469{,}093{,}347$) has the 81-bit subgroup $r = 2{,}417{,}851{,}639{,}230{,}796{,}216{,}685{,}689$ and cofactor 4. The genus-3 transfer and the sixteen factor-base designs target it.
- *The degree sweeps* use full-curve affine bases on three families at the same size: the Koblitz curve, a general binary curve and an 83-bit prime curve, with no subgroup filter.

18.1 Factor-base selection on the pipeline curve

The nearby grid. The ordered K1120 producer dump fixes every complete signed-Frobenius orbit prefix from K1102 to K1120; all 19 candidate manifests were reconstructed from it, and K1104, K1107 and K1120 match their published dumps byte for byte (`research/n83_nearby_full_grid_20261024/README.md`). The same-point Tier 1 screen of all 19 (84 children: 60 IC and 24 rho, each IC block bracketed by K1107 and a rho control) failed its frozen 4 GiB disk gate on the first two hosted attempts, with 2.26 and 2.93 GiB free and zero measured children (`research/n83_nearby_grid_tier1_20261024/README.md`); an AWS Spot variant was blocked by the provider's Spot instance quota (`Client.MaxSpotInstanceCountExceeded`, status `BLOCKED_PROVIDER_SPOT_QUOTA`; `research/n83_nearby_grid_aws_tier1_20261024/admission_20261024.json`). A later hosted screen run (GitHub run 40150483792) supplied the confirmation's candidate set.

The 16-point confirmation. Sixteen fresh public points, sealed before the screen result was inspected, were run on seven candidates in two eight-point Tier 1 strata (AMD EPYC 7763 and Intel Xeon Platinum 8573C runners), each IC run paired with a same-point rho (`research/n83_nearby_grid_confirm16_20261024/RESULT.md`). The independent verifier admitted 112 of 112 rows, and a local replay of the raw artifacts reproduced the hosted aggregate byte for byte. Table 18.2 gives the panel. K1115 has the lowest geometric-mean online time (47.114 ms) and K1111 the lowest mean cold process (159.263 s), but the frozen sign-flip rule with Bonferroni $\alpha = 0.05/21$ returned BOUNDED_UNKNOWN for both metrics (K1115 against K1111 online, $p = 0.959$; K1111 against K1116 cold, $p = 0.773$). No base was selected. The rho/IC column is an online reading with set-up excluded and a table-less rho, the comparison §15 retired; on the same panel the cold IC process takes 159–166 s, while the same-point table-less rho online times at point 0 are 1,455–3,715 ms.

A local five-base screen. On one new public point, five finalists each ran two fresh cold processes in forward and reverse order (`research/n83_local_five_base_screen_20261024/RESULT.md`). Median cold times were 66.2 s (K992/root11, 164,672 points), 62.7 s (K1024), 65.8 s (K1107), 119.7 s (K2304) and 383.6 s (K3300/wide3), with pair spreads of 4.7–19.1%; the two same-point rho controls took 2.015 and 0.989 s with different seeds. Every hash, base, rank, relation and scalar replayed, and no winner was selected because the K1024–K1107 gap (3.1 s) is smaller than the within-arm spread.

Group	Dirs	What it established	Status
Nearby grid K1102–K1120	4	19 exact prefix bases; 16-point confirmation admitted 112 of 112 IC/rho rows; no base selected	Tier 1 cloud; BOUNDED_UNKNOWN selection
Current-source K-series	10	K2304–K5500 bases reconstruct exactly; one K2304 run completes; K3596 and K4405 capped	local controls
Index capacity layouts	7	four-byte, split and fingerprint root indices; K5500 in 25.61–28.61 GiB of allocations	allocations calculated, not measured
Cross-family bridge	2	16 sealed points; never dispatched (Spot quota)	no timing
Precomputed-rho control	1	rho faster online in all three matched arms	Table 32.2
Decomposition engines	7	chained-S_3 SAT decides $m = 2$–9 on a 167-point base with tables and a meet	stage diagnostic
Degree certificates	7	first fall 2; exact top-form degrees 28 and 111–609; no regularity computed	producer certificates
Genus-3 image	3	executable transfer; Prym coprime to r; degree-2 fibres carry no information	structural, negative

Table 18.1: The 41 $n = 83$ study directories of 24 October–1 November, by group. "Tier 1" is the benchmark-contract host tier of `docs/IC_BENCHMARK_TIERS.md`. Sources: the `research/n83_*` directories cited in this chapter.

K	base points	online, ms (geo. mean)	cold, s (mean)	median rho/IC online
1103	183,098	142.358	162.320	16.632
1107	183,762	72.916	160.806	28.414
1109	184,094	99.303	165.435	14.045
1110	184,260	77.205	165.329	28.296
1111	184,426	48.226	159.263	45.384
1115	185,090	47.114	165.754	27.967
1116	185,256	58.977	159.395	56.961

Table 18.2: The nearby-grid confirmation on the pipeline curve: 16 fresh public points, two Tier 1 strata, 112 admitted IC/rho rows. Base points are $166K$ subgroup-usable points in K folded columns. The rho/IC column is online only, against a rho with no table, set-up excluded. Source: `research/n83_nearby_grid_confirm16_20261024/RESULT.md`.

The K-series, re-identified. The large bases of 14–21 October were reconstructed from current-source parent dumps: the first 2,304, 3,300, 3,450 and 3,500 orbits of the K3596 dump (596,936 points) and the first 4,300 and 4,405 of the K5500 dump (913,000 points) reproduce the historical K2304 (382,464 points), K3300 wide3 (547,800), K3450 wide3 (572,700), K3500 root4 (581,000), K4300 (713,800) and K4405 (731,230) bases exactly, by ordered-dump and point-set SHA-256 and the producer's BLAKE3 base hash (`research/n83_k2304_current_source_prefix_20261025/`, `research/n83_k3300_current_source_prefix_20261025/`, `research/n83_crossfamily_current_20261025/README.md`, `research/n83_k4300_k4405_current_source_prefix_20261025/`). Full-process controls on an earlier public point then gave (Table 18.3) one completed large base and three that did not finish on the hosts available.

Index capacity. Seven directories made the root index fit larger bases. Each is an opt-in layout with an exact-key replay rule (a candidate slot reconstructs and compares the full 83-bit root before a hit is accepted), and each stops at calculated allocations for K5500:

- *Exact four-byte slots* (31-bit state index and a root selector in one `u32`): principal allocations of 32,062,766,258 bytes, 29.86 GiB, for K5500 (`research/n83_k5500_fourbyte_20261025/README.md`).
- *ID31 four-byte slots* at the K5500/K4549 split: 12 GiB of tables instead of 15, and state, table and Bloom arrays of 27.86 GiB against 30.86 GiB for five-byte ID31 (`research/n83_fourbyte_id31_20261027/README.md`).
- *A relative-ID split* (primary 2^{31} and extension 2^{30} slots, 1,255,377,750 eleven-byte states): 25.61 GiB of principal allocations, 4.25 GiB below monolithic ID31 (`research/n83_k5500_split4_20261025/README.md`).

Base	Host and cap	Wall, s	Peak RSS	Outcome
K2304 root11	local, 1,200 s, 12 GiB	438.5	5.93 GB	completed; independent replay passed; target stage 420.2 ms
K1107 root4	local macOS ARM64, 1,200 s	438.9	9.94 GB	completed; replay passed; online 46.3 ms
K3596 root4	busy 48 GiB macOS, 1,200 s	792.5; 1,201.0	0.80; 9.58 GB	SIGTERM after CPU starvation; time cap
K4405 wide3	local, 1,200 s	873.2	11.68 GB	stopped by hand under host swap
K4405 wide3	Linux Pod, 64 GiB cgroup	—	—	not started; approved GPU pair out of capacity

Table 18.3: Full-process controls on the pipeline curve, one earlier public point each (fixture zero), outside the sealed bridge points; correctness and resource controls, not Tier 1 pairs. K1107 root4 uses a fixed 8 GiB root table for 101,708,823 distinct roots. Sources: `research/n83_k2304_current_source_control_20261026/README.md`, `research/n83_root4_small_control_20261025/README.md`, `research/n83_k3596_current_source_full_20261025/README.md`, `research/n83_k4405_current_source_full_control_20261026/README.md`, `research/n83_k4405_linux_control_20261026/README.md`.

- *Fingerprint sidecars.* A two-bit sidecar brings the ID31 layout to 28.61 GiB, still 2.25 GiB below five-byte ID31, with the packing and capacity arithmetic proved in Lean 4.34.1 (`research/n83_fourbyte_fp2_20261027/README.md`). A one-byte sidecar kept every digest identical and built the index faster in ABBA order on a shared ARM64 host: geometric-mean baseline over filtered ratios of 1.831 (K128), 1.347 (K512) and 1.505 (K1024), with a K2048 baseline that drifted 42% between bookends (`research/n83_k5500_fp8_20261025/README.md`).

Two whole-process ABBA controls on the K1107 base read the sidecars: the one-byte sidecar gave baseline over fingerprint 1.677 for the full process and 2.157 for one-target online, with a 15.3% baseline drift between bookends (`research/n83_fp8_e2e_control_20261025/README.md`); the two-bit sidecar gave whole-process ratios of 1.017 and 1.294 in the two orders on a host whose load average ran from 22 to 276 (`research/n83_fourbyte_pair_20261027/README.md`). Both are local diagnostics that the Tier 1 gate would not admit.

The bridge that did not run. The cross-family bridge sealed 16 new public points for a same-point Tier 1 comparison of the small-base and wide-base selections, with its nomination rule frozen before either panel reported (`research/n83_crossfamily_bridge_20261025/README.md`). Its current-source panel (K3450 wide3 and K3500 root4, two `g5g.4xlarge` Spot instances, a $4.20 cap) moved from `us-east-1d`, which failed its placement-score gate, to `us-west-2c`; the README still records that the panel has not dispatched or selected a winner (`research/n83_crossfamily_current_20261025/README.md`, `research/n83_crossfamily_current_20261025/AWS_WEST2_ADDENDUM.md`).

The control that did run. The matched precomputed-rho reproduction of 28 October on K1107/root11 found rho faster online in all three matched arms (Table 32.2), every interval above 1, over 16 targets with every scalar replayed (`research/n83_k1107_bl_reproduction_20261028/RESULT.md`). Its plain table-less rho, a positive control only, reads rho/IC 10.7 $[6.33, 88.8]$ online, the same kind of reading as the rho/IC column of Table 18.2.

18.2 Decomposition at $n = 83$ after the correction

The engines run at $n = 83$ are described in §12.8; this section records the decomposition results after the field-table correction.

After the 24 October correction (§15.5) chained-S_3 full-width SAT was the only decomposer that ran at $n = 83$, and it decided only $m = 2$. Six preregistered follow-ups of 31 October–1 November extended it on the pinned pipeline-curve fixture, with the factor base the Frobenius union of span$(1, z)$, 167 points (`research/n83_chain_sat_meet_20261101/PROTOCOL.md`). Each cell has three public targets (not

Study	Change	Outcome
order (31 Oct)	all-summand ordering; root-finder tables	$m = 2$: conflicts halved (median public 13,461 → 7,195; 12.8 → 6.2 s); $m = 3$: tables refute every public target with 0 conflicts and find both planted; $m = 4$: public refuted, no planted witness in 120 s; $m \geq 5$: capped
meet (1 Nov)	a meet-in-the-middle lookup for the middle intermediate	$m = 5$–8: public 3/3 refuted with 0 conflicts (median 0.80–76.7 s), planted 0/2; $m = 9$: capped; builds 0.68 s ($m = 5, 6$), 26.1–27.2 s ($m = 7, 8$), 958.8 s ($m = 9$)
rebuild (1 Nov)	rebuild the planted witness from the lookup hit	$m = 5$–8: planted 5/5 with 0 solver calls (median 0.91–81.5 s)
indexed (1 Nov)	indexed lookup, rebuild by peeling	$m = 8$: 78.4 → 39.7 s; $m = 9$: from cap to 3/3 and 5/5 (54.1 s), depth-5 set of 372,406,144 fingerprints in 999 s
parallel (1 Nov)	4-thread back enumeration	$m = 8$: 41.7 → 11.4 s; $m = 9$: 57.0 → 13.7 s; verdicts identical
parbuild (1 Nov)	4-thread depth-5 build	1,006.8 and 993.4 s → 287.7 and 283.3 s, 3.50×

Table 18.4: Chained-S_3 SAT on the pipeline curve, $m = 2$–9, 167-point base; decomposition stage only, L0 four-vCPU hosts. Sources: `research/n83_chain_sat_order_20261031/README.md`, `research/n83_chain_sat_meet_20261101/README.md`, `research/n83_chain_sat_rebuild_20261101/README.md`, `research/n83_chain_sat_indexed_20261101/README.md`, `research/n83_chain_sat_parallel_20261101/README.md`, `research/n83_chain_sat_parbuild_20261101/README.md`.

expected to decompose) and two or five planted ones. Table 18.4 gives the sequence. Every run had zero soundness flags: no planted target was refuted and every witness re-adds in the group.

The sequence turns SAT into a meet-in-the-middle with a SAT residue, and then removes the residue: once the lookup decides the middle intermediate, the public targets are refuted with zero conflicts because the lookup domain is empty on every one of them and non-empty (10–250 entries) on every planted one. The meet study states the cost as about $|B|^{m-d}$ per query after a $|B|^d$ build. That is the generic floor, and the improvements in Table 18.4 are constants of the enumeration (ordering, tables, peeling, threads). The matched rho on this curve is 1.51 s mean online. With 167 points and a 53-bit subgroup, a decomposition over this base is rare, so these are stage measurements: no relation collection, no complete logarithm and no rho ratio.

18.3 Certified degree bounds

The descended $n = 83$ systems were characterised exactly, not solved. Seven directories, all producer certificates with fresh-process replay and independent scientific review pending, give the following.

Solvers do not finish. The balanced degree sweep (Koblitz, general binary and prime; $m = 2$–9; candidate domain $2^{\lfloor 83/m \rfloor}$; a planted and a natural target per cell, two computations each) ran 96 Singular attempts with 40 s and 3 GiB per solver: 53 hit the time limit, 43 the memory limit, and no top-form or affine basis degree completed (`research/n83_ic_degree_sweep_20261027/README.md`). A fixed-base rerun at $\ell = 27$, on full-curve bases that enumerate all $2^{27} = 134{,}217{,}728$ candidate x-values per curve (`research/n83_fixed_factor_base_20261028/README.md`), completed 4 affine bases (all $m = 2$, Koblitz and binary), with 63 time and 29 memory exits (`research/n83_fixed_degree_sweep_20261028/README.md`).

First fall is 2 on binary curves. On every retained input the Koblitz and general binary systems have certified first fall degree 2 at every $m = 2$–9, planted and natural (32 exact witnesses; all 48 replays passed), consistent with Corollary 6.6. On the prime curve the balanced presentation has the proof-assisted lower bounds 24, 17, 14, 12, 10, 9, 9, 8 for $m = 2$–9 (`research/n83_ic_degree_sweep_20261027/README.md`). The fixed $\ell = 27$ catalogue certifies 48 first falls and 32 degree-two closures: at $m = 2$ the binary systems have first and last fall 2 (a restricted global certificate; in each planted case V_2 has 53

m	certified bounds, Koblitz and binary	exact top-form degree
2	$28 \le d_{\mathrm{reg}} \le 55$	28 (prime: 42)
3	$82 \le d_{\mathrm{reg}} \le 165$	111
4	$109 \le d_{\mathrm{reg}} \le 275$	194
5	$136 \le d_{\mathrm{reg}} \le 385$	277
6	$163 \le d_{\mathrm{reg}} \le 495$	360
7	$190 \le d_{\mathrm{reg}} \le 605$	443
8	$217 \le d_{\mathrm{reg}} \le 715$	526
9	$244 \le d_{\mathrm{reg}} \le 825$	609

Table 18.5: Top-form degrees of the fixed $\ell = 27$ descended systems at $n = 83$, both targets. Bounds: certificates of 28 October; exact values: the overlap proof of 29 October, which closes every interval at $111 + 83(m - 3)$ for $m \ge 3$. Neither is an affine solving degree. Sources: `research/n83_top_quotient_20261028/README.md`, `research/n83_candidate_pdp_20261028/README.md`.

independent linear constraints on 54 variables), at $m = 3$–9 the last fall is unresolved with lower bound 2, and the prime systems have first fall at least 5 (`research/n83_fixed_fall_catalog_20261028/README.md`). A later degree-four witness census raised the certified last-fall lower bound of the twelve retained $m = 3$ Koblitz and binary systems from 3 to 4, with 249 strict degree-four coordinate witnesses each (`research/n83_candidate_pdp_20261028/README.md`).

Top-form degrees are large and exact. The top-form graded quotient was bounded for all 48 fixed-base systems (`research/n83_top_quotient_20261028/README.md`) and the binary chain bounds were then closed exactly by a written overlap proof with native checks (`research/n83_candidate_pdp_20261028/README.md`). Table 18.5 gives both. For the prime systems at $m \ge 3$ the top quotient is positive-dimensional, with no finite first vanishing degree in the saved presentation; the source says this is not a claim of infinite regularity or about the affine ideal. At $m = 2$ the binary systems have FFD = LFD = 2 and top-form degree 28, which separates those quantities explicitly for these presentations.

Encodings and candidate bases. For the prime candidate interval a radix census over all 3,010 partitions of 27 (35 exact Hilbert records and 2,975 certified lower bounds) found the unique minimum exact first vanishing degree, 42, in the all-binary presentation, which excludes improvements within that presentation family (`research/n83_candidate_pdp_20261028/README.md`; the original radix protocol's unequal-arm accounting mismatch is recorded in a correction file). The same directory certifies $7 \le \mathrm{FFD} \le 17$, conditionally, for 42 prime $m = 3$–9 systems. A separate design study lists sixteen factor-base constructions for the primary $a = 0$ curve with a preregistered protocol and proof obligations, and no measurement (`research/n83_factor_base_designs_20261031/README.md`).

Reading. The binary systems fall at degree 2 and their top-form degree grows by 83 per summand. These are certified properties of specific presentations, not attack costs. They say nothing for or against a low affine solving degree, but they show that the descended $n = 83$ systems are far outside the reach of the solvers that were run, which matches the 96 and 92 unfinished attempts above.

18.4 The genus-3 image transfer

A same-field degree-3 cover $\phi : C \to E$ with C of genus 3 descends to $\mathbb{F}_2$: $E : y^2 + xy = x^3 + 1$ and $C : v^2 + u^2 v = u^7 + u$, with equal point counts over $\mathbb{F}_2$, $\mathbb{F}_4$ and $\mathbb{F}_8$ (`research/n83_jacobian_projector_20261031/RESULTS.md`). Three studies of 31 October–1 November asked whether pulling the $n = 83$ instance into $\mathrm{Jac}(C)$ gives anything rho does not have.

- *The transfer executes* (`research/n83_genus3_image_solver_20261031/README.md`). The exact $n = 83$ pullbacks of P and $Q = [37]P$ are reduced Mumford classes of exact prime order r; typed certificates prove $\phi_* \phi^* = [3]$, so the pullback is injective on the subgroup. Rho and BSGS in the Jacobian image solved toy instances at $n = 9, 13, 19, 23$ without the planted scalar, all 400 destination attempts verified ($n = 7$ failed and is retained). The source and destination operation counts have no certified common unit, so no advantage is promoted; the full-Jacobian genus-3 index calculus does not apply to an image of order $r \approx q$, and its ambient $q^{4/3}$ ledger loses to source rho by about 70.17 exponent bits.

- *The Prym shares nothing with the target* (`research/n83_jacobian_projector_20261031/RESULTS.md`). With the closed-point norm, $e = \phi^*\phi_*$ satisfies $e^2 = [3]e$, and on r-torsion $p = [3^{-1} \bmod r]e$ is a checked idempotent. At $n = 83$ the Prym has Frobenius polynomial $X^4 + q^2 = (X^2 - 2^{42}X + q)(X^2 + 2^{42}X + q)$, with elliptic factors of traces $\pm 2^{42}$, neither equal to E's trace, and $\gcd(r, q + 1 \mp 2^{42}) = \gcd(r, q^2 + 1) = 1$. There is no r-primary route through the two-dimensional complement. The identities are checked in Lean.
- *Degree-2 fibres are tautological* (`research/n83_image_closed_degree2_20261101/README.md`). On every odd extension tested a generic rational fibre has closed points of degrees $1 + 2$ whose elliptic labels are exactly P and $[2]P$, so the projected identity is $[3]P = P + [2]P$ and carries no logarithmic information. The frozen information gate fails, so no collector or timing run is admitted; ordinary ambient genus-3 collection and fixed bounded-degree branching also fail the comparison with source rho. A growing-degree or adaptive target-local router is left open.

This is the binary counterpart of the cover results of §26: the map is real and certified, and the destination carries the same r-part and nothing that makes it cheaper.

19 Weil descent and degree, continued

§16.1 measures the degree of Weil-descended summation systems to 24 October: first and last fall degrees stay at 3–4, the refutation degree tracks ℓ within one of semi-regular, closure algorithms refute a degree or two earlier but rise past 4 at $m = 3$, $\ell = 7$, and guessing one or two variables restores degree 4 there. §31 adds the first $\ell = 8$ runs: four guessed variables bring all four unsatisfiable $(22, 8)$ draws back to degree 4, one three-variable set fails on three of four, and the hybrid beats degree 5 at the priced ECC2K-130 cell only if at most six guesses suffice. This chapter records what followed:

- which guessed coordinate matters at $\ell = 8$, not only how many (§19.1);
- the first measured degree-5 closure cell (§19.2);
- the per-attempt criterion and the companion repository's closure and SAT pilots (§19.3);
- a direct check of Semaev's 2015 first-fall statement on his own cells (§19.4);
- the ceiling on learned SAT guidance (§19.5);
- the finding that F6-IC with a pair index is the meet-in-the-middle scan (§19.6).

Every result here is exact algebra or a counted stage diagnostic on toy fields, run on unisolated 4-core, 16 GB containers (the RunPod degree-5 cell excepted). None has a validation file. One carries a proof-carrying claim, recorded as unverified. Closure outcomes do not depend on the host; timings are indicative only.

19.1 Guessing at $\ell = 8$: which coordinate, not how many

The closure and guessing instruments are described in §14.1; this section records what they measured at $\ell = 8$.

All studies in this section use the frozen $m = 3$, $(n, \ell) = (22, 8)$ draws of 22 October ($N = 46$ unknowns, 44 equations, surplus -2; unsatisfiable draws 0, 3, 5, 6) or fresh draws from the same generator, the MutantXL-style closure `dreg_closure` or its guess variants, and a registered reading of each branch as refuting at degree 4 (the constant 1 appears) or not (degree 4 saturates without it). Table 19.1 lists them.

What the series shows. At equal guess count, guessing a coordinate of the swap invariant $s = x_1 + x_2$ refutes at degree 4 on more draws than guessing a raw summand coordinate: preregistered on fresh draws, $p = 0.031$ one-sided, and 6 of 6 discordant draws over both studies ($p = 0.016$, secondary). S3 suffices on 7 of 12 draws in all and X3 on 1 of 12. This contradicts the summand-dimension reading proposed after the three-guess study: S3 leaves both summands at dimension 8 and suffices more often than X3, which lowers x_1 to dimension 7. Putting a second invariant guess on the same link (D3) is never better than the raw guess on x_3; putting it on the second link's invariant $e_1 + x_3$ (L3) is as good as the raw guess on these 12 draws. The D3 failures all sit at branches 0 and 4, where both top coordinates of s vanish. Four guesses $\{7, 15, 23, 24\}$ still suffice on every draw measured, and S3 fails on 5 of 12.

What it does not show. Every row is one cell, $m = 3$, with one guess set per arm. The unspecialised $\ell = 8$ system is not measured at degree 4 (179,447 columns); "does not refute" means a closure degree of at least 5, and degree 5 is not measured here. The null study's 12 of 12 unspecialised $\ell = 7$ failures over $N = 40$, 42, 43 and surplus -2, 0, $+1$ remove the one confound the earlier $\ell = 7$ registration kept (a non-positive surplus), but cannot distinguish "3 of 4" from "4 of 4" against the three-guess $\ell = 8$ result. The second-link study's own reading is that one invariant guess per link extends along the chain, so $m = 4$ is its natural test; $m = 4$ at $\ell = 8$ has not run. The preregistered high-arity discriminator that would run it (`research/dreg_high_arity_20261101/README.md`) went through four protocol versions, each superseded before data by a source or manifest change on `main`; v4 is frozen and its launch is blocked.

The price, unchanged. The guess-then-degree-4 oracle costs $2^k\binom{N-k}{\le 4}$ per target against $\binom{N}{\le 5}$ for degree 5. At the priced ECC2K-130 cell ($m = 4$, $\ell = 33$, $N = 2 \cdot 131 + 4 \cdot 33 = 394$, $w = 1$) the ratio $\binom{394}{\le 5} / (2^k \binom{394-k}{\le 4})$ is 78.2 at $k = 0$, 5.09 at $k = 4$, 1.30 at $k = 6$ and 0.66 at $k = 7$, recomputed in the three-guess study. Both registered growth rules give $k \approx 105$–108 there. Nothing in the series measures k at $m = 4$, so the +8.43 to +58.27-bit range of §31 stands.

Study	Guess set (all set to 0)	Outcome	Verdict
guess (22 Oct)	$x_7, x_{15}, x_{23}, x_{24}$: one top coordinate per summand and one intermediate	64 of 64 branches refute at 4; suffices on 4 of 4 draws	four suffice (an upper bound)
guess3 (23 Oct)	x_7, x_{15}, x_{23}	draws 0, 3, 5 fail at branch 0 (closure rank 129,221, 128,213, 129,893 of 136,698); draw 6 suffices	three do not suffice for this set
null (28 Oct)	none, at $(22, 7)$, $N = 43$	4 of 4 draws do not refute at 4	12 of 12 unspecialised $\ell = 7$ draws now fail
invariant (29 Oct)	S3: $a_7 + b_7$, c_7, $e_1[0]$; X3: a_7, c_7, $e_1[0]$	S3 suffices on 2 of 4, X3 on 1 of 4; S3 never worse	mixed
invariant, more (29 Oct)	S3 against X3 on 8 fresh draws	S3 5 of 8, X3 0 of 8; 5 of 5 discordant draws favour S3, $p = 0.031$	S3 better (preregistered)
pair (29–30 Oct)	D3: $a_7 + b_7$, $a_6 + b_6$, $e_1[0]$ against S3	D3 2 of 12, S3 7 of 12; 5 discordant, all S3; two-sided $p = 0.0625$	no detected difference
second link (5 Nov)	L3: $a_7 + b_7$, $(e_1 + x_3)_{20}$, $e_1[0]$ against S3	L3 6 of 12, S3 7 of 12; one discordant draw; $p = 1.0$	no detected difference; unverified claim `PCC-20261106-607061`

Table 19.1: Guessed-variable closures at $m = 3$, $(n, \ell) = (22, 8)$ (the null row at $(22, 7)$), degree 4, unsatisfiable draws only; a, b, c are the summands' coordinates and $s = a + b$ the swap invariant of the first link. "Suffices" means every branch of the draw refutes at degree 4. Sources: `research/dreg_closure_guess_20261022/RESULTS.md`, `research/dreg_closure_guess3_20261023/RESULTS.md`, `research/dreg_closure_null22_7_20261028/RESULTS.md`, `research/dreg_invariant_guess_20261029/RESULTS.md`, `research/dreg_invariant_guess_more_20261029/RESULTS.md`, `research/dreg_invariant_pair_guess_20261029/RESULTS.md`, `research/dreg_second_link_guess_20261105/RESULTS.md`.

Other degree levers of the same weeks. A preregistered invariant arm that adds the summand-swap invariant p as a variable makes every equation quadratic but does not refute at degree 3 at $(22, 6)$, where the plain system refutes at 4; its degree-4 matrix would have 5.1–5.5× the plain system's columns (523,686–637,393 against 102,091), and at $(21, 7)$ degree 4 is unmeasured (`research/dreg_pair_invariant_20261028/RESULTS.md`). Fixing one variable usually brings the $\ell = 7$ draws back to degree 4, but the null control refuted 6 of 8 and was invalid, so the study is uninformative for its registered question (`research/dreg_closure_specialise_20261017/RESULTS.md`).

19.2 A measured degree-5 closure

The descent rows of Table 15.4 price the $m = 4$ ECC2K-130 route at degree 4 (+7.62 bits, contradicted) and degree 5 (+10.32 bits at $w = 1$, +45.71 at $w = 2$). Until November no degree-5 closure had been computed. The leads study of 24 October (`research/koblitz_ic_leads_build_20261024/README.md`; Amendment 1 post hoc, Amendments 2–7 frozen before the runs they govern) built four leads in Rust and, in Amendment 7 (4 November), measured cell C1.

- *C1 refutes at degree 5.* On $m = 3$, $(n, \ell) = (19, 7)$, draws 0 and 3, the incremental engine `pe_closure` (partial, low-first, batch 64) found 1 in the degree-5 closure, so each draw's closure degree is 5. Draw 0: 607,530 of 760,099 columns filled at refutation (0.80), 31,632 s, 52.6 GB peak resident; draw 3: 606,973 columns, 31,888 s, 52.5 GB. Host: 32 vCPU AMD EPYC 9965 with 128 GB (RunPod).
- *Memory.* The dense engine aborted on the same draws at 424 and 423 GB resident, more than twice its 217 GB projection and about 8× the incremental engine's measured peak, which was below the derived 73 GB bound.
- *Scope.* One cell, $m = 3$ not $m = 4$, two of four draws, no satisfiable control (the owner chose draws 0 and 3). The study reads it as support for the degree-5 rows of `research/notes/ecc2k130/RESEARCH_ECC2K130_DESCENT_DEGREE.md` at $\ell = 7$, and it changes no cost row. C1 cost about $53 of an $80 cap over six pods, of which only the last measured.

The other leads of the same study change no cost either. Partial enlargement (L2) does not reduce memory in general: in the registered order the basis fills 0.66–0.997 of the column space before 1 appears,

and other queue orders and batch sizes keep every run at 0.61–1.00. MXL's variable restriction is the one exception, on $m = 4$, $(11, 5)$: all four draws refute at fill 0.38–0.45 with 1.8–2.5× fewer products, none reaching the 0.25 payoff; elsewhere it does not help. The useful by-product is the incremental engine itself, which holds one row per column plus one chunk: 1.3–2.8 GB resident at degree 4 against 1.9–4.2 GB for the full order.

19.3 The per-attempt criterion and the closure pilots

The companion repository's degree pilots of 9–14 October are labelled by their authors exploratory, "not program evidence", with no run record or ledger write. They are worth recording for the criterion they state and for three measurements that §16.1 does not have.

The criterion. If a failed decomposition attempt costs $2^{\alpha\ell}$, relation collection can beat rho only if $\alpha < m/2 - 1$ (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/m3-closure-pilot/README.md`). A harvest of 411 measured rows and 1,554 per-target values from the repository found that SAT and enumeration solvers spend about $2^{m\ell}$ per failed attempt (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/ctrial-harvest/README.md`): WDSat's unsatisfiable conflict count equals $2^{3\ell}/3!$ to within 0.03–1% for $\ell = 5$–11, slope 2.9985 bits per unit ℓ [2.9979, 2.9992]. At $m = 3$ the budgets need -1.7 to -3.1 bits per unit ℓ against a measured $+3.36$; at $m = 4$, -0.12 to -0.61 against $+2.32$ for meet-in-the-middle. Extrapolated to $n = 131$ at the budget ℓ, WDSat with symmetry breaking lies at least 91 bits outside every budget line, a Riemann–Roch $L(4O)$ oracle measured at $n = 131$ at least 55, and the $t = 4$ chained meet-in-the-middle at least 52; $m = 5$, 6 had no usable cells. Only a bounded-degree algebraic refutation could meet the criterion.

The $m = 4$ SAT exponent. Run on the published Trimoska et al. [39] object and configuration, the pilot reproduces $\alpha = 2.981$ [2.973, 2.997] at $m = 3$; on chained S_3 at $m = 4$ with Gaussian elimination it measures $\alpha = 4.328$ [4.315, 4.340] (4.310 at fixed $n = 17$), against the $\alpha < 1$ the $m = 4$ budget needs (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/m4-sat-cost-pilot/README.md`). The pilot also found that WDSat's stock `-b` mode returns "SAT" with undefined bits on unsatisfiable $m = 4$ instances.

Closure degree stays at 4 on toy chains, and is local.

- $m = 3$. The degree-4 mutant closure W_4 refutes 60 of 60 unsatisfiable chained-S_3 instances at $n = 9$ and 40 of 40 at $n = 11$, with no false refutation of a satisfiable one, where plain Macaulay M_4 and M_5 refute none (plain needs degree 6 at $n = 9$; semi-regular predicts 7–11). A support-matched null is refuted and a dense null is not, so the low-degree refutation follows the sparse, block-bilinear support of the descended system, not Semaev's coefficients.
- $m = 4$, *5*. W_4 refutes 25 of 25 at $m = 4$, $n = 7$; 10 of 10 at $n = 9$; 2 of 2 at $n = 11$; 10 of 10 at $m = 5$, $n = 7$; and W_3 refutes 30 of 30 at $m = 5$, $n = 9$, $k = 2$, while plain M_4 refutes nothing (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/m45-closure-pilot/README.md`).
- *Locality.* Restricting monomials to consecutive blocks (locality 1) keeps the refutation on 124 of 135 unsatisfiable instances with 39–69% fewer columns at $m = 3, 4$; every miss refutes at locality 2, and none needed locality 3. At $m = 5$, $n = 9$, $k = 3$ locality 1 refutes none of 10 and locality 2 all 5 tried (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/block-local-closure-pilot/README.md`).

These are the toy-scale version of the MutantXL result of §16.1, and they meet the same wall: the closure stays at degree 4 while the columns stay within a host's memory, and the $\ell = 7$–8 cells of §19.1 and the degree-5 cell above are where it leaves degree 4.

19.4 Semaev 2015, §4.5, on its own cells

Chapter 6 proves that the binary $m = 2$ descent and Semaev's chained system have first fall degree 2 for every n (Corollary 6.6) and notes that Semaev states 4 in his §4.5 [36]. A frozen check of 2 November ran his own $t = 2$ cells of Table 2, $S_3(x_1, x_2, z)$ with $x_1, x_2 \in V$ and random z and B, 20 instances per cell and field polynomial, 200 in all (`research/semaev_t2_trace_20261102/REPORT.md`; exact $\mathbb{F}_{2^n}$ and Boolean-ring arithmetic in standard-library Python; source arXiv:1504.01175v1).

- *Refuted: "the first fall degree ... is always 4".* On all 200 instances the first fall degree is 2, each with an explicit trace certificate: with $\lambda = z^{-2}$, $\mathrm{Tr}(z^{-2}S_3) = \mathrm{Tr}(x_1) + \mathrm{Tr}(x_2) + \mathrm{Tr}(Bz^{-2})$, a constant combination of the coordinate equations of degree at most 1.
- *Split: the corpus inference that these rows refute $d_{F4} \le d_{\mathrm{ff}}$.* The least degree at which the Boolean Macaulay span contains 1 is 2 on 154 instances and 3 on 46, never 4 (per cell: $(15, 4)$ 18/2 and 19/1; $(15, 5)$ 20/0 twice; $(16, 4)$ 18/2 and 20/0; $(19, 3)$ 6/14 and 4/16; $(21, 3)$ 15/5 and 14/6). So the sharp first-fall assumption fails on 46 of 200 instances, not everywhere, and Semaev's reported $d_{F4} = 4$ is Magma's step degree, not what the systems require.
- *Why.* The quadratic parts depend only on $x_1x_2 \in V \cdot V$, of dimension $2k - 1 < n$, so most $t = 2$ systems are inconsistent at degree 2 or 3. None of the 200 has a solution in $V \times V$, which matches Semaev's success probabilities of 0.0006–0.0155 for these rows.

It complements the Semaev 2015 audits of Chapter 30, which tested eq. (11) and the typed-image ratio but not the first-fall statement: on these cells the sharp assumption fails on 23% of instances, and the stated first fall degree is wrong under Semaev's own definition.

19.5 The ceiling on learned SAT guidance

The SAT engines and encodings are described in §12.7; this section records the ceiling they place on learned guidance.

The SAT survey dismissed ML-guided SAT without measuring it. A preregistered study of 27 October measured the most any learned component could gain inside a SAT point-decomposition oracle before anyone builds one (`research/pdp_ml_sat_ceiling_20261027/README.md`; protocol frozen in `8a85584fd`; symmetrised S_4, $m = 3$, native CDCL+XOR, the Trimoska-corpus curves, uniform targets; cost in conflicts). It tested triage on target features or early solver telemetry, a fixed conflict cutoff, and learned static branching. All 4,200 labels agree with exhaustive sorted-triple search.

- *The search is an enumerator.* A refutation costs 0.963 ($n = 15$, $\ell = 5$; 3,000 targets) and 0.998 ($n = 17$, $\ell = 6$; 1,200 targets) conflicts per sorted factor-base triple, every refutation within 4.1% of its family median (5,781 and 45,649 conflicts).
- *The prize is real.* A perfect unsatisfiability predictor, charged its telemetry conflicts, would raise relations per conflict by 5.2–9.0×.
- *The signal is not.* Test AUC is within noise of 0.5 for every feature group on both families and both splits; the permuted-label control reads 0.45–0.54 and the positive control (labels flipped with probability 0.25) 0.72–0.79, so the learner works. The one interval that excludes 0.5 ($n = 15$ telemetry at 512 conflicts, 0.556 $[0.518, 0.599]$) includes a "decided within 512 conflicts" flag and gains 0.99×. Train-fitted gains up to 4.7× fall to about 1× held out.

The study closed by its preregistered rule. A related instrument finding is recorded in the atlas: five "unverified SAT answers" of an earlier binary solver comparison were quadratic-twist decompositions (`EV-ICPERF-390707`; `crypto-autoresearcher:knowledge/leads/ecdlp/ic-prime/LEAD-ICP-icperf-binary-pdp-solver.yaml`).

19.6 F6-IC with a pair index is the meet-in-the-middle scan

The engine itself, its gate and its other uses are described in §12.4; this section records what it established for binary decomposition.

The F6-IC gate of §10 decided a node exactly only once one summand's coordinate bits were all assigned. On 5 November it was changed to run the same exact closure as soon as a summand's assigned bits leave at most 64 base codes, and the change is now on by default (`research/f6_few_candidates_20261105/README.md`; nine Koblitz cells, 5 rounds × 3 repetitions, L0).

- Verdicts and indices are identical on all 80 target–engine pairs; gate pair lookups and group additions are identical on all 40 F6-IC targets.
- F6-IC CPU falls 2.67–10.29× (worst pairing of rounds ≥ 1.66×), word XORs 1.84–10.13×, and 204–1,049 reductions per cell become 5–16; the inherited-F4 control reads 0.87–1.05×.

Because the pair lookups and group additions do not change, the algebra was not ruling out any candidate x-coordinate on these cells: with the pair index, F6-IC works as the pair-index meet-in-the-middle scan, and the closure removes the algebra around it. The source marks the inference that it cannot beat ecbench's `mitm` oracle, being that oracle in the limit, as unverified. Two other items in the same study changed nothing: Four Russians batching of displaced rows would spend about 32 row operations per

pivot on tables against about 1,500 row XORs per node it replaces, and a parallel split of one target's tree gives 1.59–2.88× wall on cells with no decomposition and 0.94–1.11× where most targets decompose (opt-in, refused by ecbench).

The same conclusion holds from the cost side. A desk evaluation of 28 candidate point-decomposition strategies against the lead atlas found 17 that collide with existing leads, 7 partial overlaps and 4 with no match, rated least promising (`research/pdp_strategy_evaluation_20261031/README.md`). On the ladder rows the walk is 61–82% of S and the matrix at most 0.10%, and on the committed algebraic-oracle rows the oracle's share of S is exactly zero when native operations are unpriced, so an algebraic-oracle lever has a ceiling of 1.00× there.

20 ECC2K-130, ECC2-131, factor-base selection, composite degree and GHS

The named binary targets are in §15.7 (ECC2K-130) and §16 (ECC2-131), with the priced routes of Table 15.4. This chapter adds what those sections do not have:

- two independent budget tables and the isogeny-class census from the `cryptanalysis` repository, the Torelli toy ladder, and the state of the modular carrier search (§20.1);
- the ECC2-131 symmetry-breaking and stage measurements (§20.2);
- factor-base selection by a structural law and by a learned cost model, with the exact standing of the one validated claim (§20.3);
- quasi-subfield polynomials at $n = 131$ (§20.4);
- composite-degree curves (§20.5);
- binary GHS and Weil-restriction screens outside the late-October cubic catalogue (§20.6).

The `cryptanalysis` sources are read at the head of the branch `claude/cryptanalysis-repo-setup-p59kek` (22 October), not a `main` branch. None of the studies here carries a validation file in a repository read for this treatise.

20.1 ECC2K-130 beyond the priced routes

What one decomposition may cost. Two `cryptanalysis` studies turned the ECC2K-130 cost model around: fix rho at $2^{60.81}$ iterations on the $\langle -1, \tau \rangle$ classes, count one group addition as one rho iteration, and ask how much one decomposition may cost if the whole attack is to tie rho. The subspace version (`cryptanalysis:experiments/pdp-scaling/README.md`) finds no budget at $m = 3$ (linear algebra alone exceeds rho), $2^{12.2}$ iterations at $m = 4$, $\ell = 29$ (about 0.5 ms), $2^{32.8}$ at $m = 5$, $\ell = 28$ (about 12 minutes) and $2^{36.8}$ at $m = 6$, $\ell = 24$. Against these, its meet-in-the-middle decomposition costs 2^{58}, 2^{84} and 2^{72} group additions per point, gaps of +46, +51 and +35 bits, and the fitted algebraic costs (SAT, F4) are near 2^{180} at $m = 4$; SAT grows about 4 ($m = 3$) and 7 ($m = 4$) bits per extra dimension of the base. The Frobenius-quotient version (`cryptanalysis:experiments/frobenius-quotient-m4/README.md`) uses the one quotient that remains at $n = 131$, the orbit quotient of the non-linear weight base $F_w = \{P : 1 \le \mathrm{HW}(x(P)) \le w\}$ in a normal basis (§17.2), with columns $B/131$ credited in full. Table 20.1 gives its best weight per arity.

The quotient raises the budget from $2^{12.2}$ to $2^{31.1}$ at $m = 4$, which is the factor 131 on relations and 131^2 on linear algebra that rho already collects through $\sqrt{2 \cdot 131}$. It does not remove the obstruction. A decomposition that uses only the group law needs $|L_1||L_2| \ge \#E \approx 2^{131}$, at least $2^{65.5}$ per attempt; a method that uses the x-coordinate algebra must reach the required exponent column, which is below $m/2$ for every m, against m (exhaustive, WDSat, FES), $m - 1$ (hybrid) and $\lceil m/2 \rceil$ (MITM, floored at $2^{65.5}$). At $m = 4$ the required exponent is 0.98: one decomposition would have to cost less than listing the base once. The same study measured $m = 4$ on the weight base at toy size with CryptoMiniSat (chained S_3, weight as a cardinality bound): every completed instance at $n = 9$–15 returned a lifted, re-added decomposition with no spurious model and no oracle disagreement, and the descriptive slope of median CPU on $\log_2 B$ over four uncensored cells is 1.34.

The isogeny class, by level. The class of $E_0 : y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{131}}$ has four endomorphism levels, conductors 1, 263, p and $263p$ with $p = 146{,}505{,}763{,}881{,}528{,}721$ (`cryptanalysis:experiments/ecc2k130-isogeny-class/README.md`):

- level 1 (E_0): native and effective rho $2^{60.81}$;
- level 263, 262 floor curves: native rho $2^{64.33}$, effective $2^{60.81}$ through a verified 263-isogeny to E_0 costing at most $2^{15.6}$ $\mathbb{F}_q$-multiplications (the 262 j-invariants independently rederived by Vélu from $E_0[263]$, 262 of 262 matching);
- level p, $p + 1 \approx 2^{57.0}$ curves, and level $263p$, $262(p + 1) \approx 2^{65.1}$ curves: native $2^{64.33}$, effective in $[2^{60.81}, 2^{64.33}]$.

No curve in the class is easier than E_0. The p-level interval would close only with a characteristic-2 higher-dimensional (Kani) isogeny pipeline, and constructing any such curve costs about 2^{70} $\mathbb{F}_q$-multiplications,

m	w	$\log_2 B$	$\log_2$ budget/attempt	required exponent in B	MITM exponent	MITM gap, bits
2	—		none: linear algebra alone exceeds rho at every w			
3	6	31.6	0.5	0.01	2	+65.0
4	6	31.6	31.1	0.98	2	+34.4
5	5	27.2	40.6	1.49	3	+44.0
6	4	22.5	45.3	2.01	3	+25.3
8	3	17.5	50.3	2.87	4	+23.7
12	2	12.1	52.9	4.38	6	+25.6

Table 20.1: ECC2K-130 on the Frobenius-orbit quotient of the weight base: the most one decomposition may cost if the attack is to tie rho ($2^{60.81}$), and the meet-in-the-middle floor against it. Arithmetic only (rows $m = 7, 9, 10, 11$ omitted). Source: `cryptanalysis:experiments/frobenius-quotient-m4/README.md`.

more than solving natively on it. The treatise's 262-curve figure and the 11.45× negation-only penalty of §15.7 are the level-263 row.

A Jacobian carrier: the toy ladder. §15.7 leaves one hypothetical route open, a genus-130 Jacobian isogenous to the simple 130-dimensional abelian variety that carries $\langle G \rangle$, and quotes the Torelli codimension 8,128 against it. A frozen toy test of that rarity heuristic (`research/ic_next10_20261014/10-torelli-toy/README.md`) enumerated curves over $\mathbb{F}_2$ whose Frobenius polynomial is divisible by the trace-zero part A_n of $\mathrm{Res}_{\mathbb{F}_{2^n}/\mathbb{F}_2}(E_a)$:

- $n = 5$*: negative, with low power.* No genus-5 Jacobian (hyperelliptic and trigonal completely, canonical curves by a uniform sample of 2^{21} nets), and no plane quintic or hyperelliptic curve of genus 6–8, carries A_5; but the heuristic predicts only about 0.42 classes per sign at genus 5.
- $n = 7$*: the literal reopen criterion is met, by structure* $n = 131$ *lacks.* 1,536 hyperelliptic models of genus 7 (mass 0.5) per sign have Jacobians isogenous to the full Weil restriction, against a prediction of 2^{-14}–2^{-18}. Here $\mathrm{ord}_7(2) = 3$, A_7 splits into two 3-dimensional factors, and genus $7 = 2^3 - 1$ is what GHS gives with magic number 4 (the attribution to GHS is the author's interpretation).
- *Calibration.* The heuristic is right to within $2^{-3.5}$–$2^{+1.2}$ for the A_3 control at excess genus at most 3, and under-predicts every structured class that occurs, by up to 2^{19} at $n = 7$.

So the ECC2K-130 rarity argument rests on the absence of such structure at $n = 131$ (no small magic numbers, a simple A), not on the number 8,128. The atlas lists $n = 11$ and 13 as the next rungs (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-ecc2k130-jacobian-carrier.yaml`).

The modular carrier search. A separate line looks for a low-genus or modular carrier of the relevant CM factor through an explicit modular bridge (`research/koblitz_modular_ic_20261018/README.md`). It has run 26 rounds of exact constructions with Lean identities and no index-calculus rank. Round 22 computed the integral equivariant recovery on norm-zero targets (eight Smith factors 1, kernel rank 24); round 26 shows by exact diamond norms that the invariant quotient does not carry the intended CM factor under the retained bridge, and records the next Riemann–Roch space (degree 128, dimension 68 under the genus-61 bridge) whose basis is still required.

The ranking of 23 October. A leads survey of that date ranked six degree and closure leads and three rho-accounting items by planning cost against best-case effect, both judgements and not measurements (`research/ecc2k130_leads_survey_20261023/report.typ`). Its first lead, the matched null at $(22, 7)$, ran on 28 October (§19.1). It quotes the unconverted distances +7.62 bits ($m = 4$, degree 4), +10.32 (degree 5) and a free-oracle floor of $2^{56.46}$, and no lead moves a priced row below rho.

20.2 ECC2-131

ECC2-131 is a random binary curve over $\mathbb{F}_{2^{131}}$ ($z^{131} + z^{13} + z^2 + z + 1$, cofactor 2, $a \notin \mathbb{F}_2$, so no Frobenius saving), and 131 is prime, so there is no Weil descent to a subfield. §16 gives its program, gates and degree falsifiers. Two measurements were added.

Family	Curve	n	base M/s	sym M/s	sym/base	IC/rho step rate
binary	ECC2-131	131	0.18	5.24	28.7×	1.20
binary	calibration	127	0.36	7.59	20.8×	1.74
binary	calibration	61	0.39	4.52	11.6×	1.04
prime	ECCp-131	131	0.17	0.09	0.6×	0.02
prime	calibration	61	0.37	0.19	0.5×	0.03

Table 20.2: Decomposition-oracle step rates at $m = 2$, single thread: the per-point baseline, the symmetric e_1 oracle, and the symmetric step rate over the rho adding-walk step rate on the same host. A step-rate ratio, not a cost ratio. Source: `research/certicom131_symlp_20261025/REPORT.typ`.

Symmetry breaking and double large primes at $m = 2$. With the base $V_\ell = \{x : \deg x < \ell\}$ and a large-prime domain $V_{\ell'}$, the binary S_3 linearises in $e_1 = x_1 + x_2$ to $t^2 + t = e_1^2 + b/x_R^2$, and because $V_{\ell'}$ is closed under addition the symmetric oracle can sweep e_1 (each unordered pair once, one batched inversion) instead of lifting each x_1 with its own inversion (`research/certicom131_symlp_20261025/REPORT.typ`; single thread, seed 20261025; IC step and rho step in the same field code and process). Table 20.2 gives the step rates. For ECCp-131 the interval $[0, B)$ is not closed under addition and each step needs two square roots, so the symmetric oracle is slower than its baseline. Direct relations (both abscissae in V_ℓ) are about $5 \cdot 10^{-7}$ of decompositions on the binary calibration curves; the two-large-prime graph is what makes the bulk usable, and the planted pipeline recovers a logarithm through it on both real 131-bit groups.

The study's derived cost model (every row computed, not measured) puts an $m = 2$ solve at about $2^{2n/3}$ decomposition steps at the balancing dimension $d = (n + 1)/3$, which agrees with the merged algebraic ladder's measured trial slope 0.647 $[0.639, 0.655]$, against rho's $\sqrt{\pi r/4}$: $\log_2$ steps 87.3 against 64.8 for ECC2-131 and 65.3 for ECCp-131, a distance of about 2^{22} steps before the linear-algebra floor, which the program gates place above rho for $m \leq 3$. Both accelerators are constants on the $2n/3$ exponent; they would matter at $m \geq 4$, which the study does not reach.

Stage rates. On one isolated sibling pair of an Intel Xeon 6975P-C host, five samples per cell (`research/certicom_131_stage_bench_20261023/REPORT.md`):

- rho state steps per second: 207,822 (ECC2K-130), 207,602 (ECC2-131), 87,055 (ECCp-131);
- full-field S_3 quadratic-root queries per second: 48,080, 48,186 and 9,522.

The two binary curves share the field polynomial and differ by 0.11% in point updates; ECCp-131's generic rho step is 0.419× and its root query 0.198× ECC2K-130's. These are generic-binary CPU stage ratios, not solve costs; the optimised clients of §15.7 and §16 walk 8.5–13.2 million steps per second.

20.3 Factor-base selection

Named subspace properties (greedy coverage, cosets, geometric progressions, normal against polynomial bases) are exhausted at scale (§16, §31). Two further results choose the base by the solving degree rather than by the yield.

A structural law for $m = 2$. Over 656 factor bases (7 families, 9 fields) on $y^2 + xy = x^3 + 1$, the closure solving degree of the two-summand system is

$$D = 2 + [e < -3.5] + [e < -8.5], \qquad e = n - \dim V - \dim V^{(2)},$$

with $V^{(2)}$ the span of pairwise products, at mean absolute error 0.06 degrees, 0.08 with each field size held out (`cryptanalysis:experiments/pdp-degree-heuristics/README.md`). For prime n, $\dim V^{(2)} \geq 2\ell - 1$, and below $n - 1$ only geometric progressions $c \cdot \mathrm{span}\{1, g, \ldots, g^{\ell-1}\}$ attain it; the literature's $\mathrm{span}\{1, z, \ldots, z^{\ell-1}\}$ is one. Where the rule puts a random base a degree higher, the progressions are 15–40 times cheaper per query and per relation on average, up to 100; elsewhere 1.5–2 times; the yield is the same. Two further findings matter for pricing: the homogeneous degree of regularity of these systems is exactly $\ell + 1$, because the two-summand top parts are bilinear across the variable blocks, while the solving degree is 2–4, so the semi-regular prediction misses by 0.5–1.2 degrees on average; and at $m = 3$ the formulation decides whether the base matters (in the symmetric formulation the progression is 15–40× cheaper per query at $\ell = 3$ at equal degree, and 500–1,500× where the random base needs one more degree).

A learned cost model. The lead (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-learned-cost-model-factor-base-selection.yaml`, last reviewed 6 November) fits a gated log-linear (ridge) model of per-relation cost on structural features of V (predicted exact yield, solving degree, trace and product-space statistics) and compares its pick from a pool with the hand rule "lowest expected solving degree first". The label producer is the structural-law experiment above. As the atlas records it:

- *Stage.* 1.82 bits per relation $[1.36, 2.26]$ better than the hand rule at $\ell \le 4$ (13 cells), 1.13 $[0.45, 1.62]$ at $\ell = 5$.
- *Across curves, frozen, no refit.* $+1.84$ $[0.95, 2.68]$ on $a_2 = 1$ Koblitz curves; $+0.92$ $[0.54, 1.34]$ on ordinary curves, not confirmed on 16 fresh ordinary curves at $\ell = 4$ ($+0.43$ $[0.15, 0.74]$ against a 1.08 threshold), where the hand rule already sits near the pool best.
- *End to end.* 0.93 bits $[0.36, 1.48]$ less counted cost per verified logarithm over 9 cells ($n \le 41$). At $\ell = 4$ on ordinary curves, $n = 19$, the complete-logarithm check is a null with the sign reversed: -0.45 $[-0.98, +0.04]$ bits at $w = 0$.
- *Native ecbench on the isolated EKS route*, 5 November: K1 ($a_2 = 1$) Koblitz curves, $\ell = 3$, `ic.pipeline` with explicit binary-subspace bases, 4 targets $\times$ 5 rounds per size, 192 of 192 records verified. $\log_2(S_{\text{rule}}/S_{\text{ridge}})$ in counted complete-logarithm cost is $+1.83/+0.80/+0.80$ bits at $n = 17$ and $+0.96/+2.47/+2.47$ at $n = 19$ for $w = 0/0.1/1$, every interval's lower end above 0. The ridge pick stays behind `rho.signed_frobenius_strong` by 1.8/14.8/17.7 bits ($n = 17$) and 3.3/15.1/17.9 ($n = 19$). The producer's first "about 9 bits, identical across the band" was a pinned-calibration point and was corrected.

Standing, exactly. The atlas lead on crypto-autoresearcher `main` (commit `8d61579904`, 6 November) records the EKS result as "independently validated 2026-11-06 with VALIDATION.json, 160 of 160 replays reproduced on a second CPU". The experiment, its protocol, its validator report and that `VALIDATION.json` live in the external repository `aburan28/ml-cryptanalysis` (`experiments/fb-cost-model-curves-eks`), which this session could not open; the same atlas entry's evidence note says the validation is "in a follow-up PR". Nothing about the study is on `main` of `crypto` or `crypto-autoresearcher` except the atlas text. So for this treatise the validated claim is an unverified assertion recorded by the atlas: if it holds, it is the family's only validated, measured improvement over a binary index-calculus baseline, about 1–2.5 bits at toy size, $n = 17$, 19, with its scaling in n not yet measured (K1 $n = 23$ was frozen on 6 November). It moves no row toward rho by more than the 1.8–17.9 bits it leaves.

20.4 Quasi-subfield polynomials at $n = 131$

§16 records the $\mathbb{F}_{2^7}$ census that showed the HKPYY Lemma 4.1 bound [24] tight. The cost side is closed by theorem (`crypto-autoresearcher:analysis/qsp-ecc2k130/README.md`; the atlas records it as a theorem from Euler and Petit's work):

- every linearised quasi-subfield polynomial has $\beta \ge 3/4$, so under the HKPYY cost model the exponent is at least 0.93, at least 2^{122} at $n = 131$ against rho's $2^{60.9}$;
- with $\mathbb{F}_2$ coefficients only $X^2 + X$ ($n' = 1$) and $L_{\Phi_{131}}$ ($n' = 130$, $\beta = 0.99994$) exist at $n = 131$;
- multiplicative quasi-subfield polynomials give $|V| = 264$ or $p + 1$, with $\beta \ge 1.02$;
- the cost model needs $\beta \approx 0.0958$ to reach $2^{60.9}$.

At toy size the yield is no better than random: at $n = 31$, $k = 15$, the QSP base $t^{15} + t^7 + t^3 + t + 1$ ($\beta = 0.964$) yields 0.214 (0.190–0.240) against 0.204–0.230 for random subspaces, and the route's kill criterion fired (`research/ic_next10_20261014/08-quasi-subfield/README.md`).

20.5 Composite degree

Composite n is the one binary setting where index calculus can be priced below rho on time alone. Chapter 8 records the lift-dense twisted-line base on c2pnb208w1 and c2pnb368w1 (+0.50 bits, modelled), and §26 the GHS reachability of the Oakley curves.

c2pnb: Gaudry–Diem $n = 4$, as a cost model. The five ANSI X9.62 example curves c2pnb176v1, 208w1, 272w1, 304w1 and 368w1 have coefficients in $\mathbb{F}_{2^{16}}$ and live over $\mathbb{F}_{Q^4}$, $Q = 2^{4k}$, $k = 11, 13, 17, 19, 23$. A zero-experiment derivation (`crypto-autoresearcher:analysis/ecdlp2m-iteration-20261009/c1-gaudry-n4-c2pnb/README.md`) prices Gaudry–Diem $n = 4$ [10, 20] with a base of points with $x \in \mathbb{F}_Q$

reduced modulo negation and π_{16} ($N = Q/(2k)$ orbits) and double large primes [22], against matched rho of 78.10, 93.98, 125.78, 141.71 and 173.57 bits. Under three metrics:

- *time only:* below rho on all five rows for every per-decomposition cost $c_{\text{trial}} \in [2^{15.5}, 2^{32.5}]$, by +1.46, +5.80, +14.19, +18.34 and +26.60 bits at $c_{\text{trial}} = 2^{30}$;
- *time × memory against van Oorschot–Wiener:* worse on every row, by 35–56 bits at $c_{\text{trial}} = 2^{30}$;
- *mesh area-time:* better only at 304w1 (+1.28) and 368w1 (+0.97), and only for $c_{\text{trial}} \le 2^{27}$.

The verdict turns on c_{trial}, never measured at toy size, the per-worker FGLM memory (D^2 with $D = 4096$, brief-supplied) and the linear-algebra constant; a toy ladder over $\mathbb{F}_{2^4}$ lifted to $\mathbb{F}_{2^{4k}}$, $k \in \{5, 7, 9, 11, 13\}$, about $2 \cdot 10^4$ decompositions, would pin all three and has not run. The atlas adds that the $\mathbb{F}_{2^{16}}$-rational base is structurally empty in G and that a naive trace-zero extrapolation is a regime-invalid trap (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-composite-degree-c2pnb.yaml`).

Oakley groups 3 and 4: $n = 5$. The twisted line exists on both Oakley EC2N curves: a u with $T(u) = T(c/u) = 0$ was verified in $\mathbb{F}_{2^N}$ ($N = 155$ and 185, $n = 5$), and all 200 sampled elements of $u\mathbb{F}_q$ lift against 104 and 99 of 200 for the standard line, so as a Gaudry–Diem $n = 5$ base each decomposition trial would have $2^5 = 32$ times the relations of the standard base; on c2pnb176v1, 272w1 and 304w1 ($\mathrm{Tr}(a) = 1$) none of 200 lift (`research/oakley_lift_dense_20261027/REPORT.typ`). This is a per-trial yield statement. Three approved Oakley contracts have no runs; the atlas prices the GHS genus-16 transfer at +1.1 bits (group 3) and +10.9 bits (group 4) of advantage after the 11 October magic-number correction, by formula (`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-oakley-quintic-tower.yaml`).

20.6 Binary GHS and Weil-restriction screens

The late-October cubic GHS catalogue over $\mathbb{F}_4$, $\mathbb{F}_8$ and $\mathbb{F}_{16}$ belongs to the transfer record (§26). Four binary screens outside it bear on index calculus.

Decomposition coverage at $n = 2$ is 3/8, not Poisson. For generic $E/\mathbb{F}_{q^n}$ and the base $x \in \mathbb{F}_q$, the share of points that are a sum of n base points was computed exactly over the whole group (`research/weil_descent_angles_20261018/REPORT.md`). In odd characteristic at $n = 2$–5 it matches a random x-set of the same size (paired residual $|z| \le 0.94$). Binary $n = 2$ is the exception: coverage tends to 3/8, not the Poisson $1 - e^{-1/2} = 0.3935$ ($z = -4.72$), and the cause is proved. Every $x \in \mathbb{F}_q$ has $\mathrm{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_2}(x) = 0$, so the base lies in one coset of $2E$; the descended relation map is $\mathbb{F}_2$-affine with image in the hyperplane $\mathrm{Tr}(w/x_3^2) = 0$; so a point decomposes only if $\mathrm{Tr}(x_3) = \mathrm{Tr}(a)$, half the group, and inside that half the multiplicities are exactly $\mathrm{Bin}(2, 1/2)$. Because attack targets lie in the odd-order subgroup inside $2E$, the relevant coverage is that of $2E$, exactly 3/4 against 0.3935; the study corrected its first reading ("neutral to slightly negative", measured over the whole group) on this point. The identity holds on all 21 curves.

IPsec group 3. An archived search over the isogeny class of RFC 2409 Oakley group 3 ($\mathbb{F}_2[z]/(z^{155} + z^{62} + 1)$, $N = 12r$) screened 24,492 Frobenius orbits and found no low-genus candidate in the implemented tests; the whole class was not enumerated and no resistance claim follows (`research/ghs_ipsec_isogeny_class_20261023/README.md`).

Generalised binary GHS, explicitly. Full nonzero-c Hess screens at absolute degrees 6, 8, 10 and 12 (63, 255, 1,023 and 4,095 values) gave explicit maps and attack controls (`research/ghs_generalized_20261024/REPORT.md`). At degree 12, 147 profiles have genus at most 2, including 15 of genus 1 over $\mathbb{F}_{2^4}$, and the Weil bound ($\#\mathrm{Jac} \le 25$ and ≤ 81) rules out a homomorphism from the order-199 subgroup; the other 232 profiles are index-two genus-2 profiles that a subfield change maps to one canonical form, presenting P as $T - P$ for the 2-torsion point T. Every exact instance (subgroups of order 19, 67, 43 and 199) verified end to end, and all belong to known GHS [21] and Hess families.

A degree-73 descent at $n = 37$. A separable degree-73 isogeny of $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{37}}$ (74 rational cyclic kernels; a squarefree degree-36 kernel polynomial dividing the degree-2,664 division polynomial) was tested as a presentation change for the Boolean stage (`research/koblitz_isogeny_descent_37_results_20260925/RESULTS.md`). All 128 systems completed and every planted decomposition

verified on both curves, but the preregistered criterion (at least a 10% F5 matrix XOR reduction in every cell, with no completion-degree increase) failed: the neighbour cut XORs by at least 10% in 4 of 8 cells, regressed up to 2.00×, and raised the completion degree in 7 of 64 paired systems. The committed raw archive was found not to be gzip and was restored from the original workflow artifact; the result is unchanged.

21 Prime fields

Over $\mathbb{F}_p$ there is no subfield to descend to and no Frobenius to fold, so index calculus must find structure in the factor base itself (§1). The programme tried known-answer ladders, root-finding and pair-table oracles, the Petit–Kosters–Messeng Kummer tower, structured bases on the P-256 field, isogeny-class walks on the NIST curves, lattice reformulations, and miniatures of Curve25519. No prime-field arm comes within a single-digit factor of rho except on the smallest Table B rows, and the gap grows with size. The section ends with Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$, the one odd-characteristic extension field the programme measured end to end. Every study is listed in the programme record [3].

21.1 Known-answer ladders and leaderboard rows

Over prime fields the ledger tracks staged known-answer ladders (`docs/ic/boundary_targets.json`, regime `prime`). On $j = 0$ curves the staged driver recovers a 20-bit logarithm ($p = 1{,}048{,}609$) in 56.8–64.9 s of IC wall against 150–153 ms of rho on the identical public point, with split timers and three seeds; running the relation search on 14 threads made that stage 3–4.4× faster in wall time with counted work unchanged (`research/j0_parallel_relations_20261016/`). On the generic $a = -3$ shape (P-256 and CryptoPro-B class), a Semaev-S_4 three-decomposition family with Cantor–Zassenhaus root finding yields one relation per trial at $p = 65{,}519$ with a 120-point base; at fixed base size, median trials per relation grow from 2 to 28 to 379–506 at 16, 20 and 24 bits, about 16× per four bits. The prime `vs_rho` row is *not achieved* at any measured size.

Table B of `docs/ic/LEADERBOARD.md` has eight prime rows, $\log_2 r = 9.7$–23.4, one target, $m = 2$, folded pair table with walked targets: IC is 6.34, 5.47, 6.43, 6.71, 5.82, 8.3, 10.5 and 13.5× matched rho, the ratio rising from about 18 bits. All eight passed the validation gate on 24 October (`research/ic_validation_gate_20261024/`). In ecbench the root-finding pair table is 16–20× matched rho in counted GAE, and 7.46× [5.82, 9.63] with a lane collector, all lower bounds (`research/ecbench_rootfinding_lanes_20261013/`); a planner that picks the base, stop rule and oracle beat every one-change alternative on six curves at $w = 0$, the root-find-and-subtract variant by 171–439× (`research/ic_planner_validation_20261020/`). The planner's three weight rules, which scaled the base or stop for $w > 0$, failed out of sample on 26 October and were withdrawn (rule set `ic-planner/4`; `research/ic_planner_exhaustive_20261026/`). The frontier campaign's two-summand prime arm goes from 10.5× rho at 20 bits to 102× at 40 bits (§31.1).

Root finding against Buchberger on NIST shapes. Per query, root finding of Semaev's S_3 was 3,021× (P-256 shape) to 5,802× (P-224) cheaper than the comparison arm at a six-point base (`research/semaev_rootfind_nist_20261012/`); the arm was `groebner_f4::buchberger`, not F4, so the ratio is against Buchberger (§4.5). It is an oracle ratio with no complete solve. A host emulation of a GPU decomposition kernel passed six configurations on 40–61-bit P-256-class curves (`research/gpu_rootfind_decomposition_20261012/`); no device ran it. Three frozen SMT formulations of a 20-bit decomposition all returned UNKNOWN at 120 s, where the native reference needed 146 probes (`research/prime_field_smt_20261005/`).

Earlier, in crypto-autoresearcher. The first end-to-end prime-field index calculus of the laboratory predates this repository's: an imported Autolab round of 1 June 2026 solved logarithms with an $m = 3$ Semaev index calculus over a factor base of the smallest valid abscissae, at a fitted cost exponent of 0.889 against rho's 0.5, IC/rho growing from 21.5 to 205.7 over 10–18 bits (`ar:experiments/EXP-ALPF-032/analysis.md`). Structured bases there weakened rather than supported a relation excess (`ar:ledger/evidence/EV-PFDR-f6b76c.yaml`), and on one 13-bit curve an apparent early fall of the degree fired identically when the Semaev form was replaced by a random polynomial but not when the factor-base constraints were, so it came from the factor base, not from the summation polynomial (`ar:experiments/EXP-PFDR-4bfc6f/analysis.md`). A separate repository asked the underlying question directly—is there any coordinate-defined subset of $E(\mathbb{F}_p)$ that interacts with addition better than a random subset?—and found none on toy curves (`ecdlp-prime-field-smallness`, `README.md` at `4f20be9b36`).

21.2 The Kummer tower

Petit, Kosters and Messeng's base $V = \mu_{2^t}$ (the roots of $x^{2^t} - 1$) gives prime-field summation systems extra structure for an algebraic solver. Its claim rests on the solving degree of those systems staying bounded as $N = m\ell$ grows; the programme measured that degree for four weeks.

The solving degree. The pilot found F4's solving degree at 4–5 for $m = 2$ through $N = 18$ and read the preregistered "grows like N" as failing at toy size (`research/pkm_tower_pilot_20260924/`). Round 2 showed it is not bounded: at $m = 2$ it is 5 to $N = 18$ and 6 at $N = 20, 22$, at $m = 3$ it reaches 7 at $N = 15$, and the null family behaves the same (`research/pkm_tower_round2_20260925/`). At $m = 4$, $N = 16$ finished at $D = 7$, as at $N = 12$ (`research/pkm_tower_round3_20260925/`); at $m = 3$ the line reads $6, 6, 7, 7$ over $N = 9$–18 (`research/pkm_tower_round7_20261006/`). Signature engines agreed with F4 but needed one or two degrees more and 1.7–28× the memory, so F4 stayed (rounds 4–5); a full-echelon exit cut multiply-adds 39.8% at $m = 4$, $N = 12$ without moving D (round 6). The torus and split Dickson presentations give the same D in every cell, and the Kummer D does not depend on the 2-adic valuation (`research/pkm_dickson_torus_20261015/`). At $m = 3$, $N = 21$ two engines finished every degree-7 step without reaching the unit ideal and stopped inside degree 8 on memory—the native `f4_fp_tower` at a 317 GB matrix cap after a 67 GB peak, msolve at 99.6 GiB on 16 threads (`research/pkm_lead1_m3n21_20261016/`, `research/pkm_m3_n21_msolve_20261018/`). That implies $D \geq 8$ at $N = 21$, which both studies read as inconclusive because no degree-8 step completed. The Certicom-131 programme's gate needs $D \leq 3$ at $m \geq 4$; the toy plateau at 7 is above it, at 2^{89}–2^{98} (generous) and 2^{129}–2^{139} (plain) at 131 bits (`docs/ic/CERTICOM131_IC_PROGRAM.md`). The exponent sum of decompositions on the tower is indistinguishable from random at $k = 1$–6 (`research/ic_exponent_falsifiers_20261013/`).

In ecbench. We ran the tower end to end with the root-finding oracles on five curves of 20–28 bits (896 runs, all verified, all replay audits OK; `research/pkm_tower_ecbench_20261013/`). Index calculus costs 34.0, 41.9, 29.4, 96.6 and 140.6× rho-negation in S (95% intervals in Figure 21.1), growing by 0.26 ± 0.28 bits of S_{IC}/S_ρ per bit of r, consistent with the $p^{1/6}$ the floors predict. The tower base costs the same as a plain base of the same size to within 1% (ratios 0.999–1.012): PKM's structure helps an algebraic solver, not a table. An isogeny-tower variant costs 1.00–1.33× the plain base once its auxiliary-curve search is charged, and every arm is 17–137× rho (`research/pkm_isogeny_tower_ecbench_20261015/`); for a P-192-sized tower that search would cost about 2^{74} GAE, or 11.2 core-years, while a supersingular auxiliary curve gives the needed point in 7 ms with the same solving degree (`research/pkm_isogeny_tower_p192_20261015/`, `research/pkm_supersingular_aux_20261015/`). A 30-bit row of 24 October is 158.9× rho (§31.5). Swapping in an A11 presentation could not be priced: ecbench has no method that builds a tower summation system (`research/a11_pipeline_pricing_20261019/`).

21.3 Structured bases on the P-256 field

Twenty-four rounds between 4 and 6 October (`research/icv1-fp256-t89188191154553853111372247798585809583-f188c491_affine_bitbox_degree_round1_20261004/` and its 23 successors; cluster "icv1-fp256") searched for factor bases on the P-256 field whose summation systems have lower degree. They found degree reductions on split systems: Dickson-torus branching lowered the solving degree of components from 4 to 3, image trees for S_9 and S_{17} reached local degree 2, and an indexed quadratic image cut one stage 498×. At the P-256 modulus the images hit their generic size ceilings (width ratio 1.000000), so the degree reduction transferred and the compression did not (round 14). No round found a relation for a target that was not planted: planted length-17 relations were recovered, hash-public targets were not, and a log-transport round found zero relations among 14.4 million images (round 21). Projected end to end, the best schedule is 16.2–19.4 bits above rho in the same unit, after a symmetric projection of $2^{144.625}$ was withdrawn for exceeding its domain (round 20); the final geometric bases are 392–404× rho, and a known-log constant of 0.964× rho in round 24 is a lower bound that the study itself says is not an attack.

21.4 Small-abscissa bases and isogeny classes

Two-point small-x bases have an exact pair yield of $2B^2/(n-1)$ with no pair-sum collisions: on P-192 a base of 8,192 points needs 2^{178} trials for enough relations against rho's 2^{96}, and on all 33 SECG curves the route loses to rho by at least $2^{40.9}$ in trials (`research/prime_small_x_p192_20261014/`, `research/secg_small_x_20261014/`). At $p = 2^{61} - 1$, 1,228,800,000 target trials over 1,024 curves of a horizontal isogeny class found zero relations, while a 20-bit positive control found 33 and recovered the scalar (`research/prime61_factor_base_20261016/`, `research/prime61_disk_extend_20261018/`). Native walkers enumerated 1,064,000 distinct curves isogenous to P-256, 20,000-curve classes of P-224 and P-192, and verified isogenies of degree up to 10,453 (P-192) and 1,471 (P-224); no preregistered structural screen fired on any walked curve, so no factor-base test was ever selected (`research/p256_isogeny_million_20261006/`, `research/p192_p224_large_degree_isogenies_20261009/`). By Lemma 26.1 a homomorphic transfer could change only bounded constants in any case. The class of a 61-bit curve has a certified order of 358,094,304 with adjacency still open (`research/p61_terminal_closure_20261023/`).

21.5 Lattices

Every lattice intrinsic to $(E, P, [k]P)$ has rank 2 (4 over $\mathbb{Z}[\tau]$), and relation lattices have a derived ceiling of $p^{5/6}$ (`research/notes/ecdlp-general/RESEARCH_LATTICE_ECDLP_ROUTES.md`). A Coppersmith reformulation of the $m = 4$ oracle via S_5 reaches $\delta^* = 1/80$ against about $1/5$ needed and is closed (`research/lit2026_w10_s5_coppersmith_20261011/`); on S_3, a frozen falsifier recovered no planted decomposition in any of 96 cells with $\beta \geq 0.20$ at 24–56 bits, so the $\beta = 1/4$ ceiling stands at multiplicity ≤ 3 (`research/coppersmith_s3_theta2_20261023/`).

21.6 ECCp-131

ECCp-131 ($\log_2 r = 130.19$, rho $2^{64.92}$ GAE) has exact parameters wired into the BSGS and IC command paths; a three-point small-x probe found zero natural relations and rank zero at a 128-point base, which blocks any full-parameter IC solution and is an open problem record (`research/eccp131_bsgs_ic_20261020/`, `problems/eccp131-full-parameter-index-calculus.md`). Structured prime bases for the Certicom programme were a null result (N/E 0.969–1.000 against a random control of 0.974). One CPU core walks ECCp-131 rho at 1.23 million steps per second with a 32-bit Montgomery batched walk, and 15.06 million with an AVX-512 eight-lane walk (`research/certicom131_rho_ic_bench_20261022/`, `research/certicom131_rho_parity_20261023/`).

21.7 Curve25519 miniatures

Between 2 and 5 November thirty-one studies measured index calculus on miniatures of Curve25519. The real Curve25519 was not attacked; nothing ran at 255 bits. Each target is a miniature built by Curve25519's own selection rule. The prime p is the largest below 2^k with $p \equiv 5 \pmod 8$. The coefficient A is the smallest $A \equiv 2 \pmod 4$ with $\#E = 8r$, $\#E' = 4r'$ (r, r' prime) and a cyclic 2-part. The base point is the smallest abscissa u of order r. At $k = 255$ this rule gives $A = 486662$, $u = 9$; the 255-bit search itself was not rerun, as it needs SEA point counting. Table 21.1 lists the four main curves; later studies added $k = 16, 18, 22$ and four sibling primes per size (`research/curve25519_ic_baseline_20261102/`). Costs are in the section's unit $S = (\text{counted GAE} + w \cdot \text{unpriced native operations})/\sqrt{r}$ at $w = 0, 0.1, 1$. Against rho with negation (`rho-neg`), every arm is ranked only where its order holds across the band.

Generic track. The cofactor costs what it should: an unprojected small-x base needs 7.66–8.38× the trials (8 predicted, 1.00 on a cofactor-1 control). Projection removes the penalty. With the penalty gone, the baseline gap still grows with k (Table 21.1). The first ratchet round found nothing: a retuned base size was null (0.996 $[0.944, 1.051]$ and 0.977 $[0.941, 1.015]$); the closed cofactor fiber and a T_0-closed base were dominated; and the Kummer tower is blocked, because $p \equiv 5 \pmod 8$ forces $t \leq 2$ (`research/curve25519_ic_ratchet_20261102/`). The T_0 symmetry halves the work of a blind base (1.944–2.016×), but projection already beats it by 3.33–4.00× (`research/curve25519_t0_symmetry_20261102/`). Three-term sums improved on the baseline at $k = 20$ (0.758 $[0.716, 0.802]$) and $k = 24$

k	p	A	r ($\#E = 8r$)	baseline `proj-a` / rho-neg ($w = 0$)
20	1,048,573	926	131,203	7.8×
24	16,777,213	318	2,097,349	10.6×
28	268,435,157	5,330	33,557,809	18.1×
32	4,294,967,197	27,842	536,884,639	24.7×

Table 21.1: Curve25519 miniatures and the generic baseline. The baseline is a projected-abscissa factor base with negation-folded meet-in-the-middle (4,032 records, all verified, replay audit passed). The gap is 7.8–7.9×, 18.1–18.4× and 24.3–24.7× across $w \in \{0, 0.1, 1\}$ at $k = 20, 28, 32$. Source: `research/curve25519_ic_baseline_20261102/README.md`.

(0.596 [0.575, 0.619]). The gain did not hold at $k = 28$, where its order across the w band is not fixed (`research/curve25519_m3_generic_20261102/`).

The autolab and the collision floor. A 35-round automated ratchet (`research/curve25519_autolab_20261104/`) moved the generic incumbent from four-term sums to coefficient-box sums and then to a walk table over two seed abscissae (`wwalk`). Rounds 22–35 sit at the baby-step giant-step boundary. They count as index calculus only by the owner's ruling of 4 November. The fixed-table distances of 1.43–1.55× rho were single table draws (correction R4-I1). Averaged over 16 walk seeds they are 1.82× ($k = 32$) and 1.86× ($k = 28$). With a fresh table per run, the incumbents reach 1.94×, 1.81× and 1.82× matched rho at $k = 24, 28, 32$. Over five family curves per size the mean is 1.57, 1.78, 1.80 and 1.91× at k = 20–32, with per-curve spread about ±0.2. Every autolab result is a statement claim with no proof attached, so its standing is *unverified*. A derivation in the generic group model, checked by Monte Carlo and explicitly not a formal theorem (`research/curve25519_rho_parity_bound_20261104/`), explains the plateau. With F unknown base logarithms, a collision-based index calculus costs at least $\Gamma(F + \frac{3}{2})/(\Gamma(F + 1)\Gamma(\frac{3}{2}))$ times rho. That is 1.5× for $F = 1$ and 1.875× for $F = 2$, where Monte Carlo gives 1.889×. `wwalk` has $F = 2$, so its 1.8–1.9× is this floor, not an implementation constant. Parity needs $F = 0$, which is rho under another name.

Algebraic track. Semaev summation decided by F4 (`research/curve25519_algebraic_ecbench_20261102/`) ran end to end at $k = 20$ at 18.2×, $9.55 \cdot 10^5$× and $5.85 \cdot 10^6$× rho for $w = 0, 0.1, 1$. A timing smoke of the frozen algebraic ladder showed that ladder would take days. Its interrupted session is citable only for its four sealed records and is pinned as defective (`docs/ecbench/DEFECTIVE_SESSIONS.md`). The protocol was amended before data to measure F4 cost per call instead (`research/curve25519_algebraic_ladder_20261102/`). The later gains came from selecting the frame, the seed, the tower depth t and the guess count G, and from F4 trace replay, which is a port of the Joux–Vitse technique, not a new method.

Two studies passed independent validation:

- The best seed at $t = 5$, $G = 4$ costs 0.0578 [0.0466, 0.0716] of the seed-0 incumbent at $w \geq 0.1$ (`research/curve25519_t10_winner_ecbench_20261104/`).
- A $t = 4$, $G = 3$ arm with trace replay costs 0.2837 [0.2413, 0.3335] of that winner at $w = 0.1$, but 1.567× more at $w = 0$ (`research/curve25519_t4_candidate_ecbench_20261104/`). It is still 27.9×, $1.83 \cdot 10^4$× [$1.55 \cdot 10^4$, $2.15 \cdot 10^4$] and $1.13 \cdot 10^5$× behind rho at $w = 0, 0.1, 1$.

Against an exhaustive grid on the same base, the F4 arm is 42× too expensive, and even the pair table on that base trails the generic baseline by 2.39–3.32×. Several presentations and variants gave no gain:

- Edwards coordinates: the same solving degree in 8 of 8 pairs.
- The T_0 w-form: equal degree and F4 cost, and 1.2–3.0× worse per relation.
- Three-term Semaev sums: always worse than two-term sums on this track.
- Cofactor lifts: about 8× the F4 work for 5 hits against 0, so no per-relation gain.

Sources: `research/curve25519_edwards_semaev_20261102/`, `research/curve25519_t0_algebraic_20261102/`, `research/curve25519_cofactor_algebraic_20261102/`, `research/curve25519_leads_wave4_20261104/`.

Scaling. Fitted to modelled F4 work, the algebraic cost grows as $r^{1.03}$ against rho's $r^{1/2}$, so the gap grows as $r^{0.53}$; the optimal t is 4 at every k (`research/curve25519_ic_atlas_20261104/`). The 5 November scale-ups ($k = 24, 28, 32$; all standing unverified, no validator report) agree:

- F4 cost per target is the same at every k (1.000–1.019).
- At $k = 24$ the $t = 10$ arm finished 2 of 6 solves. At $k = 28$ it finished none, and the $k = 28$ gap is only an F4-cost estimate.
- At $k = 32$ the engine refuses $p \geq 2^{31}$.
- The cofactor-lift and three-term conclusions of $k = 20$ carry over.

Sources: `research/curve25519_scale_guess_trace_20261105/`, `research/curve25519_algebraic_ecbench_scale_20261105/`, `research/curve25519_scale_lifts_20261105/`, `research/curve25519_scale_m3_t0_20261105/`.

What this settles. On every Curve25519 miniature, track, size and weight measured, rho with negation is ahead of every index-calculus arm. The generic gap grows with k. Collision-based variants sit at a derived floor near 1.9×, and the algebraic gap grows as about $r^{1/2}$. No study makes or implies a statement about the 255-bit curve, and none weakens its security. No Curve25519 row is on the leaderboard or in `benchmarks/`. Most runs were local, on a shared host, without a second-host replay. Three leads remain open:

- the (w, D) presentation, which is untested;
- digit-set bases, at 0.438–0.479× per relation on one arm only;
- an odd-Kummer design, which has no data yet.

21.8 Audits of standard curves

Several studies examined the NIST, SECG and CryptoPro curves themselves. None found an ECDLP weakness. The NIST prime-curve seeds were regenerated as 1997–98 `drand48` output, with the skip rule and base points still unexplained (`research/nist_seed_provenance_20261016/`, `research/nist_basepoint_provenance_20261017/`); a claimed P-256 $n - 1$ outlier was not established after controls, and the published b sits at the median of every statistic over a million seed samples (`research/nist_cyclotomic_controls_20261011/`, `research/p256_trapdoor_round2_20261012/`). The $j = 0$ SECG curves have sextic twists of weak order (rho $2^{65.5}$ for secp256k1's), a fact about twists, not about the curves (`research/safecurves_trapdoor_slots_20261017/`). Special prime shapes change neither the distribution of $\#E$ nor the embedding degree, so only small-embedding-degree pairing curves open an SNFS-type route (`research/snfs_prime_families_20261023/`, `research/solinas_distribution_20261023/`). A P-192 "BREAKS" verdict applies to a singular nodal cubic, an implementation weakness (`research/p192-weak-curve-20261006/`), and the P-224 directory is an invalid-curve attack on an unvalidated ECDH oracle (`research/p224_attack/`). The CryptoPro-B and CP6-782 work is constructive: GLV-2 scalar multiplication 1.25–1.50× faster via isogeny-chain endomorphisms (`research/cryptopro_b_chain_sweep_20261006/`, `research/endo_native_bench_20261016/`). Cheon's attack with auxiliary inputs beats rho from 24 bits up, but only for the DLP with auxiliary inputs (`research/torsion_auxiliary_inputs_20260918/`).

21.9 Literature claims that did not survive

The lit2026 programme (`research/lit2026_speedup_program/`) ran each finding of a 2025–26 search as a measured change against an interleaved baseline binary. Workstream W12 implemented the zero-minor guess-and-determine attack (Mahalanobis, arXiv 2607.09814v2): it solved 48/48 planted targets on 16–26-bit curves, but its search phases scale with slope 0.905 $[0.769, 1.021]$ in $\log_2 r$ and its cost over rho grows from 260× to 6,715×—exhaustive in r, as the source's own table says. A second flagged ePrint (Hu, 2025/015) did not specify an implementable algorithm and is recorded as blocked. On the generic side, W11 cut BSGS from 16.75 to 3.71 multiplication equivalents per visited point with paired steps (wall 1.15–1.67×), which moves rho's and BSGS's constants, not index calculus.

21.10 Gaudry's $k = 4$ index calculus over $\mathbb{F}_{p^4}$

Over $\mathbb{F}_{p^k}$ with a base of points whose abscissa lies in $\mathbb{F}_p$, a relation needs a k-decomposition solved by a Macaulay system. We made the `gaudry_quartic` solve generic in the number of unknowns and faster (median whole solve 1.96/4.19/6.08× at $d = 4, 5, 6$, 37.3 → 6.1 s at $d = 6$, as recomputed by the validation sweep from the committed records on a shared host; the S_5 solve from about 30 minutes to 164–203 s;

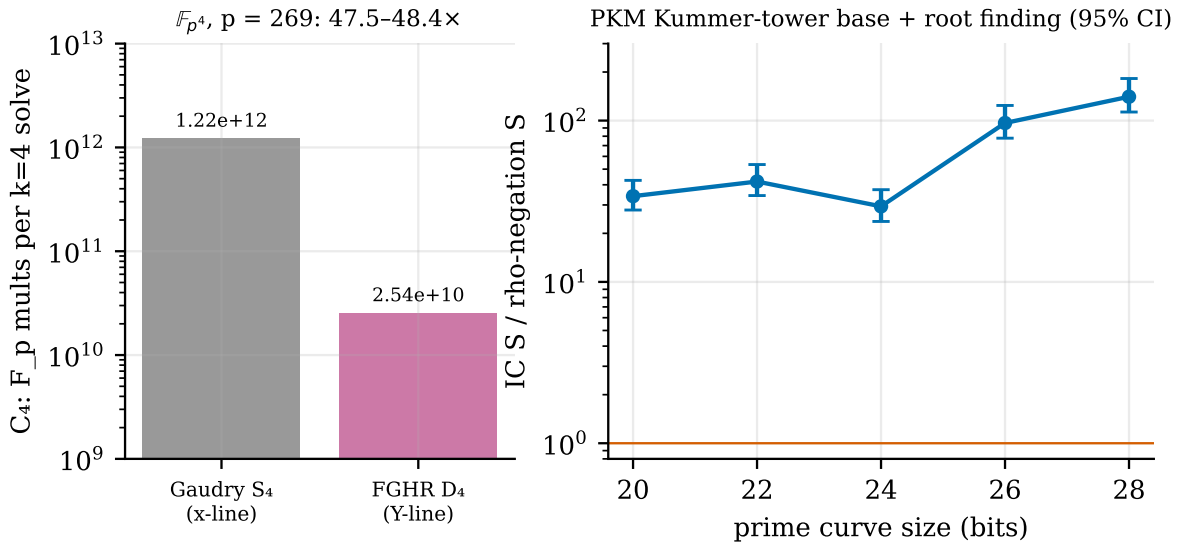


Figure 21.1: Left: cost C_4 of one $k = 4$ decomposition over $\mathbb{F}_{p^4}$, $p = 269$, Gaudry's S_4 system on the x-line against FGHR's D_4 system on the Y-line (`research/gaudry_quartic_d4_20261013/`). Right: index calculus with the PKM tower base and root finding over rho with negation, eight one-target workloads × three rounds per size, paired 95% cluster-bootstrap intervals (`research/pkm_tower_ecbench_20261013/`).

every count and solution identical). Then we built FGHR's D_4 variant (`research/gaudry_quartic_d4_20261013/`): on a curve with rational 2-torsion T, the coordinate $Y = (x - x_0 - c)/(x - x_0 + c)$ satisfies $Y(P + T) = -Y(P)$, so S_5 in Y is invariant under the Weyl group D_4 (order 192). In the invariants $\sigma_k = e_k(Y^2)$, $\pi = \prod Y_i$, the system has Bézout number 512 instead of 4,096, Macaulay degree 14 instead of 29, and a 13,412 × 11,628 matrix instead of 41,780 × 40,920.

At $p = 269$, the cost C_4 of one decomposition falls from $1.21–1.23 \cdot 10^{12}$ to $2.54–2.55 \cdot 10^{10}$ $\mathbb{F}_p$ multiplications, 47.5–48.4×; a solve takes 4.2–5.6 s instead of 164–203 s. Decompositions agree with an independent meet-in-the-middle enumeration on 19 of 19 solved targets; one target in 20 is degenerate (quotient dimension 517) and is discarded, adding about 5% to the relation phase. The Y-line base decomposes about 1/8 as often (pooled ratio 0.133 over 8,000 targets; prediction $1/8 \pm 30\%$ held on one of two curves), but each column covers $\{\pm P, \pm(P + T)\}$, halving the unknowns.

Inserted into the repository's crossover model (`research/notes/index-calculus/RESEARCH_RESIDUAL_WALKS.md` §11.16), the measured C_4 and rate move the $k = 4$ crossover with rho from $2^{149.4}$–$2^{152.7}$ to $2^{135.1}$–$2^{135.8}$ (cofactor $h = 2$, forced by the 2-torsion), and the asymptotic ratio past it from 0.34–0.63 to 0.12–0.22 of rho. This is the one place in the programme where a crossover moves; it moves in an *extrapolated model* ($n^{1/4}$ relations, $n^{1/2}$ linear algebra and rho) fed by a measured constant, so it is classed as ENGINEERING, not an advance. The validation sweep notes that this crossover was not re-derived against a rho with the negation map; against the negation walk, a later note puts the plain prime-order $k = 4$ handover at $2^{156.1}$ and a ψ-folded one at $2^{146.0}$ (`research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`). The measured characteristic-two $k = 4$ ladder is in §31.3. Extension fields of degree four are not the fields deployed curves use.

22 Prime fields: the generic track

Chapter 21 gave the prime-field programme as it stood in mid-October: the known-answer ladders, Table B, the Kummer tower, the structured P-256 bases, the isogeny-class walks and the lattice reformulations. This chapter and the next two record what happened after the owner's decisions of 29 October split prime-field index calculus into two counted tracks (`research/prime_ic_program_20261029/DECISIONS.md`, decision R1; the split itself is stated in Chapter 32, Table 32.3). The *generic* track holds every oracle built from tables, pair tables, root finding, subtraction or collision search. Its improvements count and ratchet its own baseline, but they carry the label "bounded by $\sqrt{r}$; cannot beat rho". The *algebraic* track, the subject of Chapter 23, holds relations from a solver that receives the factor-base constraint inside its system.

The chapter has one result that matters for the distance to rho and one correction. The residual collectors of §22.3 are the prime-field arms closest to rho that the programme measured: 1.3–1.7× matched rho at $w = 0$ on 36–42-bit curves, with no growth of that distance over the measured sizes. They are generic collision searches with rho's exponent, not index calculus with a better one. Their large gains at $w > 0$ were a pricing artefact, withdrawn when the owner chose option B on 1 November (§22.3.1).

Standing of the numbers. Under the rule of 4 November every producer is assumed to be possibly sabotaging the work, and a result is cited by its proof-carrying claim and standing or labelled an unverified assertion. None of the generic-track studies in this chapter has a `pcc` claim record. Four evidence directories have a `VALIDATION.json` with a replay certificate and a validator report: Table B (Chapter 21), the 20-bit $N = 45$ baseline (Chapter 32), the first $m = 4$ cells of §22.1, and the packed membership filter of §22.5. The residual collector study of 1 November has a replay certificate, a validator report and a `VALIDATION.json`, but is not an admitted row (§22.3). Every other figure below is a producer-reported value from a committed record, re-read from the source for this chapter but not independently replayed, and is so labelled where it is used. Costs are in $S = (\text{counted GAE} + w \cdot \text{unpriced native operations})/\sqrt{r}$ at the band weights $w = 0, 0.1, 1$ of §4.3, and two arms are ranked only where their order holds across the band.

The programme of 29 October. The decisions file that created the tracks also set the programme's execution plan (`research/prime_ic_program_20261029/DECISIONS.md`). Its entries are, in brief:

- `D-EKS`: compute for the published algebraic reference arm, a derived estimate of about 49 CPU-hours (range 15–150) on EKS Spot, approved;
- `D2`: the Table B incumbent may be ranked against that reference, which amends R1 so that arms of the two tracks may be ranked on the same instances, in the same unit and across the band, while each track ratchets only its own baseline;
- `D3`: a pinned, vendored msolve counter patch is allowed, its hash bound into the candidate identity;
- `D-PREFETCH`: the hash-table layout measurement of §22.5, approved;
- `D-FD7` and `D-FD8`: two amendments of the prime fall-degree study, the two-cap bracket for $t \geq 12$ with 56/112 GiB jobs and the $m = 4$ extension, adopted;
- `D-MAC`: the fall-degree study's raw runs, which lived only in one worktree on the owner's laptop, are to be recovered rather than declared lost; until then its $t \leq 10$ reproduction stays at median level;
- `D-ROUTE`: a command route for non-ecbench jobs on the lab, approved for an administrator to implement;
- `D-REPLAY`: a second-host replay of the 20-bit holdout session, then a claims record, approved.

The same file records why root finding is generic: at $m = 2$, root-find-and-subtract fixes $x_1 = c$ for each $c \in V$, solves the remaining quadratic $S_3(c, X, x_R)$ by its roots $x(R \mp P_c)$ and tests each root against a hash index of V. It returns the solution set of a Gröbner solve of $\langle S_3, F_V(x_1), F_V(x_2)\rangle$, but its cost never depends on F_V, and it runs identically on a random V.

Table 22.1 lists the prime-field arms of this chapter and of Chapter 21 that have a distance to matched rho, from the closest to the farthest.

Arm	Bits	IC/rho ($w = 0$)	Standing
Residual collector `res1eq` (§22.3)	36–42	1.32–1.72	validator report (36–37 bits); producer
Residual collector `res1dp`	36–42	1.92–2.27	as above
Residual collector `res2`	36	2.6	improvement-log row, no validator
Residual collector `res4`	20.6–32	3.78–4.94	producer
$k = 5$ sum table (§22.2)	20–48	5.4–11.0	unqualified
Table B pair table (Chapter 21)	10–24	5.47–13.5	validated
20-bit pair table, $N = 45$ (Chapter 32)	20	7.75	validated
Cofactor-cleared base, $h = 32$ (§22.4)	census	9.80	`invalid_controls`
$k = 4$ sum table (Chapter 32)	20–40	5.3–13.1	no validation
Kummer tower, P-224 shape (§23.1)	20–28	18–78	producer
Kummer tower (Chapter 21)	20–30	28–159	producer
Packed pair table with filter (§22.5)	32	31.47	validated with caveats

Table 22.1: Prime-field index calculus against matched negation rho, greater than 1 meaning rho is cheaper, as each source states it. Ranges are over sizes or curves; each row's band behaviour is in its section. The residual rows are at rho's exponent; every other row's gap grows with size where more than one size was measured.

22.1 What a generic oracle can buy

The arity gate. Let $s = \log_2 p/m$, so that a factor base of the size index calculus needs has $|V| \asymp 2^s$. Suppose a target-independent setup costs 2as and one natural decomposition query costs 2^{es}. Collecting $\Theta(2^s)$ relations costs $2^{as} + 2^{(e+1)s}$ up to polynomial factors, so beating rho's $p^{1/2}$ needs both

$$a < \frac{m}{2} \qquad \text{and} \qquad e < \frac{m}{2} - 1.$$

At $m = 2$ the second condition is $e < 0$, impossible for a nontrivial query, so exact pair enumeration is optimal at that arity up to polynomial factors; the first arity that can discriminate is $m = 3$, with $a < 1.5$ and $e < 0.5$. The derivation is the arity and solver-model correction of 31 October and 1 November (`DEC-20261031-b51282`, `DEC-20261101-26f1fa`; `research/prime_ic_lead_review_20261031/PDP_ARITY_CORRECTION.md`), which found that every accepted theorem and completed measurement of the prime factor-base degree study was a two-summand result. The same pair of gates is the "complete-cost" row of the six-obligation chain of §22.7.

A wrong gate and its replacement. An audit of 4 November of the programme memo *Alternatives to Semaev summation polynomials for prime-field ECDLP index calculus* (`research/semaev_alternatives_audit_20261104/`) found that the memo's oracle gate, "$e < 1$ at $m = 4$", is not implied by its own cost model. A generic 2+2 meet-in-the-middle oracle passes it, with measured $s = 1.93$ on $T = B^s$ and $e = 0.4$ at $B = p^{1/5}$, while costing $p^{2/3}$, more than rho. The audit's replacement is $s < m - 3$, that is $e < (m - 3)/4$. Seven Lean 4 theorems in `research/semaev_alternatives_audit_20261104/lean/` (`Routes.lean`, `Roots.lean`) cover arities 2 and 3, the $m = 4, 5$ gate, the meet-in-the-middle route and a two-roots bound; the audit's own ledger counts only these theorems as "ruled out". Its measured numbers are a single local run with no replay certificate, and its README says to treat them as unverified until re-run under the validation gate. The memo's "Idea 10" measured one squarefree collision for all $B \le 2^{16}$; Ideas 2 and 3 are $m = 2$ objects and cannot change the exponent.

The $m = 4$ planted solve. The gate correction has a measured counterpart. A direct 2+2 meet-in-the-middle at $m = 4$ over a random base of size $B = \text{round}(p^{1/5})$ (`research/ic_m4_mitm_scaling_20261031/`; stage diagnostic, producer-reported) recovered every planted four-term decomposition with a fitted exponent of 0.412 in $\log_2 p$ ($R^2 = 0.9986$, eight sizes to 52 bits), 0.404 on a three-rung ladder to 60-bit primes and 0.396–0.450 on an arbitrary-precision ladder to 70 bits. The planted solve is not a relation collector. A uniform target decomposes with probability about B^4/p, so one relation costs about $p/B^2 = p^{3/5}$: the 24–40-bit fit of the uniform arm is 0.469, but the local slope from 40 to 44 bits is $\log_2(1.08 \cdot 10^9/2.09 \cdot 10^7)/4 = 1.41$, and the arm was stopped there. The README's own reading is that the $m = 4$ solve "is not an end-to-end sub-1/2 relation collector", which is the audit's point: an oracle exponent below 1/2 in the planted regime says nothing about the cost of a relation.

Study	Bits	Ratio to its incumbent ($w = 0$)	To `rho-neg`	Exponent
$k = 5$ vs $k = 4$	40	0.74 (0.71 at $w = 1$)	9.7×	0.547 [0.543, 0.554]
	48	0.67	11.0×	($k = 4$: 0.565)
$k = 5$ vs mitm incumbent	20	0.701 [0.661, 0.744]	5.4×	
(same instances)	24	0.453 [0.433, 0.475]	6.3×	
	28	0.331 [0.320, 0.344]	7.2×	
	32	0.218 [0.211, 0.226]	7.3×	
$k = 5$ vs re-tuned incumbent	20	0.700 [0.682, 0.718]	6.0×	0.496 [0.487, 0.510]
(16 fresh curves)	24	0.446 [0.432, 0.461]	5.8×	(incumbent:
	28	0.324 [0.318, 0.331]	7.3×	0.635 [0.627, 0.645])
	32	0.217 [0.212, 0.222]	7.3×	

Table 22.2: Prime-field k-sum tables after 28 October (generic track; all unqualified, producer-reported). Ratio: paired ratio of total cost, cluster-bootstrap 95% intervals. "To rho": mean S over matched negation rho at $w = 0$. Exponent: fitted cost exponent in r at $w = 0$; the pooled study's rho-neg fits 0.457 and `bsgs-neg` 0.501 on the same sessions. Sources: `research/prime_ksum_k5_20261028/`, `research/prime_ksum_vs_incumbent_20261029/`, `research/prime_ksum_pooled_20261029/`.

The first $m = 4$ cost cells. The blind-spot programme of 18 October priced arity $m = 4$ end to end on 16–28-bit primes (`benchmarks/20261018-blindspots-prime-m4-cells/`; validated, with a negation-map rho added to the same binary and the index-calculus rows reproduced exactly in it). Full index calculus costs 8.2–29.5× the negation-map rho at $w = 0.1$ and 6.5–35.6× across the band, dearer than both rhos at every cell and weight, 16 of 16 solves per cell; the arm's own F4 grows as $|\mathcal{F}|^{11.03}$. The verdict is REGRESSES at the primitive level.

Exponents on the bounds frontier. The sealed bounds frontier (`docs/bounds/FRONTIER.md`, toy tier, 17.7–24.0 bits, 96 runs) gives the folded pair-table `ic.pipeline` a fitted exponent of 0.841 [0.646, 0.953] and the subtraction oracle 1.005 [0.880, 1.077], against 0.504 [0.316, 0.585] for interleaved baby-step giant-step. These are fit-version-2 bounds with wide intervals at four sizes; the campaign exponent 0.620 [0.557, 0.663] of Table 31.2 is the better-determined estimate of the same pair table.

Arity sweep. The crypto-autoresearcher index-calculus sweep of 27 October (`research/ic_arity_sweep_20261027/`; 234 rows, every logarithm checked) ran $m = 2, \ldots, 9$ on generated prime-order curves at 12–18 bits with exhaustive and meet-in-the-middle engines. Its unit is the tool's own, S_3 root solves against rho walk additions, which the tool says are different units. The best exhaustive arity is $m = 2$ at 10.6, 16.2, 20.4 and 60.5× rho at 12, 14, 16 and 18 bits; the best meet-in-the-middle arity is $m = 5$ or $m = 3$ at 2.8, 3.3, 5.8 and 4.3×. Odd arities have the lower fitted exponents (0.63 at $m = 3$, 0.60 at $m = 9$) because the default table split leaves one more level on the search side at even m. None of this is in the counted ecbench unit, and none of it is below rho.

22.2 k-sum tables beyond $k = 4$

Chapter 32 gave the $k = 4$ sum table of 28 October: a negation-folded table of every signed 4-sum of a smaller base, S 83.7 → 11.6 at 40 bits, 13.1× `rho-neg` there. Its generic law is $2^{k-1}\binom{A}{k} + c_k Ap/(2^k\binom{A}{k})$, with optimum exponent $k/(2k-1)$, falling toward 1/2 (baby-step giant-step) as k grows and never below it. Three studies followed (Table 22.2). All three are registered before data, every run verified and replay-audited, and none has a second-host replay, a `VALIDATION.json` or a `claims/*.json` record; the two later studies call themselves *unqualified*. The benchmark index lists them only as "not admitted to the results table until it has an independent-host replay and its own `VALIDATION.json`".

$k = 5$. A table of every signed 5-sum, with indices packed 12 bits each into one word and the insert order unchanged so that group-operation counts match, lowers S below the $k = 4$ table at 36 and 40 bits at every band weight, the registered success test (`research/prime_ksum_k5_20261028/`; 1,920 runs, all eight test sessions reproduce every replay; $k = 4$ reproduces the earlier per-run counts on 288 of 288 runs). At 40 bits S is 8.51 against 11.56. The fitted exponent over 36–48 bits is 0.547 [0.543, 0.554] against 0.565 [0.560, 0.571], finite-size fits near the generic optima $5/9 \approx 0.556$ and $4/7 \approx 0.571$. The

Quantity (40 bits, $w = 0$)	Predicted	Measured
`kl-ksum4` set-up S	5.3	5.01
`kl-ksum4` online S	0.10	0.117
equal-size preprocessing rho, set-up S	1,594	1,668
equal-size preprocessing rho, online S	0.0006	0.0008
equal-precomputation preprocessing rho, online S	0.19	0.33
`kl`/equal-size crossover (targets)	15k	14.3k

Table 22.3: Registered predictions against measurement for the known-log $k = 4$ table and preprocessing rho at 40 bits (class: control and accounting). Producer-reported. Source: `research/prime_precompute_regime_20261028/`.

table is still 7.6× baby-step giant-step at 40 bits and 10.6× at 48, so the method sits above its own generic limit at every size.

Against the incumbent. The tracks rule requires a same-instance ranking before a generic change replaces the incumbent. On the incumbent's own four holdout curves, 48 targets, method hashes and measurement seed (`research/prime_ksum_vs_incumbent_20261029/`; 3,456 runs) the $k = 5$ table costs less than the negation-folded pair table of `research/prime_base_oracle_tuning_20261029/` at every size and every weight, and the registered primary test holds in all three of its cells (24, 28, 32 bits). The generic baseline on those curves falls by 1.4× at 20 bits and 4.6× at 32. The widening gap is what the two optima predict, 5/9 against about 2/3; one curve per size gives no exponent. The study compared against the recorded incumbent, whose 32-bit base was null by construction, so a second study re-tuned the incumbent on separate train curves by the same rule as the k-sum families and tested both on 16 fresh curves (`research/prime_ksum_pooled_20261029/`; 13,824 runs). The ranking survived: $k = 5$ is below the tuned incumbent on all 16 curves and in the pooled test at all three weights. Its toy-tier exponent of 0.496 is a finite-size fit over 20–32 bits; the generic law does not allow a limit below 1/2. Tuned for $w = 0$, the incumbent is 8–13% cheaper, and $k = 5$ is still 0.25–0.49 of it. A train-frozen rule chose $k = 5$ at every size; at 20 bits $k = 4$ is cheaper at $w = 0$ and 0.1 and the two are unresolved at $w = 1$.

The precomputation regime. The $k = 4$ table is built before any target arrives, which invites two accounting questions (`research/prime_precompute_regime_20261028/`; class: control and accounting, not an improvement). At equal memory against preprocessing rho (Bernstein and Lange [4]), a known-log variant `kl-ksum4` answers each target with one probe and no matrix. At 40 bits the equal-size preprocessing rho's online S is below `kl-ksum4`'s by 142, 157 and 229× at $w = 0, 0.1, 1$, and its precomputation costs 333–339× more. Preprocessing rho sits on $E \cdot T^2 \approx N$ (6.96 at $w = 0$); the known-log table sits about $N/E \approx 1.4 \cdot 10^5$ above that line, as the Corrigan-Gibbs–Kogan bound $ST^2 \gtrsim N$ [8] allows. Table 22.3 compares the registered predictions with the 40-bit measurement. The amortised crossover between the two falls at 14.3k, 13.1k and 9.5k targets at 40 bits ($w = 0, 0.1, 1$), against a registered prediction of 15k. Against preprocessing rho at equal precomputation there is no positive crossover at 36 and 40 bits. These are multi-target and preprocessing numbers; no single-target cost in the study falls below rho's.

Known-log targets. A target-blind known-log two-sum method of 21 October (`research/ecbench_known_log_targets_20261021/`) recovered public targets without receiving their logarithms. On 16-, 20- and 24-bit fixtures its cold counted cost was below the abscissa pipeline's on both the primary and holdout panels at every band weight; at 24 bits the pipeline/known-log ratio is 15.6–17.0 on the primary panel. Negation rho remained cheaper at every size (S 1.16 against 2.70 at 24 bits), the 12-bit comparison is unresolved inside the band, wall measurements are L0 and the receipts are same-host replays. Table 22.4 gives the primary panel.

22.3 Residual collectors

A partial decomposition $R = P_u \pm P_v$ with $P_v \in \mathcal{F}$ leaves a residual $x(R \mp P_v)$ that misses $\mathcal{F}$. The residual collector (`ic_framework::residual`, `targets=residual`) keeps every residual and turns residual colli-

Curve ($\log_2 r$)	Abscissa S	Known-log S	Rho S	Abscissa / known-log, $w = 0, 0.1, 1$
`icv1-fp12-tm53` (11.807)	27.78	25.06	2.92	1.109, 1.049, 0.755
`icv1-fp16-t295` (15.697)	11.10	7.57	1.76	1.467, 1.447, 1.348
`icv1-fp20-t727` (19.101)	12.09	2.96	1.36	4.081, 4.295, 5.336
`icv1-fp24-t1059` (23.974)	42.23	2.70	1.16	15.65, 15.96, 16.98

Table 22.4: Target-blind known-log two-sum recovery against the abscissa pipeline (32 unsigned pairs, folded two-sum meet-in-the-middle) and negation rho, cold, eight targets per curve and five rounds; S at $w = 0$. Producer-reported, same-host replays, no validation. Source: `research/ecbench_known_log_targets_20261021/`.

Arm	Curve	$w = 0$	$w = 1$	Source and standing
`res4` (base 4)	20.6–32 bits	3.78–4.94	3.6–4.8*	residual_lp; producer
`res2`	p36 (36.0 bits)	2.6	2.6*	planner_exhaustive; log row 2.59×
`res2`	p37 (36.25 bits)	2.7	2.7*	planner_exhaustive; producer
`res4f` (fused)	p36 / p37	2.39 / 2.51	—	residual_fused; producer
`res1eq`	p36	1.54	2.66	residual_dp_eq; validator report
`res1eq`	p37	1.64	2.83	residual_dp_eq; validator report
`res1dp`	p36 / p37	2.13 / 2.03	2.22 / 2.07	residual_dp_eq; validator report
`res1eq`	p36 / p40 / p42	1.32 / 1.72 / 1.39	2.27 / 2.99 / 2.42	residual_scale; producer
`res1dp`	p36 / p40 / p42	2.15 / 2.27 / 1.92	2.25 / 2.28 / 1.93	residual_scale; producer
`res1eqc` (chained)	p36	1.591	2.73	res1eq_chain; producer

Table 22.5: Residual collectors: distance to matched negation rho (cost over rho, greater than 1 means rho is cheaper). * With the table-byte counter excluded, as option B prices the band (§22.3.1); the as-published $w = 1$ values were 41–58× (`res4`) and 23–24× (`res2`). The `eq` and `dp` rows are nearly unchanged by option B. Sources under `research/`: `ic_prime_residual_lp_20261026`, `ic_planner_exhaustive_20261026`, `ic_residual_fused_20261031`, `ic_residual_dp_eq_20261101`, `ic_residual_scale_20261101`, `ic_res1eq_chain_20261101`; validator report `benchmarks/20261101-ic-residual-dp-eq/validator-report.json`.

sions into relations; it is the $|D| = r$ limit of the large-prime variation. The programme ran it on prime curves from 26 October to 2 November, under the atlas lead `LEAD-ICP-residual-partial-relations` and hypothesis `H-PFDR-cb9606`. Table 22.5 gives the arms nearest rho. A residual collision is a birthday collision on $\mathbb{Z}/r$ with a base attached; the collector's cost exponent matches rho's, and on a curve-free carrier it costs the same as on the curve. That is the content of `KeyedInvariance.lean` (`research/ic_planner_exhaustive_20261026/lean/`): an algorithm that sees elements only through a keyed labelling cannot use the curve.

The first measurement. On six registered prime curves of 20.6–32 bits, against the `ic plan` recipe and `rho.negation` (`research/ic_prime_residual_lp_20261026/`; 24 of 24 runs verified per arm and curve), the collector with a four-point base was 2.3–8.4× cheaper than the plan in counted S (`res4`/plan 0.119–0.426 at $w = 0$, every interval below 1). As published, the same arm cost 1.33–3.89× the plan at $w = 1$, so the order reversed inside the band at every size and the two were not ranked. The least-squares exponents over six sizes are 0.470 for `res4` and 0.480 for rho, against 0.634 for the plan; at any fixed base the collector grows as about $r^{1/2}$. On the curve-free carrier ($\mathbb{Z}/r$ with a random-bijection key, same seeds) all nine intervals contain 1. The collector buys its group-operation savings with a table probe and an insert on every step, so the native work it adds is real work; what option B later removed from the $w > 0$ figures was a byte count, not those probes (§22.3.1). The k-sum study derived that large primes do not help the *two-summand pair table*: with two summands a large-prime table of M entries has the same law $M + cp/\sqrt{M}$ (`research/prime_ksum_table_20261028/`). The residual collector escapes that law because it gives up the pair table and becomes a collision search.

The planner re-test. The study extrapolated a crossing of the plan near 35 bits. The planner re-test of the same day (`research/ic_planner_exhaustive_20261026/`; 14 new sessions, 3,328 records, all verified; 2,428 band verdicts restated in integers and checked by Lean's `decide`) measured it. At 36.0 and 36.25 bits `res2` and `res4` are cheaper than the plan at every band weight with intervals below 1 (`res4`/plan 0.888 $[0.817, 0.965]$ and 0.865 $[0.802, 0.933]$ at $w = 1$, the risky registered prediction R3). Over all ten sizes the $w = 1$ ratio falls steadily, from 4.92 at 15.7 bits to 0.865 at 36.25. The curve-free null held (15 of 16 intervals contain 1) and the fitted slope over ten sizes is 0.460 for `res4` against 0.619 for the plan. At 36 bits the collector moved the framework's cheapest one-target arm from 41–47× rho in counted S to 2.6–2.7×. The leaderboard improvement log carries this as "Residual collector res2 vs rho.negation, fp36: 2.59×" (`docs/ic/LEADERBOARD.md`), the $w = 0$ value; the study's own decision notes that no validator has run on it, so it is an unvalidated row.

Fused steps, distinguished points and the equivariant walk. The fused residual step of 31 October (`research/ic_residual_fused_20261031/`; four sessions of 512 records, all verified, A/A 1.0000) cut group additions per step from 2.005 to 1.004 and gave fused/baseline 0.56–0.92 at $w = 0$ on all 12 (curve, base) cells. The fused walk needs 1.1–1.3× more steps per harvested relation, so priced native work ate the saving: the order is mixed at $w = 0.1$ and reversed at $w = 1$, and the preregistered rule C1 was not met. Distinguished points and an equivariant walk followed on 1 November (`research/ic_residual_dp_eq_20261101/`; two sessions of 704 records, all verified, A/A 1.0000; it supersedes the $w > 0$ figures of `research/ic_residual_dp_20261101/`, whose per-step distinguished-point test was unpriced). The best arms are `res1eq` at 1.54, 1.75 and 2.66× matched rho on p36 ($w = 0, 0.1, 1$) and 1.64, 1.86, 2.83 on p37, and `res1dp` at 2.13, 2.14, 2.22 and 2.03, 2.04, 2.07. Between `eq` and `dp` the order is unresolved inside the band: `eq` is lower at $w = 0$, `dp` at $w = 1$.

Scale. On p36, p40 and p42 (`research/ic_residual_scale_20261101/`; three sessions of 288 records, 863 verified, one baseline run exhausted at 10^8 trials) the distance of `res1eq` to matched rho is 1.32, 1.72 and 1.39 at $w = 0$ and 2.27, 2.99, 2.42 at $w = 1$; of `res1dp` 2.15, 2.27, 1.92 and 2.25, 2.28, 1.93. There is no growth of the distance with r over this range, as a method with rho's exponent predicts. Chained restarts, which restart each walk at the previous start plus one random jump instead of two scalar multiplications, were null: 0.994 $[0.953, 1.037]$ at $w = 0$ (`research/ic_res1eq_chain_20261101/`).

Batches and many targets. A residual batch of L targets against pooled multi-target rho at p36 (`research/ic_residual_batch_ecbench_20261102/`; 320 of 320 verified, local L0, L2 replay owed) holds the frozen parity rule at $L = 64$, 1.0077 $[0.9811, 1.0350]$, and fails it at $L = 16$, 1.0465 $[0.9886, 1.1078]$. Per target both batch methods are 3.3–3.4× ($L = 16$) and 6.6–6.7× ($L = 64$) cheaper than single-target rho; the matched reference for a batch is multi-target rho, and against it the collector is at parity, not ahead. An exploratory shared-table collection on a 20-bit curve (`research/ic_largeprime_multitarget_20261102/`; not preregistered) gave a per-target cost at $T = 64$ targets 4.4× below $T = 1$. That is multi-target rho reproduced. The same note sets aside triple and quadruple large primes as a generic k-sum search already measured in §22.2.

22.3.1 Correction: the byte counter

The independent validator of the 1 November study (`benchmarks/20261101-ic-residual-dp-eq/validator-report.json`, finding F1) found that the band as built priced `residual_table_bytes_uncharged`, a byte count of the residual table, as native operations at $w > 0$. The audit of the same day (`research/ic_byte_counter_pricing_20261101/`) confirmed it: the counter is emitted in `ic_framework/residual.rs`, every `*_uncharged` counter is appended to the unpriced list, and `priced_gae(w)` charges each listed unit w GAE, so each byte cost w GAE. The analogous `native_table_bytes_uncharged` is classed as shape, not work. The counter was 83–99% of the residual arms' priced S at $w = 1$ and 40–98% at $w = 0.1$, and it is the whole reason their $w > 0$ distances to rho read 17–58× instead of 2.6–5×. On the first residual study, `res4`/rho at $w = 1$ falls from 41.2–57.6 as published to 3.6–4.8 with the counter excluded; on the planner re-test, `res2`/rho at p36 falls from 23.3 to 2.6.

The owner chose option B on 1 November: the band prices operations, not bytes. `band.rs` now lists the byte counters as `NON_OPERATION_COUNTERS` and does not price them; memory is reported on its own axis. No session record was edited; the four affected studies carry a dated note and the re-run comparisons under `analysis/exbytes/`. What changed:

- *The large $w > 0$ gains are withdrawn.* They were the byte price. On the planner re-test, 36 of 136 weighted comparisons go from "unresolved inside the band" to resolved, 12 stay unresolved and 88 keep their direction; the $w > 0$ distances are flat in w.
- *The rule C1 of the scale study is met in 2 of 8 cells, not 6 of 8.* The equivariant arm fails all four cells at $w = 1$, where its interval reaches or crosses 1 (p40 `res1eq` 1.181 $[0.859, 1.625]$).
- *An unreconciled tally.* The 1 November errata give the 12-cell C1 tally as 9 of 12; the study's own `analysis/summary.json` gives 8 of 12 as built. Nobody has reconciled the ninth cell.
- *What stands.* $w = 0$ results, the leaderboard's 2.59×, every verification, audit and A/A result, and the absence of growth of the distance with r.

A second byte counter, `factor_base_vec_bytes_floor_uncharged`, is priced the same way in the Koblitz sessions, at 0.4–12% of S at $w = 1$; no Koblitz verdict changes direction. The 1 November evidence directory has a replay certificate, the validator report and a `VALIDATION.json`, but is not an admitted row: its raw records live in the artifact store, which the CI gate cannot read (`benchmarks/20261101-ic-residual-dp-eq/README.md`).

What the collectors are. The residual collectors are the nearest the programme came to rho on a prime field: 1.3–1.7× at $w = 0$ and 2.2–3.0× at $w = 1$ for `res1eq`, and about 2× at every weight for `res1dp`. They are measured constant improvements of the generic track over the pair-table incumbent, at rho's exponent. By the curve-free null and `KeyedInvariance.lean` they are birthday searches on $\mathbb{Z}/r$, and the Shoup bound applies to them; the remaining factor is a constant, and closing it would produce rho under another name, as the parity bound of §21.7 shows for the analogous collision-based Curve25519 arms.

22.4 Cofactor-cleared bases

On prime curves with cofactor h, a small-abscissa base wastes a fraction $1 - 1/h$ of its trials on points outside the prime subgroup. A preregistered A/B test of 3 November on the census V2 prime curves with $h = 32$ (`research/ic_cofactor_clear_ab_20261103/`; 448 matched pairs on 112 workloads per comparison, cloud VM graded L1, counted metrics only) compared the raw base with `prime-projected-abscissa`, which clears the cofactor. The pass line was met: the S gain of clearing is $G_0 = 6.180$ $[5.915, 6.456]$, $G_{0.1} = 6.692$ and $G_1 = 9.087$, and the $h = 1$ negative control gives 0.989. The positive control, rho-plain over rho-neg with a pre-registered lower limit above 1.2, failed on the V2 session at 1.227 $[1.149, 1.311]$. By the frozen decision rule the verdict is `invalid_controls`, and the producer declined to reinterpret it. Descriptively, trials per relation fall from 5,433 to 162 (33.6× against $h = 32$), and the cleared base is 9.80× $[9.33, 10.30]$ matched rho at $w = 0$ against 60.6× for the raw base. The study is therefore not cited as a win. The same lever on random binary curves gave 1.29, 1.58 and 3.36× at $h = 2, 2, 6$ (atlas `LEAD-ICB-cofactor-projected-ladder-base`), and the Curve25519 miniatures of §21.7 use the projected base as their baseline.

22.5 Engineering record

Base size. The 20-bit baseline at $N = 45$ (Chapter 32) was tested against $N = 64$ on three new curves, 576 matched pairs, on an AMD L2 host (`research/p20_large_n_20261029/`; 4,704 records verified, 4,032 measured replays identical). The ratio S_{64}/S_{45} is 1.001 $[0.979, 1.022]$ at $w = 0$, 0.967 at $w = 0.1$ and 0.834 at $w = 1$: unresolved inside the band, so $N = 45$ stays. It is not a relabelling, since the total does not fall at $w = 0$ and rise at $w = 1$. The run also replicated the 32-to-45 step on new curves: $N = 45$ costs 0.842, 0.822 and 0.751 of $N = 32$, within a few percent of the original 0.818, 0.798, 0.726.

Where the cost goes. A phase profile on the deployed lab image (`research/ic_profile_20261029/`; 20-bit curve, $N = 45$, admitted at L2, 144 of 144 executions) puts 71.8% of priced GAE in relation collection, 27.8% in oracle setup and nothing in the factor base, with $S_{\mathrm{IC}} = 9.93$ against $S_\rho = 1.08$. Amortising setup over P targets gives at most a 28% gain ($9.93 \to 7.17$ as $P \to \infty$) and leaves the pipeline at 6.7× matched rho; the relation phase alone is 4.7× rho. To reach parity the relation phase would have to fall by about 7×, which no setup lever touches. `target_guard_probes` is uncharged in that record, so 9.93 is a lower bound.

Table build and membership filter. Three builders of the same negation-folded pair table (scalar hash insert, one batched inversion per row, and the batched arithmetic inserted into cache-sized packed regions) count the same additions and recover the same logarithms on a 32-bit curve, 456 of 456 executions identical, so S is unchanged by construction; the wall comparison is L1 and descriptive (`research/ic_oracle_setup_blocked_20261020/`; leaderboard component section). A one-bit membership filter on the packed table, frozen in the backlog item `XP-20261029-packed-filter`, is the only validated component of the generic track (`benchmarks/20261029-ic-packed-filter/VALIDATION.json`; validated with caveats, single session, not replicated; EKS L2 on an AMD EPYC 9R45). At 32 bits it lowers whole-solve wall per charged GAE to 0.9601 $[0.9569, 0.9633]$, -3.99%, at identical charged GAE. In S it decides nothing: 1.0, 1.0505 and 1.3586 at $w = 0, 0.1, 1$, so the filtered table is costlier at $w > 0$. At 24 bits it reads a regression (1.0032) and at 28 bits it is undecidable. The distance remaining to rho-neg is 31.47–31.63×. The validator re-priced the bit operations with its own code rather than re-running `ecbench work`. The filter is a component speed-up outside tables A–C of the leaderboard and moves no ranked row.

A fixed-window generator. In crypto-autoresearcher, a fixed-window generator of the $aP + bQ$ targets cut complete-solve group operations from 200,856 to 50,490 summed over 54 toy solves per arm, with a paired full-solve reference/candidate ratio of 1.161 $[1.127, 1.198]$ and an A/A of 1.044 (`crypto-autoresearcher:ledger/evidence/EV-ICPERF-92c130.yaml`; nine public curves at 12–16 bits, one machine and one session). Five of the 54 pairs are slower, and the record states that the result does not replicate across machines, runtimes or sessions; the historical generator stays the default.

Curve generation. Every `prime_search` workload had been built by counting points with a Legendre symbol at every abscissa, about p symbols per candidate: 11 s per 24-bit and 218 s per 26-bit curve, paid again by every plan, run, verify and CI replay. Stepping one point across the Hasse interval, about $4\sqrt{p}$ affine additions, answers the only question the search asks, whether $\#E$ is prime; it draws no randomness, so it returns the same curves. Planning time fell by 5× at 16 bits, 239× at 24 and 1,592× at 26 (`research/prime_order_search_20261017/`). This is tooling; it changes no counted cost.

Line layout. A cache-line layout tournament of the pair table (`research/ic_line_layout_tournament_20261031/`) has not produced a measurement. The two preregistered EKS sessions of version 1 passed their contract and build jobs and terminated in the experiment job with remote-result exit code 23 (`TERMINAL-OBSERVATION-20261101.md`); the version 5 qualification job of 6 November stopped at its qualification under stopping rule 1, with no scientific session and no seed consumed. A version 6 redesign was approved on 6 November (`DEC-20261106-662e33`) conditional on a pre-freeze check, and nothing has been dispatched under it. The outcome is inconclusive, and the failure is tracked as `problems/ic-line-layout-eks-failure-cause.md`.

Planner weight rules. Chapter 21 reports the planner's validation on six curves. The re-test of 26 October (`research/ic_planner_exhaustive_20261026/`) held out three weight rules and withdrew all three in `ic-planner/4`: the prime rule W1 ($1.41 \cdot N_0$ at $w = 1$) held its operative half on 4 of 4 curves but failed its $w = 0$ control half, because at 15.7 and 17.7 bits $1.41N_0$ is cheaper even at $w = 0$; the binary rule W2 failed at $w = 1$; the Koblitz rule W3 was dearer at every weight on every held-out curve at 36.6–47.2 bits (1.06–1.17× at $w = 1$). `ic plan -native-weight` now keeps the counted optimum at every weight.

22.6 Structured bases in measurement

The conservation results of Chapter 5 (Proposition 5.5) and Chapter 26 (Lemma 26.1) say that a factor base chosen by a coordinate rule cannot change the number of relations in expectation. Three prime-field measurements of October tested the finite-size version.

A four-family comparison. Multiplicative-coset, low-x, low-digit-weight and random bases were compared on 48 admitted prime-order curves over 24 fields at 16, 20 and 24 bits, holding the target curve fixed and matching the number of $\pm$ point classes (`research/factor_base_evaluation_20261015/`; 26,497 point-relation identities and 402 matrix ranks checked in Sage). In independent rows per CPU second at 24 bits, low-x gave 1,352, coset 1,167, random 1,144 and low digit weight 859. Coset over

random is 1.020× with a descriptive 95% interval of 0.959–1.083, so no coset advantage is established. A quadratic-algebra norm solver produced 90 rows in 229.902 CPU seconds on fresh 24-bit inputs, against 115 rows in 0.268 seconds for the exact join; the norm solver is a Sage prototype and the join native C.

Pair-sum coincidences. A structured base might win if its pair-sum coincidence excess κ, the number of four-term relations $P+Q=R+S$ among base points over that of matched random symmetric bases, grew with the base size N. A registered break-even ladder of 29 October (`research/prime_coincidence_harvest_20261029/`) measured window, rank-2 and rank-3 label-progression bases on prime-order curves of 16 to 32 bits at two base-size exponents. The slopes $d\ln\kappa/d\ln N$ have 95% intervals inside $[-0.006, +0.008]$ for all six combinations, 18–41× below the break-even exponents 1/3, 1/5 and 1/7, and every rung's κ lies inside $[0.92, 1.08]$. The $\mathbb{Z}/N$ positive control recovers slopes 0.699 and 0.649 (exact 0.70 and 0.65). A growing harvest would need $\kappa > 1.27\,cN^{1/(2h-1)}$, that is $1{,}874c$, $101c$ and $29c$ at 32 bits for $h = 2, 3, 4$, against a measured κ of at most 1.07. The ladder is registered as `exhausted_at_scale` at 32 bits; it is not a proof.

Hashed and windowed bases. Two of the skeptic-reviewed leads of §23.6 add the second moment: hashed bases (affine, dense, sparse of weight at least 5) match a size-matched random null to 1.000 ± 0.0003 in paired ratio at $p \approx 2^{14}$–2^{22}, and a shared negation-closed x-window has a yield deficit of 0.713, 0.833 and 0.916 at $m = 4$, an instance of the symmetric-base mechanism already recorded in the knowledge base. Base choice moves constants, in the direction the conservation results predict, and the structured P-256 bases of Chapter 21 are the same finding at a larger modulus.

22.7 The obligation chain

The weak-assumption programme of 2 November (`research/ic_weak_assumptions_20261102/`; reanalysis of committed evidence, no new run) restates an index-calculus security argument as a serial chain of six obligations, each of which a complexity claim must discharge:

- solution geometry: dimension and radicality of the full system, rational lifts and exceptional charts;
- solver growth: matrix dimensions, nonzeros, arithmetic and memory, with timeouts and out-of-memory exits retained;
- yield and variance: rational decompositions per natural target, including zero mass, multiplicity and solver failures;
- relation rank: rank growth, dead columns and full-rank probability, not raw row counts;
- a non-generic interface: a coordinate predicate whose useful candidate set no generic table reproduces at the same cost;
- end-to-end composition: one parameter family charging setup, failed queries, collection, verification, linear algebra and memory, with the fixed-arity gates $a < m/2$ and $e < m/2 - 1$ of §22.1.

The machine-readable chain (`obligations.json`, checked by `audit.py`) covers naive Semaev, generic pair tables, the Kummer tower, the proposed arity-three hybrid solver, Wagner filters, the hashed-base tensor route of §24.1 and xedni. Three of its reanalyses bear on the generic track. The prime $m = 2$ yield sweep tracks the size-driven $2B^2/r$ model at toy scale, with observed/model hit-rate ratios 0.764–1.108 in nine cells; exact double counting fixes a mean decomposition count, not the probability of at least one, and the adaptively stopped collector cannot estimate the target-wise variance. The same records hold 842 rows at rank 842, a correctness control and not a rank theorem, since each episode stops early and meet-in-the-middle and subtraction expose the same relation set. Its observation about degree alone is in §23.1.

What the generic track settles. On every prime curve measured the generic track is behind rho with negation, and the best arm, the equivariant residual collector, is behind it by a constant at rho's exponent. The k-sum tables improve the incumbent by 1.4–4.6× at 20–32 bits and stay 5.4–11.0× rho, above baby-step giant-step. The pair-table constants (base size, filter, builders, profile) are measured and small. The obstruction is a theorem, not a measurement: any decomposition oracle built from enumeration or table lookup is generic, so Shoup's $T \gtrsim \sqrt{N}$ and, with preprocessing, $ST^2 \gtrsim N$ bound it (atlas `LEAD-ICP-generic-table-decomposition`, closed_obstructed). Every improvement here is a success of the generic track under the iterative criterion and a distance remaining of at least 1.3×, and none of it is evidence for the algebraic track.

23 Prime fields: algebraic presentations

The algebraic track holds relations produced by a solver that receives the factor-base constraint inside its system. Its prime-field baseline was empty when the track was created on 29 October: the owner accepted the small-abscissa base with membership polynomial $F_V = \prod_{c \in V}(x - c)$ as an algebraically defined base, which makes the naive Semaev system with msolve the counted algebraic reference, but noted that $\deg F_V = |V|$ gives a Gröbner solver no low-degree structure (`research/prime_ic_program_20261029/DECISIONS.md`, D1). The reference arm was approved for EKS Spot (decision D-EKS), and the decisions file lists its implementation and execution as not yet done. The end-to-end prime-field costs nearest to that track in the treatise are the Kummer-tower rows of Chapter 21 (28–159× rho), which Chapter 32 places on the algebraic track although their oracles are root finding and tables, so that the tower base is the only algebraic ingredient; the P-224-shaped rows and the derived F4 cost of §23.1; and the F4 rows of the Curve25519 miniatures (§21.7).

This chapter collects seven lines of work, none of which produced a priced pipeline: the Kummer-tower follow-ups (§23.1), the partitioned summation polynomials of Amadori, Pintore and Sala (§23.2), the digit presentation (§23.3), split-graph and secant presentations (§23.4), the divisor-norm direct index calculus (§23.5), the SAT and CDCL(T) lane (§23.6) and the small-root reach theorems (§23.7). Each is a degree study, a correctness implementation or a theorem. The first fall degree and its relatives are defined in Chapter 6, and that chapter's §6.5 gives the Kummer-tower degree theorem (Theorem 6.14). None of the studies below has a verified `pcc` claim; where a claim id exists it is named with its standing. The fixed-arity gate of §22.1, $a < m/2$ and $e < m/2 - 1$, applies to every route here.

23.1 Kummer-tower follow-ups

The symmetric tower. Petit, Kosters and Messeng's [33] base $V = g\mu_{2^t}$ makes membership of distinct summands $x_i = gw_i$ equivalent to $Q(X) = \prod(X - w_i)$ dividing $X^{2^t} - 1$, that is $X^{2^t} \equiv 1 \pmod{Q}$. The symmetric tower of 26 October (`research/pkm_symmetric_tower_20261026/`; stage diagnostic, no S) takes the coefficients e_k of Q and of the remainders $R_j \equiv X^{2^j} \bmod Q$ as unknowns, with equations $R_j = R_{j-1}^2 \bmod Q$, $R_{t-1}^2 \equiv 1$ and S_{m+1} rewritten in the e_k. It sees unordered sets of distinct points, the same relation with $m!$ fewer solutions. Table 23.1 gives the cells. At $m = 3$ the symmetric tower is one degree lower and its widest matrix about 10× narrower, about 50× faster per system at $N = 12$ (14.8 s against 0.3 s), but its degree still climbs: $4, 5, 5, 6$ at $N = 6, 9, 12, 15$. The follow-up of the same day (`research/pkm_symmetric_lead1_20261026/`) added $N = 18$ at $m = 3$, where $D = 6$ again, so the sequence over $N = 6$–18 reads $4, 5, 5, 6, 6$: a staircase that steps up every second cell. The $N = 18$ cell cannot tell "D bounded at 6" from "D rises by one every 6 in N"; the discriminating cell is $N = 21$, which needs more than the 13 GB that host had. From $N = 15$ to 18 the widest matrix grows 3.1× and the time 39×. At $m = 4$, S_5 was written in $e_1, \ldots, e_4$ by interpolation and passed its planted check. At $N = 12$ the symmetric system runs in 8–9 ms against 66–70 s for the plain chain, with 5.9× narrower matrices; its $D = 8$ at $N = 12$ and 16 is the degree of $S_5(e)$, its input, so D cannot rank the arms at $m = 4$, and $N = 20$ aborted under a 6 GB cap.

What the tower instrument does not check. The 29 October audit (`research/prime_one_shot_20261029/`) read the source of `examples/pkm_symmetric_tower.rs` and found that it draws a, b without checking the discriminant or the group order, includes every coset abscissa in the base without checking that it lifts to $E(\mathbb{F}_p)$, and takes x_R as a uniform field element rather than the abscissa of a subgroup point. Its planted checks evaluate assignments built from S_3 roots, not signed group decompositions. The symmetric rows are therefore algebraic instrument diagnostics, not a validated natural prime-order benchmark, and the symmetric arm excludes repeated abscissae that the plain arm permits, so the comparison is not clean evidence for coefficient sharing alone. The same audit recovered a lower bound the earlier summaries had dropped: at $m = 4$, $N = 20$ both raw msolve logs complete three productive degree-8 rounds (adding 1,241, 2,507 and 3,578 basis elements) before exit 139, so $D \geq 8$ there for that solver and presentation, whatever the final verdict.

The audit sorted the tower's failures and nulls into kinds that need different treatment, and its sorting applies beyond the tower:

m	N	plain D	symmetric D	widest matrix, plain / sym	time per system, plain / sym
2	12	5	4	2,451 / 627	0.36 s / 0.09 s
2	16	5	5	9,119 / 6,281	34 s / 24 s
2	18	5	5	14,922 / 12,214	280 s / 176 s
3	9	6	5	3,285 / 248	0.6 s / 0.02 s
3	12	6	5	12,314 / 1,224	14.8 s / 0.3 s
3	15	≥ 6	6	$\geq$ 97,456 / 10,532	$>$ 900 s / 43 s
3	18	—	6	— / 33,153	— / 1,941 s
4	12	7	8	21,619 / 3,663	66–70 s / 8–9 ms
4	16	—	8	— / 13,926	— / 11.8 s

Table 23.1: Plain and symmetric Kummer towers, F4 (`f4_fp`, grevlex) on $p = 786433$, random targets; maximum over two targets. Stage diagnostics with no priced pipeline. The recorded tower-engine D of the plain tower at $m = 3$, $N = 15$ is 7 (§21.2). Sources: `research/pkm_symmetric_tower_20261026/`, `research/pkm_symmetric_lead1_20261026/`.

- *Busy hosts.* The original PKM solver study ran at load 22–25 on four cores; a single-threaded $m = 3$ job received about 19% of one core, so the first degree-7 steps at $N = 18$ and 21 took about 2,024 and 2,041 s wall against 384 and 396 s CPU. The wall-censored inputs justify a controlled retry, not a reading of the exponent. The same study planned a planted-solution control but retained only sixteen refutations with `-planted 0`, so successful root extraction was not validated there.
- *Resource censoring.* At $m = 3$, $N = 21$ the native engine stopped before productive degree-8 reduction because a dense block would need 79,330,205,871 entries, about 317 GB at four bytes, which would not fit the planned 192 GB host either. That is a matrix-memory obstruction, not a contention artefact, and it is why §21.2 reads $D \geq 8$ as inconclusive.
- *Missing raw runs.* The last-fall study's raw runs, table and draft results were never pushed and live only in the owner's laptop worktree (`research/prime_fall_degree_20261027/RESTART.md`), so its quoted medians are session-reported and not independently recoverable. A lost file is not a negative experiment; decision `D-MAC` asks for recovery.
- *Insensitive instruments.* An ecbench Lattès and isogeny study used a degree-s top product polynomial, $s = 16, 24, 32$, that set the solving degree by itself; its null at $m = 2$, $p \leq 1031$ is not a theorem about implicit tower presentations, and rerunning the same design would add little.

Degree alone is not a cost certificate. The weak-assumption reanalysis (`research/ic_weak_assumptions_20261102/`) extracted the committed PKM $m = 4$ random-target rows at $N = 8, 12, 16$. Their F4 solving degrees are $6, 7, 7$, while the maximum column count grows 1,049 $\to$ 9,452 $\to$ 76,837 and multiply-adds $4.84 \cdot 10^7 \to 8.55 \cdot 10^{10} \to 2.70 \cdot 10^{13}$: from $N = 8$ to 16 the columns grow 73.25-fold and the work 558,304-fold, with the last two degrees equal. Three cells are not an asymptotic fit, but they are a direct counterexample to reading a stable degree as a bounded cost. All six selected $m = 4$ natural rows are also inconsistent (their ideal is the unit ideal), so a zero-dimensional flag on a random target does not establish the geometry of a successful decomposition.

P-224's 2-adic shape. A study of 22 October priced the tower at P-224-shaped Solinas primes $p = 2^n - 2^k + 1$ of 20–61 bits and at the P-224 prime itself (`research/pkm_p224_kummer_shape_20261022/`; frozen before any measured run). On the solver axis, D equals the recorded D in all 89 valued tower cells at $m = 2, 3, 4$, the width and step count of every system match, and the shape-matched null gives the same D in all 65 paired cells, although k/n ran from 0.42 to 0.70 and the odd part of $p - 1$ from 6 to 128 bits. The one cell past the recorded frontier, $m = 2$, $N = 24$, was killed at 6.0 GB, and a capped rerun gives only $D \geq 6$. On price, four Solinas curves of 20–28 bits cost 18–78× rho with negation at $w = 0$ and 19.9–134× at $w = 1$, every one of 1,152 runs verified and 864 replays reproduced; at equal t the Solinas curves cost 2–32% more absolute GAE than the earlier Proth curves. The derived end-to-end F4 cost at the one cell matching a curve (20 bits, $m = 2$, $N = 20$) is at least $2.4 \cdot 10^9$ in S, about $1.9 \cdot 10^9\times$ matched rho at $w = 0$. The verdict is inconclusive on the solver axis and negative on price.

Radix gradings. The experiment `EXP-PKM-RADIX-001` (`research/pkm_radix_grading_20261104/REPORT.md`; frozen protocol, two disclosed amendments; every number is a counted metric marked `[pcc:unverified]` in the report) regraded the tower with radix 4 and radix 8 and measured refutation

Cell	radix 2	radix 4	radix 8	D (2/4/8)	widest matrix (2/4/8)
$m = 2$, $t = 5$ ($N = 10$)	$5.43 \cdot 10^6$	$2.20 \cdot 10^6$	$1.64 \cdot 10^6$	4 / 6 / 10	337 / 252 / 260
$m = 2$, $t = 10$ ($N = 20$)	$4.82 \cdot 10^{11}$	$2.59 \cdot 10^{11}$	$8.35 \cdot 10^{10}$	6 / 8 / 13	33,446 / 28,225 / 17,101
$m = 3$, $t = 4$ ($N = 12$)	$1.06 \cdot 10^{10}$	$9.59 \cdot 10^9$	$5.98 \cdot 10^9$	6 / 9 / 12	5,019 / 6,509 / 4,063
$m = 3$, $t = 5$ ($N = 15$)	$3.19 \cdot 10^{12}$	$1.10 \cdot 10^{12}$	$7.47 \cdot 10^{11}$	7 / 9 / 12	31,397 / 28,055 / 20,284

Table 23.2: Refutation multiply-adds of the Kummer tower in three gradings, one value per cell (the two random draws agree to a few parts in 10^7); D is measured in each grading's own degree. Counted metrics, `[pcc:unverified]`. Source: `research/pkm_radix_grading_20261104/REPORT.md`.

multiply-adds. Its frozen verdicts are `gain_constant` at $m = 2$ radix 8 ($r_k = 0.173$ at $t = 10$), $m = 3$ radix 4 (0.344 at $t = 5$) and $m = 3$ radix 8 (0.234), and `no_material_effect` at $m = 2$ radix 4 (0.538). No cell is `gain_growing`. The shape-matched null gives the same verdicts with ratios equal to three digits, so the gain belongs to the support of S_3 on the tower, not to the curve; $p = 786433$ replicates the $t = 10$ bands. The author's registered prediction of no material effect anywhere was falsified in three of four cells. Table 23.2 gives the end cells.

The radices raise the degree in their own grading (at $m = 2$, $t = 10$: 6, 8, 13) while lowering the work, which is the point of the previous paragraph seen from the other side.

Isogeny kernels against μ_k. A component study of 2 November tested a user's toy observation that an isogeny-kernel base, $V = x(\langle G \rangle) \subset E'(\mathbb{F}_p)$ presented as a chain of Vélu 2-isogeny x-maps, behaves differently from Petit's μ_k chain of squarings at $m = 2$ (`research/isogeny_vs_mu_factor_base_20261102/`; toy primes $p \leq 40961$, frozen protocols, 6,144 and 11,532 systems, no gate failures, local host without an isolation tier, counted units only). At equal $|V|$ the isogeny base matches a random base in yield in 15 of 15 cells (median difference -0.020 to $+0.029$). Its extra Gröbner cost comes from the presentation, not the set: writing μ_k in coordinates whose steps look like Vélu steps costs the same as the isogeny chain to within 3%. A search over coordinates cut the lex-order gap to μ_k from 6.4–11.7× to 1.9–2.6×, but found no grevlex improvement (0 of 16 cells), and none of 11,811 valid Vélu representations reaches the density of the μ_k chain. There is no rho row; the distance is stated against μ_k.

The $t = 14$ cell. The decisive cell of the prime fall-degree study (Kummer, $m = 2$, $p = 2013265921$, $t = 14$) is half done (`research/prime_fall_degree_t14_20261101/`). At cap 5 neither random draw refutes and neither is decomposable, so $d_F \geq 6$. The cap-6 cells need a 112 GiB, 86,400 s native job; no route in the lab carried such a job on 1 November, and the cells are owed. The owner approved a command route for non-ecbench jobs (`D-ROUTE`) that an administrator has to implement.

Open. Lead 1 of the tower, whether D stays bounded as N grows, is open in every presentation tried: plain, torus, Dickson, symmetric, radix-graded and P-224-shaped. In each family D either grows with N or is pinned by the input degree, and the shape-matched null matches. The atlas lead `LEAD-ICP-pkm-tower-algebraic-oracle` is `open_blocked` on memory: the cells that would discriminate ($m = 3$, $N = 21$ in the symmetric tower; $t = 14$ at cap 6) need hosts the programme did not have. Chapter 6 gives the proved part.

23.2 Partitioned summation polynomials

Amadori, Pintore and Sala [1] partition the factor base and attach labels to the summands so that one relation comes from a smaller system. The programme implemented the paper's single-relation route and five extensions between 26 October and 5 November (`research/aps_extensions_20261026/`, `research/aps_followup_20261028/`, `research/aps_independent_replay_20261029/`, `research/aps_measurement_20261029/`, `research/aps_export_20261104/`, `research/aps_leads_20261105/`). The native module builds labelled points $R = uG + vQ$ without giving the solver the planted scalar, imposes the partition membership constraints, solves with native F4 and substitution, decodes and sign-checks the summands by curve addition, and recovers $d = -A/B$ from accumulated relations $AG + BQ = O$. The extensions are fixed coefficient labels, unordered divisor and two-pair encodings, automorphism orbits, covering partitions, and expanded, split and hybrid solving; the follow-up added an early stop on the first informative relation. In detail:

- *Informative labels.* For odd arity, points $uG + Q$ have an odd signed coefficient sum, which cannot vanish modulo $n > m$; for even arity, one point from each partition takes weights $1, 2, 4, \ldots$ with $n > \sum$weights. Regrouping or orbit expansion does not inherit the guarantee, so the recovery check stays mandatory.
- *Unordered selections.* A monic cubic $H(T) = T^3 - e_1T^2 + e_2T - e_3$ dividing $f(T) = \prod_{v \in V}(T - v)$ selects three distinct base coordinates exactly when the three remainder coefficients vanish, with the symmetric S_3 rewrite $(e_2 - a)^2 - 4e_1(e_3 + b) = 0$; the study notes that the remainder equations can be much harder to solve and claims no degree reduction.
- *Automorphism orbits.* On a $j = 0$ fixture the map $\varphi(x, y) = (\beta x, y)$ propagates labels through orbits; the intrinsic relation $R + \varphi(R) + \varphi^2(R) = O$ has $A = B = 0$ and is rejected.
- *Covering partitions.* A bit-based three-colour separating family covers every distinct triple with at most $2\lceil \log_2 s \rceil (\lceil \log_2 s \rceil - 1)$ colourings, an $O(\log^2 s)$ construction; coverage does not imply a lower total cost.
- *Solver presentations.* Expanded S_4, the paper's joined-S_3 form and first-coordinate hybrid guessing share one checked relation path.

The field is bounded by 65,521 by construction, and the study states that success of its substitution backend on the fixtures is no evidence that enumeration disappears at large sizes.

What exists is correctness on tiny public fixtures. The independent replay reads the correctness JSON for three fixed fixtures and twelve arms and re-checks group order, every labelled point, partition membership, signed sums, A, B and $dG = Q$ in separate affine arithmetic. The follow-up's release suite reports 5,002 tests passed, 0 failed and 122 ignored, with seven core-only Lean lemmas. The measurement instrument `ic.aps_v1` charges setup and five online phases on one monotonic cold timeline, but fields are capped at 65,521, abscissae at 32 and attempts at 128; its `total_gae` prices only the independent affine checks, so it is a lower bound and `native_work_accounting_complete` stays false. No benchmark has launched, and the paper's 11–22-bit ladder was not reproduced. The obstruction record `problems/aps-single-relation-complete-cost.md` lists what a complete cost needs. The four leads of 5 November (a nonhorizontal line chart $U = t^2$, $V = c/t$ with $H_{U,V}(X) = X^3 - U(X+V)^2 + aX + b$; F4 trace reuse across attempts; incremental factor-base frontiers; unequal partition profiles) are hypotheses with no implementation, filed as `PCC-20261105-aca921` and `PCC-20261105-ac1e93`, both of unverified standing. The APS route is open and unmeasured.

23.3 The digit presentation

Writing each unknown abscissa in s bounded binary digits, $\ell_k = \sum_{i<s} 2^i a_{k,i}$ with $a_{k,i}^2 = a_{k,i}$, gives a prime-field analogue of Weil descent: $n = ms$ Boolean unknowns, and the system $\tilde{S} = S_{m+1}(\ell_1, \ldots, \ell_m, x_R)$ in the function algebra $\mathbb{F}_p[a]/(a^2 - a)$. The closed form of its first fall degree, $d_{\text{ff}} = m2^{m-1} + \lfloor (s - 2^{m-1})/2 \rfloor + 1$, is in Chapter 30, and its $m = 2$ case $4 + \lfloor s/2 \rfloor$ in Chapter 6. A native $\mathbb{F}_p$ ladder with an independent prime, curve and target for every draw (`research/prime_digit_presentation_20261029/`; registered before the counted runs, one amendment) measured the *last* fall degree:

- $d_{\text{last}} = 4 + \lfloor s/2 \rfloor$ on 106 of 106 real draws at $m = 2$ (60 unsatisfiable at $s = 2$–7, 46 planted at $s = 2$–8), equal to the first fall degree in every draw; at $m = 3$, the first fall degree is 13 on 8 independent instances at $s = 4, 5$, and $d_{\text{last}} = 13$ on the 4 at $s = 4$;
- a support-matched null gives $s + 3$ on 44 of 44 draws, and the real system falls strictly earlier from $s = 3$; a block-factored null of random forms on the support of ℓ_k^e reproduces the real first and last fall degrees on 44 of 44 draws, so the early fall is block structure and carries nothing curve-specific;
- the plain presentation $\mathbb{F}_p[x_1, x_2]/(x_i^p - x_i)$ in the same engine has $d_{\text{last}} = p + 2$ on 32 of 32 draws: linear in p, where the digit degree is linear in $\log p$;
- at d_{last} the degree-filtered Macaulay space spans 100, 98, 96, 83, 81 and 60% of the 2^n monomials at $s = 2$–7, and 60% at $s = 8$.

Table 23.3 gives the unsatisfiable ladder. A low degree does not make a small system at this scale, and the lane is exhausted at the measured scale ($m = 2$, $s \leq 8$; $m = 3$, $s = 4, 5$). It would reopen if $s = 9$ broke the closed form (predicted $d_{\text{last}} = 8$, about 2^{18} columns) or if the adaptive staircase arm, not run, fell below the block null. The scale limit is the engine's `u32` accumulators ($p < 2^{12}$), not mathematics; for $s \geq 8$ no unsatisfiable target remains on the 2^{16}-point grid, so $s = 8$ has only two planted draws ($d_{\text{last}} = 8$, 39,203 columns, about 11 minutes a draw). Validation draws preceded registration, and the registered binary's hash is not reproduced by a rebuild, although the rebuilt source reproduces two of panel A's cells line for line. No new theorem is claimed: `lean/NullTable.lean` checks

s	draws	distinct primes	$d_{\text{last}} = d_{\text{ff}}$	support-matched null	columns at d_{last}	columns/2^n
2	12	5	5	5	16	1.00
3	12	10	5	6	63	0.98
4	12	11	6	7	247	0.96
5	12	11	6	8	848	0.83
6	8	8	7	9	3,302	0.81
7	4	4	7	10	9,908	0.60

Table 23.3: Digit-presented Semaev system at $m = 2$, unsatisfiable real draws, each with its own prime, curve and target; none censored. "Columns" is $\sum_{j \le d_{\text{last}}} \binom{n}{j}$, derived from the measured degree. Source: `research/prime_digit_presentation_20261029/`.

the registered table's arithmetic, and a faithful proof of the closed form is blocked on Wilson's rank theorem for inclusion matrices and two further dependencies absent from core Lean. A frozen but unrun protocol for a p^3 second-digit carry experiment is in `research/p3_second_digit_carry_20261101/`.

23.4 Split-graph and secant presentations

Take a factor base of the rational intersections of E with the graph of a low-degree ordinate function $y = h(x)$ whose $h^2 - f$ splits completely, and write a two-point decomposition through the secant through the two summands. The split-graph theorem of 31 October (`research/prime_factor_base_degree_20261031/SECANT_GRAPH.md`) proves a homogeneous top-ideal degree $d+1$ for this presentation against $2d+1$ for the same base in the dense two-summand system, and a Lean continuation proves the polynomial-ring reduction with the actual highest-degree forms, including the exact $\mathbb{F}_{17}$ fixture at degree $5 \to 3$ for every target and all four sign branches (159 declarations in fourteen modules, zero failures, no native trust). The measured part (`research/prime_factor_base_degree_20261031/parallel_20261031/RESULTS.md`) ran stock F4 on 14,220 systems, of which 14,194 verified and 26 hit root-extraction caps (all at $\mathbb{F}_{127}$, degree 5). The handled F4 degree of the secant presentation is 3 against 6 for the dense presentation on all 18 $\mathbb{F}_{17}$ targets, and $d + 1$ against $2d + 2$ on $\mathbb{F}_{83}$, $\mathbb{F}_{101}$ and $\mathbb{F}_{127}$ at graph degree $d = 3$–5, with the matched non-graph secant control in between at $2d$.

The degree gain does not become a cost gain. Against the matched secant control the candidate improves degree, width and pair and row counts. Against its own dense presentation it lowers the degree but increases the branch-summed work: on the $\mathbb{F}_{17}$ toy, 976 reduced pairs and 2,102 reduced rows against 344 and 736 for the dense system. The stock engine exposes no field-operation total, so cost per relation is null. Equal-size bases have the same all-target ordered yield $K^2 - K$. A Boolean-offset successor of 1 November (`research/prime_factor_base_degree_20261031/pencil_search_20261101/`) removes the complete-splitting requirement at canonical degree 7 against a same-base dense 9, and selector recodings prove $7 \to 6$ (XOR) and $7 \to 5$ (one-hot); affine degree, yield and work are unmeasured. The atlas lead `LEAD-ICP-split-graph-secant-presentation` is `open_next_step`: a degree theorem with no cost advantage.

23.5 Divisor-norm decompositions and a direct index calculus

An m-point relation can be written as a norm: a function in $L((m+1)O)$ of the form $A(X) - B(X)y$ whose norm $A^2 - B^2 f$ is divisible by $X - x_R$ and whose other roots must lie in the factor base. Membership in a Kummer base $g\mu_d$ can then be imposed by quotient remainders or by Graeffe coefficient towers. The "one research bet" study of 29 October chose growing-arity divisor-coefficient decomposition with shared quotient and remainder membership towers, and stated that it could not deliver an original sub-exponential algorithm because the essential complexity lemma is unproved (`research/prime_one_shot_20261029/`). Its continuation (`research/prime_one_shot_20261029/continuation/`) built the pieces:

- a native coefficient solver for arity three that receives only the curve, target and base parameters, searches the two free norm coefficients, enforces membership through quotient remainders and extracts every rational selected-chart solution; on $E : y^2 = x^3 + 2x + 2$ over $\mathbb{F}_{17}$ its complete model sets agree with a separate signed-point oracle in all 90 cases (42 SAT, 48 UNSAT, 400 models);

- a mixed-target direct index calculus that recovers the logarithm of every one of the curve's 18 nonzero targets from $R = aP + bQ$, with a left-null vector of the factor-base coefficient matrix alone and no supplied logarithms, retaining 230 attempts with all 217 nonidentity solver calls charged (59,024 coefficient pairs);
- written proofs, re-derived within the session but not formally certified, of rationality, a coverage bound $\rho_g \geq 1/(192m16^m)$ for a constant fraction of admissible bases under stated finite conditions ($1/(320m16^m)$ for the conservative selected chart), and recovery without full base rank.

Each current solver call visits $p(p-1)$ coefficient pairs, so even the coefficient loop is $\Theta(2^{2n})$ in $n = \log_2 p$: the solver is exponential. The supporting reduction would need $2^{4\sqrt{n}+O(\log n)}$ oracle calls and dense linear algebra of $2^{6\sqrt{n}+O(\log n)}$ bit operations, so a proved uniform oracle bound $C(n) = \exp(o(n))$ would give a sub-exponential algorithm in that field family. No such bound is supplied. The obstruction record is `problems/prime-divisor-uniform-solver.md`, the atlas lead `LEAD-ICP-shared-coefficient-membership` is `open_next_step`, and no exponent was measured. The same continuation found the external tensor-contraction preprint of §24.1 and checked that the literal shared-remainder system translates to $O(mb^2)$ Boolean edges, too many for the contraction bound to give $\exp(o(b))$.

23.6 SAT, SMT and CDCL(T)

Chapter 21 reports three frozen SMT formulations of a 20-bit decomposition that returned UNKNOWN at 120 s. The lead review of 31 October (`research/prime_ic_lead_review_20261031/`) asked where SAT-shaped methods belong. It ranked the hashed-base tensor route of §24.1 first, as the only inspected route with an explicit claimed sub-exponential bound, and noted that this is not evidence for a CDCL solver: the source keeps an $O(mb)$-edge tensor network because an ordinary circuit-to-CNF reduction loses the size bound. For the narrower practical-solver question it drafted a fixed-arity-three CDCL(T)/$\mathbb{F}_p$ screen on a one-hot, disjoint-window, chained-S_3 predicate, with strict gates $e < 0.5$ and $a < 1.5$, deliberately not frozen until its generator, theory propagator and preflight cost exist. The shared-coefficient discriminator (`research/prime_shared_coefficient_sat_20261031/`) passed its E0 gate (18 of 18 exact model-set matches on $\mathbb{F}_{17}$; 6 SAT and 12 UNSAT targets) and, on 1 November, its Boolean E1 gate: 36 of 36 target/backend model-set comparisons and 42 of 42 independently checked UNSAT proofs. No timing or scaling was collected; the elimination campaign of §24.1 records the screen as "representation gate passed; performance implementation-blocked".

Twelve leads. On 31 October twelve prime-field leads, L01–L12, were evaluated and then reviewed by skeptic agents, whose digest is authoritative on standing (`research/ic_prime_eth_leads_20261031/SYNTHESIS.md`; the raw lead directories are on an external volume, and nothing in the synthesis was re-run). Table 23.4 gives the standing after review. The synthesis's verdict is that nothing moved the exponent question: every result is a constant, a known negative or an unfinished harness, and the SAT and propagation lane for prime fields "should be closed for exponent purposes". One item is untested: a CaDiCaL `ExternalPropagator` with $\mathbb{F}_p$ Gaussian elimination. The synthesis also records feasibility: $m = 3$ prime SAT is censored above $b \approx 12$ in every lead that tried it, so only the $b = 6$ and $b = 9$ diagonal cells are uncensored. Its closing statement names the two places where, for the prime-field exponent question, anything remained open after that day: the Kummer-tower cells at $m \geq 4$ (§23.1) and the contraction exponent $s/\sqrt{\log s}$ of the tensor route (§24.1). Neither was touched by the twelve leads.

The review corrected the lead document in eight places. Two matter beyond the lane. The statement that black-box contraction "never beat[s] rho whatever D.1's constant is" needs an outside assumption $c \geq 1$ on an unspecified constant; Lean proves $g > 16c$, and the general threshold is $\log_2 b > 256c^2$. The figure of about $10^5 b$ edges per point addition is 9–30× high: the measured network has 3,364 edges per slot-bit for the freed binary gadget and 11.9k edges per bit for the $m = 2$ prime network at $b = 131$. Neither changes an exponent argument.

23.7 Small-root reach

A windowed decomposition asks for a root of a block $f = S_3(x_1, x_2, x_3) \bmod p$ with $|x_i| < p^{\delta_i}$, the reach being $\varepsilon = \sum \delta_i$; at $(m, w) = (4, 2)$ index calculus needs $\varepsilon = 8/9$. Chapter 21 records the S_5 Coppersmith

Lead	Headline after skeptic review	Standing
L01 char-2 gadget	slope about 2.58 bits of conflicts per window bit, against a bar of 0.5; CPU 3× worse at $\ell = 7$	constant only
L03 sparse-parity bases	paired ratio to a size-matched null 1.000 ± 0.0003, $b \le 22$	constant only
L04 backdoor census	offset $2\log_2(\text{cap}) - 1$ at $m = 3$: enumeration moved into the propagator	contested; cap-bounded statement only
L05 policy pilot	reference slope 2.17 [1.93, 2.38]; renaming is null by construction	refuted as closed
L06 contraction width	MMD+ lower bound 15–51 against enumeration 2–10.7 at $b \le 16$	refuted as closed; narrow claim stands
L07 base second moment	hashed bases 1.000 ± 0.005 of null; shared negation-closed window 0.713/0.833/0.916 at $m = 4$	contested; constant only
L08 Koblitz transfer	true margin about 1.4×, not 3.42×; binary only	refuted as stated
L09 BTN census	margin -16.1 to -16.3 bits, no encoding admitted by D.1 ($\lambda \ge 2^{85}$ needed)	tool stands, yardstick downgraded
L10 Lean filings	seven gaps between the lead text and Lean, all confirmed	refuted as fileable
L11 multiplier encodings	pc/school 0.78 [0.67, 0.92] at $b = 9$ only	stands, weak
L12 clause retention	inc/fresh 0.025 [0.014, 0.040], setup uncharged	refuted; retained setup

Table 23.4: The 31 October prime-field leads after skeptic review (L02 was never run). Producer-reported, from the synthesis; no lead has a validation record. Source: `research/ic_prime_eth_leads_20261031/SYNTHESIS.md`.

reformulation ($\delta^* = 1/80$) and the S_3 falsifier at $\beta = 1/4$. Four studies of 26–29 October turned this into theorems.

Correction. The atlas lead quoted a margin of -0.048 at $(4, 2)$. It came from `tools/smallroot_reach.py`, which compared $\prod X_i^{s_i}$ with p^{s_p}, where s_p is the exponent of p in $\det L$. The Howgrave-Graham criterion compares with $p^{t \dim L - s_p}$, and for nested shift sets $\varepsilon_{\text{HG}} = \varepsilon_{\text{tool}}/n$ in the limit (`CORR-20261024-b2a799`; `crypto-autoresearcher: analysis/smallroot-reach/CORRECTION.md`). The Jochemsz–May extended reach of S_3 falls from 0.8408 to 0.2803, and the best margin at $m = 4$ from -0.0481 to -0.6086. The note is a computation by one agent; its pre-registered audit `EXP-ECDLP-64fa3d` has not run.

Any lattice, equal bounds. The optimal-reach study (`research/smallroot_optimal_reach_20261026/`) bounded every Coppersmith lattice built from $(p, f)^t$, the provably optimal shifts of Ryan included, without building one: with equal per-variable bounds the reach is at most $(n! \operatorname{vol} N)^{-1/n}$ for Newton polytope N, which is 0.3293 for S_3 in three unknowns ($\operatorname{vol} N = 14/3$) and 0.3536 in two. The bound is attained in 11 of 11 exact cases. Against the demand of 0.8889 the margin is -0.56, proved.

Any bounds, any monomial set. The follow-up (`research/smallroot_followups_20261026/`) proved that any such lattice on any monomial set M, convex or not, satisfying the Howgrave-Graham condition needs $h_N(\delta) = \max_{v \in N} \langle \delta, v \rangle < 1$. For S_3 in three unknowns this gives $\varepsilon < 3/4$ and in two $\varepsilon < 1/2$, against 8/9. An exact check found no violation in 9,600 cases. A joint lattice for the three-equation $(4, 2)$ system needs $\nu_s < s$ for some s; the computed $\nu_s/s \ge 2.08$ in the demand region for $s \le 3$ is a computation, not a proof. A third study (`research/prime_eform_reach_20261029/`; derived proof, exact counts) proved Theorem K: for a convex M and the extent W_i of f in coordinate i, every lattice gain satisfies $\sum_i \delta_i W_i \le 1$, so $\varepsilon \le 1/\min_i W_i$, which is 1/2 for S_3, the value the numerical search had only approached. In the symmetric e-variables with full extents the same argument gives $\delta_B \le 2/(m(m+1)2^{m-1})$, that is 1/6, 1/24 and 1/80 at $m = 2, 3, 4$, so the Jochemsz–May asymptotes are optimal there. The closure is conditional: at the exceptional specialisation $z = 0$ the extents are not full, and a review counterexample reaches $7/41 = 0.1707 > 1/6$ for S_3. Finally, any x-variable lattice on a convex set reaches at most $\delta = 1/21.8$ for S_4 and $1/70.8$ for S_5, against 1/4 and 1/5 needed, 5.5 and

14.2× short (`research/prime_s5_small_root_leads_20261028/`, §7c); this rests on the written proof above, which has no Lean formalisation.

The small-root family is closed for one block, for every bound shape and every monomial set by proof, and for convex sets at the e-form asymptotes. It is open for multi-target ideals, the exceptional specialisations and joint lattices beyond $s = 3$.

24 Prime fields: the tensor route and named targets

This chapter has two parts. The first is a review of an external claim: the preprint of Akshima and coauthors, *A Subexponential Time Algorithm for Elliptic Curve Discrete Logarithm*, which claims subexponential prime-field ECDLP conditional on a Boolean tensor-contraction theorem in a second external manuscript. The programme read both, ported the prototype and built instruments. *Nothing was measured*: there is no ecbench count, no wall time and no cost against rho for this route, and the programme did not verify the theorem it rests on. The second part collects the named targets not already in Chapter 21: EcGFp5 and EcMasFp5 over $\mathbb{F}_{p^5}$, a short remainder on $\mathbb{F}_{p^4}$, ECCp-131, and the audits of standard prime curves. Xedni calculus and the JKSST audit are in Chapter 8.

24.1 Boolean tensor contraction: an external claim

The claim. Version 0.2 of the Akshima et al. preprint (pinned at `bochum-crypto/subexponential-ecdl` commit `1ee44e56`, PDF SHA-256 `82e27362...`) claims a classical expected-time algorithm

$$2^{O(b/(\log b)^{1/4})}, \qquad b = \lceil \log_2 p \rceil,$$

for prime-field ECDLP with known odd prime group order, and a fixed-cofactor extension (`research/prime_one_shot_20261029/continuation/tensor-review/README.md`). Its Theorem 4 is conditional on its Theorem 2, which invokes Appendix D.1 of *The Exponential Time Hypothesis is False* (pinned at `openai/math` commit `7ef5c0cc`, PDF SHA-256 `5b1612dc...`; the entry-cost audit of 4 November found the publisher path's later commit `42907c9e` serves the same PDF). That appendix states a contraction bound of $(s + L_{\text{in}})^{O(1)} 2^{O(s/\sqrt{\log s})}$ deterministic bit work, in polynomial space, on an s-edge bipartite Boolean tensor network whose vertex predicates are charged explicitly. The source triage records that repository and author labels establish provenance only, not endorsement, peer review or correctness.

The mechanism. The preprint encodes integer multiplication with $O(b)$ Boolean edges and maximum vertex degree $O(b^{3/4})$, through two stages of exact Fourier transforms over a finite ring; a straightforward bit circuit's extra $\log b$ factor would destroy the saving. Independent affine hashes of injectively compressed points define m separate public factor bases, each membership costing $O(b)$ edges, and a complete m-point sum costs $O(mb)$ edges. The contraction theorem then gives query work $2^{O(mb/\sqrt{\log b})}$. Fresh mixed-target labels and a matrix-only dependency recover the target. With base listing, failed queries, seed restarts, root extraction, prime selection and linear algebra charged, the total is $2^{O(b/m + mb/\sqrt{\log b})}$, and $m \approx (\log b)^{1/4}$ gives the displayed exponent. The hashed bases are generic in the sense of §22.7, so the route's non-generic content is entirely in the contraction theorem.

What the programme checked. Two in-session readings of the decisive contraction proof, D.1 and the full D.4 recurrence, checked packet shrink, degree invariants, interpolation counts, description and evaluation costs, precision absorption, coefficient-ring valuations, Cauchy charts, Hermite recovery and packet factorisation, and found no concrete gap. The parent session read the preprint's multiplication construction, the point-addition encodings, the hash coverage and the complete-cost analysis, and found no transfer gap. The review itself says these are scoped proof readings, not certification of either manuscript or of any implementation. Implementation checks are all small:

- the native tensor crate passes 34 release correctness tests, including forced recursive and direct controls and a tiny complete relation pipeline;
- an independent GMP reconstruction unit passes 72 fixed polynomial cases, 308 Cauchy charts, 8,832 node-error and 1,152 final-error checks and four negative controls, with byte-identical replay; it is not an extension-ring multiplication or contraction test;
- a Rust port of the Go `ecdltensor` prototype passes all 19 Go tests with byte-identical output, and both demos recover the planted scalar (`research/ecdl_tensor_ic_rust_port_20261029/`); its counters are exponential and it does not implement the contraction algorithm.

What was not measured. The speed study (`research/ecdl_tensor_ic_speed_20261030/`) built an `ic.tensor_contraction_v1` arm in ecbench and states that the measurement has not run; it predicts S growing about like r, since one relation costs about $|F_1||F_2| \approx 2r$ additions in the prototype. The counts-only protocol v2 for the tensor and hashed-native panels (`research/hashed_ecdl_counts_20261031/`) superseded v1 before any data, and no session has run under either. The finite-size review (`research/hashed_ecdl_finite_size_20261030/`) states that the preprint's literal constants force a finite-size fallback on every practical graph, which does not refute an asymptotic theorem; it proposes search strategies below the recursion's onset without measuring any. Its ECCp-131 fixture uses a certified hash modulus $s = 3102808443644027376164572183694178153449$.

Census and elimination. The skeptic-reviewed leads of §23.6 include two that bear on this route. L09 found that no encoding in its census is admitted by D.1 (margin -16.1 to -16.3 bits; admission needs $\lambda \geq 2^{85}$), and that with D.1's constant set to 1 the ratio $s/\sqrt{\log_2 s}$ is 37× to 41,000× the enumeration exponent. L06 found that plain tree-decomposition dynamic programming on the school multiplier CNF loses to enumeration at $b \leq 16$ (lower bound 15–51 against 2–10.7). The elimination campaign of 1 November (`research/prime_ic_elimination_20261101/`) re-fetched both PDFs at their pinned commits, matched the hashes, and kept the route at rank 1 of its frozen register: the native finite fallback is rejected; symbolic guards first pass at $\lambda = 2^{480}$; semantic cuts expose 154–354 raw fanout edges but only 7–15 exact signal classes; natural ranks beat support-matched sparse nulls, while all four matched cap-1 typed topology nulls have exactly the same rank. Its state for the route is "legacy census infeasible; minimal typed global-wiring mechanism rejected; theorem/full null open". A rank-census design (`research/tensor_boolean_breakthrough_20261031/`) proposes to test whether natural boundary tensors have lower flattening rank than rewired nulls before any solver search; its E0 core is implemented and nothing affected has been measured. The source-inspection audit of implicit tensor entries (`research/hashed_ecdl_entry_cost_20261104/`) is claim `PCC-20261104-1df881`, and the structured-predicate integration report is `PCC-20261104-458dda`; both are of unverified standing.

What would test it. The rank-census design of 31 October stages the test as six gated experiments plus one SAT/BDD triangulation, none authorised to spend benchmark budget until its predecessor's gate passes:

- E0, canonical IR equivalence and mutation: every compared network regenerated, hashed and replayed with its semantics unchanged;
- E1, a boundary-tensor flattening-rank census against graph- and predicate-preserving rewired nulls, with a held-out target and null gap;
- E1S, SAT and BDD triangulation on the same pairs, to see whether exact Boolean solvers exploit the E1 signature;
- E2, local Fourier and partition-rank certificates that predict at least 20% lower held-out exact work;
- E3, certified rewrites and orders with at least a 2× held-out gain and a lower growth slope;
- E4, a structure-specific contraction engine beating the strongest exact baseline by 2× in operations and memory;
- E5, an end-to-end relation oracle improving on four increasing sizes and an unseen target, still with no asymptotic claim.

The finite-size review proposes five ways to use the hashed relation language below the recursion's onset: making inverse-hash selectors the free inputs, solving for one witness instead of counting completions, width-aware encoding and contraction plans, instance-certified recursion with small packet parameters, and nested bases with partial relations. It notes that enumerating every selector tuple remains generic decomposition search. None of the seven experiments and none of the five routes has a result.

Reading. The route is the programme's best-supported *claimed* sub-exponential algorithm for prime-field ECDLP, and it is external prior art. Its correctness rests on an external result that the programme read but did not verify, its constants make it infeasible at finite size as written, and it has no measurement. It does not move a row of any table in this treatise, and the iterative success criterion does not apply to it until a counted cost exists. The norm and divisor candidate of §23.5 does not inherit its compactness: its literal translation has $O(mb^2)$ edges.

24.2 EcGFp5 and EcMasFp5 over $\mathbb{F}_{p^5}$

EcGFp5 and EcMasFp5 are curves over $\mathbb{F}_{p^5}$ with $p = 2^{64} - 2^{32} + 1$ and $\mathbb{F}_{p^5} = \mathbb{F}_p[z]/(z^5 - 3)$. EcGFp5 is a double-odd curve with one rational 2-torsion point. Chapter 30 records that the September lane on EcGFp5 symmetrisation measured no $m = 5$ ideal degree (`DEC-20260923-d86ff2`). That statement is superseded: the 1 November solve below measured $D = 2^{16}$ on EcGFp5-shaped curves over toy primes. It is still true that nothing was measured at the 64-bit p. Chapter 25 gives the Joux–Vitse $(n-1)$-decomposition at $n = 5$ over $E(\mathbb{F}_{p^5})$ for toy p.

Desk evaluation. The four atlas leads were evaluated on 31 October (`research/ecgfp5_lead_evaluation_20261031/`; desk evaluation, every cost a model from cited constants). The design note's "$\geq 2^{142}$" multiplies the large-prime system count $p^{2-2/5} = 2^{102.4}$ by a per-system cost without the $5! = 120$ inverse decomposition probability, and is not a lower bound for the Gaudry–Diem family. The cheapest member is the basic algorithm with a factor base tuned to the per-system cost c, whose total scales as $c^{1/3}$, about $2^{115.5+\log_2(c)/3}$ 64-bit word operations: 2^{126}–2^{136} for the repository's per-system bands. In raw time every variant is 32–45 bits below rho (rho $\approx 2^{159.3}$ steps, 2^{169} word operations), but the tuned base has 2^{57}–2^{63} columns and needs 2^{64}–2^{71} bytes against 2^{35}–2^{45} for rho with distinguished points; capped at a petabyte the method is about 2^{27} above rho. In a memory-aware model rho remains the better attack. The verdicts are: symmetrised Gaudry REVISE (measure c, not D); Joux–Vitse $(n-1)$ for the full logarithm [27] CLOSE ($((n-1)!\,\tilde{c}\,q^2/2 = 2^{131.6}\tilde{c}$ with $\tilde{c} \approx 2^{33.5}$–$2^{34.4}$ cycles lands within ± 2.5 bits of rho, at about 2^{71} bytes, and loses to rho for any base of at most $2^{62.5}$); the odd-characteristic cover CLOSE (genus 65 for EcGFp5 and 97 for EcMasFp5, index calculus at least 2^{206} on the best constructed cover; the atlas keeps this lead `open_blocked` until the source-freeze block is lifted); oracle-assisted static Diffie–Hellman REVISE (about 2^{63} oracle queries, and no deployed service found that exposes the oracle).

The quotient solve. The decisive experiment (`research/ecgfp5_quotient_solve_20261101/`; `benchmarks/20261101-ecgfp5-quotient-solve/`; pre-registered, amendments A1–A5) solved the $\mathbb{F}_p$-rational $(\mathbb{Z}/2)^4 \rtimes S_5$ quotient of the five-summand system on EcGFp5-shaped curves at $p' = 4111$, 262231 and 33554371 with msolve 0.6.5, on one 4-vCPU host. It found $D = 2^{16} = 65{,}536$ at all three primes, exactly the affine Bézout bound $8^5 \cdot 2$, confirming $D = 2^{(m-1)^2}$ with no reduction below Bézout; the planted decomposition was recovered at every prime. One complete solve costs $c_{\mathrm{cyc}} = 2^{46.30}$, $2^{46.56}$ and $2^{46.63}$ cycles. The evaluator's frozen gate G1 reads CLOSE when $c_{\mathrm{cyc}} \geq 2^{46}$, so all three cells close, by margins of 0.30, 0.56 and 0.63 bits. The F4 work is p-independent (the same final 5578×71114 matrix at every prime), F4 alone is $2^{44.72}$–$2^{45.29}$, and about two-thirds of c is msolve's rational parametrisation. Weighted orders and signature-based bases did not lower c. At $m = 3$ all five arms gave the predicted degrees in 30 solves, and msolve was found to crash for every prime from 65537 to 253229, a solver defect, so 65551 was replaced before any $m = 5$ run.

The block-Wiedemann finish. The finish was then replaced by a block-Wiedemann solve [7] on the BNS22 chain matrix of the reduced grevlex basis (`research/ecgfp5_quotient_bw_finish_20261102/`; `benchmarks/20261102-ecgfp5-bw-finish/`; frozen before the first cell, no amendments). The eliminant equals msolve's coefficient for coefficient, the stability check passes on 4,920 chains, and the planted point is recovered at every prime. Whole-solve CPU falls 1.83, 1.97 and 1.72× at the three primes, below the frozen 1.5× bar, so the finish is the new per-system baseline; the finish alone is 3.15–4.12× cheaper and the F4 work is unchanged. The new c_{cyc} is $2^{45.43}$, $2^{45.58}$ and $2^{45.84}$, below the CLOSE line by 0.57, 0.42 and 0.16 bits, but the frozen $[c/4, 4c]$ band straddles the line at every prime, so the verdict is unchanged (Table 24.1). The distance to the PROMOTE line of 2^{38} is 7.43, 7.58 and 7.84 bits. This is an engineering improvement of a per-system stage cost at toy primes, not an end-to-end ECDLP cost. An earlier HNF-based finish was a memory-bound null at the development prime $p = 331$: the determinant step was killed at 13.33 GiB on a 15.7 GiB host (`research/ecgfp5_quotient_hnf_finish_20261102/`).

Representations. A desk audit of 2 November (`research/ecgfp5_representation_audit_20261102/`) closed every degree-2 coordinate change and every larger symmetry group: EcGFp5 has one rational 2-torsion point and no rational 4-torsion, so the only usable group gives $D = 2^{16}$ at $m = 5$. It found one open lever, Pornin's $u = x/y$: since $u(P + N) = -u(P)$ and $u(-P) = -u(P)$, the quotient is a polynomial-ring form in $s_k = e_k(u_i^2)$ and $\pi = \prod u_i$, with measured $D = 2^{(m-1)^2}$ at $m = 3$ and $m = 4$ (16 and 512) and the planted point recovered. That corrected a ledger statement:

p'	W0 CPU-s	BW CPU-s	speed-up	$\log_2 c_{\mathrm{cyc}}$ W0 / BW	G1 band (BW)
4111	31,004	16,912	1.83×	46.30 / 45.43	[43.43, 47.43]
262231	37,036	18,823	1.97×	46.56 / 45.58	[43.58, 47.58]
33554371	38,778	22,516	1.72×	46.63 / 45.84	[43.84, 47.84]

Table 24.1: One complete $m = 5$ solve of the EcGFp5-shaped quotient system at three toy primes, $D = 2^{16}$ in every cell: msolve with rational parametrisation (W0) against the block-Wiedemann finish (BW); one 4-vCPU host, $c_{\mathrm{cyc}} = \text{CPU-s} \times 2.8 \cdot 10^9$. The CLOSE line is 2^{46} and the PROMOTE line 2^{38}. Sources: `research/ecgfp5_quotient_solve_20261101/`, `research/ecgfp5_quotient_bw_finish_20261102/`.

`CORR-20261102-7e3a91` scopes the earlier claim that the ring form is "unavailable" for double-odd curves to coordinates with $\varphi(-P) = \varphi(P)$. Any gain from u is expected to be a constant factor. The atlas lead `LEAD-ICP-gfp5-symmetrized-gaudry` is `closed_dominated`, with the block-Wiedemann finish as a measured per-system improvement still short of the PROMOTE bar.

$\mathbb{F}_{p^4}$. Two items extend §21.10. The D_4 index calculus ran end to end in ecbench on $E(\mathbb{F}_{29^4})$, $\#E = 708{,}208 = 16 \cdot 44{,}263$, with every phase charged: $S = 7.17 \cdot 10^8$ against 1.065 for negation rho, audit OK on three of three records (`research/gaudry_quartic_d4_ecbench_20261013/`). A later leaderboard engineering row cut the plain-S_5 cost C_4 from $1.209 \cdot 10^{12}$ to $8.69 \cdot 10^{10}$ $\mathbb{F}_p$ multiplications (13.90×) with sparsest pivots, chain eigensteps and shape replay (`benchmarks/20261019-s5-plain-c4-options/`), where shape replay alone had been a 2–5% lever (`research/gaudry_quartic_replay_20261013/`).

24.3 ECCp-131

Chapter 21 gives the ECCp-131 parameters, the zero-relation probe and the rho rates. Four further studies are nulls or constructions.

- *A full-width auxiliary tower.* The Kummer tower cannot have $t \geq 2$ on the ECCp-131 field, since $v_2(p-1) = 1$. A rational eight-step 2-isogeny tower was built instead on the auxiliary curve $E' : y^2 = x^3 + x + 301$ over the exact field (`research/eccp131_full_width_20261023/REPORT.md`; of 301 auxiliary orders computed, two were divisible by 256, at $b = 278$ and $b = 301$, and the curve with $b = 301$ was accepted): 256 distinct coset abscissae, 8 of 8 Vélu maps checked, 115 abscissae lifting to the published curve, 26,450 signed pair points, 8 of 8 planted controls, and zero natural setup or target rows, rank zero.
- *A multiplicative-coset screen.* Twelve $g\mu_5$ cosets and, in a follow-up, sixteen $g\mu_{10}$ cosets (μ_{10} is the maximal smooth subgroup, $p - 1 = 2 \cdot 5 \cdot r$ with r prime) against equal-size small-abscissa bases found zero natural rows and rank zero for both seeds, with every planted control recovered (`research/eccp131_structured_ic_20261022/REPORT.md`).
- *A representation filter.* A Wagner-style coordinate-bit filter trained on true four-sum relations against a rotated null, with 3,450 configurations in each of six base/size cells (`research/eccp131_representation_filter_20261102/REPORT.md`; `benchmarks/20261102-eccp131-representation-filter/`), qualified in no held-out cell: every recorded lower confidence ratio is below 1, and the two superficially positive small-x winners at $B = 64$ and $B = 256$ are post-addition filters that need all $\Theta(B^2)$ pair sums. The verdict is `no_qualifying_filter_in_frozen_family`, with 0 rows and rank 0 at arity 2 and 3.
- *A CUDA port.* The ECCp-131 and ECC2-131 rho walks were written as per-thread functions that compile as device code (`research/certicom131_cuda_port_20261023/`). The kernels compile and assemble for sm_80–sm_120 but have never executed on a GPU; there is no device parity result and no GPU rate.

The full-parameter index calculus remains the open problem `problems/eccp131-full-parameter-index-calculus.md`.

24.4 Audits of standard prime curves

Chapter 21 lists the audits of the NIST, SECG and CryptoPro curves that found no ECDLP weakness. The later ones concern curve provenance and class-group statistics, and none bears on ECDLP hardness.

- *The NIST seed skip rule.* Of the 21 seed candidates the 1997–98 runs accepted under every known criterion, 10 were published. None of 21 new preregistered binary features explains the 11 skips, after about 60 earlier features were ruled out (`research/nist_seed_skip_filter_20261017/`). A later search found an independently checked two-test empirical separator for all 21 eligible curves, with a root-convention dependence and a calibration that blocks reading it as the historical explanation (`research/nist_skip_criterion_20261025/`). The same question for the binary B- and K-curves is a measured census (`research/nist_binary_skipped_20261020/`), and a factorisation continuation fully factored eight of the nine distinct integers left by an external DiSSECT run, leaving one 444-bit composite cofactor of a P-256 extension-twist order (`research/nist_skipped_curve_dissection_20261021/`).
- *P-224's class group.* The 2-rank of $\mathrm{Cl}(D_K)$ is exactly 6 for P-224 (4 for P-256), and the class number is estimated at $3.5185 \cdot 10^{32} = 2^{108.1167}$, with a GRH-conditional interval $[2^{108.098}, 2^{108.135}]$ under an assumed constant; the exact value is not known (`research/p224_class_number_20261022/`). A 2-rank of 6 is not unusual: P-224 sits at the 91.5th mid-rank percentile of random 224-bit primes, upper tail 12.3% $[10.3, 14.5]$ (`research/p224_two_rank_null_20261022/`).
- *Cheon's attack on P-224's shape.* The DLP with auxiliary inputs was scaled on P-224-shaped curves (`research/cheon_p224_shape_20261022/REPORT.md`). The pre-registered rho-reliability gate H3 failed, with 5 of 84 sealed rho runs outside $S \in [0.3, 4]$, so every Cheon/rho ratio and crossing in that report is withdrawn; the Cheon-only scaling stands. It is in any case not plain ECDLP.
- *Special prime shapes.* Curves built by the CM method over 192-bit Proth, cyclotomic $\Phi_d(m)$ and smooth $p \pm 1$ primes all have embedding degree $k > 4000$, like the random control; only pairing-friendly constructions give k of 10–24 (`research/special_prime_curves_20261023/`).
- *Rho on NIST curves.* Stepping to the better of $P + M_j$ and $P - M_j$ in the `gpu/nist` small-batch kernel stays disabled after an audit; the estimates are historical diagnostics, not measured solve costs (`research/nist_best2_small_batch_20261027/`).

24.5 What the three chapters settle

On every prime curve measured, rho with negation is cheaper than every index-calculus arm, generic or algebraic. The generic track's best arms are the residual collectors at 1.3–1.7× rho ($w = 0$), a constant at rho's exponent with no growth over 36–42 bits; the k-sum tables are 5.4–11.0× rho and above baby-step giant-step. The algebraic track has no priced prime baseline beyond the Kummer-tower rows, and its degree studies show growth or input-pinned degrees in every presentation, with matrices that grow when degrees do not. The one claimed sub-exponential route is external, conditional on an unverified theorem, infeasible at finite size as written and unmeasured. On EcGFp5 the measured per-system cost moved by 1.72–1.97× and remains about 7.4–7.8 bits from the promote line.

The chapters state these corrections and withdrawals:

- the residual collectors' $w > 0$ gains were the price of a byte counter, withdrawn under option B (1 November); the scale study's C1 tally falls from 6 of 8 to 2 of 8, and a 9-versus-8 tally of 12 cells is unreconciled (§22.3.1);
- the memo gate "$e < 1$ at $m = 4$" is replaced by $s < m - 3$ (§22.1);
- the cofactor-cleared prime base is `invalid_controls`, not a win (§22.4);
- $N = 64$ at 20 bits is unresolved, and the 24- and 28-bit base selections are relabellings (§22.5);
- the small-root margin of -0.048 is superseded by about -0.6 (`CORR-20261024-b2a799`, §23.7);
- the September statement that no $m = 5$ EcGFp5 degree was measured is superseded by $D = 2^{16}$ at toy primes, and the ring-form statement is scoped by `CORR-20261102-7e3a91` (§24.2);
- the symmetric-tower rows are instrument diagnostics, not natural prime-order benchmarks (§23.1);
- the Cheon/rho ratios on P-224's shape are withdrawn (§24.4).

The open items are the discriminating Kummer cells ($m = 3$, $N = 21$ symmetric; $t = 14$ at cap 6), the algebraic reference arm, a counted cost for the tensor route, the $\mathbb{F}_p$ CDCL(T) propagator, the APS complete-cost panel, and a proved uniform bound for a divisor-norm oracle.

25 Extension fields: $k = 4$ and the D_4 symmetry

Over $\mathbb{F}_{q^k}$ with a factor base of points whose abscissa lies in $\mathbb{F}_q$, Gaudry's index calculus [20] decomposes each target into k summands by Weil restriction of S_{k+1}. It costs $\tilde{O}(q^{2-2/k})$, below rho's $q^{k/2}$ for $k \geq 3$, but with a Gröbner constant that grows very fast in k. Faugère, Gaudry, Huot and Renault [14] show that on curves with rational 2-torsion the system has the larger symmetry group D_k, the Weyl group of type D, in a well-chosen coordinate. At $k = 4$ this chapter reports how far that symmetry moves the measured gap to rho. Chapter 21 (§21.10) gives the odd-characteristic engineering in detail.

25.1 The D_4 system

On E with rational 2-torsion point T, the coordinate $Y = (x - x_0 - c)/(x - x_0 + c)$ satisfies $Y(P+T) = -Y(P)$. In Y, S_5 is invariant under D_4, of order 192. Written in the invariants $\sigma_k = e_k(Y_i^2)$ and $\pi = \prod Y_i$, the system shrinks (Table 25.1).

Table 25.2 gives the engineering ladder, each step against the previous incumbent. At step 1 all 19 solved targets agree with an independent meet-in-the-middle enumeration; at step 3 all 648 of 648 solves are exact. At step 4, double large primes cut the linear algebra to 0.827× but regress the whole solve, to 1.025×.

The D_n symmetry is FGHR's [14]. Invariant rings in characteristic two, including even-power substitutions, were studied by Galbraith and Gebregiyorgis [17]. Whether the Frobenius-reduced weighting is in that work has not been checked.

25.2 Distance to rho

Odd characteristic. Over $\mathbb{F}_{p^4}$, index calculus is still $1.39 \cdot 10^6\times$ rho at the measured sizes. The modelled handover moves from 2^{150} to about 2^{103}–2^{119}, depending on the frame, but that is an extrapolation through a measured constant.

Characteristic two. Over $\mathbb{F}_{(2^m)^4}$ the gap closes with size. IC over the negation-map rho model is:

m	6	8	10	12	14
$\log_2 n$	24	32	40	48	56
IC / rho model	5,080	648	80.6	49.7	6.06
IC / measured rho	4,427	919	98.9	47.2	4.27

These are first targets; $m = 14$ is a single target. The fitted slope is -0.248 in $\log_2 n$, which is Gaudry's $n^{-1/4}$. The extrapolated crossover was $n \approx 2^{68}$, where rho takes about 2^{34} steps, so it could be tested directly.

The red-team verdict. An adversarial review tested that crossover (`research/gaudry_k4_char2_redteam_20261026/REPORT.md`). Its verdict: **not survived (unsupported), not refuted**. Table 25.3 lists its estimates. The only clean point is at $m = 15$, with two targets per arm, so no interval is possible with $n = 2$; the $m = 17$ ratio came from a killed run and is not evidence. The rho reference survives: these are generic ordinary curves with Aut $= \{\pm 1\}$. Pricing partly survives. A prime-field cross-check agrees with the fitted exponents. The corrected crossover is at or above the edge of what was measured.

Linear algebra is 0.4% of the cost at $m = 15$. Basic Gaudry linear algebra grows as q^2, like rho, so the $n^{-1/4}$ trend lasts only while the relation phase dominates. Beyond that point double large primes are required.

The red team froze a plan to settle the question: $m = 14$–17, eight fresh targets per arm, a strong rho with batched inversion and charged canonicalisation, counted and hardware units, about 30 h of wall time. It has not run. This is the cheapest experiment in the programme that could yield a measured crossover of index calculus below rho on an elliptic curve. It would be a crossover in a weak family: $\mathbb{F}_{(2^m)^4}$ is not a field deployed curves use.

System	Bézout number	Macaulay degree	Matrix
before invariants	4,096	29	$41{,}780 \times 40{,}920$
D_4 invariants	512	14	$13{,}412 \times 11{,}628$
Frobenius-reduced, char. 2		11	384×448

Table 25.1: Size of the S_5 decomposition system at $k = 4$: before and after rewriting it in the D_4 invariants σ_k, π, and in the Frobenius-reduced form with weights $(2, 2, 1, 1)$ in characteristic two (ladder step 5 of Table 25.2; the source gives no Bézout number for it). Matrix: rows × columns of the Macaulay matrix. Sources: `research/gaudry_quartic_d4_20261013/`, `research/gaudry_quartic_d4_char2_20261015/`.

	Step	Scope	Result
1	D_4 invariants	$p = 269$	$C_4 = 2.54 \cdot 10^{10}$ instead of $1.21 \cdot 10^{12}$, a factor of 47.5–48.4
2	The Y-line on a complete ladder	$p = 101$–1033 (21 curves)	0.0548 $[0.048, 0.061]$ of plain S_4, at the same exponent
3	Sparsest pivots, the weighted grading $\sigma_4 = \pi^2$, and shape replay		C_4 falls to $6.89 \cdot 10^7$, by $369.8\times$ $[367.9, 371.6]$; ecbench S falls 398–406×
4	First-dependency stop		0.979 $[0.973, 0.986]$ of the previous step
5	A Frobenius-reduced form in characteristic two	$m = 8$	one decomposition $481\times$ $[479, 483]$ below the plain form

Table 25.2: The $k = 4$ engineering ladder, each step against the previous incumbent. C_4: $\mathbb{F}_p$ multiplications per decomposition. Result: the step's cost as the source states it, either a factor below the previous incumbent or a fraction of it. Sources: `research/gaudry_quartic_d4_20261013/` (step 1), `research/gaudry_quartic_d4_cost_20261014/` (step 3), `research/gaudry_quartic_d4_char2_20261015/` (step 5).

25.3 Other extension-field routes

- *Joux–Vitse $(n-1)$-decomposition.* The $(n-1)$-decomposition [27] at $n = 5$ over $E(\mathbb{F}_{p^5})$ completed verified logarithms at $4.5 \cdot 10^7$–$1.8 \cdot 10^8\times$ rho, with a crossover extrapolated to groups near 2^{291} (`research/joux_vitse_n1_20261015/`).
- *Joux–Vitse cover.* The cover of $E(\mathbb{F}_{p^6})$ is in Chapter 26.

Part	Quantity	Estimate	Range or note
Data	IC/rho at $m = 15$, recorded	6.30	two targets per arm
	IC/rho at $m = 15$, corrected	5.2	2.4–15.0 over unit-price variants
	IC/rho at $m = 17$	0.72	killed run; not evidence
Rho reference (survives)	floor	0.886	about 3.5% generous to rho
Pricing (partly survives)	ratio moved by the unit price	0.64–0.89	plausible range 0.31–3.1
	ratio moved by unpriced rho work	about 0.93	
Fitted exponents	IC relations	1.07	$[1.01, 1.13]$
	IC linear algebra	2.03	
	rho	2.00	$[1.71, 2.29]$
Corrected crossover	fall of IC/rho per unit of m	about $2^{-0.93}$	
	crossover m^*	≈ 17.6	range 16.3–19.2

Table 25.3: The red team's estimates for $k = 4$ over $\mathbb{F}_{(2^m)^4}$, grouped by the part of the verdict they bear on. Estimate: the point value; Range: the interval or range the review gives, with what it runs over. The corrected IC/rho applies the corrections that favour index calculus. Exponents are of cost in $q = 2^m$. Source: `research/gaudry_k4_char2_redteam_20261026/REPORT.md`.

26 Covers, isogenies and other transfers

A recurring hope was that some other representation of the same instance—an isogenous curve, a curve at another level of the isogeny volcano, a Jacobian, a Weil descent—would make index calculus cheaper. These routes produced the programme's clearest negatives, and one lemma that explains most of them.

Lemma 26.1 (transport conservation, `docs/ic/RESEARCH_TRANSFER_AWARE_IC_20261010.md` §1)**.** *Let $\varphi : E \to E'$ be an $\mathbb{F}_q$-rational isogeny (or any homomorphism of rational points), $\kappa = \ker\varphi \cap E(\mathbb{F}_q)$, $R \in E(\mathbb{F}_q)$, $\mathcal{F}' \subset \varphi(E(\mathbb{F}_q))$ a factor base on the codomain and $\mathcal{F} = \varphi^{-1}(\mathcal{F}') \cap E(\mathbb{F}_q)$. Then the number of ordered m-decompositions of R over $\mathcal{F}$ is $|\kappa|^{m-1}$ times the number of ordered m-decompositions of $\varphi(R)$ over $\mathcal{F}'$.*

Proof. Images of decompositions of R are decompositions of $\varphi(R)$. Conversely, for $\sum Q_i = \varphi(R)$ choose rational preimages P_i of the Q_i; then $\sum P_i - R \in \kappa$, and of the $|\kappa|^m$ choices exactly $|\kappa|^{m-1}$ sum to R (translate P_1 by the kernel element). □

So a homomorphic transfer changes nothing but three curve-specific inputs: the rational kernel symmetry $|\kappa|^{m-1}$ (bounded by the rational torsion; trivial on prime-order curves), the factor-base geometry available on the destination, and the cost of arithmetic there. Sequences of isogenies compose to one homomorphism; a loop is an endomorphism and acts on the subgroup as a known scalar, useful only as a known-log multiplier. Multi-curve factor bases glued by isogenies are single-curve bases in disguise.

Conductor gaps in the ECC2K-130 class. Jao, Miller and Venkatesan [26] prove random self-reducibility among curves with the same endomorphism ring, leaving open whether curves whose conductors differ by a large prime could differ in hardness. The ECC2K-130 class has conductor gap 263; we computed its structure (`docs/ic/RESEARCH_ISOGENY_CONDUCTOR_GAP_20261007.md`) and ran several rounds of experiments. Native counted-S index calculus on two fixed CM pairs gave floor/crater ratios 0.975 (99% $[0.603, 1.451]$) at $\ell = 8209$ and 1.127 $[0.922, 1.342]$ at $\ell = 131101$: no level effect (192 verified records, 96 exact replays; `research/ecbench_conductor_gap_20261012/`). Two-torsion symmetrisation is a quotient isogeny: at a matched base of abscissae it cut F4 row-reduction work 4.1–4.3× ($|H| = 2$) and 17.6–19.6× ($|H| = 4$) over 1,128 systems—but a correction the next day showed that at a matched number of *columns* the saving disappears; it was the cost of a redundant orbit base. A combined Weil descent used 234–529× the F4 operations of the blind system (`research/isogeny_conductor_gap_round3_20261011/`).

Fibre-aware index calculus. Isogeny fibres are torsors whose arithmetic is controlled by one cohomology class (`RESEARCH_ISOGENY_FIBERS_ECDLP.md`); relation generation is isogeny-equivariant at zero lifting cost, and a fibre-closed pair table was the best lever of the conductor-gap rounds (12.8× rho in its best cell). Measured against matched rho across sizes at fixed cofactor, the envelope IC/rho ratio grows from 8.4–8.8 to 35.1–37.0 ($h = 4$) and from 13.4–20.1 to 60.4–63.1 ($h = 72$), as $\#E^{1/6}$: the gap widens, it does not close (816 audited records; `research/fiber_ic_scaling_20261011/`).

Other structure. Symmetrised u-frame bases saved 0.84× at $n = 17$ and 0.76–0.77× at $n = 23$, but cost 4.01× at $n = 31$ (`research/koblitz_symmetrised_e2e_20260927/`). GHS and hyperelliptic covers: ECC2K-130 does map to hyperelliptic curves, but the genus that would pay is missing. In odd characteristic the cubic GHS transfer from $E_0(\mathbb{F}_{q^3})$ to a genus-3 Jacobian now runs end to end with double-large-prime index calculus, at about $q^{1.4}$ against rho's $q^{3/2}$, within 1.15× of a measured negation-map rho at $q = 2^{20}$ (§31.3). The isogeny-loop GLV/GLS work produced a measured 1.09–1.15× faster four-dimensional scalar multiplication (`research/isogeny_loop_glv_20261012/`) and a $\sqrt{\text{élu}}$ kernel cost exponent 0.822 $[0.805, 0.838]$; these speed up curve arithmetic, not the discrete logarithm.

Covers, 21–24 October. The Joux–Vitse genus-3 cover of $E(\mathbb{F}_{p^6})$ is now within 1.6–2.0× of a pooled rho constant at $p = 1009$; measuring the per-trial cost on a genus-255 binary Jacobian turned the $N = 130$ GHS counterfactual's modelled margin negative (-1.83 bits); and two censuses bound what covers exist at toy sizes (genus floor 13 for odd-order $E/\mathbb{F}_{q^4}$; a 2^{-l} genus-3 fraction over $\mathbb{F}_{2^{3l}}$). All are in §31.5. None touches a prime field or a prime-degree binary field, where the GHS genus is out of reach.

The published scale of high-genus index calculus. The largest Weil-descent hyperelliptic discrete logarithms in the literature are at genus 31. Jacobson, Menezes and Stein [25] applied GHS to elliptic curves over $\mathbb{F}_{2^{62}}$, $\mathbb{F}_{2^{93}}$, $\mathbb{F}_{2^{124}}$ and $\mathbb{F}_{2^{155}}$ and obtained genus-31 curves C_{62}, C_{93}, C_{124} and C_{155} over $\mathbb{F}_{2^2}$ to $\mathbb{F}_{2^5}$, each with $\#J = 2r$, r prime. They solved C_{62}, C_{93} and C_{124} with Enge–Gaudry (relation collection for C_{124} took 379 CPU-days on a mixed cluster) and only estimated C_{155}, at 26,290 days on one 1 GHz Pentium III. Velichka [42] solved C_{155} ($\#J \approx 2^{155}$) with Vollmer's algorithm and sieving: 136,533 relations, about 1,735 CPU-days in total, 21 days wall-clock, of which 14 days were linear algebra. No genus above 31 appears in that thesis; the journal version [41] was not available to us and is unchecked. Our genus-255 per-trial measurement (§31.5) is a cost sample, not a solved instance, at about eight times the largest published genus. The cover ECC2K-130 would need, of genus about 2^{130} (§15.7), lies about 2^{125} times beyond it.

Genus three against the strongest rho. Rounds three and four of the genus-3 work ran without a rho arm; a 23 October study supplied one (`research/ghs3_vs_rho_20261023/`; it has a frozen protocol and tables but no README, and no review is recorded in it). On round 4's instances, $q = 2^{14}$ to 2^{20}, three curves per size, in round 4's unit and binary, the reference was an ideal negation rho priced with Karatsuba products and 32 walks per inversion (R_{ideal}, the primary reference, the most favourable to rho) and a measured negation walk (R_{meas}). The frozen rule declares "IC below the strong rho" at a size only if every curve is below both references at every weight of the band. At $q = 2^{16}$ it is: IC/R_{ideal} is 0.70, 0.71, 0.70 at $w = 0$ and 0.64 and 0.38 (geometric means) at $w = 0.1$ and 1. At the other six sizes the decision is *unresolved*, with geometric means 0.97–1.52 (1.27 at 2^{20}). Three things qualify the result. The registered unit prices one subgroup multiplication as one $\mathbb{F}_q$ multiplication ($c_\ell = 1$); the protocol's own sensitivity, $c_\ell = 9$, puts every instance at 2.7–12.4× rho. The ratio carries $\sqrt{c}$ of the cofactor, because IC depends on q alone and rho on $\ell = \#E_0/c$: the three curves at 2^{16} all have cofactor 4, and a post-hoc fit gives exponent 0.474 in c (theory 0.5) and -0.081 per bit of q at fixed c (theory -0.167). And the pooled trend over all 21 instances is flat (0.034 $[-0.047, 0.105]$ per bit), so no crossover size is determined. This is the programme's only preregistered decision of index calculus below the strongest rho. It is a measurement of a known pipeline in a structured, weak class ($E_0/\mathbb{F}_{q^3}$ with $4 \mid \#E_0$, never prime order), where theory predicts $q^{4/3}$ against $q^{3/2}$; it depends on the pricing of one operation; and it is not yet reviewed. It updates the ledger paper's "1.15× at 2^{20}" and bears on that paper's prediction of a crossover at $q \le 2^{22}$ once rho is measured: below rho at one size, unresolved at the others, with no measurable pooled trend.

The same family through a plane quartic. The $m = 2$ type of the cubic GHS cover gives a smooth plane quartic over $\mathbb{F}_q$ directly. On a plane quartic, Diem's index calculus is $\tilde{O}(q)$ in theory, against $q^{4/3}$ for the hyperelliptic genus-3 route above. Before 18 October the repository's plane-quartic "pure IC" toys could not be priced: their logarithms came from enumerating the subgroup, and the degree-3 class keys merged distinct classes. A canonical Pic^0 arithmetic now gives every class a unique key (checked against brute-force class counts on 21 curves and full Cayley tables at $q = 5$). A generic addition costs 3,523 multiplications and 44 inversions, flat in q up to $2^{31} - 1$ (`research/quartic_pic_canonical_20261018/`).

On that arithmetic, a pipeline with relations on lines, a reduced factor base with one large prime, Lanczos modulo ℓ and canonical descent solved and verified 51 of 51 units. It ran on the GHS-weak family $E_0/\mathbb{F}_{q^3}$ with $4 \mid \#E_0$, for $q = 223$ to 262,111 (`research/quartic_n4_20261019/`; Table 26.1). Its reference is a matched *plain* rho, without the negation map, on the same subgroup, and the order holds across the pricing band for every $q \ge 4051$. This was a preregistered decision of index calculus below a matched *plain* rho. Its successor, N4 v2 (`research/quartic_n4v2_20261020/`, 20–25 October), took the same family and instances from $q = 2^8$ to 2^{22} (63 instances, 501 index-calculus units and 2,442 rho runs, all verified) and added three changes: double large primes (D), structured Gaussian elimination with Lanczos (L), and a negation-map rho. DL's exponent is the $\tilde{O}(q)$ of Diem's analysis [9, 11]. At $q = 4{,}194{,}271$ one DL solve took 12.3 CPU-seconds against 616 for negation rho.

The qualifications are substantial. The host was L0 (load average 22–60); the intervals resample only three units per size at $k \ge 17$; the conversion constants move D's high-range exponent from 1.009 to 1.048; the frozen B0 records do not replay at any of 11 replayed sizes; the negation rho fails the calibration rule at three sizes; and there is no validation file. The program's closing note calls the result "our measurements of a known result" (`research/plane_quartic_program_20261018/CLOSING.md`). It covers the GHS-weak family only; nothing here is about a generic curve, a prime-order curve or a standard curve.

Study	Arm	Over B0	IC/rho	IC exponent	Rho exponent	Crossover
N4			0.122 $[0.10, 0.16]$	1.083 $[1.040, 1.129]$	1.485 $[1.437, 1.525]$	wall time: only above $q \approx 3 \times 10^4$
N4 v2	D	0.081				
	DL	0.042 $[0.042, 0.042]$	0.005 $[0.004, 0.008]$	0.983 $[0.975, 0.993]$; 0.988 on $k = 17..22$	1.543 $[1.494, 1.583]$	$q^* = 555$ $[389, 752]$ (interpolated); $q^* \approx 1{,}889$ in CPU time
	DL1	0.022				

Table 26.1: Plane-quartic index calculus on the GHS-weak family $E_0/\mathbb{F}_{q^3}$, $4 \mid \#E_0$. N4 against a matched plain rho at $q = 262{,}111$, exponents fitted on $q \geq 4051$; N4 v2 against the frozen pipeline B0 and against negation rho at $q = 2{,}097{,}091$ (the top size). IC/rho in $\mathbb{F}_q$-multiplication equivalents. Exponents: fitted cost exponent in q, over the full range unless stated. Crossover: in counted units unless stated (the N4 crossover is in wall time). D: double large primes; DL: D with structured Gaussian elimination and Lanczos; DL1 is post hoc. Sources: `research/quartic_n4_20261019/`, `research/quartic_n4v2_20261020/`.

Primes	Walks	Quantity	Value
$p = 7$ to 31	exhaustive	pass fraction	0.53–0.61 (random curves: 0.1857)
		neighbour	0.74–0.98, at 1.5–3.6 steps
$p = 2^{10}$ to 2^{20}	sampled	walk-cost exponent	-0.14 $[-0.80, 0.95]$
$p \equiv 3 \pmod 4$	successful	walk-cost exponent	1.13 $[1.00, 1.26]$

Table 26.2: The 2-isogeny walk to curves on which Smith's construction succeeds. Pass fraction: share of descended curves passing Smith's full test. Neighbour: share of failing curves with a passing curve in their 2-isogeny component, and the mean distance to it. Walk-cost exponent: in p; the first is the frozen fit, which labels the walk cost $o(p)$; the second is post hoc, over successful walks only. Source: `research/smith_elliptic_walk_20261018/`.

Smith's isogeny and the walk to it. The other genus-3 type of the census is hyperelliptic. Smith's $(2, 2, 2)$-isogeny takes it to a plane quartic when his Propositions 2 and 4 pass. The walk study looked for curves on which Smith's construction succeeds by walking 2-isogenies on the elliptic side (`research/smith_elliptic_walk_20261018/`; Table 26.2). The preregistered expectation of about 0.186 for the pass fraction is falsified, upward. A labelled post hoc analysis attributes the frozen fit's $o(p)$ label to censored walks at the small primes, and reads the $p \equiv 3 \pmod 4$ exponent as $\tilde{\Theta}(p)$. At $p \equiv 1 \pmod 4$ most walks stay inside small trapped sets.

A second study, also open and not yet merged, maps divisor classes through Smith's isogeny and runs the whole route at toy sizes. The program's closing note lists what stays open (`research/plane_quartic_program_20261018/CLOSING.md`).

Joux–Vitse: two arms of one cover. The Joux–Vitse cover of $E(\mathbb{F}_{p^6})$ (full rational 2-torsion, a weak class) has been measured with two solvers, and their comparisons with rho do not agree. The *plain Nagao arm* with a sparse-FGLM linear stage is 1.59–1.98× a pooled rho constant at $p = 1009$ and closes as about p^{-2} (`research/jv_cover_p_scaling_20261024/`; §31.5); it is a lower-bound comparison and current. The *sieve arm* (`jv_cover -exp sieve`) measured 0.024 $[0.019, 0.031]$ of a strong rho at $p = 1009$ and 0.0042 at $p = 1823$ on 19 October, but its 20 October review withheld every comparison (no numeric acceptance for the controls, uncharged rho hashes, incomplete accounting; `research/jv_sieve_strong_rho_20261019/`); a replication with fixed thresholds failed its acceptance rule and is in an open pull request, and a prospective re-run is registered and has not run.

The sieve arm against a matched negation rho (2 November). A frozen protocol re-measured the sieve arm at $p = 1511$ ($\ell = 2^{61.4}$) against a matched negation-map rho: 16 new rho runs pooled with 126 earlier ones under a frozen homogeneity rule, with the sieve rows of the 19 October study unchanged and reproduced by a second build. The sieve route costs 0.00647 $[0.00545, 0.00799]$ of rho in group additions and 0.0156 $[0.0131, 0.0192]$ in textbook $\mathbb{F}_p$ multiplications (seeds 17–24, 95% bootstrap intervals). Rho is therefore 155× $[125, 184]$ and 64× $[52, 76]$ more expensive. Charging the cover's unemitted F4, descent and linear-algebra work at its upper bound leaves the ratio at 0.107–0.123 for

$w = 0.1$ and 1. The order holds across the band in every cell, for both seed sets and both rho references. An independent validator passed the comparison (`VAL-20261103-c31c76`), with these limitations:

- the replay is sampled: 1 of the 16 new rho runs, 1 of the 126 pooled runs and 2 of the 16 cover rows;
- the cover's F4, descent and linear-algebra additions are not emitted, so only an upper-bound proxy prices them;
- memory is not reported for either arm;
- the run was local, not on the isolated lab.

The validator classifies the result as accounting, not an improvement to any method: it is a matched measurement of an existing solver (`research/jv_sieve_negrho_20261102/`). It is the programme's one validated measurement of index calculus below a matched rho. It holds on the weak class only, says nothing about prime-order curves, and is not extrapolated. The roughly thirtyfold gap between the two arms is still unexplained.

A 9 October whole-route replay from a non-weak challenge curve, through an isogeny walk to a weak curve and its cover, read 0.058 of a pooled, unmatched rho at $p = 1009$ on one seed per prime, with an earlier revision of the solver whose source includes the sieve module (`research/jv_cover_e2e_30bit_20261009/RESULT.md`). The roughly thirtyfold gap between the arms at $p = 1009$ is unexplained on `main`. The cover-decomposition note still states a measured sieve crossover near $p \approx 430$ (`research/notes/index-calculus/RESEARCH_COVER_DECOMPOSITION_LEDGER.md` §11) with no erratum; it should be read through the review. Leaderboard Table D, the historical study, is withheld. On the method side, the sparse-FGLM stage made the plain arm 1.28× cheaper end to end (`research/jv_cover_sparse_fglm_20261018/`); relation collection is 99.99% of S (`research/ic_cover_leads_20261018/`); depth-1 isogeny classes hold no weak curve over 4,871,062 curves, while 86–95% of deeper classes hold one (`research/jv_weak_class_law_20261007/`); and the $(n-1)$-decomposition variant is $4.5 \cdot 10^7$–$1.8 \cdot 10^8\times$ rho (`research/joux_vitse_n1_20261015/`).

Cover censuses. The minimal cubic GHS genus over an isogeny class follows $\#E(K)$ mod 4: 13 for odd order, 5 for 2 mod 4, 3 for 0 mod 4, and genus 3 is the only value below rho's $q^{3/2}$ (`research/ghs_cubic_cover_census_20261012/`); at $n = 5$ the floors are 33, 33, 25 and 5 by 2-torsion type (`research/ghs_cover_census_n5_20261013/`). No deployed prime-field or prime-degree binary curve has a usable subfield; only the RFC 2409 Oakley groups 3 and 4 are reachable, at genus 15–16, where exponent models disagree in sign and the measured genus-16 relation phase alone is 5.2 bits slower than rho (`research/cover_route_prime_binary_20261008/`, `research/ghs_oakley_extended_20261023/`). All 15,791 cover maps in the registry were assessed: 165 binary maps have genus-3 arithmetic and the 15,626 odd-characteristic genus-2 covers lack a Jacobian implementation (`research/cover_transfer_assessment_20261013/`). An inventory of about 90 cover and decomposition studies found every route on prime, Koblitz and prime-degree binary curves closed, negative or constant-factor (`research/cover_decomposition_gap_search_20261018/`). Smith's $(2, 2, 2)$-isogeny route was reproduced natively (full success 0.18638 against a theoretical 0.18574 over 10^6 curves) and its sign bias shrinks like $1/\sqrt{p}$ (`research/smith_prop24_native_20261014/`, `research/smith_sign_scaling_20261016/`).

Volcano levels and generators. A 403,168-run census found the level effect equivalent in 33 of 35 comparisons and different in none (`research/volcano_ecdlp_census_20261017/`), and a large-sample rho study found the floor/crater ratio 0.998 $[0.986, 1.010]$ (`research/conductor_gap_rho_large_sample_20261018/`). The one real level effect is the automorphism group: $\sqrt{3}$ on the $j = 0$ crater, confirmed at 1.86–1.92× (`research/drastic_prime_conductor_gap_round3_20261012/`). At $\ell = 2$, floors give 5.8% fewer relations than craters, and 99% of the contrast is explained by the base's cofactor classes (`research/isogeny-volcano-cost-v2_20261017/`, `research/generator_effect_ic_20261019/`).

Other transfer lines. Index calculus on genus-2 to -4 Jacobians over $\mathbb{F}_p$ (the ADH–Gaudry attack, measured against rho on the same Jacobian) read below a distinguished-point rho in early rounds, but the reference was later found slow and the ratios are lower bounds, unranked under the band (`research/notes/index-calculus/RESEARCH_HYPERELLIPTIC_IC_RHO.md`); later rounds are in review. A weak-curve atlas of 922 exact models and 36 published cases ranks covering-model and pairing signals but leaves the selected-subgroup advantage unmeasured (`research/weak_curve_leaderboard_20261012/`); three complete isogeny volcano graphs (24,823 vertices) were built and their graph-wide IC surveys halted at checkpoints (`research/full_volcano_graphwide_ic_20261017/`). Weil restriction of ECC2K-130 to extension fields costs at least 2^{131}, $2^{70.19}$ times rho (cluster "transfer-boundaries").

Named obstructions. Blocked requirements are kept open as problem records (`problems/README.md`), among them the complete-cost calibration of root-finding IC (`problems/rootfinding-complete-cost-calibration.md`), with exit criteria rather than abandoned.

Part IV

The rho side

27 The generic floor and the index-calculus boundary

Every distance in this treatise is a ratio to Pollard rho. This part turns to the denominator. The present chapter asks what generic arguments say about index calculus itself: which constructions cannot reach rho no matter how they are engineered, where the generic track ends, and which extra inputs (auxiliary powers, advice tables) change the answer and by how much. Chapter 28 then asks how well the programme measured rho, and Chapter 29 how rho itself was varied.

Standing of the evidence. No proof-carrying claim in `pcc/claims/` covers a result of this part; the claims on `main` are leaderboard rows and Curve25519 autolab rounds. Every measured figure below is therefore the producer's reported value. Where a sealed ecbench bound (`ECBND1h...`), an audit receipt or a `VALIDATION.json` exists, it is named; "validated" is used only for the last. The theory has the tiers of Part II: the collision floor of §27.2 is a derivation checked by Monte Carlo, not a theorem in the repository's proof system.

27.1 The square-root floor and its scope

Shoup's and Nechaev's generic-group bound says that an algorithm which touches a prime-order group of order r only through the group law and equality tests needs $\Omega(\sqrt{r})$ group operations. The programme records it as a closed obstruction for every generic lead: smarter walks, kangaroos, baby-step giant-step variants, grumpy giants, multi-walk and GPU variants all move the constant and never the exponent (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-generic-sqrt-lower-bound.yaml`). An efficiently computable automorphism group of order A divides the constant by $\sqrt{A}$, which gives the floor $S_{\text{floor}} = \sqrt{\pi/(2A)}$ of §4.2.

The bound has a precise scope, and the treatise's other chapters sit on both sides of it.

- *Inside it* are rho, BSGS, the kangaroo and the pair claw, and also every index calculus whose oracle can be simulated from group operations: the table, pair-table and root-finding oracles of the prime generic track (§27.3) and the oracle-augmented proposals of Chapter 8.
- *Outside it* are methods that read the representation: Gröbner and SAT decompositions that receive the factor-base constraint inside the system, Weil descent, and the cover attacks of Part II. Only these can change the exponent, and none has done so on a generic prime-order curve (Chapter 32).

Measured exponents. The sealed frontier (`docs/bounds/FRONTIER.md`, frontier `ECFR1hac90861c5e8a`, procedure 2, 130 bounds) fits GAE $= C\,r^{\alpha}$ per method. No ranked generic arm fits an exponent above 0.53, and none sits below its floor beyond interval noise. On the prime toy tier ($\log_2 r = 15.7$–25.3), `bsgs.negation` fits $\alpha = 0.529$ $[0.486, 0.595]$ (`ECBND1h9d01a5c2ae56`). The rho arms fit *below* 1/2 there: `rho.negation` 0.398 $[0.315, 0.475]$ and `rho.plain` 0.431 $[0.384, 0.488]$. That is a finite-size effect of fixed set-up (§28.2), not a sub-square-root walk; on the Koblitz medium tier ($\log_2 r = 32.1$–47.2) the strong signed-Frobenius rho fits 0.451 $[0.399, 0.503]$, with 1/2 inside the interval. A sub-$\frac{1}{2}$ fit at toy size is therefore never read as an exponent gain.

27.2 The collision floor for factor bases

Chapter 2 explains why balanced two-summand index calculus loses ground to rho as $r^{1/6}$. A sharper question is whether *any* factor-base construction can reach rho's constant while staying generic. The Curve25519 parity study (`research/curve25519_rho_parity_bound_20261104/`) answers it with a counting argument in the generic group model.

The model. A factor-base element obtained by a non-group operation, such as decoding an abscissa, has an unknown logarithm; in the generic model it is a fresh element. The algorithm learns only from equalities between computed elements, and each equality is one linear equation over $\mathbb{Z}/r$ in the F unknown

F	rows needed	model	Monte Carlo	measured (`wwalk`, $F = 2$)
0 (rho)	1	1.000	1.000	
1	2	1.500	1.517	
2	3	1.875	1.889	1.82–1.94× rho-neg
3	4	2.188	2.196	

Table 27.1: The collision floor: cost of a single-target index calculus with F unknown base logarithms over rho, as modelled, as simulated (4,000 trials at $r = 131{,}203$, seed 20261104, `research/curve25519_rho_parity_bound_20261104/data/sim.json`), and as measured on the Curve25519-family autolab incumbent `wwalk`, a walk table on a base of two abscissae. The measured column spans $k = 24$–32 and two seed regimes (`research/curve25519_rho_parity_bound_20261104/README.md`, §2); unverified assertions.

base logarithms and $\log Q$. An equality that involves no base element already yields $\log Q$: that is rho. A solve that uses its base needs $F + 1$ independent equations.

Proposition 27.1 (collision floor; derived in the generic group model, checked by Monte Carlo)**.** *Let n walk points fall uniformly into the $r/2$ negation classes, so that colliding pairs arrive as a Poisson process of mean n^2/r. The k-th collision appears, in expectation, at*

$$\mathbb{E}[n_k] = \sqrt{r}\,\frac{\Gamma(k+\frac{1}{2})}{\Gamma(k)}.$$

Rho needs $k = 1$, which gives $\sqrt{\pi r/4}$, the negation-map constant. A single-target index calculus with F factor-base elements of unknown logarithm needs $k = F + 1$, and so costs at least

$$\frac{\Gamma(F+\frac{3}{2})}{\Gamma(F+1)\,\Gamma(\frac{3}{2})}$$

times rho: 1.500 *for* $F = 1$, 1.875 *for* $F = 2$ *and* 2.188 *for* $F = 3$.

Derivation. The Poisson count of pairs after n draws has mean $t = n^2/r$, so the k-th arrival time T_k is $\Gamma(k, 1)$-distributed and $n_k = \sqrt{rT_k}$. Then $\mathbb{E}\sqrt{T_k} = \Gamma(k+\frac{1}{2})/\Gamma(k)$, and the ratio for $k = F + 1$ against $k = 1$ is the displayed quotient. Since $\Gamma(F+\frac{3}{2})/\Gamma(F+1) \sim \sqrt{F}$, the floor grows as the square root of the number of unknown base logarithms (this asymptotic is ours, not the source's). Table 27.1 sets the model against the study's Monte Carlo and the lab's measured incumbent.

What it explains. The autolab's generic incumbent converged near 1.8× rho, and the floor says why: it is the $F = 2$ constant, not an implementation shortfall. Earlier fixed-table readings of 1.43–1.55× were single draws of the walk table, as the autolab's own correction I1 found. The Monte Carlo sits about 1% above the model, which the study attributes to the discreteness of pair counts.

What parity would need. The floor leaves three exits, and the study closes or prices each.

- $F = 0$. A base whose every logarithm is known, such as multiples of G, needs one collision and reaches rho's constant. It is rho or BSGS relabelled, which the success criterion excludes.
- *Rows without collisions.* An oracle that finds relations among base elements without group operations leaves the generic model, and its native work is then priced across the band. The best measured such oracle, F4 with trace replay on the $t = 4$ isogeny base, is $1.83 \cdot 10^4\times$ rho at $w = 0.1$ and its modelled cost scales as $r^{1.03}$ against rho's $r^{1/2}$.
- *Algebraic oracles, screened.* The follow-up screen (`research/curve25519_rho_parity_bound_20261104/ALGEBRAIC_SCREEN.md`) prices every known family at $k = 20$ against $S(w) = (F+1)(r/C)(1 + wc)/\sqrt{r}$, for coverage C and c native operations per trial, with set-up and linear algebra uncharged. The measured families fail on coverage before native cost counts. The cheapest design is a full-coverage $m = 2$ base of 256 points with one root-find per base point: 0.44× rho at $w = 0$, but 265× at $w = 0.1$ and $1.6 \cdot 10^3\times$ at $w = 1$. Parity would need $c \le 14$ native operations per trial at $w = 0.1$ and $c \le 3$ at $w = 1$; the whole $w = 1$ budget is about 1,000 GAE. All values are model outputs from measured or derived inputs.

Limits. The argument assumes uniform walk points. Real walks carry an overhead factor; measured negation rho sits at 0.98–1.08$\sqrt{r}$ on these curves against the model's 0.886$\sqrt{r}$, so a ratio compares like with like only when both arms carry the same factor. Linear algebra is uncharged, which favours index calculus, and dependent rows would only raise the count. The study states that the bound holds for one target and for many, but amortising base logarithms over many targets is outside its single-target question. Read with Chapter 2, the floor is the reason the leaderboard's success criterion counts each improvement over the index-calculus baseline separately from the distance to rho: a generic factor base can approach the floor, and the floor is above rho.

27.3 Where the generic track ends: the BSGS boundary

The owner's decisions of 29 October (`research/prime_ic_program_20261029/DECISIONS.md`, ruling R1) split prime-field index calculus into a generic and an algebraic track (Chapter 32). The ruling's argument for why root finding is generic is worth stating because it is the template for every later classification. At $m = 2$, `rootfind-subtract` fixes $x_1 = c$ for each c in the base, solves the remaining quadratic $S_3(c, X, x_R)$ through its roots $x(R \mp P_c)$, and tests each root against a hash index of the base. It returns the solution set a Gröbner solve of $\langle S_3, F_V(x_1), F_V(x_2)\rangle$ would, but its cost never depends on F_V and it runs identically on a random base. It is therefore a generic algorithm, subject to $\sqrt{r}$.

The k-sum law tends to BSGS. The negation-folded k-sum table (`research/prime_ksum_table_20261028/`) stores $2^{k-1}\binom{A}{k}$ signed k-sums of a base of size A, costs one probe per walk step, and has the generic cost law $2^{k-1}\binom{A}{k} + c_k Ap/(2^k\binom{A}{k})$ with exponent $k/(2k-1)$: 3/5 for $k = 3$, 4/7 for $k = 4$, falling towards 1/2 as k grows and never below it. The limit is baby-step giant-step, and the study says so. Its measured gains are in Chapter 32.

Generic relabellings measured as arms. Three studies measured generic decompositions as index-calculus arms and set each against the same-binary negation rho.

- *Target-blind known-log two-sum* (`research/ecbench_known_log_targets_20261021/`). Its cold counted cost was below the abscissa pipeline's at 16, 20 and 24 bits on both panels and throughout the band (paired ratio at $w = 0$ from 0.902 of the pipeline's at 12 bits to 0.064 at 24 bits), and the 12-bit pair is unresolved. Negation rho stayed cheaper at every size: at 24 bits rho over known-log is 0.4293, 0.3998 and 0.3025 at $w = 0$, 0.1, 1. Wall time is L0 and the receipts are same-host replays.
- *Curve25519-family $m = 3$ decompositions* (`research/curve25519_m3_generic_20261102/`, L1, counted units). On holdout targets the `ksum:m=3` oracle cut the incumbent's counted GAE by 24% at $k = 20$ and 40% at $k = 24$ across the band, and was unresolved at $k = 28$. The distance to rho stays 5.7–6.0× ($k = 20$), 6.2–6.5× ($k = 24$) and 11.2–17.7× ($k = 28$) across the band; the `mitm:m=3` oracle is worse than the incumbent at every N. No `VALIDATION.json` exists. The collision floor of §27.2 is the theory for this family.
- *The pair claw* of a sibling repository is randomised BSGS on signed-Frobenius classes; it is treated with the memory methods in §29.6.

The end-to-end census. The frontier refresh of 31 October (`research/ecbench_e2e_frontier_20261031/`) inventoried 74 live ecbench method identifiers (31 index-calculus, 32 rho or transfer, seven BSGS, two kangaroo, two claw) and eleven nested solver engines, and found that no admitted result on the generic prime, random-binary or medium Koblitz ladders beats matched cold one-target rho. Its two named exceptions are both classified away from the generic boundary:

- at $n = 29$, on one column, the Frobenius-layout pipeline is 0.366 [0.241, 0.523] of signed-Frobenius rho, but at one column the repository classifies that pipeline as Frobenius-folded BSGS, and the crossing rests on rho's fixed set-up at $\log_2 r = 15.4$ and widens with r;
- the GHS-weak plane-quartic family is below matched rho in its own comparison (Chapter 32), a frontier for that family only.

A current-binary audit reproduced only 72 of 144 historical F4/F5/F6 rows, so those solver counts stay bound to the binary that produced them. The census's 92 planning shards had not run at the time of writing.

Engineering towards parity on a toy suite. Before ecbench, a September study pursued full-cost parity with signed-Frobenius rho on ten toy cases (`research/rho_parity_20260915/`), with frozen

Arm	$w = 0$	$w = 0.1$	$w = 1$	fitted α
comb, $r - 1$ (48 pairs)	0.552 [0.455, 0.651]	0.528	0.418	0.450 [0.406, 0.473]
comb, $r + 1$ (48 pairs)	0.279 [0.216, 0.419]	0.259	0.165	0.428 [0.389, 0.460]
naive, $r - 1$ (40 pairs)	7.660 [6.33, 9.93]	6.983	3.938	0.441 [0.388, 0.497]
`rho.negation` ($r - 1$ curves)				0.496 [0.431, 0.611]
`rho.negation` ($r + 1$ curves)				0.513 [0.386, 0.636]

Table 27.2: Cheon's DLP with auxiliary inputs against matched negation rho: pooled paired ratio of sums over 36–56 bits at three band weights, and fitted exponents. The oracle price moves every pooled ratio by at most 0.003. Source: `research/cheon_ecbench_20261022/REPORT.md`; not validated.

all-cases gates. Five engineering iterations ran; fresh chart-over-rho runtime ratios after the fourth were 1.08–2.21 on a shared host, and the verdict is "rho parity is not established: the frozen all-cases gates are not met" (`research/rho_parity_20260915/RESULTS.md`).

27.4 Auxiliary inputs: Cheon's algorithm

Chapter 21 records in one sentence that Cheon's attack beats rho from 24 bits up, but only for the discrete logarithm with auxiliary inputs. That problem is the largest generic line of the programme absent from the treatise, and its boundary with the plain ECDLP is exact.

The floors. Given $G, [\alpha]G, \ldots, [\alpha^d]G$, a generic algorithm needs $\Omega(\sqrt{r/d})$ operations, and Cheon's $r \pm 1$ algorithms attain $\sqrt{r/d} + \sqrt{d}$ when $d \mid r - 1$ and $\sqrt{r/d} + d$, from $2d$ powers, when $d \mid r + 1$ (`research/torsion_auxiliary_inputs_20260918/README.md`).

The acquisition obstruction. From $(P, [x]P)$ a generic algorithm forms only $[a + bx]P$. The point $[x^2]P$ lies outside that span, and computing it is the square-DH problem, which has the same $\Omega(\sqrt{r})$ generic bound. Self-supplied powers therefore cost rho-scale work, and Cheon never touches the ordinary ECDLP generically (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-cheon-aux-acquisition.yaml`, decision `DEC-20260802-204`, closed as a theorem). A non-generic way to compute $[x^2]P$ below $\sqrt{r}$ would reopen it.

Cheon as ecbench methods. Three registered methods, `cheon.pminus1_comb`, `cheon.pminus1_naive` and `cheon.pplus1_comb`, are charged in ecbench's counted unit (`research/cheon_ecbench_20261022/REPORT.md`). The leaked powers come from a planted static-DH oracle whose queries ($d - 1$ for $r - 1$, $2d - 1$ for $r + 1$) and server work are counted outside `total_gae`; a method never sees the scalar, and a public target is refused. Bounds of querying arms live in separate *oracle domains* on the frontier, with oracle queries as an extra axis of dominance. Table 27.2 gives the pooled distances to matched negation rho over 36–56 bits (320 runs at L0, 320 of 320 replayed exactly).

The comb arms' order against rho holds across the twelve-cell band of native price and oracle price, so they are ranked below rho in operations; rho stays on every oracle domain's frontier because it makes no oracle queries. *No exponent separation is established*: the Cheon and rho exponent intervals overlap. Per size, the $r - 1$ comb is cheaper than rho with an interval excluding 1 at 36 and 44–56 bits and dearer at 32 and 40. The original sessions mix the 32-bit toy curve with medium-tier curves, so frontier procedure 2 cannot rank them; a re-run on the medium tier alone is frozen as a protocol with no results yet (`research/cheon_ecbench_medium_20261106/PROTOCOL.md`).

A retraction: Cheon/rho on the P-224 shape. A study built prime-order curves of 32–56 bits whose $r \mp 1$ smooth parts have P-224's ratio $\log d / \log r$ (0.12943 and 0.21106, within 0.015) and ran 126 Cheon and 84 rho solves on them (`research/cheon_p224_shape_20261022/REPORT.md`). Its preregistered gate H3 failed: 5 of the 84 sealed rho runs have S outside $[0.3, 4]$, so under its protocol the rho reference is unreliable, and on 23 October **every Cheon/rho ratio, crossing and advance reading of the study was withdrawn**. What stands is the Cheon-only scaling:

- fitted exponents 0.447 [0.417, 0.477] ($r - 1$) and 0.413 [0.379, 0.447] ($r + 1$), against formula values 0.435 and 0.395;

Setup	Curve	top exponent q	best d $(r-1)$	bits below rho
EIP-4844 KZG, mainnet	BLS12-381	4,095	3,648	5.74
EIP-4844 KZG, largest	BLS12-381	32,767	30,531	7.27
Zcash Sapling	BLS12-381	4,194,302	4,142,391	10.82
Filecoin	BLS12-381	268,435,454	265,113,024	13.82
Aztec Ignition	BN254	100,800,000	100,663,296	13.12
Perpetual Powers of Tau	BN254	536,870,910	536,259,126	14.32

Table 27.3: Cheon's generic floor at constant 1 for deployed powers-of-tau setups, against negation rho ($2^{127.25}$ on BLS12-381, $2^{126.62}$ on BN254). Floors, not measurements; the practical $r-1$ algorithm ran at about 20× its floor over 24–48 bits (`research/torsion_auxiliary_inputs_20260918/`), and applying that constant at 255 bits is extrapolation. Source: `research/srs_auxiliary_census_20260927/results/srs_census.md`.

- formula costs at the P-224 shape of $2^{97.50}$ $(r-1)$ and $2^{88.36}$ $(r+1$, about $2^{48.3}$ sequential powers needed) against rho at $2^{111.83}$;
- extrapolations of the free power law to 224 bits of $2^{101.71}$ $[2^{96.3}, 2^{107.1}]$ and $2^{95.59}$ $[2^{89.3}, 2^{101.8}]$, which are models, not measurements.

Which deployed setups release powers. Two censuses separate the question "how far could Cheon go" from "what does any protocol supply".

- *Standard curves* (crypto-autoresearcher run `RUN-AUXIN-fd1edc`, zero Cheon executions). The best exponent $E(r) = \min_d \log(\sqrt{(r \pm 1)/d} + \sqrt{d})/\log r$ is about 0.254 for P-256, secp256k1, BN254 and BLS12-381 (each at d between about $2^{126.8}$ and 2^{128}), 0.394 for P-224 $(r+1)$, and at most 0.452 for P-521. These values require degree-d supply; they are not vulnerabilities. Joining them with a typed supply roster is blocked by a refused protocol amendment (`DEC-20260923-a4c3d5`).
- *Powers-of-tau setups* (`research/srs_auxiliary_census_20260927/`, Table 27.3). This is a reproduction of the ethresear.ch analysis (thread #6692), not a discovery. It adds the EIP-4844 ceremony, exact setup sizes from each project's specification, and the largest admissible $d \mid r-1$ instead of the largest power of two, which gains about 0.49 bits for Zcash Sapling, Filecoin and Perpetual Powers of Tau.

Open. A toy reproduction with memory charged and CRT composition of supplied residues is approved but has no runs; an earlier attempt over $\mathbb{F}_{10007}$ could not separate the closed forms by a factor of two (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-cheon-toy-reproduction.yaml`).

27.5 Low-Hamming-weight logarithms

A logarithm of bit length m and weight w admits a meet-in-the-middle solve in $\sqrt{\binom{m}{w}}$ with large memory, and the literature (Esser and May) reaches $2^{0.65m}$ with polynomial memory. This is a different problem from the uniform ECDLP, and no index-calculus row depends on it. Three leads are open and none has a result:

- collision walks on weight-w representations, whose first stage expired its claim before finishing, which is not a negative result (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-lhw-polymem-collision-walks.yaml`);
- the residue-guess price of mod-M representation filters in a prime-order group, an unproved multiplier with an approved experiment and no runs (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-lhw-modular-filter-price.yaml`);
- a preprocessing frontier for the low-weight class, unexamined (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-lhw-preprocessing-frontier.yaml`).

The kangaroo's closure (§29.6) names low-entropy secrets as the one case that would reopen it.

27.6 Advice and query bounds that bind index calculus

Chapter 3 states the Corrigan-Gibbs–Kogan bound $S\,T^2 = \tilde{\Omega}(\varepsilon N)$ for S bits of advice and online time T [8], and Chapter 8 records that table-based decomposition obeys $ST^2 \gtrsim N$. Both matter because the index-calculus set-up is advice: a root or pair table built before any target is seen.

Where the online rows sit against the line. A derived calculation (`research/ic_preprocessing_rho_equal_advice_20261011/REPORT.md`; no run on a curve) gave Bernstein–Lange rho [4] the index calculus's own root-table entries as advice, on the folded group of size $r/(2n)$, with online cost $\approx 2\sqrt{N/S}$ walk steps.

- At $n = 73$ the index calculus's online target stage takes 457,561 probes against about 9,500 BL steps ($\approx 48\times$), and sits $2^{13.2}$ above the generic $ST^2 \approx N$ line.
- At $n = 83$ it takes 8,845,441 probes against about 2,600 ($\approx 3{,}400\times$), $2^{25.4}$ above the line.
- The same calculation corrected its parent note: the note's 3,000–6,000× at $n = 73$ used the rank stage's mean probes per relation, and BL precomputation is $\sqrt{NS} \approx 1.2 \cdot 10^{11}$ steps at $n = 73$ (days on one core), not "seconds to minutes". For one target, neither beats plain rho.

The measured equal-advice controls that later retired every online win are in Chapter 33 (Table 33.1) and Table 32.2.

Two open generic leads. Two constant-level questions about the walk itself remain unexamined and are listed here because each would move every reference at once.

- *Equivalence classes beyond* $\mathrm{Aut}(E)$. Rho could walk on classes $\{[h]P : h \in H\}$ for a subgroup $H \leq (\mathbb{Z}/r)^\times$ larger than the automorphism group and gain $\sqrt{|H|}$, but only if a class representative is cheap; for multipliers that are not automorphisms every canonicalisation costs a scalar multiplication. The proposed obligation is an $\Omega(\log h)$ per-step canonicalisation bound for $h > 6$ (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-non-automorphism-equivalence.yaml`; hypothesis `H-EWALK-29cb90`, never tested).
- *Cheaper step primitives.* Tag tracing (Cheon, Hong and Kim) and point-halving steps on binary curves would lower native cost per step without changing the counted constant; neither has been measured here (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-cheaper-step-primitives.yaml`).

28 The matched rho reference

The validation gate of §33.2 asks every row for "a matched rho reference: the strongest available, in the same unit and the same binary, priced across the band". This chapter asks how well that reference is known: whether ecbench charges rho what theory says, how large its finite-size bias is, which rho wins among the references once the band is applied to them, which earlier references were weaker than they looked, and how the strong Koblitz reference was engineered and checked. It complements §4.2 and §4.3, which define the instrument and the band, and Chapter 33, which lists the corrections that matching forced on index-calculus rows.

28.1 Calibration against analytic constants

The calibration study of 2 October (`research/ecbench_calibration_20261002/`) tested whether ecbench's accounting reproduces the constants that seven generic methods' analyses predict. Four sessions ran on one Apple host at L0 (operation counts only): 1,008 prime and 864 Koblitz runs in round 1, and two follow-ups of 256 runs each. All 2,384 executions verified, and each of the four audit receipts reports 12 of 12 replays identical. Table 28.1 gives the predictions at each family's largest curve.

Two preregistered predictions failed as written, and the failures are kept. A follow-up on fresh targets traced both to sample size.

- *Rho's spread.* A rho run's cost has a coefficient of variation near 0.5, so a 24-run mean moves about ±20% at 95%. The [0.85, 1.35] band had been set on a point estimate without that spread; the interval contained the theory constant throughout.
- *The kangaroo is deterministic per target.* The three rounds of each target were identical, so the effective sample was eight targets. Its cost tracks $|k - r/2|$, from $S = 0.13$ at distance $0.017r$ to 2.48 at $0.347r$, and the eight targets averaged $0.119r$ against a uniform $0.25r$, about -2.6σ. The target derivation itself passes a Kolmogorov–Smirnov test over 20,000 draws.

The lesson the study draws binds every later protocol: a tolerance on a mean is set from that mean's sampling spread, and a method that is deterministic per target needs targets, not rounds. The calibration establishes the counted charge for these seven methods up to $r = 2^{39}$; it establishes nothing about wall time (all L0), about index calculus (which has no closed-form constant), or above 2^{39}.

28.2 Finite-size bias and the missing prime medium tier

A calibrated charge is not a calibrated constant at every size. Rho's reference implementations pay fixed set-up (jump tables, distinguished-point bookkeeping, walk starts), which is a visible share of a toy solve.

- On the prime toy tier, `rho.negation` measures 1.453 [1.184, 1.772] times its floor with $\alpha = 0.398$ [0.315, 0.475] (`ECBND1h67e22de83dca`), and `rho.plain` 1.920 [1.633, 2.221] (`docs/bounds/FRONTIER.md`).
- On the Koblitz toy tier the strong reference is 16.8× its floor with $\alpha \approx 0.13$, because its set-up is 71% of its total at $r \le 2^{21}$ (`research/ecbench_bounds_challenges_20261006/`); every Koblitz toy method sits 5–10× above the floor.
- At 28–40 bits the same `rho.negation` measures 1.075–1.146× its floor on four fresh prime-order curves (`research/rho_negation_audit_20261020/`, L0, 64 targets per curve), and the strong Koblitz rho 1.089 [0.927, 1.261] on the medium tier (`ECBND1hcf69c27b4978`).

The toy excess therefore shrinks at medium size, as fixed set-up predicts, but this is descriptive: no prime medium-tier rho bound exists in the plain ECDLP domains of the frontier. Medium-tier `rho.negation` bounds (1.114 and 1.225× the floor at 35.6–55.6 bits) exist only inside the Cheon oracle domains, fitted on one tier of a mixed session and so unranked (§27.4). The open step is to seal `rho.negation`, `rho.plain` and `bsgs.negation` at $\log_2 r = 32$–48 and fit the toy-to-medium ratio (`crypto-autoresearcher: knowledge/leads/ecdlp/generic/LEAD-RHO-matched-rho-calibration.yaml`).

Consequence. Every prime-field distance in this treatise is paired against rho on the same curve and in the same session, so the toy excess sits in the denominator. Where both arms carry fixed set-up the paired ratio is fair at that size; it does not transfer to another size, and an exponent fitted over toy sizes

Method	curve	mean S [95%]	theory	S/theory	verdict
`bsgs.textbook`	fp26	1.466 [1.350, 1.572]	1.500	0.977	pass
`bsgs.interleaved`	fp26	1.396 [1.197, 1.587]	1.333	1.047	pass
`bsgs.negation`	fp26	0.966 [0.850, 1.073]	1.000	0.966	pass
`rho.plain`	fp26	1.306 [1.075, 1.561]	1.253	1.042	pass
`rho.negation`	fp26	0.935 [0.681, 1.198]	0.886	1.055	pass
`rho.signed_frobenius`	f2m41	0.134 [0.106, 0.166]	0.138	0.970	pass
`rho.plain`	f2m41	0.993 [0.688, 1.307]	1.253	0.792	fail
`kangaroo.vow`	fp26	1.203 [0.684, 1.761]	2.000	0.601	fail
follow-up, 64 fresh targets × 2 rounds					
`rho.plain`	f2m41	1.240 [1.109, 1.389]	1.253	0.989	pass
`kangaroo.vow`	fp26	1.972 [1.759, 2.178]	2.000	0.986	pass
`rho.negation`	fp26	0.991 [0.884, 1.109]	0.886	1.119	reference

Table 28.1: ecbench calibration: mean S against each method's analytic constant at $r = 2^{25.35}$ (fp26) and $r = 2^{39}$ (f2m41); 8 targets × 3 rounds in round 1. Selected rows; source `research/ecbench_calibration_20261002/README.md`.

Domain	v1 page	v2 page (current)
prime toy	ops leader `bsgs.negation` (1.100 [0.978, 1.225]) named on a point estimate	no ops leader: `bsgs.negation` and `rho.negation` (1.453 [1.184, 1.772]) overlap, a tie set of six bounds; `rho.plain` (1.920) no longer dominated by `rho.negation`
Koblitz toy	leader a three-size tier-subset bound	no ops leader (tie set of three); both `rho.signed_frobenius` bounds have a large tail (max/median 54.2 and 11.7)
Koblitz medium	ops leader strong rho	ops leader strong rho, 1.089 [0.927, 1.261], $\alpha = 0.451$ [0.399, 0.503], 144 runs, now separating on all six degrees; memory leader the pair claw

Table 28.2: The bounds frontier before and after procedure 2. Source: `research/ecbench_frontier_v2_20261023/README.md`, `docs/bounds/FRONTIER.md`.

inherits rho's sub-$\frac{1}{2}$ slope. This is why Chapter 32 quotes distances per size and fits gap exponents only where the source does.

28.3 The sealed frontier, version 1 and version 2

Bounds are sealed records of constant and exponent per method (§4.2). The first frontier page named leaders that its data did not support. The rebuild of 23 October under frontier procedure 2 (`research/ecbench_frontier_v2_20261023/`) is a **correction**: no session ran; each of the 46 published v1 records stayed byte-identical and gained a fit-version-2 twin; only v2 bounds fitted on four or more curves of a whole session are ranked; and a leader is named only when its interval separates from every other method's on shared curves. Table 28.2 gives the change per domain.

The v1 exponent intervals also carried degenerate ends from one-size resamples; the kangaroo's $[0.179, 2.500]$, on which "α agrees with declared" rested, is $[0.401, 0.724]$ in its v2 twin. Prime medium-tier plain-ECDLP bounds are still absent (§28.2).

The first challenge verdicts. A bound moves the frontier only through a verdict against a sealed challenge on frozen workloads. The first two (`research/ecbench_bounds_challenges_20261006/`, epoch 1, 288 of 288 replays identical in each session, L0) were the reference challenges themselves.

- *Prime toy*: `bsgs.negation` against `rho.negation` reads **trade**, 0.8250 [0.725, 0.945] of the operations at 15.79 [13.38, 19.06]× the table. No single curve's interval excludes 1; the pooled 96 pairs do.
- *Koblitz toy*: the lean `rho.signed_frobenius` against the strong reference reads **advances** at the constant level, 0.2991 [0.235, 0.402] of the operations and 0.560 of the table. It prices the strong walk's toy set-up and says nothing above $r = 2^{24}$.

Neither moves an exponent. Both say which reference is "strongest" at toy size depends on what is charged, which the band makes explicit.

28.4 The band orders the references themselves

The pricing band of §4.3 was introduced to compare index calculus with rho fairly. Applied to the rho references themselves, it changes which of them is strongest.

- *Negation against plain, prime toy.* `rho.negation` is better than `rho.plain` at the charged cost, but the order flips at 0.34–0.55 GAE per unpriced canonicalisation, inside the band, so neither dominates (`docs/bounds/FRONTIER.md`). At 28–40 bits the negation audit finds the same: the order holds at $w = 0$ and 0.1; at $w = 1$ the negation walk's S is 1.89–2.03, above plain, and the ranking is unresolved on all four curves (`research/rho_negation_audit_20261020/band.json`). A real canonicalisation is one $\mathbb{F}_p$ negation and one comparison, far below 1 GAE, but it was not priced.
- *Signed Frobenius against plain, Koblitz toy.* Dominance holds at the charged cost and flips at about 0.123 GAE per uncharged operation; the kangaroo's dominance by signed-Frobenius and negation rho flips at 0.290 and 0.417.
- *Extension-field references.* Two references built for the extension-field programme use the full symmetry: `rho.gls` ($A = 4$, GLS twists over $\mathbb{F}_{p^2}$) and `rho.frobenius_orbit` ($A = 6$, $E/\mathbb{F}_p$ on $E(\mathbb{F}_{p^3})$), each on nine curves with 8 targets (`research/ext_endo_rho_20261023/analysis/`). At $w = 0$ `rho.gls` costs $S = 0.472$–0.921 against negation's 0.812–1.165, and at $w = 1$ 2.316–4.197 against 1.613–2.312; every pair on every curve is unresolved inside the band. `rho.frobenius_orbit` costs 0.482–0.692 at $w = 0$ against negation's 0.783–1.404, and is below negation across the band on four of nine curves; on three curves `rho.plain` is below negation across the band.

The rule this implies is already in the gate's wording: a row's "matched" reference is the strongest available *across the band*, and where the references themselves are unresolved the row is ranked against each. A row ranked against one weight's winner alone overstates index calculus at the other weights.

28.5 References that lacked the negation map

The negation-map audit of 20 October (`research/rho_negation_audit_20261020/`) inventoried every rho walk the programme used as a reference. Two lacked the map: `volcano-rho`'s native walk in this repository, and crypto-autoresearcher's `index_calculus/rho.py`. Both now walk on $\{R, -R\}$ by default. This is a **correction of earlier references**: any index-calculus ratio computed against either before 20 October was against a rho $\sqrt{2}$ weaker than the negation walk.

- *The matched reference delivers the saving.* On four fresh prime-order curves of 28–40 bits (512 records, 16 exact replays), pooled plain over negation is 1.371 $[1.257, 1.491]$, with $\sqrt{2}$ inside the interval.
- *Little of it reaches wall time* in ecbench's implementation: the median paired wall ratio is 1.140 (L0, descriptive).
- *`volcano-rho`.* On a 36-bit curve, 256 solves per arm, off over on is about 1.40 pooled over five arms, at 9–16% more nanoseconds per iteration (an extra branch, a field negation and a 16-entry cycle-window scan), so wall time improves by about 1.2×.
- *crypto-autoresearcher.* 640 paired solves at 16–32 bits give plain over negation 1.353 $[1.281, 1.425]$; its adapter now labels the old walk `rho.plain` and refuses it as a matched reference.

An earlier fix in the collaborative walker (`research/generic_dlp_correctness_20261017/`) found that its negation-map 2-cycle test compared an unflipped sum with a normalised point and never fired; dead trails fell from 353 to 16 across three curves once it compared after the flip.

References that were not the strongest. A register item of the repair framework (`research/r12_rho_ns_per_step_20261026/`) listed every committed ecbench spec that pairs an index-calculus arm with a rho arm: 745 specs in 68 directories, of which 37 directories pair index calculus only with a rho other than the strong reference. On Koblitz curves, counted ratios against `rho.negation`, `rho.plain` or the tuned signed-Frobenius walk are therefore not matched to the strongest rho; on prime curves `rho.negation` is the strongest reference. The same note explains an audit's 599 against 240 ns per step: the unbatched tuned rho is 5.48, 9.34 and 9.56× slower per step than the strong rho at $n = 41, 53, 61$ (L0, Apple arm64, A/A within ±7%), and within the strong algorithm the field backend matters by 1.07–1.76× with the hardware. The 599 ns figure being the tuned rho is a hypothesis supported by the size match.

Round	change	instr./step	native time
canon (8 Oct)	longest-zero-run rotation search; scratch reuse		1.40× per walk ($m = 83$), 1.46–1.61× ($n = 41$–61)
Q4 (16 Oct)	changes declared in two addenda	1,196 → 938	1.02× ($n = 53$), not established; 1.21× ($n = 61$)
canon2 (16 Oct)	six changes in the strong-rho source	938 → 781	
step3 (16 Oct)	target-feature walk; batched walk starts	781 → 718	
batchinv (22 Oct)	fixed-width batch inversion	707.2 → 679.7[a]	not established

Table 28.3: Engineering rounds on the strong Koblitz reference; instructions per walk step at $n = 53$. Sources: `research/strong_rho_canon_20261008/`, `research/rho_assurance_20261016/`, `research/rho_canon2_20261016/`, `research/rho_step3_20261016/`, `benchmarks/20261022-strong-rho-batchinv/` (carries a `VALIDATION.json`; its validator passes the instruction reduction and its findings contradict the native speed-up). [a]From the `main` of 22 October, which already read 707.2.

28.6 Engineering the strong reference

Chapter 31 records that the strong signed-Frobenius rho fell from 1,196 to 679.7 instructions per step with no established native gain. Table 28.3 gives the rounds; every round left walks, step counts and recovered logarithms identical, so counted-unit records are unchanged and only instruction or wall comparisons move.

The assurance round's Q3 profile is the reason the rounds were worth doing: one step cost 1,196 instructions, about 57 field multiplications, against a floor near six or seven, and half of it was canonicalisation. A separate optimisation of ecbench's standing method (`research/ecbench_rho_optimized_20261014/`) cut complete-solve Callgrind instructions to 0.802155 of the trajectory-identical legacy path, with charged GAE exactly 1.0000 over 240 pairs and 720 of 720 replays identical; its native wall gate is unknown. The pattern is the same in every round: counted GAE is the comparison unit and does not move; wall-time comparisons against an older rho overstate index calculus by up to the round's factor (Table 33.3).

Prime-field and GPU walks. The batched affine step with Montgomery's simultaneous inversion is the prime-field engineering baseline. On P-256, a plain step costs 179 ns at 1,024 walks per inversion (`research/rho_walk_leads_20261014/`). A branch-free $\mathbb{F}_p$ 64-bit kernel took `prime_fast::rho_parallel` from 98.9 to 68.2 ns per step (`research/fp64_branchfree_20261016/`). An analytical model of counted operations times measured cycles per operation predicted a batched affine rho iteration to a median 1.8–5.5% on five fields at $B \geq 4{,}096$ walks after one fitted residual per field (`benchmarks/20261016-hwcost-rho-model/`). ECCp-131 throughput is in Chapter 21; GPU walks are in §29.9.

Harness overhead in small wall ratios. Until 22 October ecbench parsed its 4.7 MB curve registry on every lookup. Parsing it once cut the timed solve window of every rho, BSGS and kangaroo arm to 0.03–0.07× at 20 bits and 0.64–0.81× at 32 bits, against 0.96× for the index-calculus pipeline (`benchmarks/20261022-ecbench-harness-overhead/`, which carries a `VALIDATION.json`; counted results unchanged). Any small-size wall ratio recorded before then charged rho for about 4 ms of harness work inside a 0.3 ms solve, which flattered index calculus.

28.7 Formal assurance of the reference

The strong reference is the denominator of every Koblitz row, so the assurance round (`research/rho_assurance_20261016/`) checked its correctness formally as well as empirically.

- *Lean 4* (v4.34.1, no Mathlib): 34 theorems with no `sorry`, `admit` or `native_decide`, using only `propext`, `Classical.choice` and `Quot.sound`. They cover the least-rotation canonicaliser (and its bridge to the Rust zero-run mask), Montgomery batch inversion in one- and four-accumulator forms over any field, and the walk algebra: starts, jumps, canonicalisation and escapes preserve $P = aG + bQ$, and the collision formula returns the logarithm whenever $b \neq b'$.

- *TLA+*: a model of the solve over $\mathbb{Z}_R$ meets all 41 configurations' declared verdicts under TLC, including seven mutants that each fail the property meant to catch them.
- *Steps*: mean steps over $\sqrt{\pi r/(2A)}$ are 0.994, 0.922 and 0.943 at $n = 41, 53, 61$ on 400 fixtures each, with spread 0.50–0.51 against an ideal random mapping's 0.52.

The gaps are stated: the curve group is modelled as the scalar field, the GF(2^m) arithmetic and basis maps are unverified, and word overflow is argued, not proved. Four findings changed no answer; one (the documented fruitless-cycle length) was a documentation fix.

Two further Lean files bear on rho's invariance (§29.4). `research/conductor_gap_rho_large_sample_20261018/lean/RepresentationHashInvariance.lean` proves, in 11 theorems with no `sorry`, that transporting an r-adding walk by a bijection compatible with the operation preserves its trajectory and its cost at every budget, and that transport is a bijection of walk spaces; so over a uniformly random partition rho's cost distribution is the same on isomorphic groups, and a non-group invariant can enter only through the one fixed hash an implementation uses. `papers/isogeny-class-rho/lean/ScalarTransport.lean` proves (3 theorems) that a zero- and addition-preserving injective map transports answers. Neither formalises curves or isogenies.

28.8 Statistics version 2 and censored rho

The statistics paragraph of §4.2 describes the cluster bootstrap of the original comparisons. PR #2442 replaced it with a linearised $t(K-1)$ interval over K workloads and, more consequentially, kept timed-out and exhausted runs in the ratio as censored lower bounds instead of dropping them (`research/ecbench_statistics_v2_20261105/`; no new runs).

- *Width.* On uncensored rows the new interval is 13.1× wider at $K = 2$, 1.23× at $K = 12$ and about 1.0–1.06× at $K \geq 16$. Its simulated coverage is 0.957, 0.948 and 0.947 at $K = 2, 4, 8$; the old percentile bootstrap covered 0.858 at $K = 2$ in the study's own simulation.
- *Censoring.* Dropping exhausted runs had understated `ic-sub6-mitm` over strong rho by 92.9× (6.981 → 648.2 [419.6, 1001], a lower bound), and signed-Frobenius rho over `bsgs.negation` on a GHS panel by 60.4× (3.512 → 212.0), where 13 of the 40 workloads had exhausted rho runs.
- *The A/A gate.* Correcting a wall ratio by its A/A bias flipped one gating decision in the sample: `ic-sub6-mitm`'s ns per GAE ratio 1.248 corrects to 1.147 [0.997, 1.321], inside noise.

The rho-side lesson is the second item: an exhausted rho run is a failure of the reference, and a comparison that drops it reads the reference as cheaper than it was. Earlier "× rho" figures from sessions with censored runs on either arm are lower or upper bounds accordingly.

Independence. A same-machine replay under a different Rust toolchain changed the environment class identifier and would have passed the host-class part of an independent validation (`research/ecbench_independence_20261009/`). A separate hardware class that excludes toolchain, kernel and policy fields now rejects such a receipt; the fingerprints remain self-reported, so an external runner's custody record is still required.

29 Walk design, memory and many targets

This chapter collects the variations of rho the programme measured: walks on automorphism classes, the iteration function, rho across isogenous curves and on Jacobians, the memory-heavy generic methods, many-target and advice-table rho, and rho at deployed scale. Each matters to index calculus in the same way: it moves the reference.

29.1 Automorphism classes

A walk on the classes of an efficiently computable automorphism group of order A saves $\sqrt{A}$ in steps, which is the ceiling for any automorphism walk (§27.1); classes beyond $\mathrm{Aut}(E)$ are the open lead of §27.6. The Koblitz signed-Frobenius reference ($A = 2n$) is the subject of Chapter 15. On prime curves the gains are smaller, and their measurement needed a correction.

$j = 0$ and $j = 1728$, with an erratum. Round 2 of the conductor gap had matched its crater rows against negation rho only. A follow-up ran `rho.cubic` ($j = 0$, $A = 6$) and `rho.quartic` ($j = 1728$, $A = 4$) on the same twelve crater curves (`research/prime_rho_automorphism_20261015/`). Its **erratum of 24 October**: the binary left the class set-up (16–58 GAE per solve) out of each record's total. Corrected,

- `rho.cubic`/`rho.negation` is 0.647 $[0.543, 0.770]$ (was 0.611), with the prediction 0.577 inside;
- `rho.quartic`/`rho.negation` is 0.819 $[0.722, 0.929]$ (was 0.791), whose interval no longer contains the prediction 0.707.

Against the automorphism rho every crater IC/rho ratio of round 2 grows by about 1.55× ($j = 0$) and 1.22× ($j = 1728$); that adjustment applies pooled ratios and is not a joint re-run. The class-representative map is unpriced, so the ratios do not bound full cost. The volcano-census withdrawal of Chapter 33 is the same correction found from the index-calculus side.

The crater is where the fold lives. The volcano study (`research/isogeny-volcano-rho/`) measured that the automorphism-folded walk needs $1.89 \pm 0.09\times$ fewer operations on a $j = 0$ crater ($\sqrt{3}$ predicted) and $1.32 \pm 0.06\times$ on a $j = 1728$ crater ($\sqrt{2}$); in wall time that is 1.33× faster on $j = 0$ and 2.3× *slower* on $j = 1728$, where the tool had no tuned walk. Across a whole $j = 0$ class the folded walk cost 0.439 $[0.401, 0.481]$ of the plain walk against $1/\sqrt{6} = 0.408$, and on the Koblitz crater the fold wins 1.7× (τ) and 2.2× (τ and negation) in wall time although each folded step costs 41 squarings more (`research/isogeny-class-rho/REPORT.md`). Every curve the isogenies lead to has lost the fold (§29.4).

GLS and $\mathbb{F}_{p^3}$ references. The extension-field references and their band behaviour are in §28.4. On E25's ten GLS curves, Karatsuba products with Montgomery-batched norm inversions cut the matched $\langle \pm\psi \rangle$ walk's affine addition from 97 to 66.3–66.7 $\mathbb{F}_p$ multiplications, making it 1.453–1.463× cheaper while the index-calculus side moved only 1.0048–1.0057×; every row moved further from rho, to $6.39 \cdot 10^3$–$4.06 \cdot 10^5\times$ at $w = 0$ (`benchmarks/20261030-e27-walk-karatsuba/`, validated: independent replay 11 of 11 runs, with errata in the validator report). It is a correction to the reference, not an improvement.

29.2 The negation map and fruitless cycles

A negation walk creates fruitless 2-cycles $P \to P \pm R_i \to P$, and every negation reference needs an escape. The walk-leads study (`research/rho_walk_leads_20261014/`, Modal CPU, $N \approx 2^{32}$–2^{40}) measured the cycle rate and two representative rules.

- With a fixed or per-point rule the negation walk takes 1.37× $[1.31, 1.43]$ and 1.33× $[1.28, 1.39]$ fewer steps than plain, with 487 fruitless 2-cycles per 10^6 steps against the predicted $1/(2r) = 488$ for $r = 1{,}024$ jump points.
- Changing the rule per epoch gives 0.83× $[0.79, 0.87]$: worse than no negation map.
- Without an escape, every $\pm R$ arm repeats on a useless collision after 821–1,974 steps against the plain walk's 69,210 to a useful one at $N \approx 2^{32}$: 0 or 1 of 200 solved.

The distributed pool of the sibling ecc-challenges repository walks plain rho on all 23 non-ECC2K-130 campaigns. Its negation-class walk took 0.700 $[0.673, 0.728]$ of the old walk's steps with 0 of 1,800 trapped trails and about 6% per-step overhead, on toy curves of 2^{23}–$2^{34.5}$; it is in an open pull request as a post-hoc amendment, default off. These figures are as the lead atlas reports them; that repository was not available to this draft (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-pool-class-quotient.yaml`).

29.3 The iteration function

Chapter 31 records the best-of-$P \pm R$ lead: 1.129× fewer steps under distinguished points, offset by its cost per step. The same study tested three more walk changes, and the in-degree argument behind the first is the useful part.

- *Best of k candidates.* Choosing the best-scoring of k images gives a point of score rank u the in-degree ku^{k-1}; since a random mapping's collision time scales as $1/\sqrt{\mathbb{E}[\lambda^2]}$, the predicted saving is $\sqrt{k^2/(2k-1)}$, 1.155 for $k = 2$ and 1.512 for $k = 4$. A positive control hits it (1.151, 1.511); under distinguished points best of four gives 1.463 $[1.401, 1.527]$ but costs 2.4–2.6× a plain step, a net of about 0.6.
- *Lazy reduction before the distinguished-point test*: no change (0.998 $[0.995, 1.001]$); indexing the table on a lazy x costs 0.733×, about the predicted $1/\sqrt{2}$.
- *Correlated walks.* A shared index sequence makes useful collisions impossible (0.073×, with 176 of 1,000 trials censored); a precomputed pair table $R_i + R_j$ gives 1.027 $[0.982, 1.073]$, since one addition still advances one step.

Mixing. Teske's r-adding walks are near random for r of 16 to 20. In the isogeny-class study the 32-adding walk ran at 1.0097 of $\sqrt{\pi n/2}$ on 4,400 solves against an expected 1.016; a planted 4-adding walk read 1.291 against Teske's 1.137, and served as the study's positive control (`research/isogeny-class-rho-n45/`). On hyperelliptic Jacobians, mixing the divisor hash through a finaliser did not reach its registered gains (§29.5).

A compiled store-all schedule. A theoretical current-plus-archive walk $X_{t+1} = X_t + X_{t+1-d_t}$ with offsets compiled once per group order (`research/tiered_rho_compiled_walk_20260921/`) has, exhaustively over all 65,537 target logarithms at 2^{16}, an exact capped mean of 19,027,631/65,537 additions against 19,114,086/65,537 for the matched unguided schedule: 0.4523% fewer. It makes no wall-time claim.

29.4 Rho across isogeny classes and volcano levels

Chapter 8 proves that transport by an isogeny conserves index-calculus cost, and Chapter 26 records the volcano census's level effects for index calculus. The generic twin is a measured null: rho's cost does not depend on which isogenous curve, stratum or level carries the instance.

One instance on every level. `volcano-rho` carried one instance (p of 40 bits, $n = 272{,}748{,}451{,}513$) by verified Vélu 3-isogenies to all 80 curves of a depth-3 volcano and ran 100 paired-seed solves on each of 29 curves and three isomorphic controls (`research/isogeny-volcano-rho/`, benchmarks row `20261016-isogeny-volcano-rho`; 32-adding walk, no negation). Table 29.1 gives the levels.

A control of 5,000 solves found that redrawing the secret, the generator or both, or taking $k \le 2^{16}$, matches the walk-seed spread (every $|z| < 0.8$, Kolmogorov–Smirnov $p > 0.6$).

Whole classes. The isogeny-class study (`research/isogeny-class-rho/`, protocol frozen before the run) ran 85,425 solves on 200 curves of four classes: 104 curves of a 347,004-curve prime-field class (12 endomorphism-ring strata, reached through ℓ-isogenies up to $\ell = 61$), 45 curves of a $j = 0$ class, 40 curves of a binary class and the Koblitz class over $\mathbb{F}_{2^{41}}$. No curve, stratum, level, isogeny degree or kernel-field degree left the preregistered band, and the spread of means between curves is 1.1–1.2× what the single-solve spread predicts. The 409- and 1,721-isogenies of the Koblitz curve, whose division polynomials have degrees 83,640 and 1,480,920, were found through torsion abscissae, and the curves they reach cost 0.978–1.009 of the reference by stratum. The detector works: a planted negation fold read 0.63–0.77 of

Level	curves	solves	mean iterations ± 95%
0 (crater)	2	200	700,483 ± 47,849
1	6	600	681,766 ± 26,525
2	10	1,000	637,158 ± 20,410
3 (floor)	11	1,100	659,589 ± 20,993
all isogenous	29	2,900	659,263 ± 12,422
isomorphic control	3	300	674,042 ± 40,816

Table 29.1: Plain rho on one instance carried to every level of a 3-volcano (curve A, ramified crater), against $\sqrt{\pi n/2} = 654{,}547$. Per-curve means span 572,846–733,141, the spread of 100-trial means of a quantity with single-solve standard deviation about 341,000; per-step cost is 641–661 ns on every curve. Source: `research/isogeny-volcano-rho/README.md`.

the unfolded mean (expected 0.71). A 46-bit replica (4,400 solves on 38 curves) found pooled 1.0097 against 1.016 and Kruskal–Wallis $p = 0.38$; one generator family flagged there (KW $p = 1.2 \cdot 10^{-4}$) **did not replicate** on a new draw (KW $p = 0.97$), and generator-level heterogeneity is recorded as unresolved (`research/isogeny-class-rho-n45/README.md`; L0, no ecbench records, so it supports no comparison claim).

In ecbench. A preregistered ecbench study on six 24-bit volcanoes found rho's floor-over-crater ratio 1.022 $[0.983, 1.062]$ (90%) and BSGS identical on every curve and model; its v2 found rho 1.004 (`benchmarks/20261017-isogeny-volcano-cost/`, `benchmarks/20261017-isogeny-volcano-cost-v2/`). The large-sample floor/crater ratio of Chapter 26 agrees. The Lean result of §28.7 is the reason no effect was expected: over a random partition, isomorphic groups give rho the same cost distribution.

29.5 Rho on Jacobians

Genus-2 and genus-3 transfer rows (Chapter 26) are ranked against a distinguished-point rho on the same Jacobian, so that reference needed its own calibration. Three rounds on 18 hyperelliptic instances (genus 2–4, N from 1,321 to 16,790,591, 40 seeds each) tested it in counted group operations.

- *Negation fold.* Folded over plain is 0.67–0.89 at $N \le 6 \cdot 10^4$ but a median 0.968 at $N \ge 10^5$; the registered ≤ 0.80 was falsified. Plain S_{walk} was 1.82 and 2.10 at the two largest rows against an ideal 1.2533 (`benchmarks/20261018-hec-rho-negation/`).
- *Hash mixing.* Passing the divisor hash through a finaliser left the plain median at 1.406 and fold over plain at 0.960; both registered gains were falsified (`benchmarks/20261018-hec-rho-hash-mixing/`).
- *Branch width.* Full-width branch coefficients brought the median S_{walk} at $N \ge 10^5$ to 1.390 and the largest row from 2.100 to 1.481 (`benchmarks/20261019-hec-rho-branch-width/`).

The third round is an accounting correction against index calculus: the 8-bit reference walk was slow at large N, so the early genus-2 to -4 ratios that read below rho are lower bounds (Chapter 26).

29.6 Kangaroo, BSGS, the pair claw and store-all schedules

Kangaroo. On every sealed domain `kangaroo.vow` is dominated: 2.203 $[1.803, 2.527]\times$ its floor on the prime toy tier (`ECBND1h2e9e2024d8f3`) against negation rho's 1.453, and 15.558 [13.742, 17.597] on three Koblitz medium sizes against signed-Frobenius rho's 1.188 (`docs/bounds/FRONTIER.md`). Its calibration artefact of §28.1 came from its determinism per target. It is the right tool only for a known short interval or a low-entropy secret (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-kangaroo-interval.yaml`).

BSGS: fewer operations, more memory. `bsgs.negation` is in the prime toy tie set at 1.100 $[0.978, 1.225]\times$ the floor, the best point estimate, but with $0.502\sqrt{r}$ table entries, which at 256 bits is 2^{127}. The challenge verdict of §28.3 reads it as a trade: 0.825 of rho's operations at 15.8× its table. The frontier names no operations leader on the prime toy domain because the intervals overlap, and the memory rules BSGS out at cryptographic size.

The pair claw is BSGS. The four-point signed-Frobenius pair claw of cryptanalysis #175 was ported into ecbench and run cold against the strong rho on 48 one-target workloads over six Koblitz curves of $2^{32.1}$–$2^{47.2}$ (`research/ecbench_pair_claw_20261003/`; 960 runs verified, 12 of 12 replays identical, L0). Every factor-base logarithm is known, so it is a randomised BSGS on signed-Frobenius classes, of cost $(c + 1/c)/\sqrt{n}$ against rho's $\sqrt{\pi/(4n)}$.

- At its cost-minimising shape ($c = 1$) it costs 2.195 $[1.897, 2.546]$ strong rho, and at the shape the pull request ran at $n = 83$ it costs 32.65 $[25.29, 41.04]$. The ratio does not shrink with r.
- Without the Frobenius fold, `bsgs.negation` costs 6.969 $[6.063, 8.024]$ strong rho on the same workloads.

On the Koblitz medium frontier the claw at the pull request's shape is the memory leader ($0.004\sqrt{r}$ entries, 35.6× the floor), and the strong rho the operations leader; the claw's other sealed shape (2.390× the floor) is dominated by strong rho across the band. This is the $F = 0$ exit of §27.2 in practice.

Grumpy giants. Bernstein and Lange's two grumpy giants and a baby solve more targets per operation than BSGS or a random walk (0.71875 against 0.6753 of targets at $1.5\sqrt{\ell}$). A certified exhaustive ladder for its exact average-case constant ($\mathbb{Z}/\ell$, 2^{10}–2^{20}) has not run; its first stage met its support outcome at $\ell = 101,257$ (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-grumpy-store-all-schedules.yaml`). A certified constant below `bsgs.negation`'s would be a new toy operations leader, and could not beat rho at cryptographic size, because memory stays $\Theta(\sqrt{\ell})$.

29.7 Many targets: batch rho

The one-target rule of §4 exists because many-target figures were the easiest to misread. The Koblitz collection thread had quoted every "vs rho" as the per-target cost of 32 targets solved together, divided by rho solving one, and had counted each rho step and each stored table pair as one addition. The repository's multi-target margin rule makes batch rho at the same k the reference for such a figure, after Kuhn and Struik. A study frozen before it ran measured that reference and priced both units (`research/ic_rho_koblitz_20260923/`).

- *Batch rho.* k targets solved in sequence by the tuned signed-Frobenius walk with one shared distinguished-point table. Per-target cost at $k = 32$ over $k = 1$ is 0.210, 0.239 and 0.190 at $n = 41, 53, 61$, each interval covering the Kuhn–Struik 0.199; all 1,960 targets were recovered and verified.
- *Prices in the thread's unit.* A canonical step costs 2.74–2.92 units, Bailey et al.'s step 1.38–1.41, and a stored pair 4.5–6.1 (6.14 on the declared base).
- *The closest figure.* The thread's 1.17× reads 6.51× against batch rho as counted, 6.38× with rho's step and the table build both priced, 12.6× with Bailey's step on the canonical walk's count (a model), and 43.7× for one target alone.

This is an **accounting correction**: no index-calculus count changed. Its successor is Table C of the Koblitz leaderboard, priced against batch rho on the same 32 targets at 3.62–14.9× and still pre-gate (§15.6).

29.8 Advice tables

Bernstein–Lange precomputation [4] is the incumbent for every online comparison; its use as the equal-advice control that retired the online wins is in Chapter 33, and the September decisions on tables re-selected from the batch are in Chapter 30. The variants since then (`crypto-autoresearcher:knowledge/leads/ecdlp/generic/LEAD-RHO-advice-table-variants.yaml`) add little.

- A table of $T/2$ entries re-selected from the batch's solved walks falls short of a static table of T by -0.064 $[-0.080, -0.049]$ at 2^{24}, contradicting its claim; a re-scoped selection at 0.65–$0.75\,T$ met its target in three of four toy cells (preliminary).
- A basin-size instrument for table coverage failed its comparability gate.
- Carrying a table across isogenous curves is exact and costs at most 1.3% of a solve, but is never faster than solving at the source (`benchmarks/20261017-volcano-ecdlp-census/`).

29.9 Rho at scale

ECC2K-130. The programme's GPU rho on ECC2K-130 is published separately (`paper/ecc2k130/`); Chapter 15 prices the index-calculus routes to it (§15.7). For the reference the summary is this. The expected cost is $2^{60.9}$ iterations of a Frobenius-class rho, and the walk constant was measured at $2^{60.91}$–$2^{60.92}$. The σ-walk runs at 14.64 billion iterations per second on one RTX PRO 6000 in the audited campaign build and 15.88 in the best later build (a paired 1.0239× over 15.51, A/A drift below 0.1%, 300 of 300 reports replayed; `ecc2k130/SIGMA-TABLE.md`), about 230× the 2010 GTX 295. A roofline puts the ALU pipe at 91% of its peak and the carry-less speed of light at 27.4 billion per second on that card (`ecc2k130/ROOFLINE.md`). The paper separates the σ-walk from the table walk, whose 20.08 was measured under a cycle rule later found to leave cycles inconsistently; the exact rule runs at 8.24. A campaign stored 486 million distinguished points and $2^{56.68}$ iterations (5.4% of the expected work) before its fleet was reclaimed. Two iteration-function changes are derived and not rolled out, because each starts a new distinguished-point corpus:

- signed σ steps (`SIGMA_SIGN=2`) need 0.958–0.967 of the iterations at $n = 41$ at a rate of 0.9965, for 0.961–0.970 of the time per solve (`benchmarks/20261014-ecc2k130-signed-sigma/`);
- a table walk with every fruitless cycle caught is projected at 0.81–0.86× the σ-walk's cost per solve, unmeasured on the current kernel (`ecc2k130/WALK-CONSTANT.md`).

An audit of the live walk measured $2^{28.41}$ steps per completed walk at the $\mathrm{HW}(x) \leq 32$ cutoff and added duplicate-point, seed-reset and fruitless-cycle guards (`research/ecc2k130_rho_audit_20260925/`). None of this improves on the generic bound, and the paper says so.

NIST P-256 and P-384 on a GPU. The same levers applied to the `gpu/nist` engine took P-256 from 1.075 to 1.632 and P-384 from 0.210 to 0.548 billion iterations per second on one RTX PRO 6000 (1.52× and 2.61×), through PTX carry chains and a batch width $W = 32$ with by-reference inversion (`benchmarks/20261013-nist-gpu-rho-ladder/`; single-run sweep cells, no isolation receipt). An inversion-free Jacobian seeding ladder cut P-256 walk seeding from 11.53 to 0.66 ms per seed (17.4×), a set-up change with the step unchanged (`benchmarks/20261017-gpu-nist-jac-seed/`). At these rates P-256 remains about $6 \cdot 10^{21}$ card-years away (`paper/ecc2k130/`). No GPU arm is admitted to ecbench, because the matched-workload contract lacks PCIe telemetry and an admission rule (`problems/ecbench-gpu-matched-workload-contract.md`), and a P-256 best-of-two distinguished-point deficit remains unattributed, with its 0.947 cost estimate derived, not measured (`problems/nist-best2-dp-bias-and-gpu-validation.md`).

Reading. At scale the reference is an engineering artefact as much as an algorithm: its constant is measured, its throughput is set by a pipe, and each walk change has a corpus cost. None of it moves the generic exponent. All of it moves the denominator of every index-calculus row, which is why the gate of §33.2 names the reference per row and why a row ranked against a reference that has not adopted a change overstates index calculus by that change's factor.

Part V

The frontier

30 The summer record: July–September 2026

The October campaign measured in the rest of this part did not begin from nothing. Between 19 July and 30 September the crypto-autoresearcher ledger recorded 1,402 coordinator decisions on ECDLP and index calculus:

- July: 74 decisions on 20 experiments, 977 run directories;
- August: 364 decisions on 92 experiments, 321 run directories;
- September: 964 decisions on 449 experiments, 776 run directories.

Each decision names a verdict, the hypotheses and experiments it rules on, and the evidence it cites. The dated index (`research/ecdlp_jul_sep_index_20261029/`: `INDEX.json`, `INDEX.tsv`, and the generator `build_index.py`) lays these out together with the run-data path of each experiment. A decision is counted as ECDLP work if its goal is one of 19 ECDLP goals or, when it names no goal, if a target or evidence identifier carries an ECDLP family token. That scope rule is a choice and is stated in the generator. Seven decisions filed under late-September names carry October dates and are left out of the counts above. Twenty-three decision files did not parse and are not indexed.

Most of this activity was process (Figure 30.1). Of the 1,402 decisions, 561 approve a contract, 267 revise one and 124 expand one. About 157 carry an outcome on a hypothesis (support, reject, weaken, inconclusive, supersede, pause or replicate). This chapter reports those outcomes by theme, with decision identifiers so that each can be traced. All results are at toy or derivation tier unless a scale is given. None moved an exponent on a prime field.

30.1 July: the first baseline and two retracted wins

The ledger's first ECDLP decision, `DEC-20260719-001`, expanded a toy Semaev S_3 Gröbner study (`EV-SEMAEV-001`). It had twelve runs at 8–12 bits with a factor base of 14 points. The degree proxy stayed at most 4 and the wall time stayed flat near 0.16 s. The decision itself noted that three bit sizes and two seeds cannot establish a trend.

Two supports followed on 26 July, and both were overturned the next day:

- *Index calculus beats enumeration* (`H-IC-001`, 748 runs at 8–36 bits) was weakened by `DEC-20260727-008`. The criterion had been revised after the outcome. There were no decompositions at 20 bits or above, so the measured descent time was the cost of a failed attempt. The relation cost was 2287× the group order at 36 bits. The decision records: "no asymptotic exponent moves".
- *A structured factor base induces sparsity* (`H-STR-002`) was weakened by `DEC-20260727-009`. Three headline instances had no run records. The one-variable ablation gave $\alpha = 27,52,197,386$ against $27,52,199,387$ for a random base, so the factor base induced nothing.

In the same week a generic-group classifier (`EXP-GGM-001`) had its support revoked (`DEC-20260727-007`). It was a lookup table, and its nine runs were byte-identical. Fixed-curve preprocessing showed no robust gain over 72 runs (`DEC-20260726-001`, paused). A subresultant degree exponent was measured at $\beta_{\deg} = 0.5985$ against a gate of 0.3 (`DEC-20260728-003`, weakened).

Xedni-style lifting was closed in exact form (`DEC-20260724-021`, reject_scoped): $P_{\text{lift}} = (1 - 2/p + p^{-3})/(2p^3)$, so $\alpha = 3$. This is the result restated in Chapter 8.

30.2 August: yield, monodromy, oracles and transfers

Yield matches the heuristic from below. Semaev decomposition yield at $m = 2$ reached 0.80, 0.98 and 1.00 of the $B^m/(m!N)$ heuristic at $p = 1009$, 4001 and 9001 (`EV-YIELD-e1adbf`). `DEC-20260804-9da614` weakened super-heuristic yield at $m = 2$–4. Structured bases gave small-x 0.845, arithmetic progression 0.658 and random 0.563, with no super-heuristic yield (`DEC-20260804-2bfc30`). These measurements came before the conservation proposition of Chapter 5, which explains them: at matched base size the mean is fixed. A complete $m = 2$ index calculus solved a discrete logarithm at $p = 1009$ (`DEC-20260804-db75bb`). A Fourier-flatness constant for the factor base was fitted as $C(p) \sim p^{0.055}$–$p^{0.079}$ and recorded as an open problem (`H-PSEUDO`). Eight proof routes were closed as

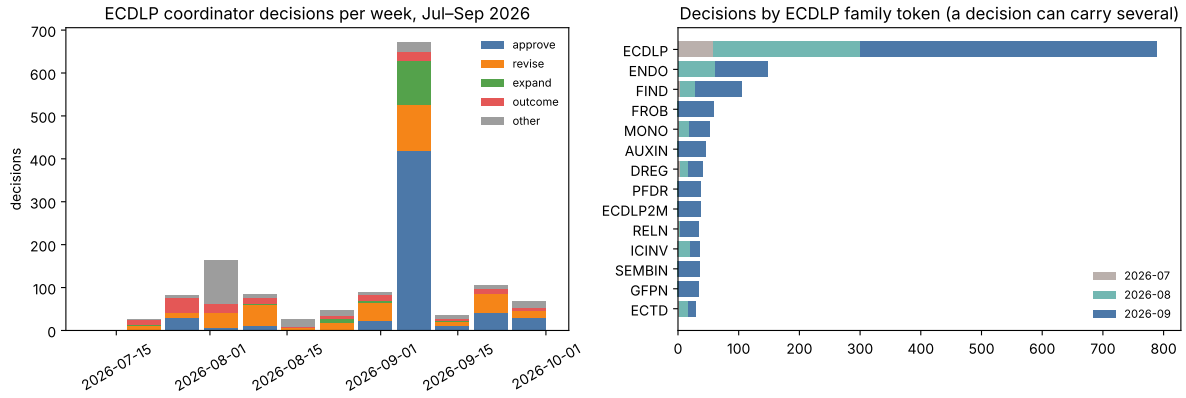


Figure 30.1: ECDLP coordinator decisions, July–September 2026. Left: decisions per week by verdict class. The class *outcome* groups support, reject_scoped, weaken, inconclusive, supersede and pause. Right: decisions per family token, by month; a decision can carry several tokens. The week of 7 September holds 672 decisions, most of them approvals of frozen stage-0 contracts.

non-proving. A frozen re-test on 4 November (`HPSEUDO-SYMNULL-20261104`) explained the recorded offset $R \approx 1.28$–1.34 as an artefact of the null. The factor base is closed under negation, but the null was an asymmetric uniform subset. Against a negation-closed null, R is consistent with 1: the upper 95% bound is 1.09 at 2^{20} and 1.02 at 2^{22}, and the pooled 56 curves give 1.005 ± 0.008. A residual of up to about 9% at 2^{20} is not excluded (`crypto-autoresearcher:docs/reports/hpseudo-symnull-20261104/`).

BKK and meet in the middle: constant factors. A BKK sparsity argument promised a speedup of at least $(m+1)/2$ (`DEC-20260804-1825ea`). It measured 1.99988 against 2.0 at $m = 3$ on a smoke test. `DEC-20260806-26c0e8` then found the multi-target window empty at the planned sizes and the gain about 1.0–1.8×. Meet-in-the-middle structure gave 18×, which rho dominates (`DEC-20260808-030b1f`). A degree-split claw lost its advantage to a matched null, $R = 0.0472$ against 0.0311 (`DEC-20260801-005`). The decision reads: "the instrument, not the hypothesis, has been refuted".

Monodromy. The work that became Chapter 5 was done in August:

- the $m = 3$ derivation (`DEC-20260802-a51c82`);
- the all-m theorem, supported as scoped and "not closed" (`DEC-20260807-64df6a`, `KN-FIND-a8990a`);
- containment in $(\mathbb{Z}/2)^{m-2}$ at $m = 4, 5$ on eight curves, with zero exceptions in an exhaustive census of 2,915,514 triples at $p = 103$ (`DEC-20260830-0f8def`);
- monodromy D_4 for the $m = 3$ symmetric-base cover (`DEC-20260830-1fb2e3`).

The fibre census was ruled blind to relation rate as an instrument (`DEC-20260810-2f86db`, reject_scoped), two months before the census corollary proved why.

Oracles and transfers. Two analysis-only results bounded the oracle leads. Oracles C and D are simulable in the generic group model, and A and B give no sub-birthday advantage (`DEC-20260804-3b4258`, support_scoped). The doubling oracle was then shown equivalent to an x-coordinate oracle (`DEC-20260806-08b9ed`). Three transfer and invariant routes were closed:

- Isogeny transfer: the anomalous, MOV and Weil-descent families are invariants of the isogeny class (`DEC-20260804-2fae6a`).
- Elkies augmentation: with $\ell \leq 23$ it matched a random-bijection null to within 0.004 in all eight cells at 20–28 bits (`DEC-20260804-d0c452`).
- CM: complex multiplication changes the lifted-height obstruction "by exactly nothing" (`DEC-20260810-2c7e3c`).

Within one isogeny class the over-dispersion of ECDLP-relevant statistics traced to the 2-torsion type (`DEC-20260811-a3398a`, `KN-FIND-ddd982`).

Degree of regularity. The boolean chained $m = 3$ system was run at $n = 21$, $D = 5$, and stopped at rank 265,950 of a predicted 268,674 (`EV-DREG-4500cd`). The record calls this "not evidence about d_{reg}

in either direction". All six $n = 15$ runs then exceeded memory (DEC-20260828-653923). The certified fall-degree results of Chapter 6 later replaced this line.

Late August was spent on governance: budget ceilings, pauses and zero-run successors. DEC-20260820-0c8ffb paused at the budget ceiling. The "fund every goal" decisions of 28 August lifted the limits.

30.3 September: tables, lifts, audits and the first binary costing

Distinguished-point tables re-selected from the batch. The claim that a table of $T/2$ entries re-selected from solved walks matches a static Bernstein–Lange table of T was contradicted at $a = 1/4$. It reached 0.65–$0.69\,T$ at steady state, with intervals excluding 1, while paying 2.6–$8.9\,T$ of working storage (DEC-20260906-1f5b71, weaken: "a tradeoff point, not an improvement"). At $N = 2^{24}$ the feasibility crossing was then bracketed between $a = 1/8$ and $a = 3/16$ (DEC-20260907-9b1b02, DEC-20260907-6d9427). The result is a ceiling only, with no online arm (DEC-20260910-34e654).

The canonical lift carries no logarithm. The p-adic lift statistic D_1 took exactly the values $\{0, 8\}$ on 1,035 pairs at $p = 1009$, and the same in a third independent implementation and on a second instance. It tracked precision $r = 4, 8, 12$ exactly (DEC-20260903-256c7c). The digit identity held in 1360/1360 cases, and the Teichmüller contrast showed no inflation: all three ratio intervals contain 1 (DEC-20260911-74c9dc, DEC-20260912-2a626b). Both results were recorded as "no attack".

Natural statistics are not biased enough. An L^1 Fourier registry over 406 cells and primes 4099–16,777,259 falsified two of its four predictions (DEC-20260921-4516a0, weaken):

- interval statistics: $\lambda = 0.0806 \pm 0.0020$, against a prediction of at most 0.05;
- cosets and quadratic residues: 0.50, against a prediction near 0.

Semaev degree and the Semaev 2015 / Nagao audits. The successful-subset loophole in the Semaev degree law was closed: $\beta_{\text{work}} = 0.797$ worst case over 36 cells and 1,152 queries (DEC-20260929-780751, reject_scoped, EV-SDEG-4a1c3d). The audits of Semaev's 2015 paper and Nagao's programme (Chapter 6) recorded these verdicts:

- eq. (11)'s λ is right to 1% at $t = 2$, but $P = 1 - e^{-\lambda}$ fails by 18–22% in the low-degree window (DEC-20260913-3b51c8);
- the typed-image ratio is not $m!$ at any of seven cells (DEC-20260913-2386b5);
- trace-parity confinement explains the whole deficit at $t = 2$ (DEC-20260913-92aa48);
- Nagao's memory claim fails at $n = 571$, and 81 of the 120 flagged cells were an artifact of an incoherent memory pairing (DEC-20260914-537000);
- a published measurement claim, P3, fails under the corrected XOR product: the Semaev median is 2 in all eight cells, with the null at least one above it (DEC-20260929-e37785, reject_scoped).

Prime-field degree versus yield. A closed form for the first fall degree of a digit-map presentation, $d_{\text{ff}} = m2^{m-1} + \lfloor (s - 2^{m-1})/2 \rfloor + 1$, survived 26,000 instances with zero deviations. Its companion fall-dimension clause was refuted by a certificate at $p = 13$ (DEC-20260904-1e27a2). The generator degree is $m2^{m-1}$, not $2m$, and the "cells below rho" of a cost table had zero rows at D_0, so they were rejected (DEC-20260904-d47cd2).

Frobenius-stable factor bases. The stable subspaces are exactly the ideals of $\mathbb{F}_q[T]/(T^n - 1)$, and the orbit count is exact on 14 of 14 cells (DEC-20260922-5796b8, EV-FROB-d336b0). The orbit gain is n, a constant factor. At $n = 131$ there are only four stable subspaces, and the orbit trick *loses* 57.0–96.0 bits for $m = 2$–5; at $n = 163$ it loses 72.7–121.7 bits. Whether WDSat is blind to basis choice was never measured, because 19 stage-0 launches failed.

The first costing of descended systems against rho. The ICPERF lane priced the descended algebraic normal form against the whole rho budget on standard binary curves (DEC-20260922-925da8, EV-ICPERF-784b25). Forming the system alone exceeds rho by +6.3 bits at $n = 233$ and by +40.8 bits at $n = 131$ (ECC2K-130). Below $n \approx 300$ the obstacle is building the system, not solving it. The excess turns negative at $n = 409$ and 571 for $m \geq 4$, and $m = 2$ loses at every degree. This is the first September result priced against rho. It is a model with measured inputs, not a solve.

Lanes without a result. Several September lanes produced approvals and redesigns but no measured outcome:

- SAT on summation systems, trapdoor detection, invariant calibrators and endomorphism decomposition, mostly frozen stage-0 contracts approved on 13 September;
- the auxiliary-input census, whose run was ruled invalid as evidence because its controls were inadmissible and its escalation was generic to 256-bit primes (`DEC-20260923-ce06a6`);
- EcGFp5 symmetrisation, where no $m = 5$ ideal degree was measured by September (`DEC-20260923-d86ff2`), although the audit found that the curve's self-reported 323 bits recomputes to 322. A quotient solve on 1 November measured $D = 2^{16}$ at all three primes it ran (`research/ecgfp5_quotient_solve_20261101/`);
- characteristic-2 quasi-subfield work, which found zero violations of its bound in about 340,000 candidates and produced the derivation $\beta > 1/2$ for every completely splitting L (`KN-FIND-617d78`).

The certified binary refutations of Chapter 6 begin here: W_4 refutes 62/62 at $n = 17$ and 400/400 at $n = 19$ (`DEC-20260924-e61f3b`, `DEC-20260926-901ea1`).

30.4 What the summer established

Table 30.1 collects one outcome per theme. Four patterns run through the 1,402 decisions:

1. *No measured improvement over rho survived review.* Every July and August support that claimed a gain was weakened or revoked within days. The surviving gains are constant factors at toy scale.
2. *The theory of Chapter 5 was found by measurement first.* The yield ratios at or below 1, the census ruled blind, and the exhaustive monodromy checks all preceded the proofs that explain them.
3. *Most closures were instrument repairs.* A large share of the weaken and reject decisions say that a control, a metric or a criterion was wrong, not the hypothesis. This is the origin of the matched-null and replay rules of Chapter 4.
4. *The binary line reached a priced verdict only in late September.* The ICPERF costing and the W_4 certificates opened the fall-degree and Koblitz work that the October campaign measured.

These four patterns are the author's reading. The decisions cited above are the record.

month	theme	decision	verdict	recorded result
Jul	IC beats enumeration	`DEC-20260727-008`	weaken	relation cost $2287\times$ the group order at 36 bits
	structured factor base	`DEC-20260727-009`	weaken	$\alpha = 27, 52, 197, 386$ vs. random $27, 52, 199, 387$
	subresultant degree	`DEC-20260728-003`	weaken	$\beta_{\mathrm{deg}} = 0.5985$ vs. gate 0.3
	xedni lifting	`DEC-20260724-021`	reject_scoped	$\alpha = 3$
Aug	Semaev yield, $m = 2$	`DEC-20260804-9da614`	weaken	0.80, 0.98, 1.00 of heuristic at $p = 1009, 4001, 9001$
	BKK speedup	`DEC-20260806-26c0e8`	supersede claim	1.0–1.8×, not $(m + 1)/2$
	degree-split claw	`DEC-20260801-005`	weaken	$R = 0.0472$ (null) vs. 0.0311
	monodromy, $m = 4, 5$	`DEC-20260830-0f8def`	support	0 exceptions in 2,915,514 triples at $p = 103$
	Elkies augmentation	`DEC-20260804-d0c452`	reject_scoped	within 0.004 of null, 8 cells, 20–28 bits
	boolean d_{reg}, $n = 21$	`DEC-20260815-6b806c`	supersede	rank 265,950 of 268,674; not evidence
Sep	re-selected DP table	`DEC-20260906-1f5b71`	weaken	0.65–$0.69\,T$ at $a = 1/4$
	canonical lift D_1	`DEC-20260903-256c7c`	support	values exactly $\{0, 8\}$ on 1,035 pairs; no attack
	natural-statistic bias	`DEC-20260921-4516a0`	weaken	$\lambda = 0.0806 \pm 0.0020$ vs. ≤ 0.05
	Semaev degree loophole	`DEC-20260929-780751`	reject_scoped	$\beta_{\mathrm{work}} = 0.797$, 36 cells
	Frobenius-stable bases	`DEC-20260922-5796b8`	support	loss of 57.0–96.0 bits at $n = 131$
	ICPERF costing	`DEC-20260922-925da8`	support	+6.3 ($n = 233$) to +40.8 ($n = 131$) bits over rho

Table 30.1: The summer record at a glance: one outcome decision per theme, July–September 2026, with the number it recorded. Every row is discussed above. Verdicts are the ledger's, shortened: "weaken" includes `weaken_h_yield`, and "supersede claim" replaced the earlier BKK support. Index: `research/ecdlp_jul_sep_index_20261029/INDEX.json`.

31 What improved, and where the frontier stands

31.1 The measured frontier: one campaign, every cost priced

The frontier synthesis of 13 October assembled ratios from studies that used different curves, sizes and accounting. To replace it we ran one campaign (`research/ic_frontier_campaign_20261015/`, protocol frozen before any measured run). Every method family ran on the same fixed curve ladders: eight Koblitz curves chosen by subgroup size ($\log_2 r \approx 11$–39, $n \in \{13, 17, 19, 23, 37, 41, 43, 47\}$), six NIST-shaped $a = -3$ prime-order curves of 20–40 bits, and the five 2-adic curves of the Kummer-tower ladder. Each curve had 16 public hash-to-subgroup targets nobody knows the logarithm of, 2 measured rounds and a warm-up, and every answer was checked outside the solver. That is 4,848 executions, 3,232 of them measured: 3,178 verified and 54 exhausted, all of the latter in the shared-rank arm at $n = 41$ and $n = 47$. Every session is audited by re-executing every measured run and requiring identical counts. Index-calculus parameters (compact-orbit columns, negfold subspace dimension, prime base size) were chosen per size on a disjoint set of training targets, with grids widened until the optimum was interior. Every cost is reported across the pricing band of §4.3.

What the campaign shows.

- *No index-calculus arm beats generic rho on any curve, at any weight.* Signed-Frobenius rho is below every index-calculus arm on all eight Koblitz curves and across the whole band; rho with negation is below every prime-field arm on all eleven prime curves (Table 31.1). The one unresolved pair in the campaign is compact orbit against the *strong* rho reference at $n = 13$, whose scalar-multiplication set-up dominates on an 11-bit subgroup; the cheaper signed-Frobenius walk is below both.
- *Compact orbit is the closest index calculus.* It sits 3.8–15× the best rho with native work free and 4.9–17.8× at $w = 0.1$; the ratio does not shrink with size.
- *Prime-field index calculus falls further behind with size.* The two-summand pair-table arm goes from 10.5× to 101.8× rho (at $w = 0.1$) between 20 and 40 bits, the $r^{1/6}$ widening its $r^{2/3}$ cost against rho's $r^{1/2}$ predicts; the Kummer-tower arm is 28–159× rho.
- *Negfold root finding stops scaling.* It is 12–45× rho up to $n = 23$, about 1,900× at $n = 37$, and does not complete within $5 \cdot 10^7$ trials from $n = 43$ on (PROTOCOL A4b).
- *Inherited F4 is not competitive once its work is priced*: S rises from 16 (word XORs free) to about $2 \cdot 10^4$ at $w = 0.1$ at $n = 13$. The band weighs one 64-bit word XOR like a table lookup, which overstates F4 (its pinned calibration is about 0.002 GAE per XOR); even with XORs free it is above rho.
- *BSGS has the lowest time on several curves* because it spends $\sqrt{r}$ memory; it is reported on the time–memory Pareto set, not ranked against constant-memory methods.

Cost exponents. Table 31.2 fits total cost $\propto r^{\alpha}$ per arm over each size tier with `ecbench bound fit -weight` (two-stage bootstrap, 95% intervals; a tier needs at least four sizes for an exponent). For the prime pair table, index calculus grows faster than rho (the predicted 2/3 against 1/2), but under the band rule the difference is not established at every weight. On $n = 13$–23 both Koblitz intervals are wide because the automorphism count $2n$ changes with the size.
The rho arms are rho with negation (prime) and signed-Frobenius rho (Koblitz). No index-calculus arm has a fitted exponent below rho's at any weight, so no crossover with rho is supported by these data, extrapolated or otherwise. The fitted rho exponents (0.435–0.444) lie below the theoretical 1/2, which reads as a toy-size effect (fixed per-run overheads and the size-dependent automorphism count) rather than a property of rho; so the measured widening of the gap is a fit on small sizes, and the conclusion that it widens as $r^{1/6}$ rests on the generic-oracle bound, not on these fits alone. The sixty-three fits are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

Limits. Operation counts only (isolation L0); wall time and a second-host replay certificate need the planned RunPod phase. Parameters were selected on four training targets per size. The shared-rank arm

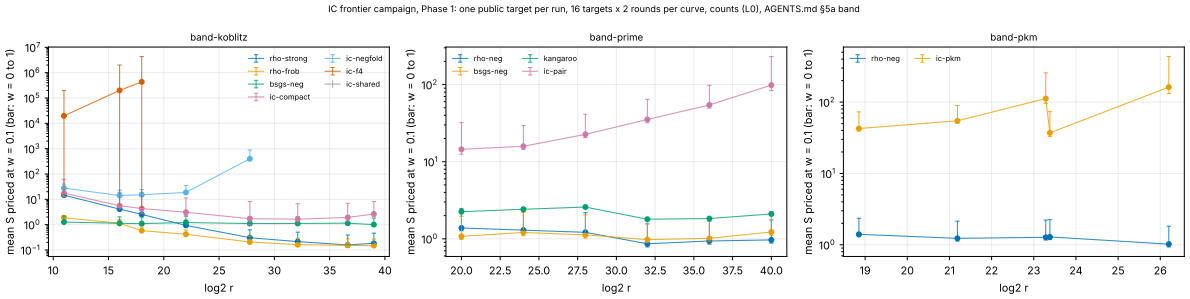


Figure 31.1: The campaign frontier: mean S per arm and curve at $w = 0.1$ GAE per native operation, the bar spanning $w = 0$ to $w = 1$; one public target per run, 16 targets × 2 rounds per curve, operation counts. Sources: `research/ic_frontier_campaign_20261015/analysis/band-*.json` (`ecbench band`).

Curve	$\log_2 r$	Arm	$w = 0$	$w = 0.1$	$w = 1$
Koblitz $n = 13$	11.0	compact orbit	6.9	9.3	27.1
		negfold	14.4	14.8	17.5
Koblitz $n = 17$	16.0	compact orbit	3.8	4.9	11.8
Koblitz $n = 19$	18.0	compact orbit	4.9	7.5	23.6
Koblitz $n = 23$	22.0	compact orbit	5.5	7.4	18.4
Koblitz $n = 37$	27.8	compact orbit	5.2	8.3	23.3
		negfold	1806	1927	2512
Koblitz $n = 43$	32.1	compact orbit	7.4	10.2	23.1
Koblitz $n = 47$	36.6	compact orbit	9.9	12.7	25.3
Koblitz $n = 41$	39.0	compact orbit	15.0	17.8	30.3
prime 20 bit	20	pair table	9.8	10.5	13.5
prime 24 bit	24	pair table	12.1	12.2	12.9
prime 28 bit	28	pair table	18.6	18.6	18.9
prime 32 bit	32	pair table	40.9	40.9	41.1
prime 36 bit	36	pair table	58.1	58.0	57.5
prime 40 bit	40	pair table	95.2	101.8	131.4
PKM 20 bit	18.9	Kummer tower	30.3	30.4	31.0
PKM 28 bit	26.2	Kummer tower	141.3	158.7	238.7

Table 31.1: Index calculus over the cheapest constant-memory generic arm on the same curve (signed-Frobenius rho on Koblitz curves, rho with negation on prime curves), mean S ratio at three band weights. Every run verified (32/32 per cell). Source: `research/ic_frontier_campaign_20261015/analysis/ratios_to_rho.tsv`.

leaves three categories of work uncounted and is reported as a lower bound only. Fitted cost exponents per arm and tier are in `research/ic_frontier_campaign_20261015/analysis/bounds/`.

31.2 The engineering record

Table 31.3 collects the measured improvements of the programme on the boundary each was measured on. Ratios are baseline/candidate; "stage" means one oracle call or one solver run; "complete" means a verified discrete logarithm with the stated phases.
Derived entries: $53.0 = 1/0.018861$; $17.5 = 1/0.057047$; $1.21 = 1/0.828$ with interval from $[0.783, 0.863]$; $18.3 = 1/0.0545$ with interval from $[0.047, 0.064]$; $1.77 = 300{,}021.9/169{,}069.8$; $1.20 = 1/0.830$ with interval from $[0.697, 0.962]$; $1.87 = 18.714/9.985$ (one session; the compact arm's S is a lower bound, so the ratio is an upper bound); 1.69 and 25.2 compare means from different sessions with different seeds and are descriptive; F5 and F6-IC ratios are reciprocals of the recorded candidate/baseline word-XOR ratios.

Arm	Tier	w	IC α	Rho α	Intervals
Prime pair table		0	0.620 [0.557, 0.663]	0.435 [0.331, 0.501]	separated
		0.1			separated
		1	0.594 [0.482, 0.665]	0.444 [0.334, 0.517]	overlap
Koblitz compact orbit	$n = 37$–47	0	0.589 [0.518, 0.743]	0.436 [0.358, 0.508]	
		0.1, 1			overlap
	$n = 13$–23	0	0.25	0.29	overlap
		0.1, 1			overlap
Kummer tower			0.717 [0.358, 0.940]	0.438 [0.332, 0.599]	overlap

Table 31.2: Fitted cost exponents α (total cost $\propto r^{\alpha}$) of each index-calculus arm and the cheapest constant-memory rho on the same curves, by size tier and band weight w; 95% intervals from a two-stage bootstrap. Intervals: whether the IC and rho intervals are separated or overlap at that weight. Rows without exponent values report only the interval comparison; no tier or weight is stated for the Kummer-tower row. Source: `research/ic_frontier_campaign_20261015/analysis/bounds/`.

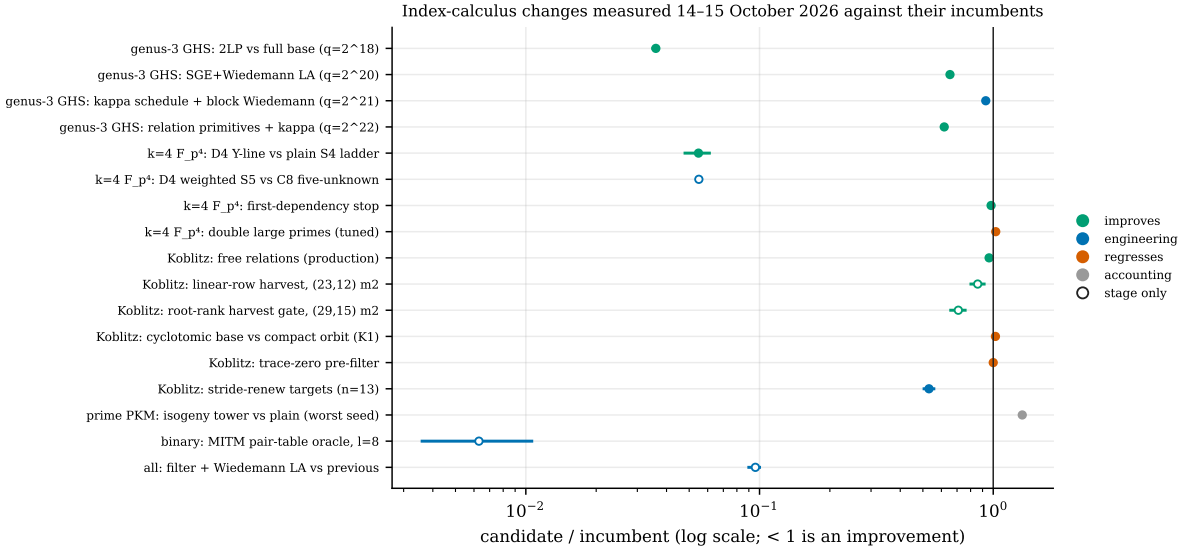


Figure 31.2: Candidate over incumbent for the changes of 14–15 October, with 95% intervals where the study reports one (the binary oracle's range is the spread over three runs). Filled: whole solve; open: one stage. Values: `papers/ecdlp-index-calculus-program/figures/ladder.tsv`; sources in Table 31.4.

31.3 The improvement ladder, 14–15 October

The repository's owner fixed the criterion for success on 15 October: a verified, measured improvement over the current index-calculus baseline is a success; each one becomes the new baseline; the gap to the matched rho is reported as the distance remaining, not as a pass/fail gate; and relabelling or accounting changes do not count. In the two days before that, two improvement rounds and a next-steps campaign landed about twenty measured changes, each paired against its own incumbent under a protocol frozen before data. Figure 31.2 and Table 31.4 list them, with the distance to rho that remains after each one. None is in the frontier campaign of §31.1, which predates them, and none was priced across the band; the boundaries are the ones the studies used, which differ from row to row.

Genus three: an exponent below rho's. The cubic GHS transfer [21] maps the DLP in $E_0(\mathbb{F}_{q^3})$ to the Jacobian of a genus-3 hyperelliptic curve over $\mathbb{F}_q$, where double-large-prime index calculus [22] costs $\tilde{O}(q^{2-2/g}) = \tilde{O}(q^{4/3})$ against rho's $q^{3/2}$. That is known; what is new is a priced, end-to-end, verified run of it on a family the repository can build ($4 \mid \#E_0(\mathbb{F}_{q^3})$, split arm, $q = 2^8$–2^{22}, every logarithm checked against the planted scalar). Adding double large primes changed the exponent from $q^{1.910}$ to $q^{1.385}$ $[1.379, 1.391]$, and keeping the large primes as columns, merging them with structured Gaussian elimination and finishing with Wiedemann moved it to $q^{1.380}$ and the constant to 0.653×. This was the

Change	Bnd.	Cl.	Ratio	Scope
Decomposition oracle: root finding				
S_3 roots, binary pair table vs matrix F4	St	E	383–1,535	$\mathbb{F}_{2^9}$, $\mathbb{F}_{2^{13}}$, $m = 2$
S_3 roots, prime vs `groebner_f4`	St	E	3.5–4.7·10^4	16–24-bit, $\lvert\mathcal{F}\rvert = 6$
S_3 roots vs Buchberger, NIST curves	St	E	3,021–5,802	P-192…P-521
`rf-mitm` vs PS-0 oracle	St	E	3,007–115,160	binary $\ell = 8..14$
Root-finding DLP vs msolve DLP	Cx	E	50–105	16–24-bit prime
Pair-table DLP vs F4 DLP	Cx	E	122	binary $n = 23$ (F4 wins $n \leq 13$)
Pair table vs subtraction	Cold	E	53.0	20-bit prime (bounded)
Frobenius pair table vs subtraction	Cold	E	17.5	Koblitz $n = 17$, priced
Root-finding K13 arm, first to best	Cold	E	1.69	38.82 → 22.97, across sessions
Windowed $m = 3$ scan	Cold	E	1.21	$[1.16, 1.28]$
Negfold root finding vs XOR-priced F4	Cold	E	18.3	$[15.6, 21.3]$, K13
Dense root index K1798 vs K1500 (probes)	On	E	1.77	$n = 83$, one point
Pair-table size 64 vs 32 abscissae	Cold	E	1.20	$[1.04, 1.43]$
Gröbner engines				
Inherited vs matrix F4 (matrix work)	St	E	198.6–222.1	K13, R13
F5 vs matrix F4 (word XORs)	St	E	1.55–1.56	K13, R13
F6-IC vs inherited F4 (word XORs)	St	E	1.22–1.30	$m = 13, 19$
`f4_fp` new vs previous	St	E	29.8–54.1	square systems, = msolve
Four-Russians routing	St	E	1.75–3.22	oracle matrices
Eager vs lazy inherited rewrite	St	E	1.40–1.62	dim 8–9
Quotient-curve $m = 2$ model	St	E	1.34–1.64	five cells
Factor bases and symmetry				
Compact orbit vs full orbit	Cold	E	≤ 1.87	Koblitz $n = 17$; compact is LB
FGHR D_4 vs Gaudry S_4, C_4	St	E	47.5–48.4	$\mathbb{F}_{269^4}$
`gaudry_quartic` solve	St	E	1.96–6.08	$d = 4..6$ (sweep recompute)
One-target online				
Compact-orbit IC vs strong rho	On	E	13.94	$n = 61$; rho without a table; retired (§31.3)
Accounting (costs that went up when priced)				
F4 word XORs priced	Cold	A	1/25.2	16.75 → 421.4, K13
F6-IC batched geometry priced	Cold	A	1/64.6–1/1,353	$m = 13, 19$; F4 unpriced (about 1/2.5 priced)

Table 31.3: Measured improvements, July–October 2026. Class: E = ENGINEERING, A = ACCOUNTING. Boundary: St = stage, Cx = complete solve outside ecbench, Cold = ecbench cold one-target S, On = online window after retained set-up. Sources: Appendix A.

Change	Cl.	Ratio [95%]	To rho
Genus 3 after the cubic GHS transfer, $E_0(\mathbb{F}_{q^3})$, cost in $\mathbb{F}_q$ multiplications			
Double large primes vs full base (04)	Ie	0.036 $[0.035, 0.036]$ at 2^{18}; $q^{1.910} \to q^{1.385}$	2.58× negation rho model at $q = 2^{20}$
Large primes as columns, SGE, Wiedemann (01)	Ie	0.653 $[0.647, 0.660]$ at 2^{20}; $q^{1.398} \to q^{1.380}$	—
$\kappa(q)$ schedule + block Wiedemann (06)	T	0.929 at 2^{21}	1.15× measured negation rho at 2^{20} (ref.)
Cheaper relation primitives + re-tuned κ (r3-01)	Ic	0.617 $[0.616, 0.619]$ at 2^{22}	no matched rho in this unit
$k = 4$ over $\mathbb{F}_{p^4}$, operation counts			
D_4 Y-line vs plain S_4, $p = 101$–1033	Ic	0.0548 $[0.048, 0.061]$	handover 2^{126}–2^{137} (d)
D_4-weighted S_5, four unknowns, vs five	E	1/18.5–1/17.8	handover $2^{135.6} \to 2^{118.9}$ (d)
First-dependency stop (`d4-ed`)	Ic	0.979 $[0.973, 0.986]$	no $k = 4$ run below rho
Double large primes, tuned	R	1.025 $[1.016, 1.037]$ (LA 0.827)	—
Koblitz, compact orbit and Gröbner oracle			
Free relations in production, $n = 41, 53, 61$	Ic	0.960 $[0.946, 0.976]$	3.8–15× rho-frob ($w = 0$, §31.1)
Linear-row harvest, $(23, 12)$, $m = 2$	Ic	0.858 $[0.803, 0.916]$	oracle stage; default on
Root-rank harvest gate, $(29, 15)$, $m = 2$	Ic	1/1.41 $[1/1.52, 1/1.32]$; end to end 1/1.20	oracle stage; default on
Gated fall-row augmentation	Ic	1/11.9 on barrel $n = 13$; 1 in production	production no-op
Equivariant Macaulay oracle	T	4.0 ($n = 5$) … 1/15.5 ($n = 13$)	not applicable to production
`stride-renew` targets, $n = 13$	E	0.531 $[0.506, 0.557]$	4.7× rho-strong at $n = 19$
Cyclotomic γ-weight base, K1	R	1.022 $[1.021, 1.023]$	yield-neutral
Trace-zero pre-filter	R	1.00003	vacuous on odd order
Prime fields and shared stages			
PKM isogeny tower, search charged	A	1.002–1.331	17–137× rho
Binary MITM pair-table oracle, $\ell = 8$	E	1/277–1/94 per target	no exponent claim
Filter + Wiedemann LA, $D = 2^{18}$	E	1/10.40; 1/1.93 vs CADO-NFS	LA is 0.1–1.9% of runs

Table 31.4: Changes measured 14–15 October 2026 against their own incumbents. Ratio: candidate/incumbent, < 1 better. Class, against the index-calculus incumbent (not the paper's ADVANCE, which is against rho): Ie = improves the exponent, Ic = improves the constant, E = ENGINEERING on one stage, R = regresses, T = trades, A = ACCOUNTING. "To rho": the distance left to the matched rho on the study's own boundary; "d" marks a derived or extrapolated value. Sources under `research/`: `ic_next10_20261014/`, `ic_improve_20261014/`, `ic_improve_20261015/`, `ic_improve3_20261015/` and the named study directories.

first index-calculus family in the programme whose fitted exponent lies below rho's (the $k = 4$ binary family of §31.4 is the second), and the only one where measured IC beats a rho arm: the item-7 rho, which walks at $1.75\sqrt{\ell}$ steps, at $q = 2^{15}$ (0.69 [0.55, 0.83]). Two further rounds kept going. A frozen schedule for the factor-base fraction $\kappa(q)$ plus block Wiedemann brought the whole solve to 0.605× the first double-large-prime arm at $q = 2^{21}$. Cheaper relation primitives (one-inversion Cantor addition, a Stickelberger pre-filter, reused exponentiations) then cut the relation phase to 0.47× without changing a single relation, and with κ re-tuned the whole solve to 0.617 [0.616, 0.619] at $q = 2^{22}$, in a unit that also charges inversions. The distance to rho shrank from 2.58× the negation-map rho model at $q = 2^{20}$ (first double-large-prime arm) to 1.15× a measured negation-map rho there after the second round ($2.31 \cdot 10^{10}$ against $2.01 \cdot 10^{10}$ $\mathbb{F}_q$ multiplications; a reference on overlapping but not identical curves). The third round's 0.62× has no matched rho in its unit yet, so whether it crosses is not established. The local exponents sit at 1.38–1.41, still above 4/3, because the linear algebra grows faster than $q^{4/3}$ and the tuned κ falls with q. The family is narrow: the transfer needs the right 2-power torsion, and $\mathbb{F}_{q^3}$ curves are not deployed. *Next improvement*: a negation-map rho measured in the third round's unit on the same instances, then the run past $q = 2^{22}$ that would show a crossover or rule it out.

The $k = 4$ chain. The D_4 Macaulay system of §21.10 entered a complete ladder ($p = 101$–1033, 21 curves): 18.2× cheaper than plain S_4 at the same exponent, with the handover with negation rho extrapolated to 2^{126}–2^{137}. Writing the D_4-symmetrised S_5 in four weighted unknowns cut the per-solve cost another 17.8–18.5×, to $1.42 \cdot 10^9$ $\mathbb{F}_p$ multiplications, 854× below the plain form; in the 2-torsion frame that moves the derived handover from $2^{157.9}$ to $2^{118.9}$. Double large primes cut the linear algebra (0.827×) but regress the whole solve; stopping at the first dependency gains 2.1%. No $k = 4$ run below rho exists; every handover is an extrapolation of fitted exponents through a measured constant. *Next improvement*: put the four-unknown solve into the complete ladder, then keep double large primes' linear-algebra gain without its relation-phase cost.

Koblitz. Free relations—root-table self-collisions used as homogeneous rows—are now the production default (0.960× counted, 0.941× wall); the linear rows the Gröbner oracle used to discard are harvested (0.86–0.93× where the gate fires, and a root-rank gate then made it fire on more cells: 1/1.41 at (29, 15), 1/1.20 end to end); the gated fall-row and equivariant passes pay only on overdetermined systems the production pipeline does not build. The cyclotomic base and the trace-zero filter were refuted. A census closes the free-relation lever at the rungs run: at $n = 61$, $K = 600$ the self-collisions span 49 independent relations of the 599 needed, and full rank would need $K \approx 2{,}050$. The distance to signed-Frobenius rho is still 3.8–15× cold at $w = 0$, and the fitted Koblitz ratio grows as about $r^{1/6}$ (slope 0.165 [0.117, 0.213] per $\ln r$ on eight public curves, every native priced). *Next improvement*: relation yield per probe, which none of these changes touched.

The online result is retired. An equal-advice audit gave Bernstein–Lange preprocessing rho [4] a distinguished-point table no larger than the index-calculus pair table and ran both on the same 64 unseen targets at $n = 41$, 53 and 61. Rho with the table is 53–317× faster online (at $n = 41$, with exactly equal advice, 317 [137, 739]), and even at 1/1,024 of the bytes it stays 7.7–32× faster. The 13.9× online win of §15 and the later 7.3× and 12.3× hosted single-target points at $n = 71$ and $n = 83$ compared index calculus *with* a table against rho *without* one. They are measurements of the preprocessing model, not index-calculus improvements, and the online frontier belongs to preprocessing rho.

The reference moved too. Two rho studies shortened the distances above. On prime curves, choosing the better of $P + R$ and $P - R$ at each step takes 1.129× fewer steps under distinguished points (1.081–1.184); each step costs more, so the predicted net gain is 1.03–1.12× with 8–64 walks per inversion and a net loss (0.95) at 1,024. On ECC2K-130, signed σ steps cut GPU time per solve to 0.961–0.970 (derived; not rolled out). Neither is yet in a campaign arm. A comparison against a rho that has not adopted them would overstate index calculus's position by that much.

31.4 The ladder continued, 16–20 October

Work that reached `main` between 16 and 20 October, after the programme record was first assembled (some of it measured on 14–15 October), added a genus-3 round, two $k = 4$ oracles, a long binary-oracle ladder, Gröbner-engine work and a repository-wide validation sweep (Figure 31.3, Table 31.5). The

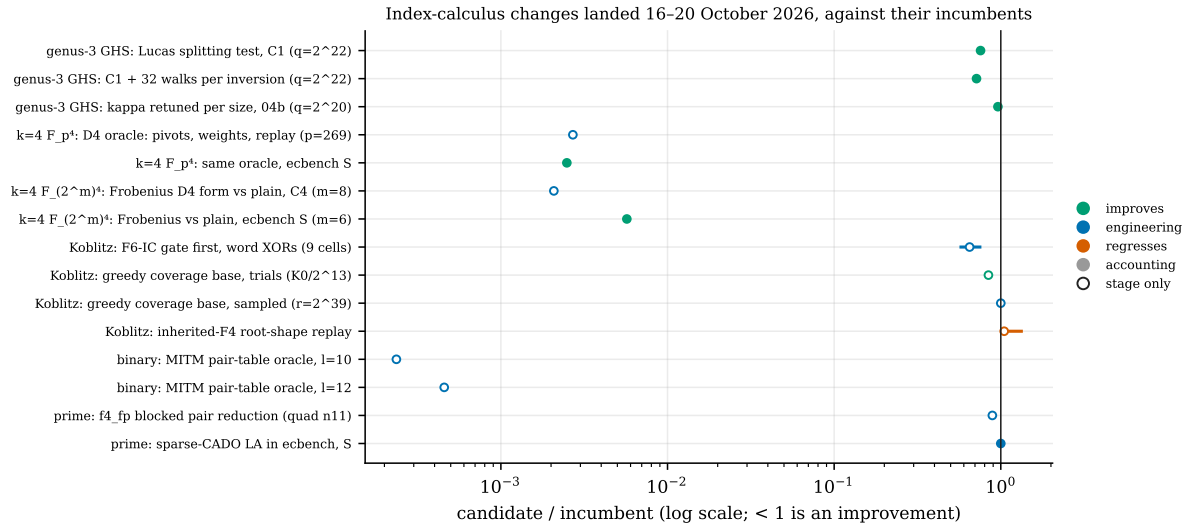


Figure 31.3: Candidate over incumbent for the changes that landed 16–20 October, as in Figure 31.2. Values: `papers/ecdlp-index-calculus-program/figures/ladder2.tsv`; sources in Table 31.5.

reading rules of §31.3 apply: each row is against its own incumbent, on its study's boundary, unpriced across the band.

Genus three, round four. A field-level splitting test (a Lucas ladder on the Cardano resolvent's cubic character, with the same relations) removes most of the per-trial log factor (9.75 to 2.71 units per bit) and moves the whole solve to 0.756× at $q = 2^{22}$; with 32 walks sharing one inversion, to 0.715×, with local exponent 1.356 (fitted 1.368 over 2^{16}–2^{22}), 0.023 above 4/3. Chained with round three, the whole solve is about 0.44× the second round's at $q = 2^{22}$ (derived, in the inversion-charging unit). Rounds three and four ran no rho arm; a 23 October study then measured round 4 against the strongest rho at $q = 2^{14}$–2^{20}: under the registered pricing it is below rho at 2^{16} at every band weight and unresolved at the other sizes (geometric means 0.70–1.52), and above rho under the protocol's heavier sensitivity pricing; §26). Re-tuning κ on training again steepened the schedule and raised the exponent. A separate rebalance study measured IC at 0.16× (0.25× renormalised) a rho on Jac(C) at $q = 2^{14}$: that rho walks on the Jacobian (four walks per size, at 1.6–2.8 steps per $\sqrt{\ell}$ against an ideal 1.25), not on $E_0(\mathbb{F}_{q^3})$, so it is not the matched reference and the ratio is not a distance to rho. Memory now limits the ladder: $q = 2^{22}$ peaks at 3.9 GB of a 4.5 GB cap.

$k = 4$: two oracles and a closing gap. Over $\mathbb{F}_{p^4}$, sparsest pivots, the weighted grading $\sigma_4 = \pi^2$ and shape replay cut one D_4 decomposition from $2.55 \cdot 10^{10}$ to $6.89 \cdot 10^7$ $\mathbb{F}_p$ multiplications (370×; 648/648 solves exact) and ecbench S by 398–406×; index calculus is still $1.39 \cdot 10^6$× rho at the measured sizes and the model handover moves from 2^{135} to 2^{103}. In characteristic two, a Frobenius-reduced form of the same system (weights $(2, 2, 1, 1)$, Macaulay degree 11, a 384×448 matrix) is 481× below the plain form. End to end on $E(\mathbb{F}_{(2^m)^4})$ the ratio of IC to the negation-map rho model falls from 5,080 at $m = 6$ to 6.06 at $m = 14$ ($n \approx 2^{56}$; 4.3× a measured rho), with fitted slope -0.248 in $\log_2 n$—Gaudry's $n^{-1/4}$—and an extrapolated crossover near $n \approx 2^{68}$ (2^{63}–2^{69} under the model). After genus three, this is the nearest any index calculus in the programme has come to rho, and its extrapolated crossover lies where rho itself takes about 2^{34} steps, so the extrapolation can be tested directly. A Joux–Vitse $(n-1)$-decomposition at $n = 5$ on $E(\mathbb{F}_{p^5})$ completed verified DLPs at $4.5 \cdot 10^7$–$1.8 \cdot 10^8$× rho, with a crossover extrapolated to groups near 2^{291}.

Binary oracles and their floor. The MITM pair-table oracle on a non-Koblitz curve reached $\ell = 12$ over $\mathbb{F}_{2^{37}}$ (2,474–4,244× over pairs-and-solve at $\ell = 9, 10$; 928–2,321× at $\ell = 11, 12$, the last as recomputed by the validation sweep; 0 disagreements on 256 targets). Against that floor, no algebraic oracle (inherited F4, F6-IC, SAT, split S_4 hybrids, a wide chained S_3 solver) grows more slowly in n: the cheapest is $3 \cdot 10^2$× the floor at $n = 13$ and $3.8 \cdot 10^4$× at $n = 31$ for $m = 3$, and the $m = 4$ chain grows at slope 0.805 against the floor's 0.333. A greedy coverage base needs 1.08–1.18× fewer trials on

Change	Cl.	Ratio [95%]	To rho
Genus 3 after the cubic GHS transfer			
Lucas splitting test (round 4, C1)	Ie	0.756 [0.755, 0.756] at 2^{22}; local 1.408 → 1.364	no rho arm since round 2
C1 + 32 walks per inversion	Ic	0.715 at 2^{22}; local 1.356	as above
Factor base retuned per size (04b)	Ic	0.961 at 2^{20}	0.16× rho on Jac(C) at 2^{14}, not on E_0
$k = 4$ over $\mathbb{F}_{p^4}$ and $\mathbb{F}_{(2^m)^4}$			
D_4 oracle: pivots, weighted form, replay, $p = 269$	E	1/370 [1/371.6, 1/367.9]	—
Same oracle, ecbench S	Ic	1/402 [1/406, 1/398]	$1.39 \cdot 10^6\times$ rho
Frobenius-reduced D_4 form, characteristic 2, $m = 8$	E	1/481 [1/483, 1/479]	—
Same, ecbench S ($m = 6$)	Ic	0.0057	6.1× rho model, 4.3× measured, at $n \approx 2^{56}$
Koblitz and binary oracles			
F6-IC gate-first order	E	1/1.73–1/1.34 word XORs	default on
Greedy coverage base, $K_0/2^{13}$	Ic	0.843 (expected trials)	gain 1.000 at $r = 2^{39}$
Inherited-F4 root-shape replay	R	1.015–1.328	switch stays off
Binary MITM oracle, $\ell = 9..12$	E	1/4,244–1/928 per target	relation collection $\Theta(2^n)$
Algebraic $m = 3, 4$ oracles vs MITM floor	R	slope 0.53–0.81 vs 0.21–0.36	$3 \cdot 10^2$–$3 \cdot 10^4\times$ floor
Prime fields and shared stages			
`f4_fp` blocked pair reduction	E	0.78–0.89 (wall, quad $n10, 11$)	beats msolve there
Sparse-CADO LA inside ecbench	E	S 1.000; LA wall 0.25–0.72	149× rho
S_3 solve on SECG small-x bases	—	table 2,600–313,000× cheaper	52–280 bits over rho
Column-touch conversion K (ECC2K-130 $m = 4$)	A	+7.62/ + 10.32 → +9.78/ + 12.48 bits ($w = 1$ floor; +45.71 at $w = 2$, unconverted)	accounting

Table 31.5: Changes that landed 16–20 October 2026, against their own incumbents; columns and classes as in Table 31.4. Sources under `research/`: `ic_improve4_20261015/`, `ic_improve_20261014/04b-genus3-2lp-rebalance/`, `gaudry_quartic_d4_cost_20261014/`, `gaudry_quartic_d4_char2_20261015/`, `joux_vitse_n1_20261015/`, `binary_mitm_oracle_l9_l10_20261015/`, `binary_mitm_oracle_l11_l12_20261016/`, `ic_m3_algebraic_vs_mitm_20261014/`, `secg_small_x_20261014/`, `greedy_factor_base_yield_20261014/`, `f6_gate_first_x86_20261016/`, `pdp_shape_replay_20261015/`, `f4_fp_blocked_pairs_20261016/`, `ic_linalg_pipeline_20261019/`, `k_conversion_20261018/`.

small Koblitz curves, but its gain falls to 1.0000 at $r = 2^{39}$. On the 33 SECG curves an S_3 solve over a small-x base, with no table, is 52–280 bits over rho.

The reference moved again. The strong signed-Frobenius rho of every Koblitz comparison executed 718 instructions per walk step after three rounds, down from 1,196; a fourth round of batched inversion took main from 707.2 to 679.7 Callgrind instructions per step at $n = 53$, with no established native-time gain (outputs identical; `research/rho_batchinv_20261022/`; steps per ideal 0.92–0.99 on 400 fixtures), and ecbench's standing `rho.signed_frobenius_strong` runs at 0.802× its former instruction count with identical GAE. On hyperelliptic Jacobians the programme's rho was slow at large N until full-width branch coefficients brought S_{walk} from 2.10 to 1.48; the negation fold saves only 0.97× at $N \geq 10^5$. Counted-unit comparisons (S) are unaffected by the instruction counts; wall-time comparisons against the older rho overstate index calculus by up to the same factor.

Corrections from the validation sweep. A repository-wide validation sweep (19–20 October) contradicted five recorded results, all now struck through in their sources with dated notes. Three bear on this paper. (i) The one-point $n = 83$ online ratio (18.5×, §9) and the hosted $n = 71/n = 83$ panels (7.3×, 12.3×) compared against rho with an empty table; with a Bernstein–Lange table matched to the index calculus's set-up time or memory, precomputed rho is faster in every matched $n = 83$ pair (3.7× time-matched and 40.6× memory-matched against $K = 3300$; 82–396× against $K = 600$). They join the $n = 61$ point as retired. (ii) The root-finder session's replay-all, recorded as 120/168 identical with the 48 mismatches confined to inherited F4, replays 72/168 on current `main`, with mismatches in all four IC arms, root finding included. The sweep later isolated the cause: an x86-only BMI2 rewrite in inherited F4 charges a different word-XOR count from the portable path, and with it disabled x86 and emulated aarch64 both replay 168/168 (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`). F4 counts are therefore comparable across instruction sets only when the path is stated, and until the method identity records it the root-finder rows rest on the recorded receipt. (iii) The equal-bytes preprocessing control at $n = 73$ and $n = 83$ is built and frozen but not run. *Next improvement* (from the 18 October leads review): a large-memory host. The two exponent-level questions named then—the Weil-descent cell $(m, \ell) = (4, 6)$ and the PKM tower at $m = 3$, $N = 21$—have since run on rented hosts: the $(4, 6)$ cell refuted at degree 4 at $N = 54$, and the PKM cell finished degree 7 and stopped on memory inside degree 8, implying $D \geq 8$ but read as inconclusive (§§16.1,21.2). The $(3, 7)$, $\ell = 8$ and $m = 4$, $N = 20$ cells remain.

31.5 The ladder continued, 21–24 October

Work that reached `main` between 21 and 24 October falls into four groups: factor bases closed under cheap known-logarithm maps, a paired re-run of the Koblitz leaderboard recipe, a guess-and-determine attack on the Weil-descent degree question, and transfer measurements, among them a third family whose measured gap to rho is now single-digit (Table 31.6). The reading rules of §31.3 apply: each row is against its own incumbent and on its study's boundary. The companion ledger (`papers/ecdlp-ic-improvement-ledger/`) records the same period as a ratchet of incumbents; this section keeps to what moves the frontier or corrects it.

One mechanism, four factor bases. The week's measured gains on the index-calculus side are all the same idea: put a point and its images under maps of known logarithm—the Frobenius τ on a Koblitz curve, the CM endomorphism ϕ, small integer multiples, the cofactor $[h]$—into one column with a known coefficient, so the linear-algebra stage needs fewer relations. On Koblitz curves the $\mathbb{Z}[\tau]$ base needs $K+1$ relations instead of about one per orbit: child CPU of the complete process falls 1.9–4.4× at $n = 41$–61 and 6.2–10.5× at $n = 71$. These are measured against the repository's own tuned producer, on a host at load about 20 (child CPU is the less affected number), with three targets up to $n = 61$ and one at $n = 71$. The prime CM base ($j = 0$, $j = 1728$, $D = -7, -8$) cuts S 10–20×, and the integer-multiple base on random binary curves 3–4×; both at 36–40-bit or degree-26–28 sizes, where the best bases have 3–10 columns over 512–3,480 points and, as their sources note, tend to a baby-step giant-step table over multiples as the column count falls. The cofactor-projected base (LEAD-19) is the only one measured on leaderboard rows, with 20 repeats and the order stable across the band: 1.29–3.36× in cold S, leaving 6.1–14.4× to matched rho. Every source states that the exponent is unchanged; the gain is bounded by the size of the multiplier set and shrinks as linear algebra's share of the cost shrinks. The known-coefficient

Change	Cl.	Ratio [95%]	To rho
Factor bases closed under known-log maps			
$\mathbb{Z}[\tau]$ small-norm multipliers, Koblitz $n = 41$–61	Ic	1/1.9–1/4.4 child CPU (3 targets, loaded host)	not measured
Same, wide path, $n = 71$	Ic	1/6.2 equal points; 1/10.5 retuned (1 target)	not measured
$\mathbb{Z}[\phi]$ multipliers, prime CM curves, 36–40 bit	Ic	1/10–1/20 in S	7–12×
Integer-multiple closure, random binary, degree 26–28	Ic	1/3.1–1/3.9 vs plain	6–12×
Cofactor-projected base (LEAD-19), $\mathbb{F}_{2^{15}}, \mathbb{F}_{2^{21}}, \mathbb{F}_{2^{27}}$	Ic	1/1.29 [1/1.52, 1/1.12], 1/1.58 [1/1.82, 1/1.35], 1/3.36 [1/4.02, 1/2.80]	6.1, 11.5, 14.4×
Koblitz leaderboard recipe (Table A, $n = 41$–61)			
Rule v4 recipe vs v3, paired, six sizes	Ic	0.953 $[0.904, 0.998]$ at $n = 57$; five sizes not separated	9.9–43× at the canonical step
Weil descent, degree question			
Hybrid guess, $(m, \ell) = (3, 7)$, $k^* \leq 2$ guesses	E	1/116–1/260 of projected degree-5 closure	—
Implied ECC2K-130 $m = 4$ row (derived)	N	+8.43 (flat k) to +58.27 bits (linear k)	vs +10.32 ($w = 1$), +45.71 ($w = 2$)
Prime fields			
Kummer tower, 30 bit (new row)	N	—	158.9× $[121.6, 221.4]$
Coppersmith lattice on S_3, $\ell \leq 3$	N	0/10 recovered in all 96 cells with $\beta \geq 0.20$	ceiling $\beta = 1/4$ stands
Transfer and covers			
Joux–Vitse cover of $E(\mathbb{F}_{p^6})$, $p = 53$–1009	N	slope −2.05 $[-2.28, -1.83]$ in $\ln p$	1.59–1.98× at $p = 1009$ (LB)
Measured trial cost, genus-255 binary Jacobian	A	103 curve additions per trial ($2^{6.69}$)	$N = 130$ margin +4.71 → −1.83 bits
GHS covers of $E/\mathbb{F}_{q^4}$, odd q, census	N	genus floor 13 on odd-order curves	model exponent gap 0.154

Table 31.6: Changes that landed 21–24 October 2026, against their own incumbents; columns and classes as in Table 31.4 (N = null or diagnostic, no cost row). Sources under `research/`: `ic_improve_20261023/` (`README.md`, `PRIME_README.md`, `BINARY_README.md`), `lead19_projected_base/`, `ic_tool_program/rule/v4/`, `dreg_hybrid_guess_20261024/`, `jv_cover_p_scaling_20261024/`, `hcdlp_trial_cost_20261024/`, `ghs_cover_census_n4_20261024/`, `coppersmith_s3_theta2_20261023/`; and `docs/ic/CERTICOM131_IC_PROGRAM.md`.

columns are those of GGMP's Frobenius-orbit bases [18] (§15); what is new here is extending them from automorphisms to small-norm and integer multipliers. Whether that extension is in the literature has not been checked. All four are ENGINEERING, and none is measured against a published index calculus; the $\mathbb{Z}[\tau]$ source notes that the summation-polynomial methods have no table-based counterpart at these parameters.

Koblitz: the recipe is near its optimum, the distance is larger. Rule v4 retuned columns, collection window and unit size on disjoint training seeds and re-ran Table A paired against rule v3 (32 pairs per size, 16 seeds, A/A and a positive control that detects a 10–15% slowdown). One size improves (0.953 $[0.904, 0.998]$ at $n = 57$) and five are not separated, so it is not a Table A improvement under its own rule. The free-relation lever costs 5–9% in this pipeline. More consequential is the reference: with rho priced in the same process at the canonical batched step, the cold distance is 9.9–43×, 1.76–1.94× the as-built figure that rule v3 used. The ledger's rule v8 reads 4.67–18.07× on the same sizes in its own unit; the two are different domains in the sense of Definition 4.1 and are not combined here.

Weil descent: a hybrid closes one cell, the degree question stays open. The $(m, \ell) = (3, 7)$ cell had no refutation at degree 4. Fixing one or two variables restores degree 4: all branches refute with $k^* \leq 2$ on 10 of 10 unsatisfiable draws, at 1/116–1/260 of the projected degree-5 cost, in 3–6 GB (planted-solution and placebo controls passed). For the ECC2K-130 $m = 4$ row this changes less. Its often-quoted "+10.32 bits over rho if degree 5 holds" prices one column touch per Macaulay column ($w = 1$), a floor no elimination beats; at $w = 2$, the exponent the measured closure tracks, the same row is +45.71 bits, and the measured conversion $K = 4$–16 rho steps per touch raises both (§31.4). With the hybrid, the row is +8.43 bits if the number of guesses stays flat in ℓ and +58.27 bits if it grows linearly; one value of ℓ cannot separate them. Pre-registered $m = 3$ runs at $(n, \ell) = (22, 8)$ have since completed. Four guessed variables bring all four unsatisfiable draws back to degree 4, and one three-variable set fails on three of the four draws. That makes the count 0 at $\ell = 6$, about 1 at $\ell = 7$ and at most 4 at $\ell = 8$, which still does not decide whether it stays bounded or grows. At the priced $m = 4$, $\ell = 33$ cell the hybrid beats degree 5 only if the count is at most 6 (`research/dreg_closure_guess_20261022/`, `research/dreg_closure_guess3_20261023/`). In no scenario does the row go below the degree-4 floor of +7.62 bits, so the route does not reach rho.

Prime fields: further behind. A new 30-bit Kummer-tower row in ecbench (160/160 verified, loaded host) is 158.9× $[121.6, 221.4]$ rho at $w = 0$, against 141× at 28 bits in the campaign; its source calls it "a measurement, not an improvement". Two falsifiers for the Certicom 131-bit programme (random-b binary degrees of regularity; structured bases) were null. A frozen-protocol test of a Coppersmith lattice on S_3 recovered no planted decomposition in any of 96 cells with $\beta \geq 0.20$ at 24–56 bits, with positive controls passing below, so the $\beta = 1/4$ ceiling stands at multiplicity ≤ 3.

A third family crosses rho: the Joux–Vitse cover. The genus-3 cover of $E(\mathbb{F}_{p^6})$ over $\mathbb{F}_{p^2}$ with Nagao decompositions (full rational 2-torsion; a weak class, not prime-order curves) ran at $p = 53$–1009, three seeds per prime, every logarithm checked. The gap to rho falls from a geometric mean of 620× at $p = 53$ to 1.73× at $p = 1009$ (1.59–1.98), with fitted slope -2.05 $[-2.28, -1.83]$ in $\ln p$ against the cost model's -2; the line meets rho near $p \approx 1330$ ($[1170, 1540]$, an extrapolation). Two limits keep it a lower-bound comparison: the cover arm does not emit its unpriced F4 and Wiedemann work, so it cannot be ranked under the band; and rho cannot be run at $p \geq 503$ (subgroups of 2^{52}–2^{58}), so those ratios use the pooled rho constant $S = 1.344$ measured at $p \leq 251$. If that constant kept falling at its (insignificant) point trend, the ratio at $p = 1009$ would be about 20% higher. A separate sieve arm on the same cover was later measured against a matched negation rho at $p = 1511$ and validated. It is 0.0065× rho (155× $[125, 184]$ below it), and the order holds across the band (§26).

Covers: censuses and a repriced counterfactual. Measuring the cost of one relation trial (a Cantor addition plus a smoothness test) on a genus-255 binary Jacobian gives 103 additions of a point over $\mathbb{F}_{2^{1131}}$ ($2^{6.69}$), which turns the $N = 130$ GHS counterfactual's modelled margin over rho from +4.71 bits to −1.83 bits; only a trial below about 29 additions would preserve it, and the smoothness test alone costs 16.5. Two censuses fix what covers exist at toy sizes: for $E/\mathbb{F}_{q^4}$ in odd characteristic the genus floor is 13 on odd-order curves ($2 - 2/g$ against rho's 2 leaves an exponent gap of 0.154, constants excluded), and for binary curves over $\mathbb{F}_{2^{3l}}$ the genus-3 class is a 2^{-l} fraction (0.78% at $l = 7$).

The validation gate. On 24 October the leaderboard's gate ran for the first time: all 13 Table B rows (eight prime, five random binary) passed—every counted field replays identically, every scalar checks in an independent verifier, and IC stays above rho at every weight of the band (4.05–54.76×)—and all nine Table C rows failed; Table A was not in scope (`research/ic_validation_gate_20261024/`). The study staged its manifest rather than installing it, and the leaderboard now marks the Table B rows validated. The ledger paper, current to 23 October, records that no row had passed; for Table B that is now out of date. Validation says the numbers are what the archive says and the reference is fair, not that index calculus approaches rho.

What is still not measured. No arm in the programme has yet been compared with a published index calculus at the same (q, n, m): FGHR-style symmetrised summation polynomials over the standard base $\{P : x(P) \in \mathbb{F}_q\}$, solved by msolve, or WDSat for binary decompositions. (WDSat has been run as a solver on its authors' published instances, §16, but not inside an index calculus with relations, linear algebra and a rho comparison.) Every ratio in Tables 31.4–31.6 is against rho or against the programme's own previous arm. Defining and running that reference arm (open problem 10) is the precondition for any claim of improving on the state of the art.

31.6 Three frontiers

Cold, one target, set-up included. Unchanged: generic rho. Every measured index-calculus arm is above its matched rho (Figure 31.4); the closest complete count is root-finding negfold at 1.55× strong signed-Frobenius rho at $n = 13$; the closest lower-bound arms are 1.08× (compact orbit with a counted Frobenius table) and 6.97× (matrix F4/F5 against rho-negation at K13). Prime-field arms are 12–239× rho and diverge with size (the 30-bit Kummer-tower row of §31.5 is 159×). Off the prime and prime-degree binary fields, three families are within a single-digit factor, all in weak classes: genus three after cubic GHS (below the strongest rho at $q = 2^{16}$ under the registered pricing, unresolved at the other sizes to 2^{20}, above it under a heavier sensitivity pricing), $k = 4$ over $\mathbb{F}_{(2^m)^4}$ (4.3× a measured rho at $n \approx 2^{56}$, closing as $n^{-1/4}$), and the Joux–Vitse cover of $E(\mathbb{F}_{p^6})$ (plain Nagao arm; 1.59–1.98× a pooled rho constant at $p = 1009$, closing as about p^{-2}; a lower-bound comparison).

Online, after retained set-up. Generic again: Bernstein–Lange preprocessing rho. The admitted compact-orbit point at Koblitz $n = 61$, 13.9× $[9.8, 19.1]$ over strong rho without a table (cold median 83.4× rho), is 53–221× slower online than preprocessing rho given equal or less advice (§31.3), and the $n = 83$ points lose every pair run with rho given a table matched on set-up time or memory (§§31.4,15.5).

Stage. Moved substantially: the decomposition oracle is two to five orders of magnitude cheaper than at the start of the programme on the workloads where root finding applies, the $\mathbb{F}_p$ F4 is at msolve parity, the Boolean F4 family is one to two orders of magnitude cheaper in counted matrix work, and the $k = 4$ extension-field solve is 48× and then a further 370× cheaper over $\mathbb{F}_{p^4}$ (481× below the plain form in characteristic two). None of this, by itself, moves the cold frontier: the stages it leaves untouched—relation yield, factor-base size, linear algebra, table set-up—now dominate.

31.7 What would move the cold frontier

From the frontier synthesis and the problem records: (i) register the torsion-symmetrised bases and the $n = 83$ dense-root engine in ecbench under immutable method identities and emit cold and online records; (ii) resolve the inherited-F4 cross-host counter drift before using that counter in a bound; (iii) run a four-size native challenge against the standing rho bound—two sizes cannot establish an exponent; (iv) obtain an isolated cold session at $n = 83$ with every set-up phase and retained byte charged; (v) price F6-IC's batched geometry with a pinned conversion. Since then three items have moved to the front: (vi) run $k = 4$ over $\mathbb{F}_{(2^m)^4}$ at $m = 15$–17 against a measured negation-map rho, where the fitted trend predicts parity near $n \approx 2^{68}$; (vii) measure the matched rho on $E_0(\mathbb{F}_{q^3})$ in the genus-3 rounds' unit (done on 23 October: below rho at 2^{16} under the registered pricing, unresolved elsewhere, §26); (viii) obtain a large-memory host for the $(4, 6)$ Weil-descent cell and the PKM tower. The 21–24 October work adds two more and re-weights one: (ix) measure rho on the Joux–Vitse cover family at $p \geq 503$ (or a size where it can be run) and emit the cover arm's unpriced counters, so that the p^{-2} trend can be ranked

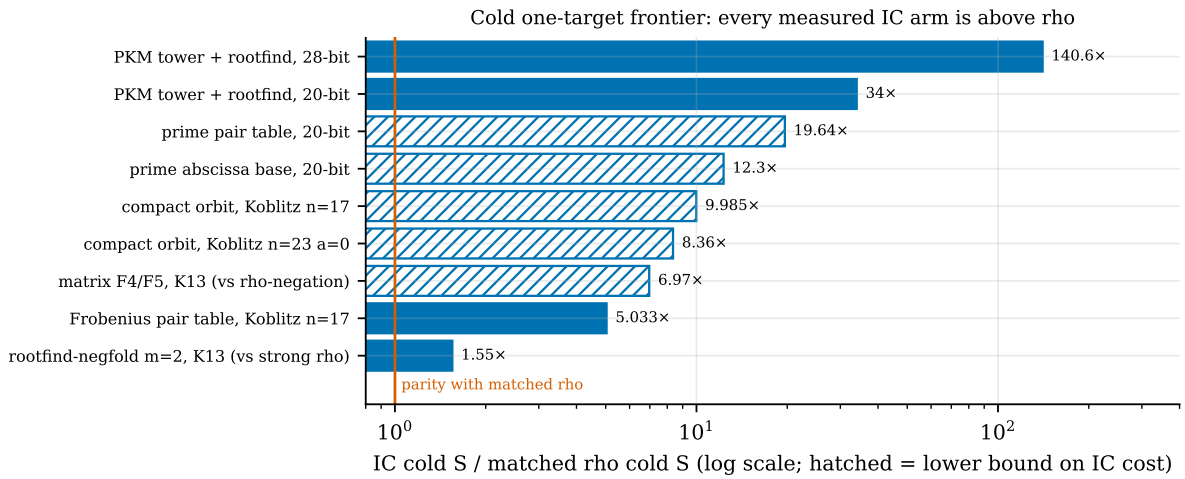


Figure 31.4: Cold one-target S of index-calculus candidates over their matched rho arm (each study's own estimator; curves differ, so bars are not ranked against each other). Hatched: lower bound on IC cost (unpriced native or solver work). Sources: `research/ecbench_frontier_20261013/`, `research/ecbench_rootfinding_assessment_20261012/`, `research/pkm_tower_ecbench_20261013/`, `research/ecbench_rootfind_faster_20261014/`.

under the band and tested past $p \approx 1330$; (x) define and run a published-index-calculus reference arm (open problem 10); and (viii) is worth less than it looked for the binary question, since the $m = 4$ row at $w = 2$ is about +46 bits over rho even if degree 5 holds (§31.5). A result in the cold domain that beats rho at four sizes would be the first ADVANCE of the programme.

32 The distance to rho, family by family

This chapter gives, for each curve family, the current distance between index calculus and a matched rho. Each figure carries its reference, its boundary and its validation status. Here a ratio is IC over rho, so greater than 1 means rho is cheaper. Chapter 31 gives the frontier campaign, the improvement ladders of 14–24 October, and the three frontiers (cold, online, stage). This chapter adds the work of 25–29 October and states the position as it stands at the end of the programme.

32.1 Koblitz curves

The leaderboard's Table A measures a Koblitz descent: aimed $m = 3$ collection, then an $m = 2$ or $m = 3$ descent. It is measured cold on one unseen public point per process, with 64 points per size, at six sizes from $2^{36.6}$ to $2^{47.2}$. Figure 32.2 shows the rule at three baselines, and Table 32.1 gives its ratios. From v3 to v7 the rho stayed the same and index calculus became 1.52–1.89× faster; the board carries the v7 figures, validated (§33.2). In v8, rho's signed-Frobenius canonicalisation, batched 16 lanes at a time in the same binary, raised the gap by 1.30–1.58× at every size. All six v8 values lie inside ranges declared before the rows were taken, and the 408 rows of each run all pass the `vs_rho` check (`research/ic_tool_program/rule/v8/README.md`). The v3→v7 improvement lowered the constant, not the growth.

The priced campaign. The closest Koblitz arm, compact Frobenius orbits, is 3.8–15× signed-Frobenius rho at $w = 0$. Fully priced against fully priced rho, its best cell is 1.566 $[0.992, 2.347]$; the ratio is never below 1 and grows as $r^{0.165\,[0.117,0.213]}$.

$n = 83$. The cold pipeline uses the base K1107/root11, which confirmed a saving of 1.195 s $[0.957, 1.433]$ on a 115 s process (16 public points, GitHub Tier 1 host). The last cold IC/rho ratio is the K1120 holdout's 55.4, from five points. A matched precomputed-rho reproduction on 28 October gave the median online IC over rho of Table 32.2, for three ways of matching rho's precomputation. Cold, IC took 155.5 s against 36.96 s for plain rho over all 16 targets. In operation counts the order flips across the pricing band, so no ranking is made.

32.2 Prime curves

Generic track. The prime-field programme's owner decisions of 29 October split prime-field index calculus into two counted tracks (`research/prime_ic_program_20261029/DECISIONS.md`):

- a *generic* track, with table, pair-table and root-finding oracles, labelled "bounded by $\sqrt{r}$; cannot beat rho";
- an *algebraic* track, whose prime baseline is empty.

Table 32.3 gives the distances on the generic track. The campaign pair table's exponent against rho's is the $r^{2/3}$ against $r^{1/2}$ of Chapter 2. The k-sum oracle (`research/prime_ksum_table_20261028/`) is a negation-folded table of every signed 4-sum of a smaller base. It lowers mean S at $w = 0$ by 1.9× at 20 bits, rising to 7.2× at 40 bits (83.7 → 11.6), and its fitted exponent falls from 0.620 $[0.586, 0.650]$ to the tabulated value, against the theory value $k/(2k-1) = 4/7$. 2,016 runs were verified and replayed; the source notes that the method is generic, so its limit is baby-step giant-step. The new 20-bit baseline (`research/prime_base_oracle_tuning_20261029/`) raises the factor base from 32 to 45, which gives 0.818 $[0.773, 0.866]$ of the incumbent at $w = 0$; a second host (L2) replayed all 1,008 records identically and gave 0.836 $[0.806, 0.866]$. The same rule at 24 and 28 bits was a relabelling, and at 32 bits null.

Algebraic track. The Kummer tower is 28–159× rho (158.9 $[121.6, 221.4]$ at 30 bits). No algebraic prime baseline exists. Its solving degree is the subject of §6.5.

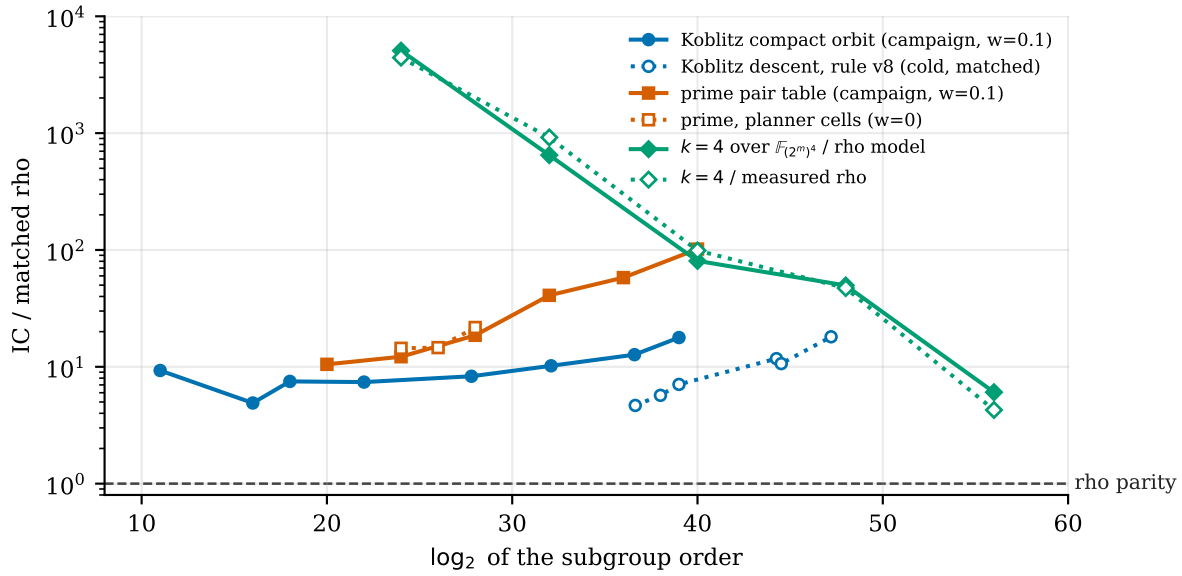


Figure 32.1: Distance to matched rho by family and size. Koblitz and prime curves: the frontier campaign's priced arms ($w = 0.1$, 16 targets × 2 rounds); open markers are the Koblitz rule at v8 (cold, 64 targets) and the prime planner cells ($w = 0$). $k = 4$ over $\mathbb{F}_{(2^m)^4}$: the characteristic-two ladder against the negation-map rho model (filled) and a measured rho (open); the $n \approx 2^{56}$ point is one target. Curves and units differ between families, so the lines are not ranked against each other. Sources: `figures/data-ledger.json`.

32.3 Binary curves of prime degree

Table B's random-binary rows are 10.6–38.2× matched rho at $\mathbb{F}_{2^{15}}$–$\mathbb{F}_{2^{27}}$, and they are validated. A cofactor-projected base cuts three of them by 1.29–3.36×, leaving 6.1–14.4×. That base is not yet a board row. For Certicom ECC2-131, the $m = 3$ floor with a free oracle is already above rho (Chapter 16).

32.4 Weak extension-field families

Only in these families (Figure 32.3, Table 32.4) does index calculus approach rho or go below it. The genus-3 hyperelliptic row is the programme's only preregistered decision of IC below the strongest rho (Chapter 26). The plane-quartic row uses double large primes, structured elimination and Lanczos; its 501 IC units and 2,442 rho runs are all verified, and it is a measurement of a known result (Diem, Diem–Thomé).

32.5 Summary

On every prime-order curve of cryptographic shape that the programme measured, rho is cheaper. Index calculus goes below rho only in weak extension-field families, which is where theory said it would. There the programme's contribution is matched, priced, verified measurement of known algorithms.

Rule	Rho	IC/rho	Slope	Status
v3	same as v7	5.23–20.97	0.176	
v7	same as v3	3.43–11.5		on the board; validated
v8	matched, same binary	4.67 [4.34, 5.07], 5.70, 7.07, 11.75, 10.65, 18.07 [15.98, 20.58]	0.168 [0.128, 0.208]	inside pre-declared ranges
v8	canonical step (model)	6.45–23.32		in v8's unit
v4	canonical step (model)	9.9–43		§31.5

Table 32.1: The Koblitz rule (leaderboard Table A) at each baseline. IC/rho: cold IC over cold rho on the same unseen public point, one point per process, 64 points per size; a range runs over the six sizes $2^{36.6}$ to $2^{47.2}$, and the v8 row lists all six values in the order the source gives them. Slope: of the gap in $\ln r$. The canonical-step rows are a model of rho, not a measured rho: rho priced at the canonical batched step. Source: `research/ic_tool_program/rule/v8/README.md`.

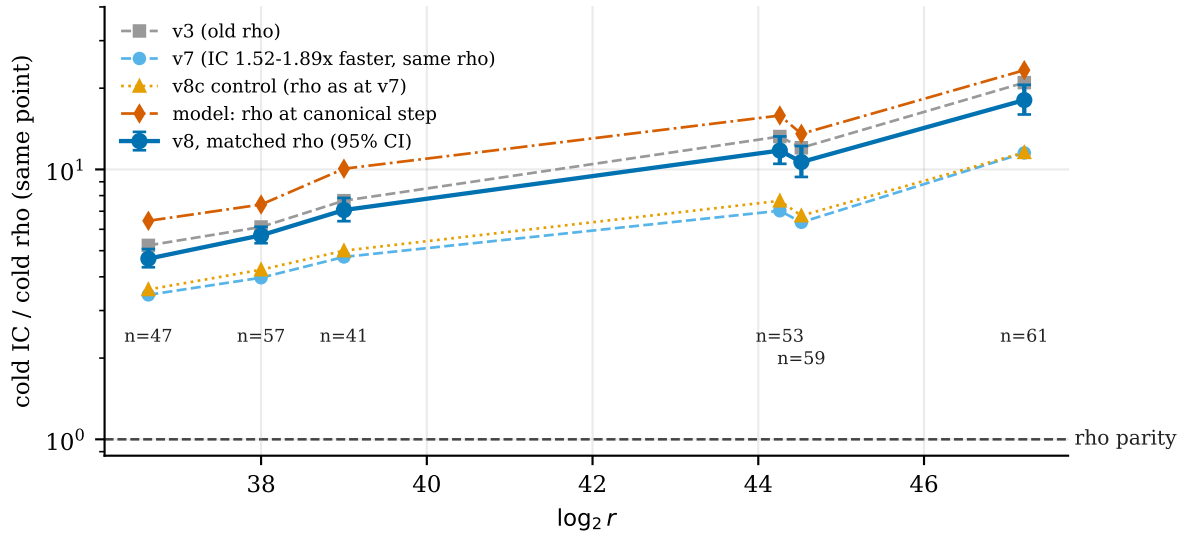


Figure 32.2: The Koblitz rule at three baselines: cold IC over cold rho on the same public point, 64 targets per size. From v3 to v7 the IC improved against the same rho. v8 runs the same IC against a matched, faster rho; its same-binary control, v8c, reproduces v7. The dash-dot line prices rho at the canonical step. Source: `research/ic_tool_program/rule/v8/README.md`.

Rho precomputation	IC/rho	Interval
650 s table walks	40.9	[7.58, 141.9]
walks matched to IC set-up	4.10	[1.58, 24.8]
table matched to IC memory	33.1	[6.60, 74.4]

Table 32.2: Koblitz $n = 83$, base K1107/root11: median online IC over a precomputed rho, by how rho's precomputation is matched; matched reproduction of 28 October. Source: `research/n83_k1107_bl_reproduction_20261028/RESULT.md`.

Arm	Bits	IC/rho	Exponent	Status
Table B prime rows	10–24	5.47–13.5		validated
Campaign pair table, $w = 0.1$	20	10.5	0.620 [0.557, 0.663]	
	40	101.8		
k-sum table (28 October)	20	5.3	0.521 [0.506, 0.538]	L0; no validation
	40	13.1		
New 20-bit baseline	20	7.75 [6.92, 8.70]		validated; was 9.47
Planner cells		12.6–25.8		

Table 32.3: Generic-track prime-field index calculus. IC/rho: cost over rho, as each source states it (Table B against matched negation rho; the new baseline against negation rho); greater than 1 means rho is cheaper. Exponent: fitted cost exponent in r; the campaign's rho with negation fits 0.435. Sources: leaderboard Table B; the campaign (§31.1); `research/prime_ksum_table_20261028/`; `research/prime_base_oracle_tuning_20261029/`.

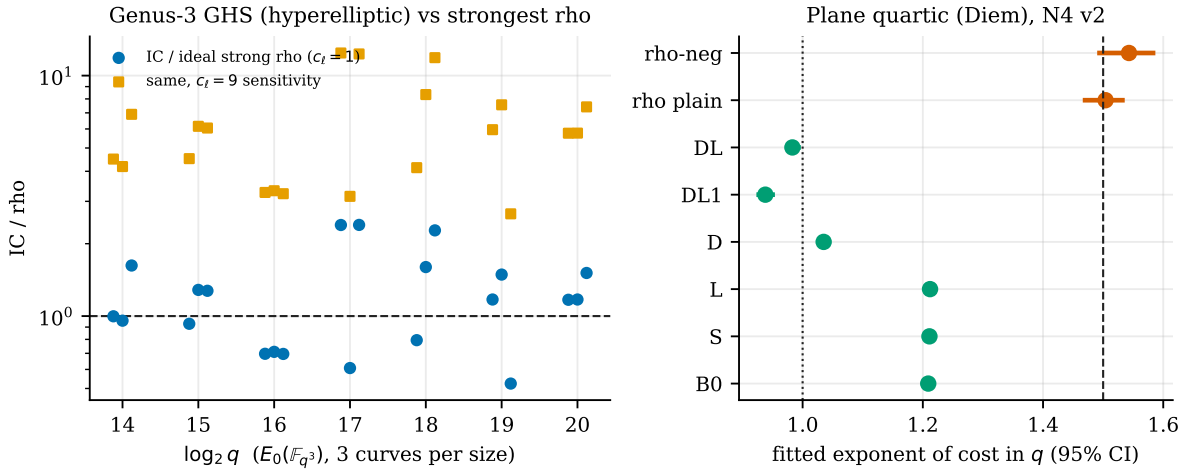


Figure 32.3: Weak families. *Left:* genus-3 GHS index calculus (hyperelliptic, double large primes, round 4) over the ideal strong rho on 21 instances of $E_0(\mathbb{F}_{q^3})$, in round 4's unit. Blue: one subgroup multiplication priced at one $\mathbb{F}_q$ multiplication, the registered unit. Orange: priced at nine, the protocol's sensitivity. Below the dashed line IC is cheaper. Source: `research/ghs3_vs_rho_20261023/TABLES.md`. *Right:* plane-quartic (Diem) index calculus, N4 v2. Fitted exponents of cold cost in q over $k = 8$–22 against plain and negation rho, at $w = 0$. Dotted line: $\tilde{O}(q)$. Dashed line: rho's $q^{3/2}$. Source: `research/quartic_n4v2_20261020/README.md`. Data: `figures/data-new.json`.

Family	Size	IC/rho	Rho	Exponents	Status
Genus 3, cubic GHS, hyperelliptic	$q = 2^{16}$	0.70, 0.71, 0.70	ideal rho	pooled slope 0.034 $[-0.047, 0.105]$ per bit	preregistered; at $c_\ell = 9$ every instance 2.7–12.4
Genus 3, cubic GHS, plane quartic (Diem)	$q = 2{,}097{,}091$	0.005 $[0.004, 0.008]$	negation rho	0.983 $[0.975, 0.993]$ against 1.543	interpolated crossover $q^* = 555$ $[389, 752]$; L0; no validation file
Joux–Vitse cover of $E(\mathbb{F}_{p^6})$	$p = 1009$	1.59–1.98	pooled rho constant		closing as about p^{-2}; lower bound
$k = 4$ over $\mathbb{F}_{(2^m)^4}$	$m = 15$	5.2			after corrections; crossover unsupported
GHS covers over $\mathbb{F}_{4096}$		549–2,039			exact genus-3 covers found by search; full transfer cost

Table 32.4: Weak extension-field families. IC/rho: cost over the stated rho reference; greater than 1 means rho is cheaper. Exponents: fitted cost exponent in q, IC against rho. Genus-3 hyperelliptic: at six other sizes the decision is unresolved; $c_\ell = 9$ prices one subgroup multiplication at nine $\mathbb{F}_q$ multiplications. Plane quartic: three caveats, the host was L0 under heavy load, there is no validation file, and the frozen baseline's records do not replay. Sources: Chapter 26; `research/quartic_n4v2_20261020/`; Chapter 25; `research/ghs_q16_genus3_subcovers_20261028/REPORT.md`.

Family	IC/rho	Qualifier
Plane quartic via cubic GHS, $q \approx 2^{21}$	0.005	negation rho; weak family; L0; no validation file
Genus-3 hyperelliptic via GHS, $q = 2^{16}$	0.70	ideal strong rho; one size of seven; pricing-sensitive
Joux–Vitse cover, $p = 1009$	1.59–1.98	pooled rho constant; lower bound
$k = 4$, $\mathbb{F}_{(2^m)^4}$, $m = 15$	5.2	two targets; crossover unsupported
Koblitz, fully priced orbit (best cell)	1.566	never below 1; grows as $r^{0.165}$
Koblitz rule v8, $2^{36.6}$	4.67	cold, matched, 64 targets
Prime, k-sum table, 20 bits	5.3	generic track; not validated
Prime, Table B, 12 bits	5.47	validated
Random binary, Table B	10.6	validated
Koblitz $n = 83$	55.4	cold, five points

Table 32.5: Smallest measured distances to rho, by family.

33 Corrections and validation

A ratchet is worth only as much as its pawl. Several recorded results were corrected once the reference was matched, native work was priced, or the evidence was replayed. Each correction is recorded in the source with a dated note; frozen evidence is never rewritten. Figure 33.1 shows the before and after values. Every correction moved IC's position against rho *down*, and several moved it from apparently ahead of rho to behind it.

33.1 Claims that did not survive

1. Every online win is retired. The one-target online ladder compared IC *with* a retained table against rho *without* one (Table 33.1).

Given a distinguished-point table no larger than IC's pair table, Bernstein–Lange preprocessing rho [4] is 53–317× faster online at $n = 41, 53, 61$ on the same 64 targets (benchmarks row `20261014-ic-next10-09-equal-advice`). At $n = 83$, rho with a table matched to IC's set-up time or memory wins every pair: 3.74× time-matched and 40.61× memory-matched against K3300, and 81.5–395.5× against K600 (benchmarks row `20261016-n83-precomputed-rho-control`). Cold, the hosted $n = 83$ point is 133.7× rho. Those online rows measured what a table buys, and generic rho buys more per byte. The within-IC result survives: K1798's probe reduction over K1500 at $n = 83$ (from 300,021.9 to 169,069.8 probes per relation).

2. Pricing native work moved two arms. Table 33.2 gives both before and after.

The pricing change itself (#1774, 0.002304 GAE per word XOR) made every earlier F4 S a lower bound. A separate algorithmic change (#1862) raised counted work (`research/f4_pricing_jump_20261018/README.md`).

3. Matching the rho reference. Table 33.3 lists the changes to the rho side.

4. A replay regression, and its root cause. The root-finder ecbench session was recorded with a replay certificate of 120 of 168 identical records; the 48 mismatches were all in inherited F4's word-XOR counts. On `main` of 19 October it replayed only 72 of 168, with mismatches in all four index-calculus arms.

- *First diagnosis* (20 October). A cross-ISA replay found that with the x86-only BMI2 branch disabled, x86 reproduces all 168 of 168 records, and so does an aarch64 build under emulation (`research/validation_sweep_20261019/D-known-issues/item1_cross_isa_20261020/RESULT.md`).
- *Cause, stated precisely* (26 October addendum). At the recorded commit, BMI2 availability decided whether F4's row specialisation was *eager* (every kept row rewritten at once and charged per row) or *lazy* (rewritten on first read). The two differ in the quantity charged, for example 301,961 against 300,531 word XORs. Since #2043 every host is eager and the two eager paths charge identical counts. Sessions now record the version of the F4 accounting they were made under (`research/validation_sweep_20261019/D-known-issues/ADDENDUM-20261026-item1-accounting.md`).

Archived sessions recorded with the lazy charge cannot replay on `main`. That is a change of definition, not a dependence on the host. Scalars, trial counts and charged group operations are unaffected.

5. Reported cells that compared unlike statistics. The u64-rootbatch rank and per-target cells compared a fastest run with a median, and were corrected from 2.95× and 6.09× to 2.47× and 5.00× (Chapter 15). The $n = 83$ stacked producer's "360 → 94.4" cold-gap history mixed a ratio from another run; paired within the run it is 305.8 → 94.4 (Chapter 15).

6. Results resting on faulty code or wrong references.

- *Plane-quartic IC.* Its S values of 650,896 to $3.79 \cdot 10^7$ rest on a degree-3 reduction in `research/ghs_cubic_quartic_ic_20261012/quartic_ic.rs` that merges distinct classes (reported in an open pull request). Recoveries are still checked against the planted scalar, but relation counts and S are unverified.

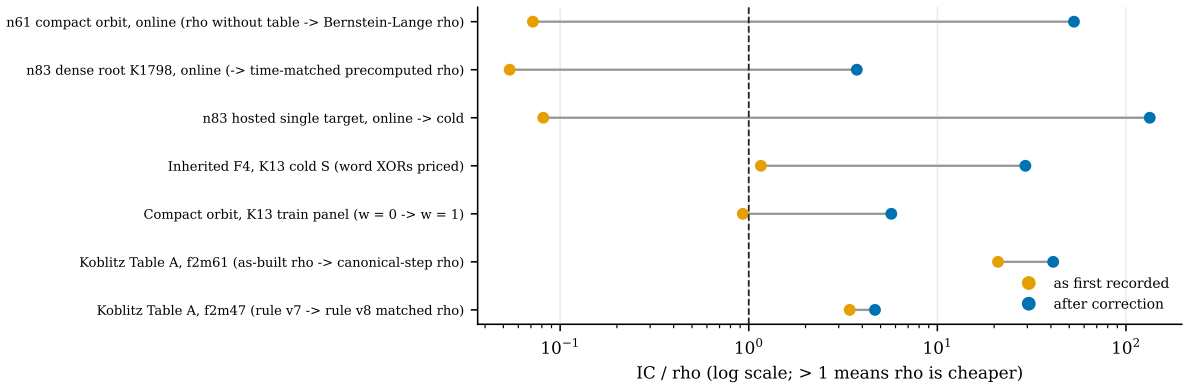


Figure 33.1: IC/rho as first recorded (orange) and after correction (blue), on the boundary the correction used. Values left of the dashed line had claimed IC ahead of rho. Sources: `figures/data-ledger.json`.

n	Reading	IC online over rho
61	strong rho, 64 targets	13.94× [9.81, 19.13]
83	dense root index	18.53×
71	hosted single target	7.29×
83	hosted single target	12.29×

Table 33.1: The retired online wins: IC's online speed-up over rho with no table, IC keeping its retained table.

- *Volcano census crater.* A claim that IC is 22–25% cheaper on the volcano crater was withdrawn, because the crater's rho reference should have been the cubic-automorphism rho. Against it, IC is 84.9× rho at $w = 0$ (`research/volcano_ecdlp_census_20261017/REPORT.md`, erratum A3).

Reading. None of these corrections removes an IC improvement against its own incumbent. They correct the *distance to rho*, which is the quantity most exposed to an unmatched reference or an unpriced counter. This is why the gate of §33.2 asks for a matched, band-priced rho on every row.

33.2 Validation status

Since 19 October no row may enter the index-calculus leaderboard or `benchmarks/README.md` without three pieces of evidence, named by the evidence directory's `VALIDATION.json` (`docs/VALIDATION_GATE.md`):

- an independent replay certificate with verified scalars;
- an independent validator report that recomputes every headline number;
- a matched rho reference: the strongest available, in the same unit and the same binary, priced across the §5a band.

On 29 October, **19 of the 28 leaderboard rows are validated** (`docs/ic/leaderboard.json`, `validation_gate.status_counts`).

Table 33.4 gives the status by leaderboard table.

Validation says the numbers are what the archive says, and that the reference is fair. **No validated row is below its reference.**

The theory of Part II has a different standard of evidence:

- written proofs, which still need expert review;
- independently checked certificates;
- Lean for the finite and algebraic identities only.

Each theorem there names its tier.

Arm	Reading	Rho	IC
Inherited F4, $n = 13$	Boolean word XORs unpriced	$S = 14.4$	$S = 16.7$
	priced		$S = 421.4$
Compact orbit, $n = 13$	earlier, lower bound		1.08× rho
	strong rho, $w = 0$		0.93
	strong rho, $w = 0.1$		1.53
	strong rho, $w = 1$		5.7×

Table 33.2: The two arms moved by pricing native work. Inherited F4, whose unpriced S was close to rho's: `research/ecbench_rootfind_faster_20261014/`. Compact orbit at the three band weights w; the pair is unresolved inside the band (`research/ic_frontier_campaign_20261015/select/SELECTION.md`).

Change	Quantity	Scope	Factor
Rule v8, matched rho	Koblitz gap raised	every size	1.30–1.58×
Standing rho improved	instruction count	`rho.signed_frobenius_strong`	0.802×
	instructions per step	strong Koblitz rho	718 instead of 1,196
Better of $P \pm R$	steps cut	prime side	1.129×
Signed σ steps	GPU time cut to	ECC2K-130	0.961–0.970

Table 33.3: Changes to the rho reference and their factors. Rule v8: Chapter 32. The improved ecbench standing rho keeps identical GAE, so counted-unit comparisons are unaffected; wall-time comparisons against the older rho overstate IC by up to that factor. The last two are rho studies, neither yet in a campaign arm.

Table	Scope	Rows	Status	Rho reference	IC/rho
B	prime and random binary, one target	13	validated 24 October	matched negation-map	5.47–38.21×
				across the band	4.05–54.76×
A	Koblitz descent, one target	6	validated at baseline v7	as-built, same process	3.43–11.5×
				matched, rule v8	4.67–18.07×
C	32-target batch	9	pre-gate	none matched, no band	
—	components outside A–C	2	validated		

Table 33.4: Validation status of the leaderboard on 29 October. Table B validated on 24 October (`research/ic_validation_gate_20261024/`): 1,257 runs replay identically in a fresh process; 1,578 of 1,578 negative controls are rejected by an independent verifier; 52 headline numbers recompute to relative 10^{-9}; index calculus stays above rho at every band weight. Table A (64 targets per size) validated at baseline v7 (`research/ic_tool_program/rule/v7/`); the sweep flags the reference as only partly adequate, and the rule v8 row is from Chapter 32. Table C's replays and recomputations pass, but no matched rho with a band exists for them. The two components are the prime-field base-size tuning and the post-Gröbner linear stage (`research/prime_base_oracle_tuning_20261029/VALIDATION.json`, `research/postgb_linear_stage_20261026/validation/VALIDATION.json`).

Part VI

Outlook

34 Open problems and proposed experiments

1. The solving degree of Weil-descended summation systems. Every subexponential claim for prime-degree binary curves rests on a bound for this degree (§3); none is proved and the experimental evidence is contested. *Experiment:* measure first-fall, last-fall and solving degree of S_{m+1} restricted to ℓ-dimensional subspaces over a grid of (n, ℓ, m) with $m \geq 3$, at least four values of n per (ℓ, m), against the semi-regular prediction, with multi-graded regularity (one block per summand) as a second model. *Settles:* whether the degree grows with n at fixed (ℓ, m); a growth consistent with zero would revive the Petit–Quisquater and Semaev heuristics, a positive slope would close them for this model.

2. Cold index calculus against rho with every operation priced. No measured arm beats matched rho on the cold one-target boundary once native work is priced on both sides (§31). *Experiment:* the frontier campaign of §31 extended to the largest sizes each arm can reach, with exponents fitted per arm and the crossover reported only where the fitted exponents separate across the whole pricing band. *Settles:* whether any implemented arm has an exponent below $1/2$ in r on these families.

3. Relation yield is now the dominant stage. Root finding made the decomposition oracle cheap; what remains is the number of trials per relation, governed by the factor-base density. *Experiment:* for each factor-base family, the measured trials-per-relation law against r and base size, compared with the counting prediction $m!\, r/N^{m-1}$; the residual (e.g. the unexplained factor 5 in the ECC2K-130 extrapolation, §15) is the target.

4. Root finding beyond two summands on prime curves. The two-summand oracle is prior art in mechanism (McGuire–Mueller) and costs $\Theta(N)$ per trial; three summands cost a pair table of N^2 entries. *Question:* is there a sub-N^2 structure for S_4 membership over a prime-field base—for example from the D_m-invariant coordinates of 2-torsion curves—and does it change the $r^{2/3}$ cost of two-summand prime index calculus? A negative answer with a counting argument would be as useful as a positive one.

5. The preprocessing regime. Index calculus with a retained table beats rho without one, and loses by 53–317× to Bernstein–Lange rho with an equal or smaller table (§31.3). *Experiment:* measure the advice–time curve (S, T) of the compact-orbit arm across sizes against preprocessing rho's $S\,T^2 \approx N$; an index-calculus online result counts only if it lies below that curve at equal S.

6. Symmetry at $k = 4$, end to end. The D_4 reduction and its four-unknown weighted form move a *modelled* handover over $\mathbb{F}_{p^4}$ from about 2^{150} to 2^{119}–2^{137} depending on frame (§31.3). Since then the odd-characteristic oracle is 370× cheaper again (index calculus still $1.39 \cdot 10^6\times$ rho), and in characteristic two a Frobenius-reduced form brings the measured gap to 6.1× rho at $n \approx 2^{56}$, closing as $n^{-1/4}$ (§31.4). *Experiment:* continue the characteristic-two ladder to $m = 15$–17 with a measured negation-map rho at every rung, cold and priced across the band; a measured crossover near $n \approx 2^{68}$ would confirm the fit, and its absence would locate the term the model misses. A red-team review (Chapter 25) found the only clean point, $m = 15$, at 5.2× rho after corrections and placed the crossover at $m^* \approx 17.6$ (16.3–19.2); its frozen replication plan (eight targets per arm, $m = 14$–17, about 30 h) is the experiment to run.

7. Pricing and calibration infrastructure. Rankings now require every arm's native work to be counted (AGENTS.md §5a). *Work:* count the three uncounted categories of the shared-rank arm; calibrate every ladder size on one fixed host; record the inherited-F4 code path (BMI2 or portable) in its method identity, now that the cross-host counter drift is traced to it; register the $n = 83$ dense root index in the harness.

8. Genus three after cubic GHS, to the crossover. The one family whose fitted exponent (about $q^{1.4}$) lies below rho's $q^{3/2}$ was 1.15× a measured negation-map rho at $q = 2^{20}$ before rounds three and four cut a further 0.44× (§§31.3,31.4), and is now below the strongest rho at $q = 2^{16}$ under the registered pricing, unresolved at the other sizes to 2^{20}, and above it under a heavier sensitivity pricing, with no

measurable pooled trend (§26). What would settle it is the same comparison at $q = 2^{21}$–2^{24} on instances of one cofactor, with the subgroup multiplication's price measured rather than chosen; its local exponent is down to 1.356, still above 4/3 because the linear algebra grows faster than $q^{4/3}$, and $q = 2^{23}$ needs more than the 4.5 GB cap. *Experiment:* measure negation-map rho on the same instances in the latest unit (inversions and mod-ℓ multiplications priced), and run both past $q = 2^{22}$, in the harness and across the pricing band. A measured crossover there would be the programme's first ADVANCE against rho, on a narrow and undeployed family.

9. Plane quartics: Diem's $\tilde{O}(q)$. Genus-3 GHS covers whose Jacobian is that of a smooth plane quartic admit Diem's index calculus [9, 11] at $\tilde{O}(q)$, against $\tilde{O}(q^{4/3})$ for the hyperelliptic route and $q^{3/2}$ for rho on $E(\mathbb{F}_{q^3})$. The repository builds such quartics directly ($m = 2$ GHS, toy index calculus at $q \leq 11$) and through Smith's $(2, 2, 2)$-isogeny from the hyperelliptic case (18.6% of random curves); a four-workstream programme was scoped on 18 October (`research/plane_quartic_program_20261018/PROGRAM.md`). *Experiment:* the hyperelliptic ladder's instances, carried to plane quartics where Smith's conditions pass, with Diem's relation search and the same linear algebra, against the same matched rho. The 23 October literature round ranks this branch as its one exponent candidate (LEAD-15, `research/pdp_literature_leads_20261022/`). *Since then:* the arithmetic has been rebuilt with canonical class keys, which resolve the degree-3 merges (§4.5). On the direct quartic route, index calculus is below a matched plain rho from $q = 4051$, with fitted exponent 1.083 against 1.485, and N4 v2 extends this against a negation-map rho to $q \approx 2^{22}$, at exponent 0.983 and 0.005× rho (§26). *Still open:* a validation of N4 v2 on an isolated host with a replayable baseline; index calculus on the quartics reached through Smith's isogeny; and a record-scale run (for example $q \approx 2^{40}$, $\ell \approx 2^{120}$) where rho is out of reach and the published Diem–Thomé timings are the comparator.

10. A published index calculus as the reference arm. Every ratio in this treatise is against rho or the programme's own previous arm; none is against the state of the art in index calculus at the same (q, n, m) (§31.5). *Work:* fix, per curve and size, the standard base $\{P : x(P) \in \mathbb{F}_q\}$, the FGHR-symmetrised summation polynomial (2-torsion and, on Koblitz curves, Frobenius invariants), msolve as the solver and WDSat for binary decompositions, and run it in ecbench on the existing ladders (binary $n = 13$–37, Koblitz $n = 41$–61, $k = 4$ extension fields). *Settles:* whether any engineering result of §§9–26 improves on published index calculus, as opposed to on this programme's own baseline. Stage ratios quoted against the in-house F4 say nothing about published Gröbner engines until this is done.

11. The Joux–Vitse cover, to the crossover. The cover of $E(\mathbb{F}_{p^6})$ closes on rho as about p^{-2} and is 1.6–2.0× a pooled rho constant at $p = 1009$, with the line meeting rho near $p \approx 1330$ by extrapolation (§31.5). The cover's *sieve* arm has since been measured below a matched negation rho at $p = 1511$, by 155× [125, 184], and validated, with the order holding across the band (§26). Three questions remain:

- why the two arms differ by about 30× at $p = 1009$;
- where the Nagao arm itself crosses rho;
- whether the sieve arm's order survives once its F4, descent and linear-algebra work is emitted and priced, rather than bounded.

Experiment: emit those counters on both arms, and run both arms and rho at $p \approx 1300$–2000 on the isolated lab, across the band. On a weak, undeployed class this would turn a validated accounting result into a priced ADVANCE.

12. A growing lower bound on the last fall degree. Chapter 6 proves the first fall degree constant and certifies $d_F \geq 5$ on named binary instances, but no presentation has a proved lower bound that grows with n or k. *Proof route:* construct degree-c pseudo-solutions, i.e. linear functionals on $B_{\leq c}$ that kill $W_c(F)$ and not 1, for a family of instances, generalising the closure certificates of Proposition 6.4. The regular closure pattern observed (exactly $4n$ new cubic dimensions per round, then a stall) and the iterated-system lower-bound technique of [38] are the starting points. *Settles:* the open problem of HKY [23] and Kousidis–Wiemers [29]; a bound $\omega(\sqrt{n \ln n})$ would invalidate the subexponential conclusion of Semaev's analysis for these systems.

13. Certificates at scale. *Experiment:* certify $1 \notin W_4$ (and, where it holds, $1 \notin W_5$) on about ten random and ten polynomial subspaces per n for $n = 33$–51, publish the certificates, and extend the guarded chain to $B = 4$–8 by computation. *Settles:* whether the step $4 \to 5$ is a population property, and where $5 \to 6$ occurs; the exact d_F of the guarded chain beyond $B = 3$.

14. The prime-field Kummer tower, re-measured. The session values $d_F = 3, 4, 4, 4, 4, 5, 5, 5, 5, 6$ at $t = 1..10$ were never committed. *Experiment:* regenerate stage A at $t = 1..14$ with closure certificates and score the registered prediction $d_F = 4 + \lfloor (t-2)/4 \rfloor$; extrapolated F4 times are about 1.7 h, 17 h and 7 days at $t = 12$–14 (`research/prime_ic_program_20261029/`). *Settles:* whether the proved constant first fall degree 4 sits beside a last fall degree that keeps growing in the setting PKM introduced.

35 Conclusion

Four months of work on elliptic-curve index calculus produced four kinds of result.

Theory.

- *What is fixed for every curve.* The summation cover's monodromy is $(\mathbb{Z}/2)^{m-2}$ everywhere. Its fibres split completely where relation search operates. The mean yield at a given base size is a double count. So no choice of curve or base geometry can raise the *mean* decomposition yield at matched base size. Coverage, the cost of the membership oracle and the difficulty of the solve remain free, and the engineering of Part III works on those.
- *What is open.* For three presentations of the decomposition system the first fall degree is a constant of the presentation, and the last fall degree is not bounded by it:
  - proved for the guarded chain;
  - certified on named instances of binary Weil descent;
  - measured for prime-field Kummer towers.

  The subexponential claims for binary curves assume exactly the step these results show is not automatic. Whether the last fall degree grows without bound is unproved.

Engineering. The decomposition oracle became two to five orders of magnitude cheaper where root finding applies. The Gröbner engines improved by one to two orders in counted work. Factor bases closed under known-logarithm maps, a k-sum table, and set-up and linear algebra engineering each cut a constant. None of this changed an exponent on a prime field or a prime-degree binary field. The prime-field k-sum oracle moved its exponent from 0.62 to 0.52, but it is generic, and its limit is baby-step giant-step.

Measurement. At a matched reference, priced across a band and replayed, rho is cheaper on every prime-order curve the programme measured:

- 4.7–18× on Koblitz curves of 37–47 bits;
- 5.3–102× on prime curves of 20–40 bits;
- 10.6–38× on random binary curves.

Nineteen leaderboard rows passed a validation gate that demands independent replay, recomputation and a matched rho. None of them is below its reference. Every online win disappeared under an equal-advice audit, and the corrections forced by matching, pricing and replaying lengthened the measured distance more often than any single improvement shortened it.

Where index calculus wins. In weak extension-field families ($E_0/\mathbb{F}_{q^3}$ through genus-3 covers, Joux–Vitse covers of $E(\mathbb{F}_{p^6})$, and perhaps $k = 4$ in characteristic two) the measured cost crosses or approaches rho, as theory predicts. The plane-quartic route measured exponent ≈ 1 against rho's 1.5, at 0.005× negation rho near $q = 2^{21}$. These are known algorithms; the programme's contribution there is fair measurement.

What would change the picture. Three items, in order of what they would settle:

1. *A proved lower bound on the last fall degree that grows with n*, for any presentation of the binary decomposition system. It would answer the question left open by Huang, Kosters and Yeo and by Kousidis and Wiemers, and it is the only result that bears on the subexponential claims.
2. *More certified instances* at $n = 33$–51, with public certificates, to show whether the last fall degree reaches 6.
3. *The frozen $k = 4$ replication* at $m = 14$–17, the cheapest experiment that could put a measured crossover on record. It would be in a weak family.

Nothing here concerns a deployed curve. The open problems of Chapter 34 state each remaining question as the experiment or proof that would settle it.

A Artefacts

Every number in this treatise is copied from a committed artefact. The treatise ran one re-verification (§6.3) and no new measurement. Artefact paths are relative to the `crypto` repository unless prefixed `crypto-autoresearcher:`.

Companion documents.

- `papers/ecdlp-index-calculus-program/` [3] is the chronological record, with study catalogues for both repositories (about 1,400 directories) and the list of open pull requests. Where this treatise reuses its chapters (background, literature, measurement, root finding, Gröbner engines, Koblitz, binary and prime curves, transfers, frontier, open problems), the text is that record's as of 26 October, updated where later evidence changed a number.
- `papers/ecdlp-ic-improvement-ledger/` [2] is the ratchet of incumbents to 23 October.
- `research/paper_opportunities_20261029/` is the assessment that identified the fall-degree results of Chapter 6 as the programme's strongest theory, and that holds the certificate re-verification.

Theory (Part II).

- `crypto-autoresearcher:papers/semaev-conservation-specialization/` (Chapter 5).
- `crypto-autoresearcher:analysis/semaev-m2-last-fall-degree/`: `theorem-dossier.md`, `results/`, and `verify/` (`cert_check.py`, `b_sweep.py`).
- `research/guarded_fall_degree_20261028/`: `proofs/`, `extensions/`, `native/`, `formal/`.
- `research/prime_fall_degree_20261027/DERIVED.md`.
- `crypto-autoresearcher:papers/nagao-lemma4-note/` and `crypto-autoresearcher:docs/reports/2026-10-22-semaev-first-fall-replay/`.
- `crypto-autoresearcher:knowledge/leads/ecdlp/` (Chapter 8), `research/twist_obstruction_20261026/` and `research/lift_dense_factor_base_20261027/`.

The summer record (Chapter 30).

- `research/ecdlp_jul_sep_index_20261029/`: `INDEX.json`, `INDEX.tsv`, `SUMMARY.json`, `build_index.py`. Each row cites its `crypto-autoresearcher:ledger/decisions/` file and the `experiments/EXP-*/runs/` directories of its targets.

Late measurements (Chapters 25–33).

- `research/gaudry_k4_char2_redteam_20261026/REPORT.md`.
- `research/quartic_n4v2_20261020/` and `research/ghs3_vs_rho_20261023/`.
- `research/prime_ksum_table_20261028/` and `research/prime_base_oracle_tuning_20261029/`.
- `research/prime_ic_program_20261029/`.
- `research/n83_k1107_bl_reproduction_20261028/`.
- `research/ghs_q16_genus3_subcovers_20261028/`.
- `research/postgb_linear_stage_20261026/`.
- `research/ic_validation_gate_20261024/`, `docs/ic/LEADERBOARD.md` and `docs/ic/leaderboard.json`.

Figures. The figures of the programme record and the ledger are copied with their data (`figures/data-program.json`, `figures/data-ledger.json`); their plotting scripts live in those papers' directories. The new figures (fall degrees, weak families) are rendered by `figures/make_fig_fall_degrees.py` and `figures/make_fig_weak.py` from `figures/data-new.json`. That file holds recorded values only, each with its source.

B Corrections, retractions and withdrawn readings

This appendix collects the corrections, retractions, errata, withdrawn claims and reversed verdicts recorded in the program's evidence directories and ledgers that the main text does not already carry. Chapter 33 and §4.5 hold the corrections that moved index calculus against rho on the leaderboard (the retired online wins, the priced word XORs, rule v8, the cross-host F4 counter drift, the volcano-crater and plane-quartic withdrawals). They are not repeated here.

Each entry gives what was first read, the record that read it, what replaced the reading and when, and the source. Three conventions apply throughout.

- *Standing.* Unless a row says *validated*, the correction is itself a producer's record, dated and additive (the frozen evidence it corrects is left in place), and carries no proof-carrying-claim verdict. Under the rule that every producer may be sabotaging the work, such a row is an unverified assertion of its producer, as is the reading it replaced.
- *Direction.* A correction of the rho reference or of the pricing is an accounting change. None of the rows below is an improvement of index calculus, and none moves a validated leaderboard row below its reference.
- *Cross-references.* Seven corrections are being written into the main text (the sections are named in Table B.1). They are listed here with a pointer only.

B.1 Binary and Koblitz fields

Table B.2 lists the binary-field items. Two need a sentence beyond the table.

Torsion anchors. The merged report of `research/ic_m2_cycle_leads_20261030` (PR #3009) stated that torsion anchors "cost 1.5× ($n = 23$) to 16× ($n = 31$) on cofactor-4 curves". A frozen-protocol test at $n = 41$ found the on/off trial ratio 0.975 $[0.641, 1.420]$, and a post-hoc replicate of 32 paired targets at $n = 31$ gave 13.3× $[10.3, 17.5]$: the penalty exists at $n = 31$, is about 1.5× at $n = 23$ and is absent at $n = 41$, so the range is not to be quoted as a property of torsion anchors. A companion study on $a = 0$, $n = 37$ found a penalty of 6.3–14.5× at every cofactor-to-base ratio from 0.30 to 10.84. Its README also records that the earlier bootstrap intervals (PRs #3126 and #3138) were drawn with `mawk`'s non-reproducible `srand`; recomputed with a seeded generator they move by at most 0.02. The earlier $n = 37$ figure 0.134 $[0.082, 0.205]$ was off/on; it is the same measurement as the on/off 7.46 $[4.90, 11.92]$ and is not a reversal (`research/ic_m2_torsion_n41_20261101/REPORT.md`, `research/ic_m2_torsion_ratio_n37_20261101/REPORT.md`).

The $k = 2$ collision excess. On $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{19}}$ ($r = 130{,}873$, birthday expectation 5.004) the archived census read informative-to-birthday ratios of 3.397 and 7.593 at $k = 2$ and called them a k-specific collision excess. The census counted a colliding pair as informative unless its difference lay in the span of one kernel vector per generator, but the factor base is the union of $\langle \tau, -1 \rangle$ orbits, so negation identities and τ-shifted kernel relations were counted as informative. Replaying the archived code classifies 16 of 17 and 33 of 38 pairs as structural; the genuine-to-expected ratios are then 0.20 and 1.00 at $k = 2$ and 0.40 and 0.80 at $k = 3$, none above the band (`crypto-autoresearcher: analysis/ecdlp2m-iteration-20261009/certbin-k2-collision-audit/README.md`).

B.2 Prime fields

Table B.3 lists the prime-field items; the first needs a paragraph.

Byte counters priced as operations. The §5a band priced `residual_table_bytes_uncharged`, a byte count of the residual table, at w GAE per byte. On the residual arms it was 83–99% of S at $w = 1$, and it alone made their $w > 0$ distance to rho 17–58× rather than 2.6–5×. On 1 November the owner chose option B: the band prices operations only, and `band.rs` lists the byte counters without pricing them.

Item	Original reading	Correction	Main text
H-PSEUDO character-sum constant	Fourier-flatness offset $R \approx 1.28$–1.34, $C(p) \sim p^{0.055}$–$p^{0.079}$, open	4 Nov: the (x, y)-keyed small-x base is negation-closed and the null was not; against a negation-closed null R is consistent with 1 (pooled 56 curves 1.005 ± 0.008); verdict *explained* under a non-rejection rule, so a residual up to about +9% at 2^{20} is not excluded	Chapter 30
Xedni lifting	"closed" in exact form	1 Nov: the Jacobson–Koblitz–Silverman–Stein–Teske obstruction is conditional (Lean audit); overbroad readings refuted; algorithm-specific bridges still open	Chapters 8, 30
$\ell = 8$ hybrid-guess cells	the $\ell = 8$ guess run had not completed	22 Oct: four guessed variables bring every $\ell = 8$ draw back to degree 4; 23 Oct: three do not, for that guess set	§16.1, §31
Planner weight rules	`ic-planner/2`, `/3`: $1.41N_0$ (prime, $w = 1$), $0.71c_0$ (Koblitz, $w \geq 0.3$)	26 Oct: both failed their preregistered out-of-sample tests; `ic-planner/4` withdraws them	§21
EcGFp5 symmetrisation	no $m = 5$ ideal degree measured	1 Nov: $D = 2^{16} = 65{,}536$ at all three primes	Chapter 30
$n = 83$ Weil-descent refutations	F4, F5, F6-IC, crossbred, SAT-descent and MQ-FES "proved UNSAT" on planted relations	24 Oct: `FieldStructure` truncated the field table to one 64-bit word; builders now refuse such fields; every earlier $n = 83$ engine refutation is void	§15.5, §16
Joux–Vitse sieve arm	19 Oct comparisons withheld by the 20 Oct review; replication failed its rho A/A rule	2–3 Nov: matched negation-map rho at $p = 1511$; sieve below rho in every cell of the band; *validated* (`VAL-20261103-c31c76`)	§26

Table B.1: Corrections carried into the main text, listed here by pointer. Sources: `crypto-autoresearcher: docs/reports/hpseudo-symnull-20261104/README.md`; `research/jksst_xedni_audit_20261101/`; `research/dreg_closure_guess_20261022/RESULTS.md`, `research/dreg_closure_guess3_20261023/RESULTS.md`; `research/ic_planner_weight_rule_20261021/README.md`, `research/ic_planner_weight_rule_binkob_20261026/README.md`; `research/ecgfp5_quotient_solve_20261101/README.md`; `research/n83_msweep_20261024/README.md`; `research/jv_sieve_negrho_20261102/`.

Item	Original reading	Correction	Source
Torsion anchors, cofactor 4	PR #3009: 1.5–16× more trials	1 Nov: 0.975 [0.641, 1.420] at $n = 41$; not a general property (see text)	`research/ic_m2_torsion_n41_20261101/`
Semaev 2015, $t = 2$	first fall degree "always 4" (§4.5); corpus inference that these rows refute $d_{F4} \leq d_{ff}$	2 Nov: $d_{ff} = 2$ on 200 of 200 instances by a trace certificate; the solver needs degree 2 on 154 and 3 on 46, never 4, so the sharp assumption fails on 46/200 only	`research/semaev_t2_trace_20261102/REPORT.md`; cf. Corollary 6.12
Hamming ball B_3, $n = 19$	$R = 1.10$ against random subspaces	1.10 came from one noisy subspace denominator; against 8 matched subspaces $R = 0.27$ [0.13, 0.46] and 0.32 [0.14, 0.55]; *not decided*	`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-nonlinear-frob-stable-bases.yaml`
SAT answers in `EV-ICPERF-390707`	five unverified SAT answers	they are decompositions on the quadratic twist $y^2 + xy = x^3 + 1$; instrument not invalidated, SAT-over-Gröbner gap direction reproduced	`crypto-autoresearcher:ledger/evidence/EV-ICPERF-390707.yaml`
Certified $k = 2$ collisions	17 and 38 pairs against 5.004: a k-specific excess	instrument artefact; 16/17 and 33/38 structural (see text)	certbin audit (see text)
Learned factor-base selection	producer: "about 9 bits, identical across the band" behind rho	5–6 Nov: a pinned-calibration point; ridge pick 1.8/14.8/17.7 bits ($n = 17$) and 3.3/15.1/17.9 ($n = 19$) behind strong rho at $w = 0/0.1/1$; the lead records independent validation	`crypto-autoresearcher:knowledge/leads/ecdlp/ic-binary/LEAD-ICB-learned-cost-model-factor-base-selection.yaml`
Root linear-row harvest (#1862)	raised counted word XORs about 7% (one workload, K_0, $n = 13$)	24 Oct: the sign depends on size: S up 9.6% and 7.0% at $n = 11, 13$, down 0.3% and 4.1% at $n = 17, 19$; every priced F4 arm stays 7–1,250× strong rho	`research/f4_harvest_remeasure_20261024/README.md`

Table B.2: Binary and Koblitz corrections not in the main text. All are producer-reported except the learned-selection row, whose atlas lead records a `VALIDATION.json`; the treatise did not open that repository. The already-recorded binary corrections (September reversals, the retired $n = 61$ 13.94× and $n = 83$ 18.53× online readings, Table A's understated gap, the contradicted degree-4 floor) are in Chapter 33 and §4.5.

Item	Original reading	Correction	Source
Residual collectors at $w > 0$	res4/rho 41–58× at $w = 1$; C1 met in 6 of 8 cells	1 Nov, option B: res4/rho 3.6–4.8×; C1 met in 2 of 8	`research/ic_byte_counter_pricing_20261101/`
Small-root lattice reach	$\varepsilon(S_3) = 0.8408$, best margin -0.0481 at $m = 4$	24 Oct (`CORR-20261024-b2a799`): wrong exponent in `smallroot_reach.py`, $\varepsilon_{HG} = \varepsilon_{TOOL}/n$; $\varepsilon(S_3) = 0.2803$, margin -0.6086 at $(4, 2)$, best -0.5161 at $m = 32$	`crypto-autoresearcher:analysis/smallroot-reach/CORRECTION.md`
Oracle gate of the Semaev-alternatives memo	"$e < 1$ at $m = 4$"	4 Nov: not implied by its own cost model; a generic 2+2 MITM passes it ($s = 1.93$) at cost $p^{2/3}$; correct gate $s < m - 3$, i.e. $e < (m - 3)/4$	`research/semaev_alternatives_audit_20261104/README.md`
Cofactor-cleared prime base	$G_0 = 6.180$ $[5.915, 6.456]$, pass line met at every weight	3 Nov: verdict `invalid_controls` by the frozen rule (positive control's lower limit $1.149 < 1.2$); not to be cited as a gain	`research/ic_cofactor_clear_ab_20261103/README.md`
Multi-target Koblitz figures	closest batch figure 1.17× rho	accounting: 6.51× batch rho at $k = 32$ (Kuhn–Struik reference), 6.38× with step and build priced; one target 43.7×	`research/ic_rho_koblitz_20260923/README.md`

Table B.3: Prime-field corrections, with one Koblitz accounting row that belongs with them because it changed a rho reference rule. All are producer-reported; the small-root correction is a single-agent computation that has had no independent review.

Re-running every committed weighted comparison under the rule turned all 24 residual-versus-planner comparisons of `research/ic_prime_residual_lp_20261026/` from "unresolved inside the band" to "residual cheaper at every weight". It also resolved 36 of 136 comparisons in the planner-exhaustive study and 2 of 6 in the Koblitz weight-rule study, and reversed no direction. Its effect cut both ways: the pre-registered criterion C1 of `research/ic_residual_scale_20261101/` is met in 2 of 8 cells, not 6 of 8, because the `eq` candidate's apparent gain at $w = 1$ rested on byte counts. The $w = 0$ results and the 2.59× leaderboard row are unchanged (`research/ic_byte_counter_pricing_20261101/README.md`).

B.3 Engines and measurement

Several corrections changed the instrument, not a result (Table B.4). They matter because intervals and references quoted earlier in this treatise were computed under the older rules.

- *Accounting v2* (23 October) sized the strong Koblitz reference to r. Its mean S falls to 0.182 $[0.171, 0.196]$, 0.223 $[0.201, 0.248]$ and 0.678 $[0.499, 0.891]$ of the frozen id at $n = 13$, 19 and 37, and its fixed set-up share from 83.3%, 71.3% and 30.2% to 39.9%, 29.3% and 4.3%. Any small-n comparison against the frozen strong id read rho too expensive by up to that factor. The report states that this is a correction of the reference instrument, not a faster algorithm (`research/ecbench_accounting_v2_20261023/REPORT.typ`).
- *In-house F4 is not a published engine.* A desk review found that the repository's F4 needs 10^3–10^7 more operations than a brute grid on the same PKM systems, so root-finding-over-F4 ratios such as those of §9 (for example 383×) are not evidence about published Gröbner solvers; degree measurements made with msolve are unaffected (`research/ic_sweeper_20261024/README.md`).

Item	Original reading	Correction	Source
Strong Koblitz rho id	frozen id, set-up 30–83% of S	23 Oct: v2 sized to r, S ratio 0.182–0.678 (see text)	`research/ecbench_accounting_v2_20261023/`
Comparison intervals (PR #2442)	percentile bootstrap over verified pairs	5 Nov: linearised $t(K-1)$, 13.1× wider at $K=2$; censored runs included: `ic-sub6-mitm`/strong rho 6.981 → 648.2 (92.9×), `rho-frob`/`bsgs-neg` 60.4×, both lower bounds; point estimates over verified pairs unchanged	`research/ecbench_statistics_v2_20261105/`
Calibration, round 1	`rho.plain` 0.792, `kangaroo.vow` 0.601 of theory, outside their bands	failures stand; traced to a band set on a point estimate (rho) and 8 effective targets (kangaroo); preregistered follow-up on 64 fresh targets	`research/ecbench_calibration_20261002/README.md`
Independent validation	two `ECBENV2` classes pass as independent hosts	9 Oct: same Mac, different `rustc`; hardware class `ECBHW1` added, which rejects it	`research/ecbench_independence_20261009/`
Runner PR #2455	IC online window "about 15% lower"	23 Oct: counts identical in 10 of 10 pairs; online ratio 1.067, not distinguished; the 18.4–38.8× shorter small-arm walls are an allocator effect (`malloc_trim`, `/proc` reads)	`research/ecbench_allocator_confound_20261023/`
Bounds frontier v1	ops leaders named on point estimates; exponent intervals ending at −1.0, 0.0, 2.5	23 Oct, procedure 2: no ops leader in prime toy or Koblitz toy (tie sets); degenerate ends came from one-size resamples	`research/ecbench_frontier_v2_20261023/`
$n = 29$ generator-orbit panel	Frobenius-layout pipeline below rho-frob, 0.366 $[0.241, 0.523]$	31 Oct: at one column it is a Frobenius-folded BSGS; the crossing rests on rho's fixed set-up at $\log_2 r = 15.4$ and the gap grows with r; not an IC win	`research/ecbench_e2e_frontier_20261031/README.md`
E42 Macaulay tails	preregistered control range $[900, 960]$ re-allocations	3 Nov: the range was mis-derived (measured 245.6); E42's kill fired with W unranked; E42b re-registered before any ranking data	`research/e42_macaulay_tails_20261103/REPORT.typ`

Table B.4: Instrument and measurement corrections. All producer-reported; none changes an IC count except where the row says so.

B.4 Rho references

Table 33.3 gives the rho-side changes that moved leaderboard rows. Table B.5 adds those that moved study readings. One binary defect underlies the first two rows: the build that wrote the crater-automorphism and prime-automorphism sessions left the `rho.cubic`/`rho.quartic` class set-up (16–58 GAE per solve) out of each record's total (`research/class_setup_impact_20261024/`).

B.5 Transfers and covers

The transfer audit's v2 certificates (amendment 3, 26 October) changed the baseline itself. The model's $\sqrt{\pi r/4}$ is already rho with negation; dividing it again by $\sqrt{c}$ counted negation twice and put every baseline 0.5 bit too low. The amendment also gave rho on a binary cover's Jacobian image the Frobenius class of the baseline curve (Koblitz cover ratios had been overstated by $\frac{1}{2}\log_2(n/d)$ bits), required a margin of 8 bits before a price counts as below the baseline, and kept the verdict `refuted` only for a carried route priced as a route; formula sweeps, identity-path prices and unconstructed maps are now `unknown`. The regeneration first overwrote the v1 certificates; they were restored byte for byte in `v1/` on 27 October (`research/transfer_audit_20261011/PROTOCOL.md`). The other transfer items are in Table B.6.

B.6 Structure: conductor gaps, curve traits and instruments

Conductor gaps. The conductor-gap series has the longest correction trail. The CM-pairs study first stated that "the Lean-provable theorem" gave the null for generic methods. On 16 October a sheriff review (`SHF-20261016-3caa6f`, finding L5) noted that `lean/GenericInvariance.lean` covers only algorithms whose control flow depends on equality answers and a random tape, that is BSGS-type algorithms, while ecbench's rho and kangaroo hash the point representation. On 17 October a second review (`SHF-20261017-e4d7f6`, L-N2) withdrew the conclusion "no implemented method sees the gap", and on 28 October (`SHF-20261028-0f3f64`, PPN-1) the banner that still carried it was corrected. The open rho reading at $\ell = 8209$, 1.22 $[1.005, 1.46]$ from 4 targets, was then excluded by 12,288 paired solves under a fresh seed: any floor/crater difference is below about 1.2% there. In the easy-gap tier ($\ell \approx 2^{25}$) the frozen interval 0.917 $[0.851, 0.988]$ excluded 1, but a preregistered twin round put the decisive floor-twin/crater-twin ratio at 1.040 $[0.966, 1.119]$: the shift attaches to the coordinate models, not to the endomorphism ring (`research/conductor_gap_cm_pairs_20261010/README.md`, `research/conductor_gap_rho_large_sample_20261018/README.md`). The index-calculus side is in §26; the remaining structural items are in Table B.7.

B.7 Unconventional routes

None of the rows in Table B.8 opens or closes a route by itself. Two changed a status in the other direction from the usual one: the jet correction removed a closure that had no derivation behind it, and the JKSST audit (Table B.1) turned an exact-form closure into a conditional one. A withdrawn closure is a correction, not evidence for the route.

B.8 Papers, notes and provenance

Table B.9 lists corrections to papers and notes, and provenance flags that bar a record from being cited as evidence.

The LLL degeneracy note. The note reported two curve-specific failure modes: secp256k1 failing LLL at 256 bits where P-256 and brainpoolP256r1 succeed, and a uniform failure at 384 bits and above. Both were overflow bugs in the `f64` Gram–Schmidt step (n^4 exceeds the largest `f64`), and were withdrawn on 16 October (`SHF-20261016-3caa6f`). On the fixed code every distinct seed pair at 256 and 384 bits recovers the key, secp256k1 included (3 of 3); P-521 at $k_{\text{bits}} = 384$ recovers 0 of 3 on scaled `f64` and 3 of 3 with 2048-bit fixed-point Gram–Schmidt. On 17 October the mechanism sentence for P-521 ("catastrophic cancellation") was withdrawn as well (`SHF-20261017-e4d7f6`), because with the scaling

Item	Original reading	Correction	Source
Automorphism rho, prime	cubic/negation 0.611, quartic/negation 0.791	24 Oct: 0.647 $[0.543, 0.770]$ and 0.819 $[0.722, 0.929]$; the quartic interval no longer contains $1/\sqrt{2}$, so finding 1 holds for cubic only	`research/prime_rho_automorphism_20261015/README.md`
Volcano crater controls	$j = 0$ 0.697, $j = 1728$ 0.707	24 Oct: 0.716 $[0.598, 0.857]$ and 0.723 $[0.621, 0.843]$; both still exclude 1	`research/isogeny-volcano-cost_20261017/README.md`
Negation map	`volcano-rho` and the autoresearcher `index_calculus/rho.py` walk on points	20 Oct: neither had the map; both now walk on $\{R, -R\}$; earlier IC comparisons against them used a weaker rho	`research/rho_negation_audit_20261020/README.md`
Cheon, P-224 shape	Cheon/rho ratios and crossings	23 Oct: H3 gate failed (5 of 84 rho runs outside $[0.3, 4]$); every Cheon/rho reading withdrawn; Cheon-only scaling stands	`research/cheon_p224_shape_20261022/REPORT.md`
curve25519 `wwalk`	1.43–1.55× matched rho	R4-I1: one fixed table draw; over 16 walk seeds 1.82× and 1.86×; use about 1.9×	`research/curve25519_autolab_20261104/README.md`
Hyperelliptic rho	8-bit branch coefficients; R1, H1, H2 predicted fold gains	18–19 Oct: R1, H1, H2 falsified; full-width coefficients give S_{walk} median 1.390 at $N \geq 10^5$ (2.100 → 1.481 on the largest row): the old reference was slow	`benchmarks/README.md` rows `20261018-hec-rho-*`, `20261019-hec-rho-branch-width`
GLV-invariant matched walk (E27)	distances of §9.24	30 Oct: the same walks cost 1.45× less; every row 1.448–1.456× further from rho at $w = 0$; *validated* (`VAL-20261031-e27a1c`)	`research/e27_walk_20261030/REPORT.typ`
Order-24 walk (E29)	crossing with order 6 at $w = 1$ (§9.33)	with its 40.6 unpriced mod-n operations per step priced, order 24 loses at every weight; the crossing was a one-op-per-step artefact	GLV note, §9.39 (E34)
Verschiebung split	round 1: $\lambda = t - 1$ via V^n, ratio 0.49, "not established"	round 2: $V^n = [2^n] \circ \pi^{-1}$ is n doublings plus square roots; ratio ≥ 2.5 at $c_{\text{dbl}} = 8$, a provable loss	`research/isogeny_loop_glv_20261011/README.md`

Table B.5: Corrections to rho references used by studies. GLV note: `research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md`, whose §9.19 also records seven textual errata found by the validation gate with no result, verdict or ranking changed. All rows except E27 are producer-reported.

Item	Original reading	Correction	Source
Transfer-audit baseline	§4 baseline; `refuted` for any route not below it	26 Oct: every baseline +0.5 bit; 8-bit margin; `refuted` narrowed (see text)	`research/transfer_audit_20261011/`
Cover descent at the genus floor	846 `cover_descent` rows `formula_advantage`; earlier reading: all rest on a genus no GHS cover has	priced at the genus the construction has: 599 remain (576 at $k = 6$, 23 at $k = 14$), 41 marginal, 206 boundary; rows stay `proposed`	`research/cover_descent_genus_floor_20261015/README.md`
Hilbert stop on the weak-class cover (B0)	counted S falls 2.53–2.64×	23 Oct: relabelling; unpriced F4-driver operations rose 8.3–8.5×; the repaired form B is engineering (1.85–1.88×)	`research/ic_cover_weakclass_20261023/REPORT.md`
GHS cover census, $n = 4$	preregistered: floor 21 for $E(K)[2] = \mathbb{Z}/2$; odd-class minima in $\{13, 25\}$	both failed; derived floor 25; odd class 13 iff $j \notin \mathbb{F}_{q^2}$, else 37, never 25; one check bug fixed	`research/ghs_cover_census_n4_20261024/PROTOCOL.md`
Isogeny seed, curve25519	seed 1093 cheaper than seed 0	3 Nov: V2 failed (mean S ratio $0.641 > 0.6$); interval $[0.379, 1.084]$ includes 1; gain not established	`research/curve25519_isogeny_seed_20261103/README.md`
CM canonical lift (lane L9)	closure argument as first written	`CORR-20260810-930c42`: closes only ordinary prime-field Serre–Tate/global CM models of fixed arity 2–9; point-lifting routes give $\Pr \le p^{-1+o(1)}$, exponent 1 against rho's 1/2	`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-cm-canonical-lift-heights.yaml`
GHS magic number	legacy `ec_trapdoor::magic_number` (plain orbit)	13 Oct: MMT definition; consequences for ECC2K-130 are in §15 and §31.5	`research/weak_isogeny_classes_20261013/REPORT.md`

Table B.6: Corrections in the transfer and cover studies. All producer-reported.

Item	Original reading	Correction	Source
Conductor gap, CM pairs	"no implemented method sees the gap"; null is a theorem for generic methods	16, 17, 28 Oct: theorem covers BSGS-type arms only; banner withdrawn and qualified (see text)	`research/conductor_gap_cm_pairs_20261010/`
Rho at $\ell = 8209$; easy-gap tier	1.22 $[1.005, 1.46]$; post hoc 1.17 $[1.00, 1.38]$	18 Oct: first excluded; second attributed to coordinate models (1.040 $[0.966, 1.119]$)	`research/conductor_gap_rho_large_sample_20261018/`
Round-1 crater/floor set-up gap	prime floor/crater S ratio 3.10	round 2, A0: a closure counting defect (the floor paid 3.56× the needed set-up); ratio 1.00	`research/isogeny_conductor_gap_round2_20261011/README.md`
Drastic prime gap, $j = 0$	floor/crater rho quotient 0.5286	round 2, 32 fresh targets: 0.9468 $[0.6622, 1.3474]$ (99%); target-sampling noise	`research/drastic_prime_conductor_gap_round2_20261012/README.md`
NIST skip criterion	coefficient test separates the skipped curves	25 Oct: the smaller P-224 root rejects the published model, and the two models are isomorphic; representation-dependent, not an intrinsic weakness	`research/nist_skip_criterion_20261025/REPORT.md`
Coordinate census `EV-ECDLP-81f4d4`	strength *strong*	26 Oct (`CORR-20261026-c94668`): lowered to *preliminary*, with other records of the same sheriff case	`crypto-autoresearcher:ledger/corrections/CORR-20261026-c94668.yaml`
Weak-class instrument I1	*invalid* on all three classes	4 Nov: the A/A gate (5% on 20+20 draws) was miscalibrated; with a calibrated gate I1 is valid and blind; the paired-curve anomaly did not replicate (108+108 curves)	`crypto-autoresearcher:docs/reports/weakclass-i1-20261104/`, `crypto-autoresearcher:docs/reports/paired-anomaly-20261104/`
Polynomial-ring form (FHJRV)	"unavailable" unless b is a square	2 Nov (`CORR-20261102-7e3a91`): holds for the x (Kummer) class only; for $u = x/y$ on a double-odd curve the quotient is a polynomial ring	`crypto-autoresearcher:ledger/corrections/CORR-20261102-7e3a91.yaml`

Table B.7: Structural corrections. All producer-reported, the sheriff findings included.

Item	Original reading	Correction	Source
Jet oracle	"simulable with $C = 1$, closed at exponent 1/2", a derivation "valid at all scales"; close jet candidates by theorem	27 Jul (`CORR-20260727-003`): withdrawn, no derivation exists; closure countermanded; simulability open; `EV-GGM-002` supersedes `EV-GGM-001`	`crypto-autoresearcher:ledger/corrections/CORR-20260727-003.yaml`
Planted generic-group null	Stage 1 null on planted $\mathbb{Z}/n$, n of 16–20 bits	`EV-ECDLP-8a5106`: O-ARTIFACT, the kangaroo check M4 missed its 0.10 band; decision *refine*	`crypto-autoresearcher:ledger/evidence/EV-ECDLP-8a5106.yaml`
ETH-route leads	twelve leads as written in `LEADS.md`	2 Nov errata: L01, L05, L06, L08, L10, L12 refuted as stated; "$\sim 10^5 b$ edges per addition" 9–30× high; the 2^{256} threshold needs $c \geq 1$; char-2 gadget margin 1.4×, not 3.42×	`research/ic_prime_eth_leads_20261031/LEADS.md` §9
Learned WDSat branching	learned order 16,097 held-out conflicts against 3,092	the learned policy chose the ordinary ascending order and equals the block-core arm (16,097); every learned and control arm ran with symmetry disabled and the 3,092 arm with it on, so the comparison is confounded with the symmetry switch (derived)	`cryptanalysis:experiments/ideas_learning_20261018/REPORT.md`
Hashed-ECDL set-up hang	user report: Fourier-network construction impractical	30 Oct: the hang is the AKS branch above `u64`; attribution corrected	`research/hashed_ecdl_hash_prime_20261030/README.md`
Eigen-Séta (not ECDLP)	`walk` variant; NIST categories 1/3/5	11 Oct: `walk` insecure (5/5 ciphertexts decrypted from public data); 17 Oct: categories hold only in gate count, unqualified claim withdrawn	`research/eigen_seta_prototype_20261011/README.md`, `research/eigen_seta_assessment_20261017/REPORT.md`

Table B.8: Corrections on unconventional routes, besides H-PSEUDO and xedni (Table B.1). All producer-reported.

P-521's diagonal is exactly 2^{500} and the described subtraction is exact (`papers/lll-degeneracy-note/README.md`). The note has no index-calculus content; it is listed because it is the clearest case here of a curve "property" that was a numerical bug.

What the list shows. Three patterns recur. First, a reference or a pricing rule was corrected more often than an index-calculus count: the class set-up, the strong Koblitz id, the negation map, the byte counter and the transfer baseline each changed the distance to rho with no change to the attacking side. Second, a preregistered control or gate that failed, as in the Cheon, cofactor-clearing and E42 rows, voided the verdict or the ranking whatever the measured quantity said; the rule was applied as frozen. Third, several "properties" of curves or classes (the LLL failures, the conductor-gap rho shifts, the NIST coefficient test) turned out to be properties of a number format, a coordinate model or a representation. As in Chapter 33, no correction listed here moved a generic or leaderboard index-calculus row below a matched rho. The one row below rho is the Joux–Vitse sieve route, which is restricted to the `jv_cover` weak class (full rational 2-torsion, never prime order) and is an accounting result against a newly matched reference.

Item	Original reading	Correction	Source
LLL degeneracy note	secp256k1 and $\geq$ 384-bit curves fail LLL	withdrawn 16–17 Oct (see text)	`papers/lll-degeneracy-note/`
Structural-completeness paper	"the theorem is proven by case analysis"	17 Oct (`SHF-20261017-e4d7f6`, P-N4–N6): "argued"; §8 holds proof sketches; check counts recounted	`papers/structural-completeness/README.md`
Isogeny-class rho paper	no test fired; inferential null	19 Oct review: false (one B1 generator, two P1 cells, one P0 cell flagged; Kruskal–Wallis $p = 0.039$); KS defect, missing tie corrections; null provisional, p-values not recomputed	`papers/isogeny-class-rho/REVIEW-20261019.md`
Chung–Costello–Smith Recover	displayed Eqs. (11), (12)	22 Oct: in this implementation's convention the literal equations give 1,257 wrong signed divisors and 3 $D = 0$ cases of 1,260; corrected equations recover all 1,260	`research/kummer_recover_20261022/REPORT.md`
`secpkk1` ladder (lit2026 W11(e))	published 8/12/16-bit challenge cards	the cards use the opposite prime-search direction from the paper's own Algorithm 3	`research/lit2026_w11_generic_engineering_20261011/README.md`
Teske trapdoor evidence	`EV-ECTD-8c098d` rho-substitution prose	fabrication defect disclosed: the stated empirical justification has no supporting data	`crypto-autoresearcher:ledger/evidence/EV-ECTD-8c098d.yaml`
Kernel-of-reduction design	`DEC-20261023-26a6de` cites `KN-LIT-7593`	the citation resolves to an unrelated AES paper	`crypto-autoresearcher:knowledge/leads/ecdlp/transfer/LEAD-TRF-kernel-of-reduction-lemma.yaml`
Synthetic volcano study	volcano and ECDLP figures	invented data by its own banner; not evidence	`research/synthetic_volcano_ecdlp_20261018/README.md`
Frobenius-quotient follow-up	reported SAT runs	supplied as a note; models and logs not committed; not reviewed evidence	`research/frobenius_quotient_followup_20260926/README.md`
Scalar-3 Frobenius gate	new screen for $q = 16$	the mechanism is a rediscovery of the scalar-3/Jordan screen of `research/cubic_extension_transfer_20261025/`; only its extension to $q = 16$ is new	`research/ghs_scalar3_frobenius_gate_20261027/REPORT.md`

Table B.9: Corrections to papers and notes, and provenance flags. All producer-reported or sheriff-reported.

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